Supplement to: Estimation and inference in penalized function-on-function regression with within-curve residual dependence

Fabian Scheipl*

Department of Statistics, LMU Munich

Munich Center for Machine Learning (MCML)

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*All code and text were generated by large language models in collaboration with the author; see the statement on the use of generative AI at the end of the main text of the paper.

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This supplement accompanies the paper. S1 and S2 give the full results of the simulation and the plasmode study, S3 the comparators, S4 approaches that did not work (intervals centered at the NCV fit and smoothing-parameter uncertainty terms), S5 misspecified truths and basis size, S6 data and code, S7 the values shown in figures of the paper and of this supplement. The CL2 algebra, the descriptions of both studies' setups and the computing times are in Appendices A–C of the paper. Every number is computed from the study's summary files when the document is rendered.

Coverage, its 5% quantile over the grid (lower tail), relative error, relative half-width, detection rate and interval score are defined in Section 2.4 of the paper. In addition, a *setting* is one combination of design factors of the simulation, with 200 replicate datasets, and the *wrong-sign rate* is the share of all grid points at which an interval excludes zero on the wrong side. Monte Carlo (MC) standard errors are computed over replicates; for a grid-averaged coverage, from the replicate-level grid averages, so the correlation of coverage indicators across grid points is accounted for. EDF denotes effective degrees of freedom and, in tables and figure labels, MSE the mean squared error. Ratios of mean squared errors in this supplement are NCV to REML (below 1: NCV is better), unless stated otherwise; Section 4 of the paper gives ratios of relative errors as REML to NCV.

The lower tail is a quantile over pointwise coverages that are each estimated from a finite number of replicates, so it lies below the nominal level even for an interval with coverage 0.95 at every grid point: for independent grid points, it would then be about 0.925 with 200 replicates, 0.91 with 100 and 0.90 with 50. Lower tails are to be read against these values, not against 0.95.

The four estimands are the conditional mean $\mathrm{E}(Y \mid X)$ (*the mean*), the coefficient surface $\beta(s,t)$ (or $f(x,t)$ in the smooth-effect settings), the functional intercept $\alpha(t)$ and the effect $\gamma(t)$ of the scalar covariate (Section 2.4 of the paper). REML + CL2 and NCV + CL2 denote the REML fit or the fit by curve-blocked neighborhood cross-validation (NCV, which leaves out whole curves) with the CL2 covariance of Appendix A of the paper (a curve-clustered, leverage-adjusted sandwich plus the smoothing-bias allowance $\mathbf{V}_p - \mathbf{V}_e$) and normal critical values; model-based intervals use mgcv's $\mathbf{V}_c$ for REML fits and $\mathbf{V}_p$ for NCV fits, because, in the mgcv build of the study, $\mathbf{V}_c$ of an NCV fit holds only the smoothing-parameter correction and is not a valid covariance (S6).

The *main factorial* of the simulation crosses family, error process, number of curves G and signal level; each block of *extension settings* changes one aspect of it (Appendix B.1 of the paper). The error processes of the main factorial are independent, rough stationary (an exponential correlation with a nugget) and smooth (from DTI residual curves). The signal levels are low, default and high ($R^2 = 0.2, 0.5, 0.8$ for Gaussian responses; $\sqrt{S} = 0.5$ (default) and 1 on the link scale for Poisson and binary responses, where "low" and "default" coincide). The bases are the default and bases with 2× and 4× finer knots. The functional covariate is rich (the default spectrum) or low-rank. The plasmode truths are the fitted means of the REML fit (*REML-fitted truth*), of the NCV fit (*NCV-fitted truth*), and of a refit with every smoothing parameter at the geometric mean of the REML and NCV values (*midpoint truth*); the *truth source* is the fit that gives the truth, the *residual source* the fit (REML or NCV) whose residual curves form the errors.

S1 Simulation study

The design is described in Appendix B.1 of the paper.

S1.1 Coverage per setting

Table 1: Grid-averaged coverage (MC SE) of the mean and $\beta(s,t)$ per setting and interval. REML model / NCV model: model-based intervals of the REML / NCV fit; NCV wid.: NCV + CL2 widened in quadrature by the NCV–REML difference (S4.1(b)), with the Bayesian or the frequentist (freq.) CL2; AR(1): AR(1) working model.

block	family	error	G	signal	truth	D	basis	estimand	NCV+CL2	NCV wid.	NCV wid. (freq.)	NCV model	REML+CL2	REML model	AR(1)
main factorial	Gaussian	indep.	40	low	smooth	61	default	β(s,t)	0.984 (0.002)	0.988 (0.001)	0.924 (0.004)	0.984 (0.002)	0.990 (0.001)	0.994 (0.001)	
main factorial	Gaussian	indep.	40	low	smooth	61	default	E(Y | X)	0.946 (0.003)	0.952 (0.003)	0.910 (0.004)	0.952 (0.002)	0.956 (0.002)	0.967 (0.002)	
main factorial	Gaussian	indep.	100	low	smooth	61	default	β(s,t)	0.982 (0.001)	0.989 (0.001)	0.939 (0.003)	0.983 (0.001)	0.989 (0.001)	0.992 (0.001)	
main factorial	Gaussian	indep.	100	low	smooth	61	default	E(Y | X)	0.950 (0.002)	0.955 (0.002)	0.922 (0.003)	0.952 (0.002)	0.961 (0.002)	0.968 (0.002)	
main factorial	Gaussian	rough	40	low	smooth	61	default	β(s,t)	0.898 (0.007)	0.978 (0.003)	0.972 (0.003)	0.648 (0.010)	0.925 (0.004)	0.635 (0.009)	
main factorial	Gaussian	rough	40	low	smooth	61	default	E(Y | X)	0.883 (0.005)	0.943 (0.003)	0.937 (0.003)	0.546 (0.007)	0.929 (0.003)	0.617 (0.005)	
main factorial	Gaussian	rough	100	low	smooth	61	default	β(s,t)	0.924 (0.004)	0.982 (0.002)	0.976 (0.002)	0.682 (0.009)	0.941 (0.003)	0.671 (0.008)	
main factorial	Gaussian	rough	100	low	smooth	61	default	E(Y | X)	0.920 (0.003)	0.954 (0.002)	0.948 (0.002)	0.595 (0.006)	0.942 (0.002)	0.641 (0.005)	
main factorial	Gaussian	smooth	40	low	smooth	61	default	β(s,t)	0.902 (0.007)	0.979 (0.002)	0.974 (0.003)	0.633 (0.010)	0.923 (0.004)	0.609 (0.010)	
main factorial	Gaussian	smooth	40	low	smooth	61	default	E(Y | X)	0.895 (0.005)	0.949 (0.003)	0.944 (0.003)	0.545 (0.007)	0.930 (0.003)	0.601 (0.006)	
main factorial	Gaussian	smooth	100	low	smooth	61	default	β(s,t)	0.925 (0.004)	0.983 (0.002)	0.977 (0.002)	0.667 (0.009)	0.944 (0.003)	0.658 (0.009)	
main factorial	Gaussian	smooth	100	low	smooth	61	default	E(Y | X)	0.921 (0.003)	0.956 (0.002)	0.951 (0.002)	0.586 (0.006)	0.944 (0.002)	0.631 (0.005)	
main factorial	Gaussian	indep.	40	default	smooth	61	default	β(s,t)	0.982 (0.002)	0.989 (0.001)	0.937 (0.003)	0.983 (0.002)	0.989 (0.001)	0.992 (0.001)	
main factorial	Gaussian	indep.	40	default	smooth	61	default	E(Y | X)	0.948 (0.002)	0.954 (0.002)	0.921 (0.003)	0.954 (0.002)	0.958 (0.002)	0.968 (0.002)	
main factorial	Gaussian	indep.	100	default	smooth	61	default	β(s,t)	0.982 (0.001)	0.989 (0.001)	0.951 (0.003)	0.983 (0.001)	0.986 (0.001)	0.989 (0.001)	
main factorial	Gaussian	indep.	100	default	smooth	61	default	E(Y | X)	0.950 (0.002)	0.958 (0.002)	0.933 (0.002)	0.953 (0.002)	0.961 (0.002)	0.967 (0.001)	
main factorial	Gaussian	rough	40	default	smooth	61	default	β(s,t)	0.922 (0.005)	0.982 (0.002)	0.976 (0.002)	0.682 (0.009)	0.927 (0.004)	0.638 (0.008)	
main factorial	Gaussian	rough	40	default	smooth	61	default	E(Y | X)	0.920 (0.003)	0.956 (0.002)	0.951 (0.002)	0.603 (0.005)	0.931 (0.003)	0.624 (0.005)	
main factorial	Gaussian	rough	100	default	smooth	61	default	β(s,t)	0.934 (0.004)	0.983 (0.001)	0.977 (0.002)	0.700 (0.007)	0.940 (0.003)	0.672 (0.007)	0.972 (0.002)
main factorial	Gaussian	rough	100	default	smooth	61	default	E(Y | X)	0.931 (0.003)	0.958 (0.002)	0.953 (0.002)	0.623 (0.005)	0.942 (0.002)	0.646 (0.005)	0.934 (0.003)
main factorial	Gaussian	smooth	40	default	smooth	61	default	β(s,t)	0.924 (0.005)	0.983 (0.002)	0.978 (0.002)	0.670 (0.009)	0.926 (0.004)	0.615 (0.009)	
main factorial	Gaussian	smooth	40	default	smooth	61	default	E(Y | X)	0.922 (0.004)	0.959 (0.003)	0.954 (0.003)	0.591 (0.006)	0.931 (0.003)	0.606 (0.005)	
main factorial	Gaussian	smooth	100	default	smooth	61	default	β(s,t)	0.934 (0.004)	0.983 (0.002)	0.978 (0.002)	0.684 (0.008)	0.944 (0.003)	0.657 (0.008)	0.993 (0.001)
main factorial	Gaussian	smooth	100	default	smooth	61	default	E(Y | X)	0.929 (0.003)	0.959 (0.002)	0.954 (0.002)	0.607 (0.005)	0.944 (0.002)	0.634 (0.005)	0.972 (0.002)
main factorial	Gaussian	indep.	40	high	smooth	61	default	β(s,t)	0.981 (0.001)	0.989 (0.001)	0.948 (0.003)	0.983 (0.001)	0.984 (0.001)	0.988 (0.001)	
main factorial	Gaussian	indep.	40	high	smooth	61	default	E(Y | X)	0.948 (0.002)	0.956 (0.002)	0.930 (0.002)	0.954 (0.002)	0.957 (0.002)	0.966 (0.002)	
main factorial	Gaussian	indep.	100	high	smooth	61	default	β(s,t)	0.979 (0.001)	0.990 (0.001)	0.959 (0.002)	0.980 (0.001)	0.979 (0.001)	0.983 (0.001)	
main factorial	Gaussian	indep.	100	high	smooth	61	default	E(Y | X)	0.950 (0.002)	0.960 (0.002)	0.941 (0.002)	0.954 (0.002)	0.958 (0.002)	0.963 (0.001)	
main factorial	Gaussian	rough	40	high	smooth	61	default	β(s,t)	0.932 (0.003)	0.984 (0.001)	0.978 (0.002)	0.695 (0.007)	0.927 (0.004)	0.637 (0.008)	
main factorial	Gaussian	rough	40	high	smooth	61	default	E(Y | X)	0.927 (0.003)	0.959 (0.002)	0.955 (0.002)	0.626 (0.005)	0.932 (0.002)	0.628 (0.005)	
main factorial	Gaussian	rough	100	high	smooth	61	default	β(s,t)	0.936 (0.003)	0.982 (0.001)	0.976 (0.002)	0.703 (0.007)	0.941 (0.003)	0.670 (0.006)	
main factorial	Gaussian	rough	100	high	smooth	61	default	E(Y | X)	0.934 (0.002)	0.958 (0.002)	0.954 (0.002)	0.638 (0.005)	0.942 (0.002)	0.651 (0.004)	
main factorial	Gaussian	smooth	40	high	smooth	61	default	β(s,t)	0.931 (0.004)	0.984 (0.002)	0.978 (0.002)	0.684 (0.008)	0.926 (0.004)	0.617 (0.008)	
main factorial	Gaussian	smooth	40	high	smooth	61	default	E(Y | X)	0.928 (0.003)	0.960 (0.002)	0.956 (0.003)	0.610 (0.005)	0.932 (0.003)	0.611 (0.005)	
main factorial	Gaussian	smooth	100	high	smooth	61	default	β(s,t)	0.938 (0.003)	0.984 (0.001)	0.978 (0.002)	0.693 (0.007)	0.943 (0.003)	0.653 (0.007)	
main factorial	Gaussian	smooth	100	high	smooth	61	default	E(Y | X)	0.933 (0.003)	0.959 (0.002)	0.955 (0.002)	0.621 (0.005)	0.944 (0.002)	0.636 (0.004)	
main factorial	Poisson	indep.	40	default	smooth	61	default	β(s,t)	0.985 (0.001)	0.990 (0.001)	0.936 (0.003)	0.986 (0.001)	0.989 (0.001)	0.993 (0.001)	
main factorial	Poisson	indep.	40	default	smooth	61	default	E(Y | X)	0.947 (0.002)	0.953 (0.002)	0.916 (0.003)	0.954 (0.002)	0.958 (0.002)	0.968 (0.002)	
main factorial	Poisson	indep.	100	default	smooth	61	default	β(s,t)	0.982 (0.001)	0.990 (0.001)	0.949 (0.002)	0.983 (0.001)	0.986 (0.001)	0.989 (0.001)	
main factorial	Poisson	indep.	100	default	smooth	61	default	E(Y | X)	0.948 (0.002)	0.956 (0.002)	0.929 (0.002)	0.952 (0.002)	0.960 (0.002)	0.966 (0.002)	
main factorial	Poisson	rough	40	default	smooth	61	default	β(s,t)	0.921 (0.005)	0.980 (0.002)	0.972 (0.003)	0.705 (0.009)	0.927 (0.004)	0.684 (0.008)	
main factorial	Poisson	rough	40	default	smooth	61	default	E(Y | X)	0.910 (0.004)	0.949 (0.003)	0.943 (0.003)	0.612 (0.006)	0.930 (0.003)	0.657 (0.005)	
main factorial	Poisson	rough	100	default	smooth	61	default	β(s,t)	0.936 (0.003)	0.982 (0.002)	0.976 (0.002)	0.715 (0.007)	0.942 (0.003)	0.698 (0.007)	
main factorial	Poisson	rough	100	default	smooth	61	default	E(Y | X)	0.929 (0.003)	0.956 (0.002)	0.951 (0.002)	0.632 (0.005)	0.943 (0.002)	0.664 (0.004)	
main factorial	Poisson	smooth	40	default	smooth	61	default	β(s,t)	0.926 (0.005)	0.983 (0.002)	0.976 (0.003)	0.691 (0.009)	0.928 (0.004)	0.658 (0.009)	
main factorial	Poisson	smooth	40	default	smooth	61	default	E(Y | X)	0.914 (0.004)	0.954 (0.003)	0.948 (0.003)	0.603 (0.007)	0.934 (0.003)	0.642 (0.005)	
main factorial	Poisson	smooth	100	default	smooth	61	default	β(s,t)	0.935 (0.004)	0.984 (0.002)	0.978 (0.002)	0.706 (0.008)	0.945 (0.003)	0.682 (0.008)	
main factorial	Poisson	smooth	100	default	smooth	61	default	E(Y | X)	0.927 (0.003)	0.957 (0.002)	0.952 (0.002)	0.617 (0.005)	0.945 (0.002)	0.653 (0.005)	
main factorial	Poisson	indep.	40	high	smooth	61	default	β(s,t)	0.981 (0.001)	0.988 (0.001)	0.936 (0.003)	0.983 (0.001)	0.986 (0.001)	0.990 (0.001)	
main factorial	Poisson	indep.	40	high	smooth	61	default	E(Y | X)	0.939 (0.003)	0.950 (0.002)	0.918 (0.003)	0.948 (0.002)	0.955 (0.002)	0.966 (0.002)	
main factorial	Poisson	indep.	100	high	smooth	61	default	β(s,t)	0.976 (0.002)	0.986 (0.001)	0.948 (0.002)	0.979 (0.002)	0.979 (0.001)	0.985 (0.001)	

Table 1: Grid-averaged coverage (MC SE) of the mean and $\beta(s,t)$ per setting and interval. REML model / NCV model: model-based intervals of the REML / NCV fit; NCV wid.: NCV + CL2 widened in quadrature by the NCV–REML difference (S4.1(b)), with the Bayesian or the frequentist (freq.) CL2; AR(1): AR(1) working model. *(continued)*

block	family	error	G	signal	truth	D	basis	estimand	NCV+CL2	NCV wid.	NCV wid. (freq.)	NCV model	REML+CL2	REML model	AR(1)
main factorial	Poisson	indep.	100	high	smooth	61	default	E(Y | X)	0.941 (0.002)	0.954 (0.002)	0.930 (0.003)	0.947 (0.002)	0.955 (0.002)	0.962 (0.002)	
main factorial	Poisson	rough	40	high	smooth	61	default	β(s,t)	0.922 (0.004)	0.979 (0.002)	0.970 (0.002)	0.715 (0.008)	0.929 (0.004)	0.703 (0.007)	
main factorial	Poisson	rough	40	high	smooth	61	default	E(Y | X)	0.902 (0.004)	0.947 (0.003)	0.940 (0.003)	0.627 (0.006)	0.930 (0.003)	0.674 (0.005)	
main factorial	Poisson	rough	100	high	smooth	61	default	β(s,t)	0.933 (0.003)	0.981 (0.001)	0.974 (0.002)	0.723 (0.006)	0.940 (0.003)	0.705 (0.007)	
main factorial	Poisson	rough	100	high	smooth	61	default	E(Y | X)	0.922 (0.003)	0.953 (0.002)	0.947 (0.002)	0.643 (0.005)	0.940 (0.002)	0.672 (0.005)	
main factorial	Poisson	smooth	40	high	smooth	61	default	β(s,t)	0.923 (0.004)	0.979 (0.002)	0.972 (0.002)	0.699 (0.008)	0.930 (0.004)	0.683 (0.008)	
main factorial	Poisson	smooth	40	high	smooth	61	default	E(Y | X)	0.905 (0.004)	0.949 (0.003)	0.943 (0.003)	0.610 (0.006)	0.934 (0.003)	0.656 (0.005)	
main factorial	Poisson	smooth	100	high	smooth	61	default	β(s,t)	0.929 (0.003)	0.980 (0.002)	0.973 (0.002)	0.711 (0.007)	0.944 (0.003)	0.696 (0.007)	
main factorial	Poisson	smooth	100	high	smooth	61	default	E(Y | X)	0.923 (0.003)	0.954 (0.002)	0.949 (0.002)	0.625 (0.006)	0.945 (0.002)	0.662 (0.004)	
main factorial	binary	indep.	40	default	smooth	61	default	β(s,t)	0.973 (0.005)	0.979 (0.004)	0.893 (0.007)	0.974 (0.004)	0.973 (0.005)	0.984 (0.004)	
main factorial	binary	indep.	40	default	smooth	61	default	E(Y | X)	0.923 (0.004)	0.935 (0.004)	0.879 (0.005)	0.927 (0.004)	0.925 (0.004)	0.952 (0.003)	
main factorial	binary	indep.	100	default	smooth	61	default	β(s,t)	0.982 (0.002)	0.987 (0.001)	0.911 (0.005)	0.982 (0.002)	0.990 (0.001)	0.994 (0.001)	
main factorial	binary	indep.	100	default	smooth	61	default	E(Y | X)	0.939 (0.003)	0.945 (0.003)	0.896 (0.004)	0.942 (0.003)	0.949 (0.003)	0.963 (0.002)	
main factorial	binary	rough	40	default	smooth	61	default	β(s,t)	0.790 (0.014)	0.948 (0.005)	0.935 (0.005)	0.627 (0.017)	0.937 (0.004)	0.800 (0.008)	
main factorial	binary	rough	40	default	smooth	61	default	E(Y | X)	0.842 (0.007)	0.930 (0.003)	0.920 (0.004)	0.587 (0.008)	0.927 (0.003)	0.747 (0.006)	
main factorial	binary	rough	100	default	smooth	61	default	β(s,t)	0.901 (0.008)	0.971 (0.003)	0.958 (0.004)	0.738 (0.011)	0.951 (0.003)	0.819 (0.007)	
main factorial	binary	rough	100	default	smooth	61	default	E(Y | X)	0.876 (0.005)	0.933 (0.003)	0.922 (0.003)	0.628 (0.007)	0.940 (0.003)	0.752 (0.005)	
main factorial	binary	smooth	40	default	smooth	61	default	β(s,t)	0.785 (0.013)	0.953 (0.005)	0.944 (0.005)	0.569 (0.016)	0.926 (0.005)	0.749 (0.010)	
main factorial	binary	smooth	40	default	smooth	61	default	E(Y | X)	0.857 (0.006)	0.942 (0.003)	0.935 (0.003)	0.550 (0.008)	0.921 (0.004)	0.704 (0.006)	
main factorial	binary	smooth	100	default	smooth	61	default	β(s,t)	0.896 (0.008)	0.972 (0.003)	0.962 (0.003)	0.708 (0.012)	0.952 (0.003)	0.779 (0.008)	
main factorial	binary	smooth	100	default	smooth	61	default	E(Y | X)	0.883 (0.004)	0.939 (0.003)	0.930 (0.003)	0.601 (0.007)	0.941 (0.003)	0.722 (0.005)	
main factorial	binary	indep.	40	high	smooth	61	default	β(s,t)	0.978 (0.002)	0.985 (0.002)	0.912 (0.005)	0.980 (0.002)	0.986 (0.003)	0.991 (0.003)	
main factorial	binary	indep.	40	high	smooth	61	default	E(Y | X)	0.937 (0.003)	0.945 (0.003)	0.895 (0.004)	0.943 (0.003)	0.949 (0.003)	0.964 (0.003)	
main factorial	binary	indep.	100	high	smooth	61	default	β(s,t)	0.980 (0.002)	0.987 (0.001)	0.928 (0.004)	0.982 (0.002)	0.989 (0.001)	0.993 (0.001)	
main factorial	binary	indep.	100	high	smooth	61	default	E(Y | X)	0.943 (0.002)	0.951 (0.002)	0.911 (0.003)	0.948 (0.002)	0.959 (0.002)	0.967 (0.002)	
main factorial	binary	rough	40	high	smooth	61	default	β(s,t)	0.905 (0.008)	0.974 (0.003)	0.959 (0.004)	0.755 (0.010)	0.940 (0.004)	0.813 (0.008)	
main factorial	binary	rough	40	high	smooth	61	default	E(Y | X)	0.871 (0.005)	0.935 (0.003)	0.922 (0.003)	0.651 (0.007)	0.929 (0.003)	0.761 (0.005)	
main factorial	binary	rough	100	high	smooth	61	default	β(s,t)	0.930 (0.004)	0.980 (0.002)	0.968 (0.002)	0.780 (0.008)	0.951 (0.003)	0.820 (0.007)	
main factorial	binary	rough	100	high	smooth	61	default	E(Y | X)	0.913 (0.003)	0.949 (0.002)	0.938 (0.002)	0.699 (0.005)	0.944 (0.002)	0.766 (0.004)	
main factorial	binary	smooth	40	high	smooth	61	default	β(s,t)	0.897 (0.008)	0.974 (0.003)	0.962 (0.004)	0.722 (0.011)	0.932 (0.004)	0.765 (0.009)	
main factorial	binary	smooth	40	high	smooth	61	default	E(Y | X)	0.874 (0.005)	0.942 (0.003)	0.931 (0.003)	0.624 (0.007)	0.923 (0.003)	0.713 (0.006)	
main factorial	binary	smooth	100	high	smooth	61	default	β(s,t)	0.925 (0.005)	0.980 (0.002)	0.970 (0.002)	0.759 (0.008)	0.952 (0.003)	0.783 (0.007)	
main factorial	binary	smooth	100	high	smooth	61	default	E(Y | X)	0.907 (0.004)	0.948 (0.003)	0.939 (0.003)	0.663 (0.006)	0.943 (0.002)	0.729 (0.005)	
misreg.	Gaussian	misreg.	40	high	smooth	61	default	β(s,t)	0.937 (0.004)	0.974 (0.003)	0.956 (0.004)	0.752 (0.008)	0.949 (0.004)	0.761 (0.007)	
misreg.	Gaussian	misreg.	40	high	smooth	61	default	E(Y | X)	0.919 (0.004)	0.946 (0.003)	0.934 (0.004)	0.696 (0.007)	0.938 (0.003)	0.729 (0.005)	
misreg.	Gaussian	misreg.	40	high	rough	61	default	β(s,t)	0.820 (0.008)	0.928 (0.003)	0.901 (0.004)	0.595 (0.009)	0.945 (0.003)	0.739 (0.006)	
misreg.	Gaussian	misreg.	40	high	rough	61	default	E(Y | X)	0.885 (0.004)	0.932 (0.003)	0.918 (0.003)	0.635 (0.006)	0.934 (0.003)	0.706 (0.005)	
misreg.	Gaussian	misreg.	100	high	smooth	61	default	β(s,t)	0.937 (0.004)	0.976 (0.002)	0.959 (0.003)	0.739 (0.008)	0.951 (0.003)	0.753 (0.007)	0.875 (0.005)
misreg.	Gaussian	misreg.	100	high	smooth	61	default	E(Y | X)	0.930 (0.004)	0.952 (0.003)	0.941 (0.003)	0.706 (0.006)	0.942 (0.003)	0.730 (0.005)	0.844 (0.005)
misreg.	Gaussian	misreg.	100	high	rough	61	default	β(s,t)	0.863 (0.006)	0.934 (0.003)	0.908 (0.004)	0.628 (0.007)	0.944 (0.003)	0.729 (0.006)	
misreg.	Gaussian	misreg.	100	high	rough	61	default	E(Y | X)	0.911 (0.003)	0.938 (0.003)	0.926 (0.003)	0.665 (0.005)	0.937 (0.003)	0.706 (0.005)	
misreg.	Poisson	misreg.	40	high	smooth	61	default	E(Y | X)	0.792 (0.007)	0.910 (0.004)	0.900 (0.004)	0.486 (0.010)	0.902 (0.004)	0.634 (0.005)	
misreg.	Poisson	misreg.	40	high	rough	61	default	E(Y | X)	0.773 (0.007)	0.910 (0.003)	0.902 (0.003)	0.454 (0.009)	0.899 (0.003)	0.614 (0.005)	
misreg.	Poisson	misreg.	100	high	smooth	61	default	E(Y | X)	0.801 (0.007)	0.922 (0.003)	0.915 (0.003)	0.438 (0.009)	0.910 (0.003)	0.571 (0.004)	
misreg.	Poisson	misreg.	100	high	rough	61	default	E(Y | X)	0.780 (0.006)	0.918 (0.003)	0.912 (0.003)	0.409 (0.007)	0.906 (0.003)	0.558 (0.004)	
misreg.	binary	misreg.	40	high	smooth	61	default	E(Y | X)	0.919 (0.004)	0.932 (0.003)	0.884 (0.004)	0.901 (0.004)	0.944 (0.003)	0.945 (0.003)	
misreg.	binary	misreg.	40	high	rough	61	default	E(Y | X)	0.892 (0.004)	0.908 (0.004)	0.854 (0.005)	0.871 (0.005)	0.925 (0.003)	0.924 (0.003)	
misreg.	binary	misreg.	100	high	smooth	61	default	E(Y | X)	0.931 (0.003)	0.941 (0.003)	0.902 (0.003)	0.909 (0.004)	0.954 (0.002)	0.943 (0.003)	
misreg.	binary	misreg.	100	high	rough	61	default	E(Y | X)	0.909 (0.003)	0.919 (0.003)	0.878 (0.004)	0.882 (0.004)	0.932 (0.003)	0.918 (0.003)	
dense grid	Gaussian	indep.	100	default	smooth	241	default	β(s,t)	0.976 (0.001)	0.988 (0.001)	0.956 (0.002)	0.978 (0.001)	0.976 (0.001)	0.980 (0.001)	
dense grid	Gaussian	indep.	100	default	smooth	241	default	E(Y | X)	0.947 (0.002)	0.957 (0.002)	0.938 (0.002)	0.951 (0.002)	0.956 (0.002)	0.961 (0.001)	

Table 1: Grid-averaged coverage (MC SE) of the mean and $\beta(s,t)$ per setting and interval. REML model / NCV model: model-based intervals of the REML / NCV fit; NCV wid.: NCV + CL2 widened in quadrature by the NCV–REML difference (S4.1(b)), with the Bayesian or the frequentist (freq.) CL2; AR(1): AR(1) working model. *(continued)*

block	family	error	G	signal	truth	D	basis	estimand	NCV+CL2	NCV wid.	NCV wid. (freq.)	NCV model	REML+CL2	REML model	AR(1)
dense grid	Gaussian	rough	100	default	smooth	241	default	β(s,t)	0.917 (0.004)	0.988 (0.001)	0.987 (0.001)	0.406 (0.006)	0.940 (0.002)	0.359 (0.004)	
dense grid	Gaussian	rough	100	default	smooth	241	default	E(Y | X)	0.926 (0.003)	0.966 (0.002)	0.965 (0.002)	0.348 (0.004)	0.941 (0.002)	0.367 (0.003)	
dense grid	Gaussian	smooth	100	default	smooth	241	default	β(s,t)	0.916 (0.004)	0.989 (0.001)	0.988 (0.001)	0.391 (0.007)	0.944 (0.002)	0.350 (0.004)	
dense grid	Gaussian	smooth	100	default	smooth	241	default	E(Y | X)	0.923 (0.003)	0.967 (0.002)	0.966 (0.002)	0.333 (0.004)	0.943 (0.002)	0.353 (0.003)	
dense grid	Poisson	indep.	100	default	smooth	241	default	β(s,t)	0.978 (0.001)	0.988 (0.001)	0.955 (0.002)	0.979 (0.001)	0.978 (0.001)	0.983 (0.001)	
dense grid	Poisson	indep.	100	default	smooth	241	default	E(Y | X)	0.947 (0.002)	0.958 (0.002)	0.936 (0.002)	0.951 (0.002)	0.957 (0.002)	0.963 (0.001)	
dense grid	Poisson	rough	100	default	smooth	241	default	β(s,t)	0.915 (0.004)	0.988 (0.001)	0.986 (0.001)	0.418 (0.006)	0.938 (0.002)	0.374 (0.004)	
dense grid	Poisson	rough	100	default	smooth	241	default	E(Y | X)	0.922 (0.003)	0.965 (0.002)	0.964 (0.002)	0.357 (0.004)	0.941 (0.002)	0.380 (0.003)	
dense grid	Poisson	smooth	100	default	smooth	241	default	β(s,t)	0.915 (0.004)	0.988 (0.001)	0.987 (0.001)	0.406 (0.007)	0.943 (0.003)	0.369 (0.004)	
dense grid	Poisson	smooth	100	default	smooth	241	default	E(Y | X)	0.921 (0.003)	0.966 (0.002)	0.965 (0.002)	0.342 (0.004)	0.943 (0.002)	0.370 (0.003)	
dense grid	binary	indep.	100	default	smooth	241	default	β(s,t)	0.984 (0.001)	0.990 (0.001)	0.937 (0.003)	0.985 (0.001)	0.990 (0.001)	0.993 (0.001)	
dense grid	binary	indep.	100	default	smooth	241	default	E(Y | X)	0.951 (0.002)	0.956 (0.002)	0.921 (0.003)	0.955 (0.002)	0.962 (0.002)	0.970 (0.002)	
dense grid	binary	rough	100	default	smooth	241	default	β(s,t)	0.868 (0.010)	0.986 (0.001)	0.984 (0.001)	0.464 (0.010)	0.938 (0.002)	0.450 (0.005)	
dense grid	binary	rough	100	default	smooth	241	default	E(Y | X)	0.861 (0.006)	0.953 (0.002)	0.951 (0.002)	0.365 (0.005)	0.938 (0.002)	0.452 (0.003)	
dense grid	binary	smooth	100	default	smooth	241	default	β(s,t)	0.866 (0.009)	0.986 (0.001)	0.984 (0.001)	0.422 (0.010)	0.943 (0.002)	0.420 (0.005)	
dense grid	binary	smooth	100	default	smooth	241	default	E(Y | X)	0.872 (0.005)	0.959 (0.002)	0.958 (0.002)	0.337 (0.005)	0.941 (0.002)	0.418 (0.003)	
rough truth	Gaussian	indep.	100	low	rough	61	default	β(s,t)	0.933 (0.004)	0.938 (0.004)	0.843 (0.007)	0.933 (0.004)	0.948 (0.003)	0.957 (0.002)	
rough truth	Gaussian	indep.	100	low	rough	61	default	E(Y | X)	0.942 (0.002)	0.945 (0.002)	0.913 (0.003)	0.945 (0.002)	0.948 (0.002)	0.956 (0.002)	
rough truth	Gaussian	smooth	100	low	rough	61	default	β(s,t)	0.803 (0.006)	0.945 (0.002)	0.934 (0.003)	0.515 (0.006)	0.942 (0.003)	0.645 (0.007)	
rough truth	Gaussian	smooth	100	low	rough	61	default	E(Y | X)	0.912 (0.003)	0.948 (0.002)	0.943 (0.002)	0.579 (0.005)	0.943 (0.002)	0.627 (0.005)	
rough truth	Gaussian	indep.	100	default	rough	61	default	β(s,t)	0.953 (0.003)	0.955 (0.003)	0.894 (0.004)	0.955 (0.003)	0.961 (0.002)	0.967 (0.002)	
rough truth	Gaussian	indep.	100	default	rough	61	default	E(Y | X)	0.946 (0.002)	0.949 (0.002)	0.926 (0.002)	0.951 (0.002)	0.952 (0.002)	0.958 (0.002)	
rough truth	Gaussian	smooth	100	default	rough	61	default	β(s,t)	0.824 (0.007)	0.935 (0.003)	0.924 (0.003)	0.528 (0.007)	0.941 (0.003)	0.642 (0.007)	
rough truth	Gaussian	smooth	100	default	rough	61	default	E(Y | X)	0.920 (0.002)	0.948 (0.002)	0.944 (0.002)	0.597 (0.004)	0.944 (0.002)	0.631 (0.004)	
rough truth	Gaussian	indep.	100	high	rough	61	default	β(s,t)	0.954 (0.002)	0.959 (0.002)	0.916 (0.003)	0.959 (0.002)	0.963 (0.002)	0.969 (0.002)	
rough truth	Gaussian	indep.	100	high	rough	61	default	E(Y | X)	0.946 (0.002)	0.949 (0.002)	0.933 (0.002)	0.951 (0.002)	0.952 (0.002)	0.958 (0.001)	
rough truth	Gaussian	smooth	100	high	rough	61	default	β(s,t)	0.900 (0.005)	0.950 (0.003)	0.940 (0.003)	0.587 (0.007)	0.942 (0.003)	0.649 (0.006)	
rough truth	Gaussian	smooth	100	high	rough	61	default	E(Y | X)	0.932 (0.002)	0.950 (0.002)	0.946 (0.002)	0.614 (0.004)	0.944 (0.002)	0.635 (0.004)	
rough truth	Poisson	indep.	100	default	rough	61	default	β(s,t)	0.952 (0.003)	0.955 (0.002)	0.889 (0.004)	0.955 (0.002)	0.960 (0.002)	0.966 (0.002)	
rough truth	Poisson	indep.	100	default	rough	61	default	E(Y | X)	0.944 (0.002)	0.947 (0.002)	0.922 (0.002)	0.948 (0.002)	0.950 (0.002)	0.957 (0.002)	
rough truth	Poisson	smooth	100	default	rough	61	default	β(s,t)	0.811 (0.007)	0.932 (0.003)	0.920 (0.003)	0.530 (0.007)	0.943 (0.003)	0.670 (0.007)	
rough truth	Poisson	smooth	100	default	rough	61	default	E(Y | X)	0.918 (0.003)	0.947 (0.002)	0.942 (0.002)	0.607 (0.004)	0.944 (0.002)	0.652 (0.004)	
rough truth	Poisson	indep.	100	high	rough	61	default	β(s,t)	0.950 (0.002)	0.955 (0.002)	0.903 (0.004)	0.955 (0.002)	0.961 (0.002)	0.968 (0.002)	
rough truth	Poisson	indep.	100	high	rough	61	default	E(Y | X)	0.940 (0.002)	0.946 (0.002)	0.925 (0.002)	0.945 (0.002)	0.949 (0.002)	0.957 (0.002)	
rough truth	Poisson	smooth	100	high	rough	61	default	β(s,t)	0.860 (0.007)	0.935 (0.003)	0.924 (0.004)	0.578 (0.007)	0.940 (0.003)	0.677 (0.006)	
rough truth	Poisson	smooth	100	high	rough	61	default	E(Y | X)	0.919 (0.003)	0.945 (0.002)	0.940 (0.002)	0.617 (0.005)	0.943 (0.002)	0.661 (0.004)	
rough truth	binary	indep.	100	default	rough	61	default	β(s,t)	0.911 (0.004)	0.921 (0.004)	0.784 (0.008)	0.912 (0.004)	0.935 (0.003)	0.950 (0.002)	
rough truth	binary	indep.	100	default	rough	61	default	E(Y | X)	0.926 (0.003)	0.934 (0.003)	0.886 (0.004)	0.929 (0.003)	0.927 (0.003)	0.944 (0.002)	
rough truth	binary	smooth	100	default	rough	61	default	β(s,t)	0.799 (0.009)	0.940 (0.004)	0.923 (0.004)	0.587 (0.010)	0.948 (0.003)	0.766 (0.008)	
rough truth	binary	smooth	100	default	rough	61	default	E(Y | X)	0.872 (0.005)	0.933 (0.003)	0.924 (0.003)	0.596 (0.007)	0.940 (0.002)	0.718 (0.005)	
rough truth	binary	indep.	100	high	rough	61	default	β(s,t)	0.919 (0.005)	0.929 (0.005)	0.820 (0.008)	0.921 (0.005)	0.944 (0.003)	0.954 (0.003)	
rough truth	binary	indep.	100	high	rough	61	default	E(Y | X)	0.934 (0.002)	0.938 (0.002)	0.898 (0.003)	0.939 (0.002)	0.942 (0.002)	0.952 (0.002)	
rough truth	binary	smooth	100	high	rough	61	default	β(s,t)	0.803 (0.006)	0.931 (0.003)	0.912 (0.004)	0.598 (0.006)	0.947 (0.003)	0.767 (0.007)	
rough truth	binary	smooth	100	high	rough	61	default	E(Y | X)	0.899 (0.003)	0.941 (0.002)	0.931 (0.003)	0.655 (0.005)	0.943 (0.002)	0.729 (0.004)	
smooth effect	Gaussian	indep.	100	default	smooth	61	default	E(Y | X)	0.956 (0.002)	0.965 (0.001)	0.933 (0.002)	0.959 (0.002)	0.961 (0.001)	0.968 (0.001)	
smooth effect	Gaussian	rough	100	default	smooth	61	default	E(Y | X)	0.925 (0.003)	0.957 (0.002)	0.949 (0.002)	0.628 (0.005)	0.938 (0.002)	0.668 (0.004)	
smooth effect	Gaussian	smooth	100	default	smooth	61	default	E(Y | X)	0.925 (0.003)	0.957 (0.002)	0.950 (0.002)	0.614 (0.005)	0.937 (0.002)	0.650 (0.004)	
smooth effect	Poisson	indep.	100	default	smooth	61	default	E(Y | X)	0.947 (0.002)	0.958 (0.002)	0.921 (0.002)	0.950 (0.002)	0.961 (0.001)	0.969 (0.001)	
smooth effect	Poisson	rough	100	default	smooth	61	default	E(Y | X)	0.910 (0.003)	0.951 (0.002)	0.943 (0.002)	0.632 (0.005)	0.939 (0.002)	0.689 (0.004)	
smooth effect	Poisson	smooth	100	default	smooth	61	default	E(Y | X)	0.911 (0.003)	0.952 (0.002)	0.945 (0.002)	0.620 (0.005)	0.936 (0.002)	0.668 (0.004)	
smooth effect	binary	indep.	100	default	smooth	61	default	E(Y | X)	0.945 (0.002)	0.954 (0.002)	0.887 (0.004)	0.949 (0.003)	0.963 (0.002)	0.974 (0.002)	

Table 1: Grid-averaged coverage (MC SE) of the mean and $\beta(s,t)$ per setting and interval. REML model / NCV model: model-based intervals of the REML / NCV fit; NCV wid.: NCV + CL2 widened in quadrature by the NCV–REML difference (S4.1(b)), with the Bayesian or the frequentist (freq.) CL2; AR(1): AR(1) working model. *(continued)*

block	family	error	G	signal	truth	D	basis	estimand	NCV+CL2	NCV wid.	NCV wid. (freq.)	NCV model	REML+CL2	REML model	AR(1)
smooth effect	binary	rough	100	default	smooth	61	default	E(Y | X)	0.864 (0.006)	0.934 (0.004)	0.917 (0.004)	0.648 (0.007)	0.936 (0.002)	0.772 (0.004)	
smooth effect	binary	smooth	100	default	smooth	61	default	E(Y | X)	0.877 (0.005)	0.945 (0.003)	0.933 (0.003)	0.629 (0.007)	0.936 (0.002)	0.743 (0.005)	
families	scaled t	indep.	100	default	smooth	61	default	β(s,t)	0.982 (0.001)	0.990 (0.001)	0.954 (0.002)	0.983 (0.001)	0.985 (0.001)	0.988 (0.001)	
families	scaled t	indep.	100	default	smooth	61	default	E(Y | X)	0.950 (0.002)	0.959 (0.002)	0.936 (0.002)	0.953 (0.002)	0.960 (0.001)	0.965 (0.001)	
families	scaled t	smooth	100	default	smooth	61	default	β(s,t)	0.940 (0.004)	0.985 (0.001)	0.979 (0.002)	0.703 (0.008)	0.944 (0.003)	0.671 (0.007)	
families	scaled t	smooth	100	default	smooth	61	default	E(Y | X)	0.938 (0.003)	0.962 (0.002)	0.958 (0.002)	0.636 (0.005)	0.947 (0.002)	0.644 (0.005)	
families	beta	indep.	100	default	smooth	61	default	β(s,t)	0.979 (0.002)	0.988 (0.001)	0.947 (0.003)	0.981 (0.002)	0.984 (0.001)	0.989 (0.001)	
families	beta	indep.	100	default	smooth	61	default	E(Y | X)	0.946 (0.002)	0.956 (0.002)	0.930 (0.002)	0.953 (0.002)	0.958 (0.002)	0.966 (0.001)	
families	beta	smooth	100	default	smooth	61	default	β(s,t)	0.929 (0.004)	0.982 (0.002)	0.976 (0.002)	0.691 (0.008)	0.941 (0.003)	0.659 (0.007)	
families	beta	smooth	100	default	smooth	61	default	E(Y | X)	0.924 (0.003)	0.956 (0.002)	0.951 (0.002)	0.623 (0.005)	0.941 (0.002)	0.635 (0.004)	
families	negative binomial	indep.	100	default	smooth	61	default	β(s,t)	0.982 (0.002)	0.990 (0.001)	0.944 (0.003)	0.983 (0.001)	0.988 (0.001)	0.991 (0.001)	
families	negative binomial	indep.	100	default	smooth	61	default	E(Y | X)	0.950 (0.002)	0.958 (0.002)	0.930 (0.003)	0.954 (0.002)	0.962 (0.002)	0.968 (0.002)	
families	negative binomial	smooth	100	default	smooth	61	default	β(s,t)	0.932 (0.004)	0.983 (0.002)	0.977 (0.002)	0.699 (0.009)	0.943 (0.003)	0.669 (0.008)	
families	negative binomial	smooth	100	default	smooth	61	default	E(Y | X)	0.925 (0.003)	0.956 (0.002)	0.951 (0.002)	0.622 (0.005)	0.943 (0.002)	0.641 (0.005)	
families	negative binomial	indep.	100	default	smooth	61	default	β(s,t)	0.962 (0.002)	0.982 (0.001)	0.955 (0.002)	0.929 (0.003)	0.972 (0.002)	0.945 (0.002)	
families	negative binomial	indep.	100	default	smooth	61	default	E(Y | X)	0.934 (0.002)	0.950 (0.002)	0.932 (0.002)	0.868 (0.004)	0.952 (0.002)	0.898 (0.003)	
families	negative binomial	smooth	100	default	smooth	61	default	β(s,t)	0.913 (0.005)	0.984 (0.001)	0.981 (0.002)	0.558 (0.008)	0.938 (0.003)	0.531 (0.006)	
families	negative binomial	smooth	100	default	smooth	61	default	E(Y | X)	0.913 (0.003)	0.957 (0.002)	0.954 (0.002)	0.486 (0.005)	0.939 (0.002)	0.529 (0.004)	
exact AR(1)	Gaussian	AR(1)	100	default	smooth	61	default	β(s,t)	0.933 (0.003)	0.984 (0.001)	0.979 (0.002)	0.684 (0.007)	0.940 (0.003)	0.651 (0.007)	0.989 (0.001)
exact AR(1)	Gaussian	AR(1)	100	default	smooth	61	default	E(Y | X)	0.930 (0.003)	0.959 (0.002)	0.955 (0.002)	0.603 (0.005)	0.942 (0.002)	0.625 (0.004)	0.961 (0.002)
sign-changing	Gaussian	sign-chg.	100	default	smooth	61	default	β(s,t)	0.935 (0.003)	0.985 (0.001)	0.977 (0.001)	0.782 (0.005)	0.942 (0.003)	0.700 (0.006)	0.992 (0.001)
sign-changing	Gaussian	sign-chg.	100	default	smooth	61	default	E(Y | X)	0.915 (0.003)	0.957 (0.002)	0.951 (0.002)	0.642 (0.005)	0.942 (0.002)	0.646 (0.004)	0.963 (0.002)
sign-changing	Poisson	sign-chg.	100	default	smooth	61	default	β(s,t)	0.935 (0.003)	0.986 (0.001)	0.977 (0.001)	0.791 (0.005)	0.942 (0.003)	0.728 (0.006)	
sign-changing	Poisson	sign-chg.	100	default	smooth	61	default	E(Y | X)	0.905 (0.003)	0.951 (0.002)	0.944 (0.002)	0.653 (0.005)	0.941 (0.002)	0.670 (0.005)	
sign-changing	binary	sign-chg.	100	default	smooth	61	default	β(s,t)	0.927 (0.005)	0.977 (0.002)	0.956 (0.003)	0.841 (0.007)	0.959 (0.002)	0.885 (0.005)	
sign-changing	binary	sign-chg.	100	default	smooth	61	default	E(Y | X)	0.882 (0.005)	0.945 (0.003)	0.930 (0.003)	0.724 (0.006)	0.943 (0.002)	0.794 (0.005)	
misreg. (AR(1))	Gaussian	misreg.	100	high	rough	61	default	β(s,t)	0.863 (0.006)	0.934 (0.003)	0.908 (0.004)	0.628 (0.007)	0.944 (0.003)	0.729 (0.006)	0.860 (0.005)
misreg. (AR(1))	Gaussian	misreg.	100	high	rough	61	default	E(Y | X)	0.911 (0.003)	0.938 (0.003)	0.926 (0.003)	0.665 (0.005)	0.937 (0.003)	0.706 (0.005)	0.838 (0.004)
heteroskedastic	Gaussian	var(t)	40	default	smooth	61	default	β(s,t)	0.918 (0.005)	0.979 (0.003)	0.973 (0.003)	0.662 (0.009)	0.926 (0.004)	0.619 (0.009)	
heteroskedastic	Gaussian	var(t)	40	default	smooth	61	default	E(Y | X)	0.919 (0.004)	0.957 (0.003)	0.951 (0.003)	0.593 (0.006)	0.932 (0.003)	0.614 (0.005)	
heteroskedastic	Gaussian	var(t)	100	default	smooth	61	default	β(s,t)	0.933 (0.004)	0.983 (0.002)	0.977 (0.002)	0.686 (0.008)	0.944 (0.003)	0.662 (0.007)	0.991 (0.001)
heteroskedastic	Gaussian	var(t)	100	default	smooth	61	default	E(Y | X)	0.928 (0.003)	0.959 (0.002)	0.954 (0.002)	0.614 (0.005)	0.944 (0.002)	0.642 (0.005)	0.972 (0.002)
heteroskedastic	Gaussian	var(z)	40	default	smooth	61	default	β(s,t)	0.925 (0.004)	0.985 (0.002)	0.979 (0.002)	0.668 (0.009)	0.939 (0.003)	0.619 (0.009)	
heteroskedastic	Gaussian	var(z)	40	default	smooth	61	default	E(Y | X)	0.912 (0.004)	0.955 (0.003)	0.950 (0.003)	0.560 (0.007)	0.934 (0.003)	0.588 (0.006)	
heteroskedastic	Gaussian	var(z)	100	default	smooth	61	default	β(s,t)	0.935 (0.004)	0.984 (0.001)	0.978 (0.002)	0.683 (0.009)	0.949 (0.003)	0.660 (0.008)	0.992 (0.001)
heteroskedastic	Gaussian	var(z)	100	default	smooth	61	default	E(Y | X)	0.923 (0.003)	0.955 (0.002)	0.950 (0.002)	0.572 (0.006)	0.944 (0.002)	0.606 (0.005)	0.949 (0.003)
heteroskedastic	Gaussian	var(t,z)	40	default	smooth	61	default	β(s,t)	0.924 (0.005)	0.983 (0.002)	0.977 (0.002)	0.667 (0.009)	0.940 (0.003)	0.629 (0.009)	
heteroskedastic	Gaussian	var(t,z)	40	default	smooth	61	default	E(Y | X)	0.911 (0.004)	0.955 (0.003)	0.949 (0.003)	0.563 (0.007)	0.936 (0.003)	0.596 (0.005)	
heteroskedastic	Gaussian	var(t,z)	100	default	smooth	61	default	β(s,t)	0.934 (0.004)	0.984 (0.001)	0.978 (0.002)	0.688 (0.009)	0.950 (0.003)	0.666 (0.008)	0.990 (0.001)
heteroskedastic	Gaussian	var(t,z)	100	default	smooth	61	default	E(Y | X)	0.922 (0.003)	0.955 (0.002)	0.950 (0.002)	0.580 (0.006)	0.944 (0.002)	0.615 (0.005)	0.951 (0.002)
low-rank covariate	Gaussian	indep.	100	default	smooth	61	default	β(s,t)	0.986 (0.001)	0.994 (0.001)	0.953 (0.003)	0.986 (0.001)	0.991 (0.001)	0.994 (0.001)	
low-rank covariate	Gaussian	indep.	100	default	smooth	61	default	E(Y | X)	0.948 (0.002)	0.956 (0.002)	0.933 (0.002)	0.952 (0.002)	0.960 (0.002)	0.966 (0.001)	
low-rank covariate	Gaussian	smooth	100	default	smooth	61	default	β(s,t)	0.936 (0.004)	0.987 (0.001)	0.982 (0.002)	0.712 (0.009)	0.945 (0.003)	0.666 (0.010)	
low-rank covariate	Gaussian	smooth	100	default	smooth	61	default	E(Y | X)	0.930 (0.003)	0.958 (0.002)	0.954 (0.002)	0.607 (0.005)	0.944 (0.002)	0.633 (0.005)	
low-rank covariate	Poisson	indep.	100	default	smooth	61	default	β(s,t)	0.987 (0.001)	0.994 (0.001)	0.953 (0.003)	0.987 (0.001)	0.992 (0.001)	0.994 (0.001)	
low-rank covariate	Poisson	indep.	100	default	smooth	61	default	E(Y | X)	0.946 (0.002)	0.954 (0.002)	0.928 (0.003)	0.950 (0.002)	0.959 (0.002)	0.965 (0.002)	
low-rank covariate	Poisson	smooth	100	default	smooth	61	default	β(s,t)	0.939 (0.004)	0.989 (0.001)	0.984 (0.002)	0.735 (0.009)	0.946 (0.004)	0.693 (0.009)	
low-rank covariate	Poisson	smooth	100	default	smooth	61	default	E(Y | X)	0.928 (0.003)	0.957 (0.002)	0.952 (0.002)	0.617 (0.006)	0.945 (0.002)	0.652 (0.005)	
low-rank covariate	binary	indep.	100	default	smooth	61	default	β(s,t)	0.987 (0.002)	0.992 (0.001)	0.915 (0.005)	0.987 (0.002)	0.993 (0.001)	0.996 (0.001)	
low-rank covariate	binary	indep.	100	default	smooth	61	default	E(Y | X)	0.937 (0.003)	0.944 (0.003)	0.899 (0.004)	0.941 (0.003)	0.945 (0.003)	0.960 (0.002)	
low-rank covariate	binary	smooth	100	default	smooth	61	default	β(s,t)	0.904 (0.008)	0.977 (0.003)	0.966 (0.003)	0.734 (0.012)	0.954 (0.004)	0.789 (0.010)	

Table 1: Grid-averaged coverage (MC SE) of the mean and $\beta(s,t)$ per setting and interval. REML model / NCV model: model-based intervals of the REML / NCV fit; NCV wid.: NCV + CL2 widened in quadrature by the NCV–REML difference (S4.1(b)), with the Bayesian or the frequentist (freq.) CL2; AR(1): AR(1) working model. *(continued)*

block	family	error	G	signal	truth	D	basis	estimand	NCV+CL2	NCV wid.	NCV wid. (freq.)	NCV model	REML+CL2	REML model	AR(1)
low-rank covariate	binary	smooth	100	default	smooth	61	default	E(Y | X)	0.890 (0.004)	0.939 (0.003)	0.931 (0.003)	0.609 (0.007)	0.940 (0.003)	0.718 (0.005)	
basis size	Gaussian	indep.	100	default	smooth	61	2× finer knots	β(s,t)	0.994 (0.001)	0.997 (0.000)	0.959 (0.002)	0.994 (0.001)	0.997 (0.000)	0.998 (0.000)	
basis size	Gaussian	indep.	100	default	smooth	61	2× finer knots	E(Y | X)	0.962 (0.001)	0.970 (0.001)	0.934 (0.002)	0.964 (0.001)	0.973 (0.001)	0.978 (0.001)	
basis size	Gaussian	indep.	100	default	smooth	61	4× finer knots	β(s,t)	0.999 (0.000)	0.999 (0.000)	0.964 (0.002)	0.999 (0.000)	1.000 (0.000)	1.000 (0.000)	
basis size	Gaussian	indep.	100	default	smooth	61	4× finer knots	E(Y | X)	0.972 (0.001)	0.979 (0.001)	0.934 (0.002)	0.973 (0.001)	0.982 (0.001)	0.985 (0.001)	
basis size	Gaussian	smooth	100	default	smooth	61	2× finer knots	β(s,t)	0.952 (0.003)	0.994 (0.001)	0.990 (0.001)	0.789 (0.006)	0.945 (0.003)	0.671 (0.007)	
basis size	Gaussian	smooth	100	default	smooth	61	2× finer knots	E(Y | X)	0.933 (0.003)	0.971 (0.001)	0.965 (0.002)	0.660 (0.005)	0.947 (0.002)	0.673 (0.004)	
basis size	Gaussian	smooth	100	default	smooth	61	4× finer knots	β(s,t)	0.967 (0.002)	0.998 (0.000)	0.995 (0.000)	0.871 (0.005)	0.954 (0.002)	0.673 (0.006)	
basis size	Gaussian	smooth	100	default	smooth	61	4× finer knots	E(Y | X)	0.937 (0.002)	0.979 (0.001)	0.973 (0.001)	0.707 (0.005)	0.951 (0.002)	0.706 (0.003)	
basis size	Poisson	indep.	100	default	smooth	61	2× finer knots	β(s,t)	0.994 (0.001)	0.997 (0.000)	0.956 (0.002)	0.995 (0.001)	0.997 (0.000)	0.998 (0.000)	
basis size	Poisson	indep.	100	default	smooth	61	2× finer knots	E(Y | X)	0.961 (0.002)	0.969 (0.001)	0.929 (0.002)	0.963 (0.002)	0.974 (0.001)	0.978 (0.001)	
basis size	Poisson	indep.	100	default	smooth	61	4× finer knots	β(s,t)	0.999 (0.000)	0.999 (0.000)	0.961 (0.002)	0.999 (0.000)	1.000 (0.000)	1.000 (0.000)	
basis size	Poisson	indep.	100	default	smooth	61	4× finer knots	E(Y | X)	0.970 (0.001)	0.977 (0.001)	0.927 (0.002)	0.971 (0.001)	0.982 (0.001)	0.985 (0.001)	
basis size	Poisson	smooth	100	default	smooth	61	2× finer knots	β(s,t)	0.953 (0.003)	0.995 (0.001)	0.990 (0.001)	0.805 (0.006)	0.948 (0.003)	0.702 (0.007)	
basis size	Poisson	smooth	100	default	smooth	61	2× finer knots	E(Y | X)	0.930 (0.003)	0.970 (0.001)	0.963 (0.002)	0.666 (0.005)	0.949 (0.002)	0.697 (0.004)	
basis size	Poisson	smooth	100	default	smooth	61	4× finer knots	β(s,t)	0.970 (0.002)	0.998 (0.000)	0.995 (0.000)	0.886 (0.005)	0.955 (0.002)	0.715 (0.006)	
basis size	Poisson	smooth	100	default	smooth	61	4× finer knots	E(Y | X)	0.935 (0.003)	0.977 (0.001)	0.970 (0.001)	0.714 (0.005)	0.952 (0.002)	0.738 (0.003)	
basis size	binary	indep.	100	default	smooth	61	2× finer knots	β(s,t)	0.994 (0.001)	0.996 (0.001)	0.909 (0.004)	0.994 (0.001)	0.997 (0.000)	0.998 (0.000)	
basis size	binary	indep.	100	default	smooth	61	2× finer knots	E(Y | X)	0.954 (0.002)	0.959 (0.002)	0.895 (0.004)	0.956 (0.002)	0.965 (0.002)	0.975 (0.001)	
basis size	binary	indep.	100	default	smooth	61	4× finer knots	β(s,t)	0.999 (0.000)	0.999 (0.000)	0.908 (0.004)	0.999 (0.000)	0.999 (0.000)	1.000 (0.000)	
basis size	binary	indep.	100	default	smooth	61	4× finer knots	E(Y | X)	0.963 (0.002)	0.967 (0.002)	0.891 (0.004)	0.965 (0.002)	0.974 (0.001)	0.982 (0.001)	
basis size	binary	smooth	100	default	smooth	61	2× finer knots	β(s,t)	0.917 (0.007)	0.990 (0.001)	0.983 (0.001)	0.785 (0.011)	0.958 (0.003)	0.789 (0.007)	
basis size	binary	smooth	100	default	smooth	61	2× finer knots	E(Y | X)	0.892 (0.004)	0.959 (0.002)	0.950 (0.002)	0.641 (0.007)	0.946 (0.002)	0.753 (0.004)	
basis size	binary	smooth	100	default	smooth	61	4× finer knots	β(s,t)	0.933 (0.007)	0.997 (0.000)	0.993 (0.001)	0.845 (0.010)	0.960 (0.002)	0.795 (0.006)	
basis size	binary	smooth	100	default	smooth	61	4× finer knots	E(Y | X)	0.899 (0.004)	0.973 (0.002)	0.964 (0.002)	0.671 (0.007)	0.950 (0.002)	0.781 (0.003)	

Table 2: Coverage (MC SE) of $\alpha(t)$ and $\gamma(t)$ per setting; for the smooth-effect settings, the first group of columns holds $f(x,t)$. Interval labels as in the previous table.

								α(t) or f(x,t)				γ(t)			
block	family	error	G	signal	truth	D	basis	REML model	REML+CL2	NCV+CL2	NCV wid.	REML model	REML+CL2	NCV+CL2	NCV wid.
main factorial	Gaussian	indep.	40	low	smooth	61	default	0.954 (0.005)	0.940 (0.005)	0.937 (0.005)	0.941 (0.005)	0.958 (0.005)	0.942 (0.006)	0.935 (0.007)	0.938 (0.007)
main factorial	Gaussian	indep.	100	low	smooth	61	default	0.955 (0.005)	0.950 (0.005)	0.941 (0.005)	0.944 (0.005)	0.968 (0.004)	0.957 (0.005)	0.948 (0.005)	0.952 (0.005)
main factorial	Gaussian	rough	40	low	smooth	61	default	0.621 (0.013)	0.933 (0.007)	0.847 (0.014)	0.912 (0.008)	0.640 (0.012)	0.937 (0.007)	0.867 (0.011)	0.915 (0.008)
main factorial	Gaussian	rough	100	low	smooth	61	default	0.653 (0.011)	0.948 (0.005)	0.924 (0.007)	0.939 (0.006)	0.656 (0.012)	0.937 (0.006)	0.904 (0.008)	0.923 (0.007)
main factorial	Gaussian	smooth	40	low	smooth	61	default	0.608 (0.014)	0.930 (0.007)	0.871 (0.013)	0.919 (0.009)	0.620 (0.014)	0.945 (0.006)	0.889 (0.010)	0.926 (0.008)
main factorial	Gaussian	smooth	100	low	smooth	61	default	0.635 (0.012)	0.948 (0.006)	0.925 (0.007)	0.941 (0.006)	0.649 (0.013)	0.948 (0.006)	0.925 (0.007)	0.942 (0.006)
main factorial	Gaussian	indep.	40	default	smooth	61	default	0.956 (0.004)	0.947 (0.004)	0.940 (0.005)	0.943 (0.005)	0.959 (0.004)	0.951 (0.005)	0.941 (0.005)	0.944 (0.005)
main factorial	Gaussian	indep.	100	default	smooth	61	default	0.956 (0.004)	0.956 (0.004)	0.945 (0.005)	0.949 (0.005)	0.960 (0.004)	0.950 (0.005)	0.946 (0.005)	0.951 (0.004)
main factorial	Gaussian	rough	40	default	smooth	61	default	0.632 (0.012)	0.940 (0.006)	0.924 (0.006)	0.936 (0.006)	0.653 (0.011)	0.938 (0.006)	0.920 (0.007)	0.940 (0.006)
main factorial	Gaussian	rough	100	default	smooth	61	default	0.659 (0.010)	0.951 (0.005)	0.938 (0.006)	0.945 (0.005)	0.663 (0.012)	0.938 (0.006)	0.924 (0.006)	0.936 (0.006)
main factorial	Gaussian	smooth	40	default	smooth	61	default	0.614 (0.013)	0.937 (0.007)	0.919 (0.008)	0.936 (0.007)	0.621 (0.012)	0.946 (0.006)	0.927 (0.007)	0.946 (0.006)
main factorial	Gaussian	smooth	100	default	smooth	61	default	0.641 (0.011)	0.950 (0.005)	0.941 (0.006)	0.949 (0.005)	0.655 (0.012)	0.948 (0.005)	0.935 (0.006)	0.946 (0.006)
main factorial	Gaussian	indep.	40	high	smooth	61	default	0.954 (0.004)	0.947 (0.004)	0.939 (0.004)	0.943 (0.004)	0.959 (0.004)	0.945 (0.005)	0.935 (0.005)	0.943 (0.005)
main factorial	Gaussian	indep.	100	high	smooth	61	default	0.956 (0.004)	0.954 (0.004)	0.950 (0.004)	0.953 (0.004)	0.955 (0.004)	0.944 (0.005)	0.938 (0.005)	0.947 (0.005)
main factorial	Gaussian	rough	40	high	smooth	61	default	0.638 (0.011)	0.941 (0.005)	0.933 (0.006)	0.941 (0.006)	0.657 (0.010)	0.940 (0.006)	0.931 (0.006)	0.943 (0.006)
main factorial	Gaussian	rough	100	high	smooth	61	default	0.665 (0.010)	0.953 (0.005)	0.946 (0.005)	0.949 (0.005)	0.666 (0.011)	0.939 (0.005)	0.928 (0.006)	0.936 (0.005)
main factorial	Gaussian	smooth	40	high	smooth	61	default	0.622 (0.012)	0.940 (0.007)	0.930 (0.007)	0.943 (0.007)	0.629 (0.012)	0.945 (0.006)	0.935 (0.006)	0.949 (0.006)
main factorial	Gaussian	smooth	100	high	smooth	61	default	0.646 (0.011)	0.949 (0.005)	0.946 (0.005)	0.954 (0.005)	0.655 (0.011)	0.947 (0.005)	0.936 (0.006)	0.946 (0.005)
main factorial	Poisson	indep.	40	default	smooth	61	default	0.960 (0.004)	0.953 (0.004)	0.940 (0.005)	0.946 (0.005)	0.955 (0.005)	0.948 (0.005)	0.939 (0.005)	0.943 (0.005)
main factorial	Poisson	indep.	100	default	smooth	61	default	0.954 (0.005)	0.951 (0.005)	0.936 (0.005)	0.943 (0.005)	0.963 (0.004)	0.955 (0.005)	0.947 (0.005)	0.951 (0.005)
main factorial	Poisson	rough	40	default	smooth	61	default	0.655 (0.011)	0.935 (0.006)	0.909 (0.007)	0.937 (0.006)	0.667 (0.012)	0.937 (0.006)	0.905 (0.009)	0.932 (0.007)
main factorial	Poisson	rough	100	default	smooth	61	default	0.668 (0.011)	0.950 (0.005)	0.934 (0.006)	0.948 (0.006)	0.677 (0.011)	0.937 (0.006)	0.923 (0.007)	0.933 (0.006)
main factorial	Poisson	smooth	40	default	smooth	61	default	0.641 (0.013)	0.933 (0.007)	0.913 (0.009)	0.941 (0.007)	0.647 (0.012)	0.943 (0.006)	0.910 (0.008)	0.937 (0.007)
main factorial	Poisson	smooth	100	default	smooth	61	default	0.665 (0.012)	0.948 (0.005)	0.927 (0.007)	0.948 (0.006)	0.662 (0.012)	0.945 (0.005)	0.931 (0.006)	0.944 (0.006)
main factorial	Poisson	indep.	40	high	smooth	61	default	0.961 (0.004)	0.950 (0.004)	0.927 (0.006)	0.946 (0.005)	0.956 (0.005)	0.941 (0.005)	0.927 (0.006)	0.936 (0.005)
main factorial	Poisson	indep.	100	high	smooth	61	default	0.950 (0.005)	0.948 (0.005)	0.926 (0.006)	0.941 (0.005)	0.954 (0.004)	0.943 (0.005)	0.937 (0.005)	0.943 (0.005)
main factorial	Poisson	rough	40	high	smooth	61	default	0.677 (0.011)	0.944 (0.006)	0.898 (0.009)	0.946 (0.006)	0.669 (0.011)	0.930 (0.006)	0.899 (0.008)	0.928 (0.006)
main factorial	Poisson	rough	100	high	smooth	61	default	0.671 (0.011)	0.947 (0.005)	0.918 (0.007)	0.947 (0.005)	0.688 (0.010)	0.938 (0.006)	0.921 (0.007)	0.932 (0.006)
main factorial	Poisson	smooth	40	high	smooth	61	default	0.653 (0.012)	0.940 (0.006)	0.891 (0.009)	0.946 (0.006)	0.658 (0.011)	0.938 (0.006)	0.902 (0.007)	0.934 (0.006)
main factorial	Poisson	smooth	100	high	smooth	61	default	0.659 (0.011)	0.949 (0.005)	0.918 (0.007)	0.949 (0.005)	0.669 (0.011)	0.943 (0.005)	0.921 (0.006)	0.937 (0.006)
main factorial	binary	indep.	40	default	smooth	61	default	0.903 (0.013)	0.859 (0.014)	0.873 (0.012)	0.889 (0.012)	0.931 (0.009)	0.890 (0.011)	0.876 (0.012)	0.896 (0.012)
main factorial	binary	indep.	100	default	smooth	61	default	0.943 (0.006)	0.924 (0.007)	0.912 (0.008)	0.920 (0.007)	0.945 (0.007)	0.923 (0.008)	0.912 (0.009)	0.917 (0.009)
main factorial	binary	rough	40	default	smooth	61	default	0.712 (0.014)	0.904 (0.008)	0.844 (0.012)	0.908 (0.009)	0.724 (0.013)	0.920 (0.009)	0.799 (0.014)	0.885 (0.010)
main factorial	binary	rough	100	default	smooth	61	default	0.749 (0.011)	0.926 (0.007)	0.836 (0.013)	0.902 (0.008)	0.740 (0.012)	0.924 (0.008)	0.826 (0.014)	0.888 (0.011)
main factorial	binary	smooth	40	default	smooth	61	default	0.668 (0.016)	0.885 (0.010)	0.874 (0.011)	0.931 (0.008)	0.691 (0.015)	0.922 (0.008)	0.838 (0.012)	0.915 (0.009)
main factorial	binary	smooth	100	default	smooth	61	default	0.702 (0.013)	0.920 (0.008)	0.856 (0.012)	0.908 (0.008)	0.730 (0.013)	0.939 (0.007)	0.854 (0.012)	0.914 (0.009)
main factorial	binary	indep.	40	high	smooth	61	default	0.956 (0.005)	0.938 (0.006)	0.920 (0.007)	0.929 (0.007)	0.942 (0.007)	0.921 (0.008)	0.910 (0.009)	0.918 (0.008)
main factorial	binary	indep.	100	high	smooth	61	default	0.958 (0.004)	0.949 (0.005)	0.933 (0.006)	0.938 (0.005)	0.948 (0.005)	0.935 (0.006)	0.922 (0.007)	0.926 (0.007)
main factorial	binary	rough	40	high	smooth	61	default	0.747 (0.011)	0.907 (0.008)	0.835 (0.015)	0.910 (0.009)	0.746 (0.012)	0.929 (0.008)	0.817 (0.014)	0.892 (0.009)
main factorial	binary	rough	100	high	smooth	61	default	0.759 (0.011)	0.937 (0.006)	0.909 (0.007)	0.934 (0.006)	0.758 (0.010)	0.941 (0.005)	0.893 (0.009)	0.917 (0.007)
main factorial	binary	smooth	40	high	smooth	61	default	0.686 (0.013)	0.903 (0.008)	0.859 (0.011)	0.929 (0.007)	0.716 (0.013)	0.919 (0.008)	0.833 (0.012)	0.902 (0.009)
main factorial	binary	smooth	100	high	smooth	61	default	0.730 (0.011)	0.929 (0.007)	0.891 (0.008)	0.932 (0.006)	0.737 (0.011)	0.940 (0.006)	0.895 (0.009)	0.926 (0.007)
misreg.	Gaussian	misreg.	40	high	smooth	61	default	0.766 (0.013)	0.943 (0.007)	0.932 (0.008)	0.940 (0.008)	0.718 (0.013)	0.929 (0.007)	0.903 (0.009)	0.927 (0.008)
misreg.	Gaussian	misreg.	40	high	rough	61	default	0.754 (0.012)	0.949 (0.006)	0.924 (0.008)	0.941 (0.007)	0.678 (0.011)	0.915 (0.007)	0.872 (0.009)	0.912 (0.007)
misreg.	Gaussian	misreg.	100	high	smooth	61	default	0.766 (0.012)	0.953 (0.007)	0.948 (0.007)	0.952 (0.007)	0.719 (0.013)	0.933 (0.007)	0.920 (0.007)	0.930 (0.007)
misreg.	Gaussian	misreg.	100	high	rough	61	default	0.732 (0.011)	0.949 (0.006)	0.943 (0.007)	0.950 (0.006)	0.692 (0.013)	0.928 (0.006)	0.923 (0.006)	0.930 (0.006)
dense grid	Gaussian	indep.	100	default	smooth	241	default	0.952 (0.004)	0.951 (0.004)	0.943 (0.004)	0.949 (0.004)	0.951 (0.004)	0.943 (0.004)	0.933 (0.004)	0.939 (0.004)
dense grid	Gaussian	rough	100	default	smooth	241	default	0.382 (0.009)	0.953 (0.005)	0.941 (0.005)	0.953 (0.004)	0.397 (0.011)	0.936 (0.006)	0.919 (0.006)	0.936 (0.006)
dense grid	Gaussian	smooth	100	default	smooth	241	default	0.373 (0.010)	0.947 (0.005)	0.938 (0.006)	0.954 (0.005)	0.369 (0.010)	0.947 (0.005)	0.930 (0.006)	0.950 (0.006)
dense grid	Poisson	indep.	100	default	smooth	241	default	0.954 (0.004)	0.951 (0.004)	0.943 (0.005)	0.951 (0.004)	0.957 (0.004)	0.948 (0.004)	0.935 (0.005)	0.942 (0.004)
dense grid	Poisson	rough	100	default	smooth	241	default	0.397 (0.010)	0.952 (0.005)	0.932 (0.006)	0.956 (0.005)	0.411 (0.010)	0.936 (0.006)	0.915 (0.007)	0.934 (0.006)

Table 2: Coverage (MC SE) of $\alpha(t)$ and $\gamma(t)$ per setting; for the smooth-effect settings, the first group of columns holds $f(x,t)$. Interval labels as in the previous table. *(continued)*

block	family	error	G	signal	truth	D	basis	REML model	REML+CL2	NCV+CL2	NCV wid.	REML model	REML+CL2	NCV+CL2	NCV wid.
dense grid	Poisson	smooth	100	default	smooth	241	default	0.380 (0.010)	0.947 (0.005)	0.925 (0.007)	0.953 (0.005)	0.387 (0.010)	0.947 (0.005)	0.923 (0.006)	0.948 (0.005)
dense grid	binary	indep.	100	default	smooth	241	default	0.957 (0.004)	0.951 (0.005)	0.940 (0.005)	0.945 (0.005)	0.960 (0.004)	0.951 (0.005)	0.942 (0.005)	0.944 (0.005)
dense grid	binary	rough	100	default	smooth	241	default	0.447 (0.012)	0.919 (0.007)	0.831 (0.013)	0.928 (0.007)	0.474 (0.011)	0.941 (0.006)	0.813 (0.014)	0.913 (0.008)
dense grid	binary	smooth	100	default	smooth	241	default	0.410 (0.011)	0.920 (0.007)	0.843 (0.012)	0.938 (0.005)	0.445 (0.011)	0.943 (0.005)	0.843 (0.012)	0.932 (0.007)
rough truth	Gaussian	indep.	100	low	rough	61	default	0.955 (0.004)	0.952 (0.005)	0.941 (0.005)	0.944 (0.005)	0.952 (0.004)	0.942 (0.005)	0.939 (0.005)	0.943 (0.004)
rough truth	Gaussian	smooth	100	low	rough	61	default	0.634 (0.012)	0.947 (0.006)	0.925 (0.007)	0.940 (0.006)	0.642 (0.012)	0.942 (0.006)	0.928 (0.007)	0.938 (0.006)
rough truth	Gaussian	indep.	100	default	rough	61	default	0.956 (0.004)	0.956 (0.004)	0.945 (0.005)	0.950 (0.004)	0.954 (0.004)	0.945 (0.004)	0.940 (0.005)	0.944 (0.004)
rough truth	Gaussian	smooth	100	default	rough	61	default	0.642 (0.012)	0.950 (0.005)	0.937 (0.006)	0.947 (0.005)	0.652 (0.012)	0.947 (0.006)	0.937 (0.006)	0.943 (0.006)
rough truth	Gaussian	indep.	100	high	rough	61	default	0.957 (0.004)	0.955 (0.004)	0.948 (0.004)	0.952 (0.004)	0.954 (0.004)	0.945 (0.004)	0.938 (0.005)	0.941 (0.005)
rough truth	Gaussian	smooth	100	high	rough	61	default	0.648 (0.011)	0.950 (0.005)	0.945 (0.005)	0.950 (0.005)	0.657 (0.011)	0.947 (0.005)	0.944 (0.005)	0.947 (0.005)
rough truth	Poisson	indep.	100	default	rough	61	default	0.952 (0.004)	0.952 (0.005)	0.941 (0.005)	0.946 (0.005)	0.952 (0.005)	0.942 (0.005)	0.936 (0.005)	0.941 (0.005)
rough truth	Poisson	smooth	100	default	rough	61	default	0.666 (0.012)	0.950 (0.005)	0.936 (0.006)	0.944 (0.006)	0.670 (0.011)	0.945 (0.005)	0.930 (0.006)	0.938 (0.006)
rough truth	Poisson	indep.	100	high	rough	61	default	0.952 (0.005)	0.950 (0.005)	0.936 (0.006)	0.950 (0.005)	0.955 (0.004)	0.945 (0.004)	0.937 (0.005)	0.945 (0.005)
rough truth	Poisson	smooth	100	high	rough	61	default	0.668 (0.012)	0.948 (0.005)	0.939 (0.006)	0.952 (0.005)	0.678 (0.011)	0.943 (0.005)	0.924 (0.006)	0.937 (0.005)
rough truth	binary	indep.	100	default	rough	61	default	0.942 (0.006)	0.921 (0.007)	0.920 (0.008)	0.926 (0.008)	0.907 (0.007)	0.877 (0.008)	0.897 (0.008)	0.906 (0.008)
rough truth	binary	smooth	100	default	rough	61	default	0.708 (0.013)	0.920 (0.007)	0.862 (0.012)	0.914 (0.008)	0.720 (0.012)	0.931 (0.006)	0.839 (0.011)	0.904 (0.007)
rough truth	binary	indep.	100	high	rough	61	default	0.957 (0.004)	0.947 (0.005)	0.939 (0.005)	0.942 (0.005)	0.928 (0.006)	0.916 (0.006)	0.918 (0.006)	0.923 (0.006)
rough truth	binary	smooth	100	high	rough	61	default	0.729 (0.011)	0.932 (0.006)	0.892 (0.008)	0.935 (0.006)	0.745 (0.011)	0.938 (0.006)	0.902 (0.008)	0.927 (0.006)
smooth effect	Gaussian	indep.	100	default	smooth	61	default	0.976 (0.001)	0.971 (0.002)	0.975 (0.001)	0.983 (0.001)	0.962 (0.004)	0.952 (0.004)	0.946 (0.005)	0.951 (0.005)
smooth effect	Gaussian	rough	100	default	smooth	61	default	0.677 (0.005)	0.945 (0.002)	0.940 (0.003)	0.975 (0.002)	0.661 (0.012)	0.938 (0.006)	0.925 (0.007)	0.935 (0.006)
smooth effect	Gaussian	smooth	100	default	smooth	61	default	0.650 (0.005)	0.942 (0.003)	0.939 (0.003)	0.973 (0.002)	0.648 (0.012)	0.950 (0.005)	0.935 (0.007)	0.947 (0.006)
smooth effect	Poisson	indep.	100	default	smooth	61	default	0.978 (0.001)	0.973 (0.001)	0.967 (0.002)	0.976 (0.001)	0.959 (0.004)	0.949 (0.005)	0.942 (0.005)	0.946 (0.005)
smooth effect	Poisson	rough	100	default	smooth	61	default	0.697 (0.005)	0.945 (0.002)	0.920 (0.004)	0.964 (0.002)	0.673 (0.011)	0.938 (0.006)	0.920 (0.007)	0.932 (0.006)
smooth effect	Poisson	smooth	100	default	smooth	61	default	0.671 (0.005)	0.943 (0.003)	0.922 (0.004)	0.965 (0.002)	0.664 (0.012)	0.946 (0.005)	0.931 (0.006)	0.942 (0.006)
smooth effect	binary	indep.	100	default	smooth	61	default	0.988 (0.001)	0.982 (0.001)	0.969 (0.002)	0.976 (0.002)	0.952 (0.006)	0.930 (0.007)	0.922 (0.008)	0.929 (0.008)
smooth effect	binary	rough	100	default	smooth	61	default	0.791 (0.005)	0.949 (0.003)	0.885 (0.008)	0.958 (0.004)	0.729 (0.013)	0.925 (0.008)	0.834 (0.014)	0.889 (0.011)
smooth effect	binary	smooth	100	default	smooth	61	default	0.754 (0.006)	0.945 (0.003)	0.895 (0.008)	0.966 (0.003)	0.723 (0.013)	0.943 (0.006)	0.850 (0.012)	0.910 (0.009)
families	scaled t	indep.	100	default	smooth	61	default	0.955 (0.004)	0.954 (0.004)	0.947 (0.004)	0.950 (0.004)	0.956 (0.004)	0.950 (0.004)	0.946 (0.005)	0.950 (0.004)
families	scaled t	smooth	100	default	smooth	61	default	0.651 (0.012)	0.951 (0.005)	0.947 (0.005)	0.955 (0.005)	0.668 (0.012)	0.950 (0.005)	0.944 (0.006)	0.952 (0.005)
families	beta	indep.	100	default	smooth	61	default	0.958 (0.004)	0.950 (0.004)	0.940 (0.005)	0.946 (0.005)	0.959 (0.004)	0.950 (0.004)	0.939 (0.005)	0.946 (0.004)
families	beta	smooth	100	default	smooth	61	default	0.642 (0.012)	0.945 (0.005)	0.935 (0.006)	0.948 (0.005)	0.656 (0.012)	0.946 (0.005)	0.931 (0.006)	0.943 (0.006)
families	negative binomial	indep.	100	default	smooth	61	default	0.954 (0.004)	0.950 (0.005)	0.935 (0.005)	0.943 (0.005)	0.965 (0.004)	0.957 (0.005)	0.948 (0.005)	0.953 (0.005)
families	negative binomial	smooth	100	default	smooth	61	default	0.647 (0.012)	0.942 (0.006)	0.924 (0.007)	0.943 (0.006)	0.658 (0.012)	0.943 (0.005)	0.926 (0.007)	0.940 (0.006)
families	negative binomial	indep.	100	default	smooth	61	default	0.896 (0.007)	0.950 (0.005)	0.923 (0.006)	0.941 (0.005)	0.881 (0.007)	0.948 (0.005)	0.927 (0.006)	0.939 (0.005)
families	negative binomial	smooth	100	default	smooth	61	default	0.554 (0.011)	0.942 (0.006)	0.915 (0.007)	0.948 (0.006)	0.541 (0.012)	0.942 (0.005)	0.909 (0.008)	0.935 (0.006)
exact AR(1)	Gaussian	AR(1)	100	default	smooth	61	default	0.636 (0.011)	0.955 (0.005)	0.939 (0.005)	0.949 (0.005)	0.645 (0.012)	0.937 (0.006)	0.921 (0.006)	0.933 (0.006)
sign-changing	Gaussian	sign-chg.	100	default	smooth	61	default	0.633 (0.011)	0.951 (0.005)	0.905 (0.007)	0.941 (0.006)	0.645 (0.012)	0.944 (0.005)	0.902 (0.007)	0.935 (0.006)
sign-changing	Poisson	sign-chg.	100	default	smooth	61	default	0.656 (0.011)	0.947 (0.005)	0.885 (0.009)	0.937 (0.006)	0.673 (0.011)	0.939 (0.006)	0.882 (0.008)	0.918 (0.007)
sign-changing	binary	sign-chg.	100	default	smooth	61	default	0.768 (0.010)	0.934 (0.006)	0.844 (0.012)	0.918 (0.007)	0.790 (0.010)	0.937 (0.006)	0.810 (0.012)	0.904 (0.008)
misreg. (AR(1))	Gaussian	misreg.	100	high	rough	61	default	0.732 (0.011)	0.949 (0.006)	0.943 (0.007)	0.950 (0.006)	0.692 (0.013)	0.928 (0.006)	0.923 (0.006)	0.930 (0.006)
heteroskedastic	Gaussian	var(t)	40	default	smooth	61	default	0.630 (0.013)	0.939 (0.006)	0.922 (0.008)	0.937 (0.007)	0.632 (0.012)	0.944 (0.006)	0.925 (0.007)	0.945 (0.006)
heteroskedastic	Gaussian	var(t)	100	default	smooth	61	default	0.650 (0.011)	0.951 (0.005)	0.943 (0.005)	0.952 (0.005)	0.661 (0.012)	0.947 (0.005)	0.932 (0.006)	0.943 (0.006)
heteroskedastic	Gaussian	var(z)	40	default	smooth	61	default	0.618 (0.014)	0.930 (0.007)	0.913 (0.008)	0.935 (0.007)	0.490 (0.014)	0.926 (0.007)	0.898 (0.009)	0.931 (0.007)
heteroskedastic	Gaussian	var(z)	100	default	smooth	61	default	0.647 (0.012)	0.945 (0.005)	0.936 (0.006)	0.946 (0.005)	0.494 (0.013)	0.938 (0.006)	0.920 (0.007)	0.945 (0.005)
heteroskedastic	Gaussian	var(t,z)	40	default	smooth	61	default	0.629 (0.014)	0.935 (0.006)	0.917 (0.007)	0.936 (0.007)	0.502 (0.014)	0.925 (0.007)	0.893 (0.009)	0.930 (0.007)
heteroskedastic	Gaussian	var(t,z)	100	default	smooth	61	default	0.655 (0.012)	0.947 (0.005)	0.939 (0.006)	0.949 (0.005)	0.510 (0.013)	0.940 (0.005)	0.920 (0.006)	0.944 (0.006)
low-rank covariate	Gaussian	indep.	100	default	smooth	61	default	0.956 (0.004)	0.956 (0.004)	0.945 (0.005)	0.950 (0.005)	0.960 (0.004)	0.950 (0.005)	0.946 (0.005)	0.951 (0.004)
low-rank covariate	Gaussian	smooth	100	default	smooth	61	default	0.642 (0.011)	0.949 (0.005)	0.939 (0.006)	0.950 (0.005)	0.655 (0.012)	0.949 (0.005)	0.934 (0.006)	0.946 (0.006)
low-rank covariate	Poisson	indep.	100	default	smooth	61	default	0.952 (0.005)	0.952 (0.005)	0.937 (0.005)	0.942 (0.005)	0.963 (0.004)	0.955 (0.005)	0.948 (0.005)	0.950 (0.005)
low-rank covariate	Poisson	smooth	100	default	smooth	61	default	0.667 (0.012)	0.947 (0.006)	0.928 (0.007)	0.948 (0.006)	0.663 (0.012)	0.945 (0.005)	0.932 (0.006)	0.944 (0.006)
low-rank covariate	binary	indep.	100	default	smooth	61	default	0.942 (0.006)	0.923 (0.007)	0.917 (0.007)	0.921 (0.007)	0.944 (0.007)	0.922 (0.008)	0.910 (0.009)	0.917 (0.009)
low-rank covariate	binary	smooth	100	default	smooth	61	default	0.702 (0.013)	0.922 (0.007)	0.868 (0.011)	0.913 (0.008)	0.731 (0.013)	0.940 (0.007)	0.867 (0.011)	0.915 (0.008)
basis size	Gaussian	indep.	100	default	smooth	61	2× finer knots	0.968 (0.003)	0.965 (0.003)	0.955 (0.004)	0.961 (0.004)	0.971 (0.003)	0.963 (0.003)	0.952 (0.004)	0.960 (0.004)

Table 2: Coverage (MC SE) of $\alpha(t)$ and $\gamma(t)$ per setting; for the smooth-effect settings, the first group of columns holds $f(x,t)$. Interval labels as in the previous table. *(continued)*

block	family	error	G	signal	truth	D	basis	REML model	REML+CL2	NCV+CL2	NCV wid.	REML model	REML+CL2	NCV+CL2	NCV wid.
basis size	Gaussian	indep.	100	default	smooth	61	4× finer knots	0.978 (0.002)	0.975 (0.002)	0.966 (0.003)	0.971 (0.003)	0.976 (0.002)	0.973 (0.003)	0.960 (0.003)	0.968 (0.003)
basis size	Gaussian	smooth	100	default	smooth	61	2× finer knots	0.697 (0.010)	0.950 (0.005)	0.941 (0.005)	0.956 (0.005)	0.703 (0.011)	0.950 (0.005)	0.932 (0.006)	0.949 (0.005)
basis size	Gaussian	smooth	100	default	smooth	61	4× finer knots	0.739 (0.010)	0.954 (0.004)	0.944 (0.005)	0.960 (0.004)	0.734 (0.011)	0.950 (0.005)	0.934 (0.006)	0.955 (0.005)
basis size	Poisson	indep.	100	default	smooth	61	2× finer knots	0.968 (0.003)	0.964 (0.004)	0.946 (0.005)	0.955 (0.004)	0.971 (0.003)	0.964 (0.003)	0.951 (0.004)	0.955 (0.004)
basis size	Poisson	indep.	100	default	smooth	61	4× finer knots	0.978 (0.002)	0.975 (0.003)	0.958 (0.004)	0.968 (0.003)	0.978 (0.002)	0.973 (0.003)	0.959 (0.004)	0.964 (0.003)
basis size	Poisson	smooth	100	default	smooth	61	2× finer knots	0.716 (0.010)	0.949 (0.005)	0.927 (0.007)	0.956 (0.005)	0.723 (0.010)	0.950 (0.005)	0.928 (0.006)	0.943 (0.005)
basis size	Poisson	smooth	100	default	smooth	61	4× finer knots	0.752 (0.010)	0.950 (0.005)	0.931 (0.007)	0.963 (0.005)	0.769 (0.010)	0.953 (0.005)	0.929 (0.006)	0.947 (0.005)
basis size	binary	indep.	100	default	smooth	61	2× finer knots	0.961 (0.005)	0.944 (0.006)	0.933 (0.006)	0.938 (0.006)	0.958 (0.006)	0.936 (0.007)	0.921 (0.008)	0.926 (0.008)
basis size	binary	indep.	100	default	smooth	61	4× finer knots	0.969 (0.004)	0.956 (0.005)	0.937 (0.007)	0.945 (0.006)	0.966 (0.005)	0.949 (0.006)	0.928 (0.008)	0.934 (0.007)
basis size	binary	smooth	100	default	smooth	61	2× finer knots	0.728 (0.013)	0.915 (0.007)	0.866 (0.011)	0.921 (0.008)	0.762 (0.012)	0.942 (0.006)	0.865 (0.011)	0.923 (0.009)
basis size	binary	smooth	100	default	smooth	61	4× finer knots	0.713 (0.014)	0.899 (0.008)	0.877 (0.011)	0.936 (0.006)	0.779 (0.013)	0.946 (0.006)	0.872 (0.011)	0.929 (0.008)

S1.2 Lower tails, number of curves, signal and truth roughness

In the main-factorial settings with dependent errors, the worst setting's 5% quantile of pointwise coverage (lower tail) for the mean and $\beta(s,t)$ is 0.91 for REML + CL2 at $G = 100$ and 0.89 at $G = 40$; for model-based intervals, it is as low as 0.54. With 40 instead of 100 curves, REML + CL2 loses 1.1–1.8 pp of coverage (means per family and estimand; coverage by G in the figure under *Twenty curves*), and NCV + CL2 loses 1.0–6.9 pp. Signal strength changes REML + CL2's coverage of $\beta(s,t)$ little ($G = 100$, smooth and independent errors; last figure of this section); NCV + CL2's coverage with the rough truth and smooth errors rises with the signal. Truth roughness matters for NCV + CL2: with the rough truth and smooth errors, it covers β at 0.80–0.90 (smooth truth 0.90–0.94), and REML + CL2 covers it at 0.94–0.95.

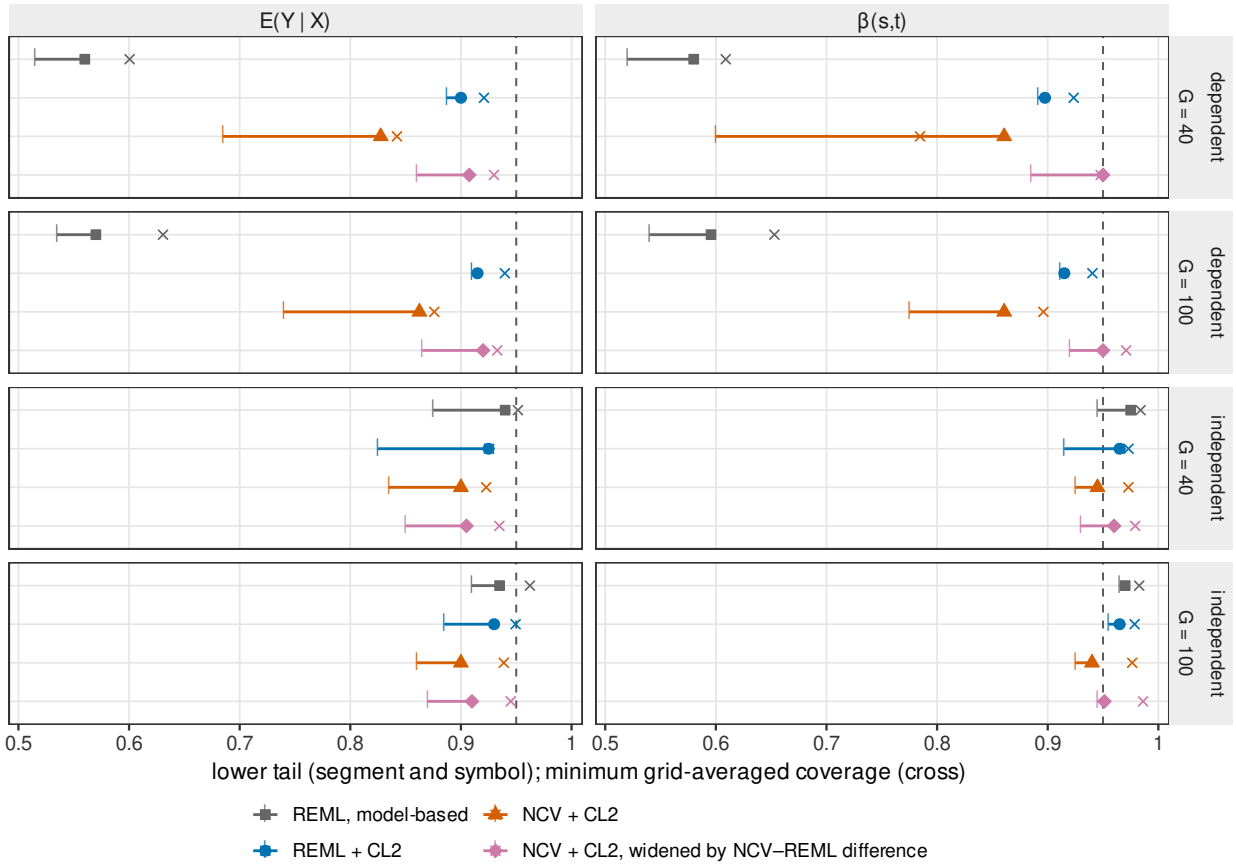


Figure 1: Lower tail (5% quantile of pointwise coverage) by dependence, G, estimand and interval, main factorial settings. Symbol: median over settings; segment: down to the minimum over settings; cross: minimum grid-averaged coverage. Dashed: nominal 0.95 for coverage. Values: S7.

Twenty curves. The main factorial was also run with $G = 20$, the first 20 curves of each replicate, so the three numbers of curves share their draws (figure below; means over each family's settings for coverage, medians for relative error and detection). Under dependent errors, REML + CL2 covers the mean, β and $\gamma(t)$ at 0.89–0.92 with 20 curves, 0.92–0.94 with 40 and 0.94–0.95 with 100, and $\alpha(t)$ at 0.85–0.91 with 20. NCV + CL2 falls to 0.78–0.90 with 20 curves, the model-based REML intervals to 0.58–0.77. Under independent errors, every interval undercovers binary $\alpha(t)$ and $\gamma(t)$ with 20 curves, the model-based one included (0.88–0.93). REML's median relative error for β under dependence is 0.86–2.43 with 20 curves, NCV's 0.29–0.74; REML + CL2 detects 0.16–0.45 of the clearly non-zero grid points of β, NCV + CL2 0.40–0.84.

With the Satterthwaite critical value (table below; default signal, as for 40 and 100 curves), REML + CL2 covers

the mean, β and γ at 0.94–0.96 under dependence with 20 curves, against 0.89–0.92 with the normal critical value and 0.91–0.93 with t_{G-1}, at 1.19–1.33 times the width with the normal critical value. Binary $\alpha(t)$ stays at 0.90 under dependence and 0.84 under independent errors, and, under independent Gaussian and Poisson errors, the intervals overcover (0.97–1.00).

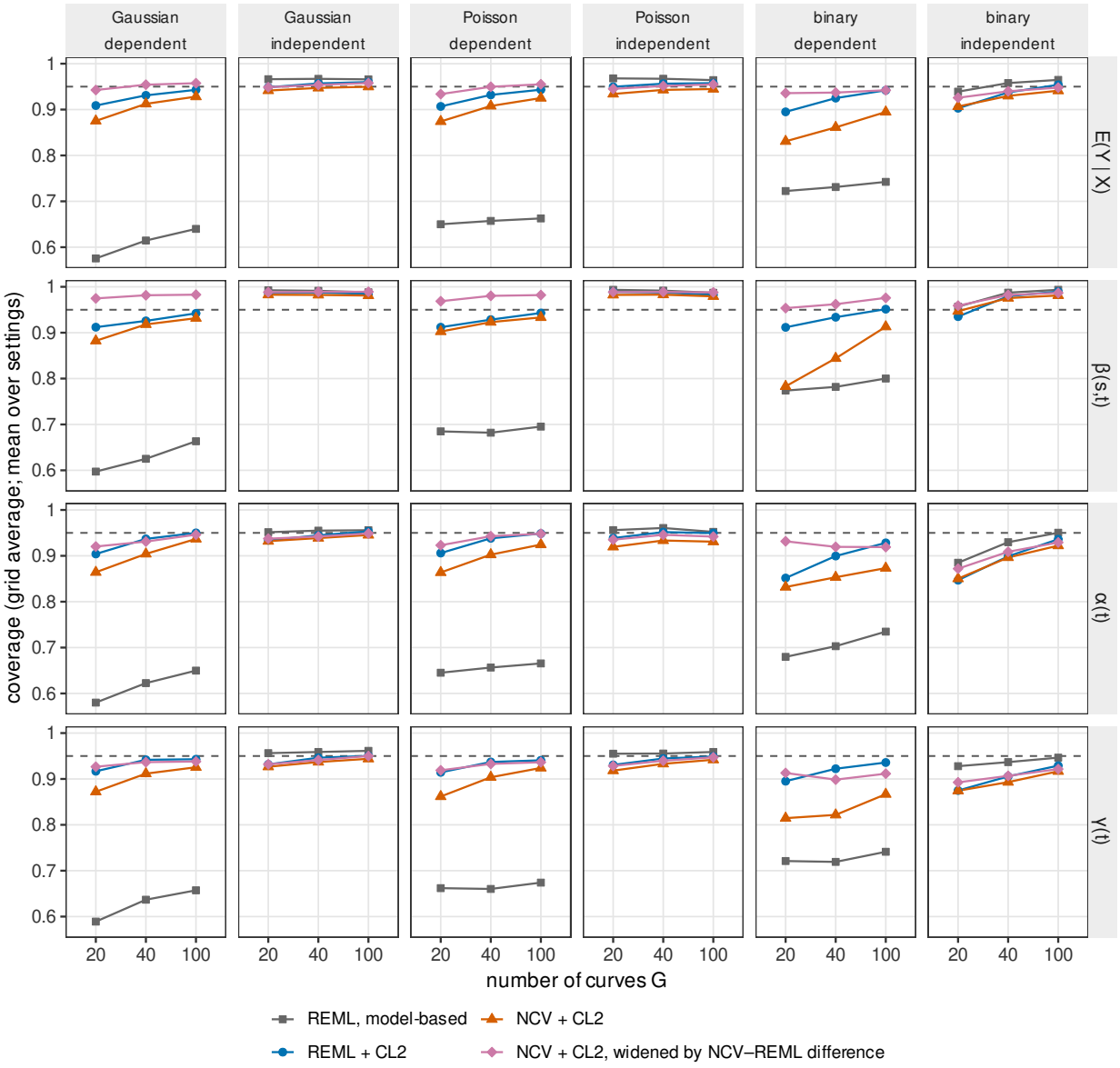


Figure 2: Main factorial by number of curves G = 20, 40, 100 (all signal levels; dependent: rough stationary and smooth errors): grid-averaged coverage (mean over settings), by family, dependence, estimand and interval. Dashed: nominal 0.95. Relative error, detection rate and all values: S7.

Table 3: REML + CL2 with the normal (z), t_{G-1} and Satterthwaite critical values by number of curves, main factorial at the default signal: grid-averaged coverage (mean over the settings of each family and dependence class), with the 5% quantile of pointwise coverage (q05) and the width relative to z of the Satterthwaite interval with 20 curves.

family	errors	estimand	z, 20	t(G-1), 20	Satt., 20	q05 Satt., 20	width Satt./z, 20	z, 40	Satt., 40	z, 100	Satt., 100
Gaussian	independent	E(Y | X)	0.952	0.964	0.982	0.965	1.24	0.958	0.973	0.961	0.967
Gaussian	independent	β(s,t)	0.989	0.993	0.998	0.990	1.24	0.989	0.995	0.986	0.989
Gaussian	independent	α(t)	0.939	0.952	0.963	0.915	1.13	0.947	0.956	0.956	0.959
Gaussian	independent	γ(t)	0.937	0.950	0.974	0.955	1.25	0.951	0.969	0.950	0.959
Gaussian	dependent	E(Y | X)	0.909	0.927	0.958	0.935	1.25	0.931	0.953	0.943	0.951
Gaussian	dependent	β(s,t)	0.913	0.930	0.961	0.938	1.26	0.926	0.950	0.942	0.951
Gaussian	dependent	α(t)	0.905	0.922	0.945	0.911	1.17	0.938	0.952	0.951	0.953
Gaussian	dependent	γ(t)	0.918	0.934	0.964	0.948	1.25	0.942	0.961	0.943	0.953
Poisson	independent	E(Y | X)	0.949	0.961	0.981	0.960	1.27	0.958	0.974	0.960	0.967
Poisson	independent	β(s,t)	0.990	0.993	0.998	0.990	1.24	0.989	0.995	0.986	0.989
Poisson	independent	α(t)	0.937	0.953	0.963	0.912	1.13	0.953	0.961	0.951	0.954
Poisson	independent	γ(t)	0.930	0.943	0.972	0.953	1.26	0.948	0.967	0.955	0.965
Poisson	dependent	E(Y | X)	0.908	0.925	0.956	0.930	1.33	0.932	0.954	0.944	0.953
Poisson	dependent	β(s,t)	0.912	0.929	0.960	0.935	1.25	0.928	0.953	0.943	0.953
Poisson	dependent	α(t)	0.904	0.923	0.941	0.891	1.15	0.934	0.945	0.949	0.952
Poisson	dependent	γ(t)	0.915	0.931	0.962	0.946	1.26	0.940	0.963	0.941	0.951
binary	independent	E(Y | X)	0.879	0.898	0.927	0.810	1.21	0.925	0.946	0.949	0.957
binary	independent	β(s,t)	0.898	0.911	0.933	0.800	1.23	0.973	0.982	0.990	0.993
binary	independent	α(t)	0.804	0.825	0.844	0.613	1.11	0.859	0.872	0.924	0.927
binary	independent	γ(t)	0.850	0.872	0.914	0.833	1.25	0.890	0.918	0.923	0.932
binary	dependent	E(Y | X)	0.895	0.915	0.945	0.913	1.19	0.924	0.946	0.940	0.949
binary	dependent	β(s,t)	0.910	0.928	0.958	0.935	1.25	0.932	0.954	0.951	0.960
binary	dependent	α(t)	0.856	0.879	0.897	0.821	1.14	0.894	0.909	0.923	0.928
binary	dependent	γ(t)	0.895	0.914	0.950	0.919	1.26	0.921	0.945	0.931	0.941

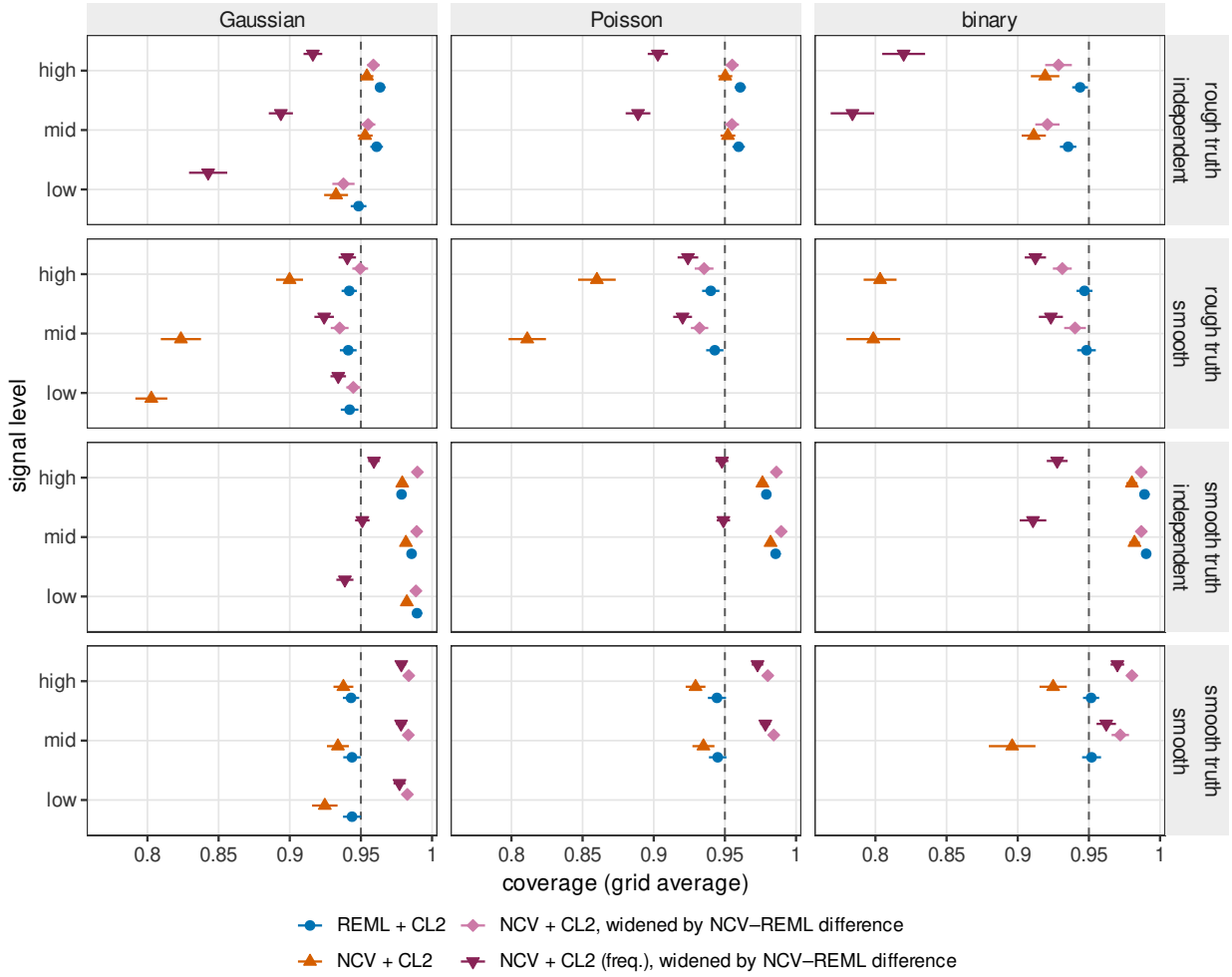


Figure 3: Coverage of $\beta(s,t)$ by signal level and truth roughness, $G = 100$, smooth and independent errors. Bars: ±2 MC SE; dashed: nominal 0.95.

S1.3 Error structures and extended families

From $D = 61$ to $D = 241$ grid points per curve, the coverage of $\beta(s,t)$ by model-based intervals drops by 30.7–36.9 pp under dependent errors, while REML + CL2 changes by -1.3 to 0.0 pp. In the heteroskedastic Gaussian settings, a variance profile along t changes no interval, while variance that depends on the scalar covariate breaks the model-based intervals for $\gamma(t)$ (0.49–0.51) but not REML + CL2 (0.93–0.94). Under misregistration, REML + CL2 covers the Gaussian estimands at 0.93–0.94; for misregistered Poisson responses, every interval undercovers the mean (REML + CL2 0.90–0.91). Under sign-changing error correlation ($G = 100$), model-based intervals cover 0.65–0.88 and REML + CL2 0.94–0.96. In the extended families (scaled t, Beta and negative-binomial responses) under smooth errors at $G = 100$, model-based intervals cover 0.63–0.67 and REML + CL2 0.94–0.95.

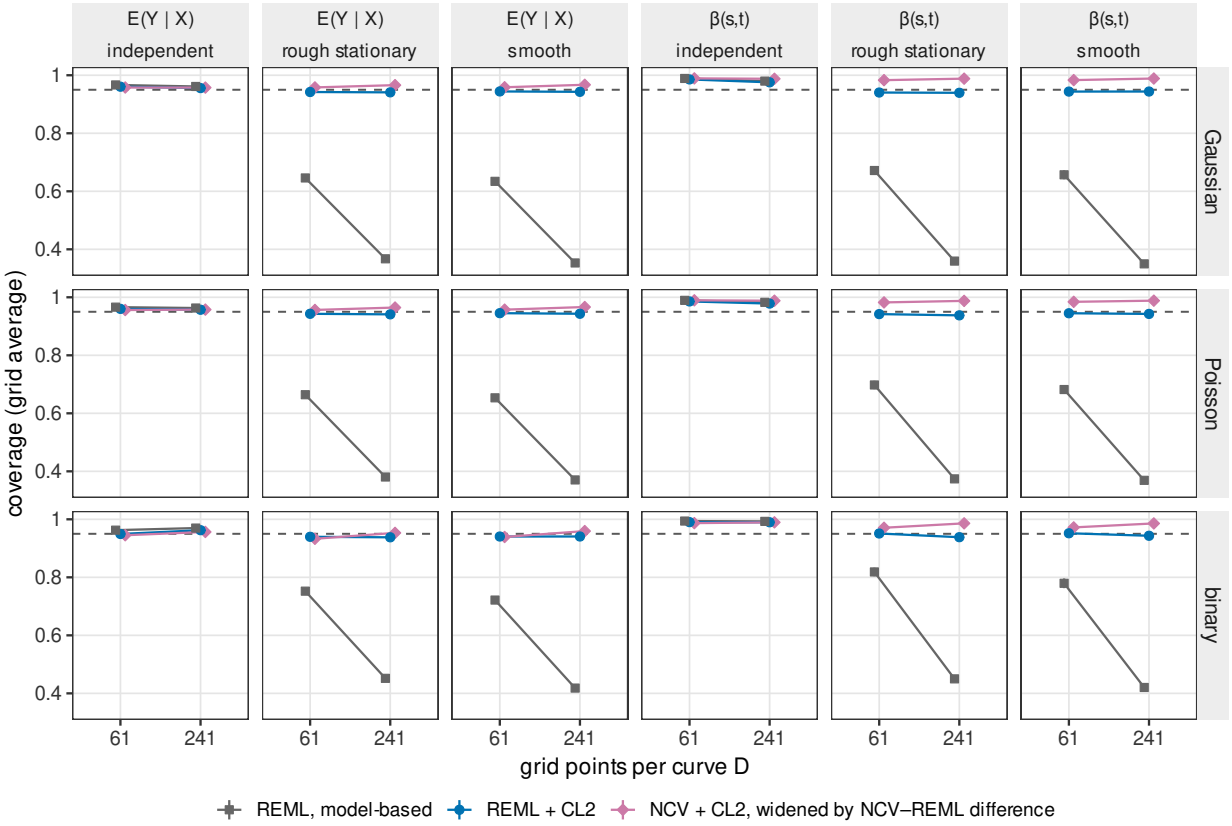


Figure 4: Coverage of $\beta(s,t)$ and the mean at $D = 61$ and $D = 241$ grid points per curve, $G = 100$, by family (rows), estimand and errors (columns). Bars: ±2 MC SE; dashed: nominal 0.95.

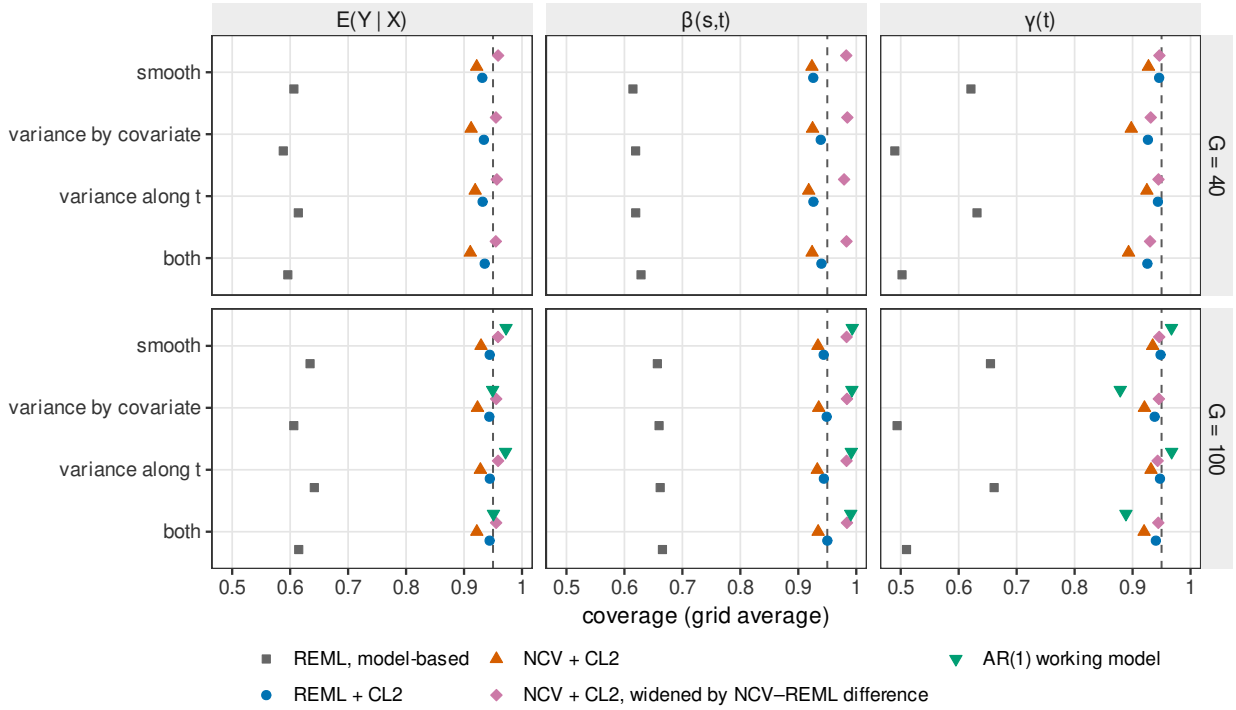


Figure 5: Heteroskedastic Gaussian errors next to the homoskedastic smooth process: grid-averaged coverage per interval. Variance along t, by the covariate, or both. The AR(1) working model was fitted at $G = 100$ only. Dashed: nominal 0.95. Values: S7.

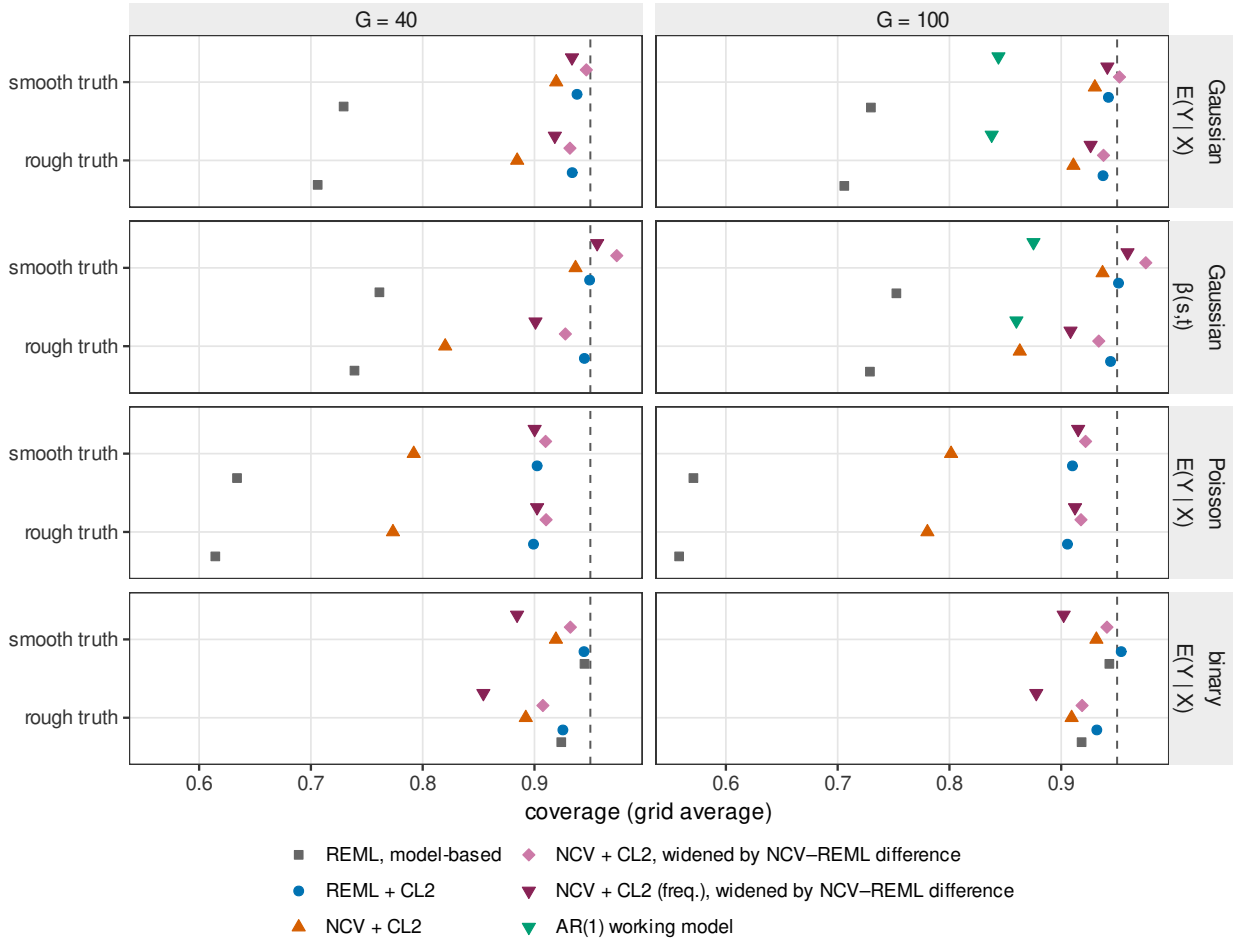


Figure 6: Misregistration settings (high signal): grid-averaged coverage of the observed-clock estimands per interval, by family, estimand, truth and G; binary and Poisson settings score the mean only. The AR(1) working model was fitted at $G = 100$ only. Dashed: nominal 0.95. Values: S7.

Table 4: Sign-changing error correlation, $G = 100$: coverage per interval.

family	error	estimand	AR(1) working model	NCV + CL2	NCV + CL2, widened by NCV–REML difference	REML + CL2	REML, model-based
binary	sign-chg.	β(s,t)		0.927	0.977	0.959	0.885
binary	sign-chg.	E(Y | X)		0.882	0.945	0.943	0.794
Gaussian	sign-chg.	β(s,t)	0.992	0.935	0.985	0.942	0.700
Gaussian	sign-chg.	E(Y | X)	0.963	0.915	0.957	0.942	0.646
Poisson	sign-chg.	β(s,t)		0.935	0.986	0.942	0.728
Poisson	sign-chg.	E(Y | X)		0.905	0.951	0.941	0.670

Table 5: Extended families at $G = 100$: coverage and NCV/REML MSE ratio (95% bootstrap interval). Negative-binomial data are also fitted as Poisson.

family	fitted family	error	estimand	REML, model-based	REML + CL2	NCV + CL2	NCV + CL2, widened by NCV–REML difference	MSE ratio NCV/REML
beta	beta	smooth	β(s,t)	0.659	0.941	0.929	0.982	0.18 [0.15, 0.22]
beta	beta	smooth	E(Y | X)	0.635	0.941	0.924	0.956	0.70 [0.68, 0.72]
beta	beta	indep.	β(s,t)	0.989	0.984	0.979	0.988	0.75 [0.65, 0.91]
beta	beta	indep.	E(Y | X)	0.966	0.958	0.946	0.956	0.98 [0.96, 0.99]
negative binomial	negative binomial	smooth	β(s,t)	0.669	0.943	0.932	0.983	0.18 [0.14, 0.21]
negative binomial	Poisson	smooth	β(s,t)	0.531	0.938	0.913	0.984	0.09 [0.07, 0.11]
negative binomial	negative binomial	smooth	E(Y | X)	0.641	0.943	0.925	0.956	0.69 [0.64, 0.74]
negative binomial	Poisson	smooth	E(Y | X)	0.529	0.939	0.913	0.957	0.55 [0.52, 0.59]
negative binomial	negative binomial	indep.	β(s,t)	0.991	0.988	0.982	0.990	0.89 [0.75, 1.05]
negative binomial	Poisson	indep.	β(s,t)	0.945	0.972	0.962	0.982	0.53 [0.49, 0.59]
negative binomial	negative binomial	indep.	E(Y | X)	0.968	0.962	0.950	0.958	1.06 [1.03, 1.09]
negative binomial	Poisson	indep.	E(Y | X)	0.898	0.952	0.934	0.950	0.90 [0.87, 0.93]
scaled t	scaled t	smooth	β(s,t)	0.671	0.944	0.940	0.985	0.22 [0.18, 0.26]
scaled t	scaled t	smooth	E(Y | X)	0.644	0.947	0.938	0.962	0.71 [0.70, 0.74]
scaled t	scaled t	indep.	β(s,t)	0.988	0.985	0.982	0.990	0.60 [0.58, 0.63]
scaled t	scaled t	indep.	E(Y | X)	0.965	0.960	0.950	0.959	0.95 [0.94, 0.97]

S1.4 Studentized errors

The studentized error $z = (\text{estimate} - \text{truth})/\text{SE}$ of the REML fit, pooled over grid points and replicates, is descriptive: an SD above one with mean near zero indicates standard errors that are too small; a shifted mean indicates bias. Pooling cannot separate a global scale error from local biases of opposite sign. In the main-factorial settings with dependent errors at $G = 100$, the SD of z with the CL2 standard error is 0.99–1.03 for $\beta(s,t)$ (model-based standard error: 1.48–2.12).

The intercept $\alpha(t)$ of binary responses behaves differently. Under independent errors at $G = 40$, its studentized CL2 error has mean 0.04 and SD 1.28 (model-based standard error: mean 0.06, SD 1.15). These values are averages over the two signal levels: the SD is 1.49 at the default signal and 1.06 at the high signal. These averages are compatible with a CL2 standard error that is too small for this term, but they do not exclude local bias. Under dependence, the studentized error is additionally shifted (mean -0.45 to -0.39 at $G = 40$ and -0.24 to -0.23 at $G = 100$, per dependent error process, averaged over the signal levels; SD 1.06 at $G = 100$).

The covariance-components runs (Appendix A.6 of the paper) add a diagnostic. Fixing the smoothing parameters at their typical value raises the coverage of binary $\gamma(t)$ under independent errors at $G = 40$ from 0.89 to 0.93, and binary $\alpha(t)$ under smooth errors from 0.88 to 0.92; binary $\alpha(t)$ under independent errors stays at 0.87 (0.86 with selected smoothing parameters). For $\gamma(t)$, and for $\alpha(t)$ under dependence, a large part of the shortfall therefore looks like variability of the selected smoothing parameters, which CL2 ignores; for $\alpha(t)$ under independent errors, it remains unexplained. Satterthwaite degrees of freedom widen these intervals only slightly (Appendix A.7 of the paper), so they cannot compensate for a standard error that is too small by about a fifth.

Table 6: Pooled studentized errors of the REML fit with the CL2 and the model-based standard error: mean, SD and share of $|z| > 1.96$ (means over settings), per estimand; the $\beta(s,t)$ columns hold $f(x,t)$ in the smooth-effect settings. Mean only for Gaussian responses.

						E(Y | X)						β(s,t) or f(x,t)						α(t)						γ(t)					
						CL2			model			CL2			model			CL2			model			CL2			model		
block	family	error	G	basis	covariate	mean	SD	|z|>1.96	mean	SD	|z|>1.96	mean	SD	|z|>1.96	mean	SD	|z|>1.96	mean	SD	|z|>1.96	mean	SD	|z|>1.96	mean	SD	|z|>1.96	mean	SD	|z|>1.96
exact AR(1)	Gaussian	AR(1)	100	default	rich	-0.017	1.031	0.058	-0.039	2.219	0.375	0.003	1.032	0.060	0.010	2.130	0.349	-0.039	0.997	0.045	-0.084	2.155	0.364	-0.042	1.022	0.063	-0.098	2.167	0.355
basis size	binary	smooth	100	2× finer knots	rich							0.003	0.962	0.042	0.005	1.582	0.211	-0.382	1.062	0.085	-0.561	1.737	0.272	-0.016	1.014	0.058	-0.017	1.693	0.238
basis size	binary	smooth	100	4× finer knots	rich							0.001	0.959	0.040	0.004	1.555	0.205	-0.594	1.043	0.101	-0.869	1.603	0.287	0.003	1.007	0.054	0.017	1.643	0.221
basis size	binary	indep.	100	2× finer knots	rich							0.010	0.626	0.003	0.009	0.592	0.002	0.025	1.014	0.056	0.044	0.937	0.039	-0.047	1.056	0.064	-0.049	0.985	0.042
basis size	binary	indep.	100	4× finer knots	rich							0.007	0.525	0.001	0.007	0.498	0.000	0.026	0.962	0.044	0.043	0.892	0.031	-0.045	1.011	0.051	-0.046	0.945	0.034
basis size	Gaussian	smooth	100	2× finer knots	rich	-0.013	1.010	0.053	-0.027	2.018	0.327	0.000	1.015	0.055	0.002	2.060	0.329	-0.023	0.999	0.050	-0.044	1.918	0.303	-0.022	0.999	0.050	-0.046	1.896	0.297
basis size	Gaussian	smooth	100	4× finer knots	rich	-0.014	0.991	0.049	-0.028	1.872	0.294	0.001	0.983	0.046	0.004	2.005	0.327	-0.031	0.982	0.046	-0.056	1.756	0.261	-0.022	0.998	0.050	-0.040	1.762	0.266
basis size	Gaussian	indep.	100	2× finer knots	rich	-0.001	0.878	0.027	-0.002	0.861	0.022	0.002	0.650	0.003	0.002	0.637	0.002	-0.007	0.936	0.035	-0.006	0.925	0.032	-0.025	0.934	0.037	-0.026	0.908	0.029
basis size	Gaussian	indep.	100	4× finer knots	rich	-0.001	0.822	0.018	-0.001	0.810	0.015	0.001	0.535	0.000	0.001	0.526	0.000	-0.006	0.884	0.025	-0.005	0.875	0.022	-0.022	0.884	0.027	-0.022	0.864	0.024
basis size	Poisson	smooth	100	2× finer knots	rich							0.001	1.004	0.052	0.000	1.929	0.298	-0.169	0.986	0.051	-0.304	1.810	0.284	-0.027	0.995	0.050	-0.046	1.804	0.277
basis size	Poisson	smooth	100	4× finer knots	rich							0.000	0.978	0.045	0.003	1.843	0.285	-0.241	0.967	0.050	-0.404	1.639	0.248	-0.019	0.983	0.047	-0.023	1.646	0.231
basis size	Poisson	indep.	100	2× finer knots	rich							0.005	0.645	0.003	0.005	0.630	0.002	0.034	0.932	0.036	0.036	0.917	0.032	-0.035	0.931	0.036	-0.036	0.903	0.029
basis size	Poisson	indep.	100	4× finer knots	rich							0.003	0.529	0.000	0.003	0.518	0.000	0.037	0.878	0.025	0.039	0.864	0.022	-0.030	0.886	0.027	-0.032	0.862	0.022
main factorial	binary	smooth	40	default	rich							0.009	1.081	0.071	0.015	1.716	0.243	-0.454	1.134	0.106	-0.657	1.902	0.323	-0.045	1.119	0.080	-0.073	1.939	0.296
main factorial	binary	smooth	100	default	rich							0.001	0.987	0.048	0.004	1.608	0.219	-0.241	1.063	0.075	-0.355	1.844	0.284	-0.019	1.029	0.061	-0.025	1.818	0.266
main factorial	binary	indep.	40	default	rich							0.011	0.835	0.021	0.010	0.767	0.013	0.044	1.276	0.101	0.064	1.149	0.070	-0.068	1.203	0.094	-0.065	1.055	0.063
main factorial	binary	indep.	100	default	rich							0.011	0.751	0.010	0.011	0.713	0.007	0.033	1.060	0.064	0.049	0.993	0.050	-0.041	1.082	0.071	-0.043	1.018	0.054
main factorial	binary	rough	40	default	rich							0.012	1.048	0.061	0.019	1.511	0.194	-0.393	1.130	0.095	-0.521	1.778	0.271	-0.062	1.113	0.076	-0.111	1.728	0.265
main factorial	binary	rough	100	default	rich							0.006	0.987	0.049	0.010	1.479	0.181	-0.232	1.058	0.068	-0.317	1.712	0.246	-0.037	1.053	0.068	-0.059	1.714	0.251
main factorial	Gaussian	smooth	40	default	rich	-0.020	1.073	0.069	-0.040	2.334	0.394	0.003	1.094	0.075	0.006	2.336	0.387	-0.041	1.055	0.064	-0.080	2.315	0.385	-0.050	1.029	0.055	-0.103	2.219	0.377
main factorial	Gaussian	smooth	100	default	rich	-0.016	1.026	0.056	-0.034	2.201	0.366	0.003	1.021	0.056	0.009	2.123	0.344	-0.032	1.003	0.051	-0.062	2.144	0.359	-0.026	1.002	0.052	-0.062	2.111	0.347
main factorial	Gaussian	indep.	40	default	rich	-0.010	0.959	0.043	-0.008	0.919	0.033	0.003	0.769	0.013	0.003	0.743	0.009	-0.027	1.014	0.055	-0.018	0.981	0.045	-0.026	1.007	0.054	-0.030	0.956	0.041
main factorial	Gaussian	indep.	100	default	rich	-0.002	0.952	0.040	-0.001	0.927	0.034	0.005	0.794	0.015	0.005	0.773	0.012	-0.011	1.008	0.047	-0.008	0.993	0.044	-0.031	0.990	0.050	-0.033	0.959	0.039
main factorial	Gaussian	rough	40	default	rich	-0.017	1.075	0.069	-0.035	2.227	0.377	0.006	1.093	0.074	0.012	2.182	0.363	-0.042	1.048	0.062	-0.081	2.214	0.369	-0.060	1.035	0.062	-0.129	2.125	0.350
main factorial	Gaussian	rough	100	default	rich	-0.016	1.032	0.058	-0.036	2.127	0.354	0.004	1.029	0.059	0.010	2.049	0.329	-0.035	1.003	0.049	-0.075	2.062	0.341	-0.046	1.022	0.062	-0.100	2.066	0.339
main factorial	Poisson	smooth	40	default	rich							0.006	1.079	0.071	0.010	2.063	0.330	-0.202	1.031	0.063	-0.383	2.080	0.353	-0.050	1.038	0.060	-0.095	2.056	0.348
main factorial	Poisson	smooth	100	default	rich							0.001	1.015	0.055	0.004	1.977	0.311	-0.112	0.997	0.052	-0.215	2.037	0.338	-0.038	1.012	0.056	-0.077	2.025	0.334
main factorial	Poisson	indep.	40	default	rich							0.009	0.769	0.012	0.008	0.741	0.008	0.030	0.991	0.048	0.036	0.960	0.039	-0.026	1.026	0.055	-0.027	0.969	0.045
main factorial	Poisson	indep.	100	default	rich							0.008	0.810	0.018	0.009	0.785	0.013	0.033	1.008	0.050	0.035	0.991	0.048	-0.043	0.997	0.051	-0.044	0.963	0.041
main factorial	Poisson	rough	40	default	rich							0.008	1.081	0.072	0.014	1.938	0.306	-0.193	1.022	0.060	-0.353	1.982	0.334	-0.071	1.066	0.066	-0.145	1.994	0.332
main factorial	Poisson	rough	100	default	rich							0.004	1.029	0.059	0.010	1.925	0.299	-0.115	1.006	0.051	-0.219	1.978	0.331	-0.065	1.028	0.063	-0.124	1.962	0.318
dense grid	binary	smooth	100	default	rich							0.003	1.026	0.057	0.013	3.578	0.580	-0.421	1.034	0.080	-1.282	3.386	0.590	-0.005	1.013	0.057	-0.028	3.410	0.555
dense grid	binary	indep.	100	default	rich							0.005	0.739	0.010	0.004	0.713	0.007	-0.007	1.004	0.049	-0.001	0.970	0.043	-0.063	1.001	0.049	-0.063	0.962	0.040
dense grid	binary	rough	100	default	rich							0.004	1.047	0.062	0.013	3.278	0.550	-0.393	1.043	0.081	-1.064	3.076	0.553	-0.029	1.036	0.059	-0.087	3.111	0.526
dense grid	Gaussian	smooth	100	default	rich	-0.011	1.028	0.057	-0.051	4.408	0.647	0.000	1.024	0.056	0.007	4.468	0.650	-0.024	1.007	0.053	-0.098	4.203	0.627	-0.016	1.010	0.053	-0.085	4.168	0.631
dense grid	Gaussian	indep.	100	default	rich	-0.011	0.970	0.044	-0.010	0.948	0.039	0.002	0.854	0.024	0.002	0.837	0.020	-0.023	0.997	0.049	-0.022	0.990	0.048	-0.027	1.009	0.057	-0.030	0.979	0.049
dense grid	Gaussian	rough	100	default	rich	-0.012	1.035	0.059	-0.055	4.186	0.633	0.002	1.038	0.060	0.013	4.293	0.641	-0.031	0.996	0.047	-0.135	3.932	0.618	-0.036	1.027	0.064	-0.162	3.984	0.603
dense grid	Poisson	smooth	100	default	rich							0.000	1.029	0.057	0.004	4.226	0.631	-0.178	0.997	0.053	-0.690	3.958	0.620	-0.020	1.007	0.053	-0.084	3.943	0.613
dense grid	Poisson	indep.	100	default	rich							0.003	0.840	0.022	0.004	0.821	0.017	0.001	0.993	0.049	0.001	0.986	0.046	-0.041	1.000	0.052	-0.042	0.965	0.043
dense grid	Poisson	rough	100	default	rich							0.002	1.045	0.062	0.010	4.092	0.626	-0.178	0.993	0.048	-0.670	3.761	0.603	-0.046	1.024	0.064	-0.178	3.783	0.589
families	beta	smooth	100	default	rich							0.002	1.031	0.059	0.007	2.112	0.341	-0.063	1.020	0.055	-0.124	2.141	0.358	-0.029	1.012	0.054	-0.070	2.096	0.344
families	beta	indep.	100	default	rich							0.004	0.801	0.016	0.005	0.774	0.011	0.021	1.023	0.050	0.024	0.995	0.042	-0.032	1.000	0.050	-0.034	0.959	0.041
families	negative binomial	smooth	100	default	rich							0.002	1.036	0.060	0.005	2.436	0.400	-0.201	1.010	0.058	-0.440	2.324	0.399	-0.029	1.020	0.057	-0.074	2.401	0.400
families	negative binomial	indep.	100	default	rich							0.008	0.825	0.020	0.009	0.882	0.032	-0.010	1.010	0.050	-0.005	1.100	0.075	-0.041	0.992	0.048	-0.050	1.097	0.077
families	scaled t	smooth	100	default	rich							0.003	1.017	0.056	0.009	2.062	0.329	-0.027	1.000	0.049	-0.053	2.090	0.349	-0.012	0.996	0.050	-0.027	2.073	0.332
families	scaled t	indep.	100	default	rich							0.003	0.802	0.015	0.003	0.784	0.012	0.004	1.000	0.046	0.006	0.992	0.045	-0.030	0.990	0.050	-0.031	0.966	0.044
heteroskedastic	Gaussian	var(z)	40	default	rich	-0.017	1.072	0.066	-0.046	2.515	0.412	0.000	1.054	0.061	0.002	2.313	0.381	-0.042	1.066	0.070	-0.084	2.331	0.382	-0.087	1.126	0.074	-0.242	3.068	0.510
heteroskedastic	Gaussian	var(z)	100	default	rich	-0.021	1.035	0.056	-0.056	2.393	0.394	0.002	1.007	0.051	0.003	2.125	0.340	-0.045	1.015	0.055	-0.105	2.154	0.353	-0.056	1.068	0.062	-0.184	2.965	0.506
heteroskedastic	Gaussian	var(t)	40	default	rich	-0.019	1.069	0.068	-0.035	2.331	0.386	0.002	1.091	0.074	0.007	2.328	0.381	-0.039	1.043	0.061	-0.070	2.292	0.370	-0.058	1.025	0.056	-0.119	2.212	0.368
heteroskedastic	Gaussian	var(t)	100	default	rich	-0.015	1.023	0.056	-0.028	2.203	0.358	0.003	1.018	0.056	0.009	2.122	0.338	-0.028	0.995	0.049	-0.046	2.133	0.350	-0.029	1.001	0.053	-0.067	2.114	0.339
heteroskedastic	Gaussian	var(t,z)	40	default	rich	-0.017	1.068	0.064	-0.043	2.506	0.404	-0.001	1.048	0.060	0.003	2.293	0.371	-0.038	1.057	0.065	-0.070	2.304	0.371	-0.090	1.122	0.075	-0.246	3.056	0.498
heteroskedastic	Gaussian	var(t,z)	100	default	rich	-0.020	1.032	0.056	-0.051	2.392	0.385	0.002	1.003	0.050	0.005	2.128	0.334	-0.043	1.008	0.053	-0.099	2.135	0.345	-0.058	1.064	0.060	-0.187	2.953	0.490
low-rank covariate	binary	smooth	100	default	lowrank							0.000	0.992	0.046	0.001	1.592	0.211	-0.230	1.082	0.078	-0.342	1.921	0.298	-0.018	1.032	0.060	-0.027	1.822	0.269
low-rank covariate	binary	indep.	100	default	lowrank							0.009	0.702	0.007	0.009	0.653	0.004	0.022	1.109	0.077	0.047	1.016	0.058	-0.051	1.118	0.078	-0.053	1.038	0.056
low-rank covariate	Gaussian	smooth	100	default	lowrank	-0.016	1.026	0.056	-0.034	2.203	0.367	0.001	1.018	0.055	0.005	2.089	0.334	-0.032	1.002	0.051	-0.062	2.143	0.358	-0.027	1.000	0.051	-0.065	2.108	0.345
low-rank covariate	Gaussian	indep.	100	default	lowrank	-0.002	0.953	0.040	-0.001	0.928	0.034	0.003	0.724	0.009	0.003	0.702	0.006	-0.011	1.005	0.044	-0.008	0.993	0.044	-0.031	0.987	0.050	-0.034	0.957	0.040

Table 6: Pooled studentized errors of the REML fit with the CL2 and the model-based standard error: mean, SD and share of $|z| > 1.96$ (means over settings), per estimand; the $\beta(s,t)$ columns hold $f(x,t)$ in the smooth-effect settings. Mean only for Gaussian responses. *(continued)*

block	family	error	G	basis	covariate	mean	SD	|z|>1.96	mean	SD	|z|>1.96	mean	SD	|z|>1.96	mean	SD	|z|>1.96	mean	SD	|z|>1.96	mean	SD	|z|>1.96	mean	SD	|z|>1.96	mean	SD	|z|>1.96
low-rank covariate	Poisson	smooth	100	default	lowrank							0.001	1.012	0.054	0.004	1.970	0.307	-0.113	0.997	0.053	-0.221	2.051	0.333	-0.033	1.004	0.055	-0.068	2.029	0.337
low-rank covariate	Poisson	indep.	100	default	lowrank							0.006	0.720	0.008	0.006	0.694	0.006	0.023	1.004	0.048	0.027	0.988	0.048	-0.044	0.981	0.045	-0.045	0.952	0.037
sign-changing	binary	sign-chg.	100	default	rich							-0.002	0.951	0.041	-0.003	1.252	0.115	-0.231	1.052	0.066	-0.296	1.587	0.232	-0.018	1.050	0.063	-0.026	1.576	0.210
sign-changing	Gaussian	sign-chg.	100	default	rich	-0.004	1.031	0.058	-0.013	2.123	0.354	0.002	1.029	0.058	0.006	1.917	0.300	-0.012	1.007	0.049	-0.030	2.160	0.367	-0.023	1.016	0.056	-0.052	2.129	0.355
sign-changing	Poisson	sign-chg.	100	default	rich							0.001	1.027	0.058	0.004	1.801	0.272	-0.117	1.015	0.053	-0.226	2.058	0.344	-0.023	1.023	0.061	-0.050	2.013	0.327
rough truth	binary	smooth	100	default	rich							0.000	1.007	0.052	0.002	1.654	0.234	-0.246	1.057	0.074	-0.362	1.833	0.282	-0.005	1.045	0.065	-0.026	1.847	0.268
rough truth	binary	indep.	100	default	rich							0.002	1.044	0.060	0.001	0.989	0.048	0.070	1.070	0.066	0.084	0.995	0.051	-0.023	1.205	0.103	-0.035	1.120	0.082
rough truth	Gaussian	smooth	100	default	rich	-0.016	1.029	0.057	-0.033	2.213	0.369	0.002	1.031	0.058	0.007	2.164	0.355	-0.030	1.004	0.051	-0.058	2.149	0.359	-0.021	1.010	0.054	-0.065	2.117	0.350
rough truth	Gaussian	indep.	100	default	rich	-0.003	0.995	0.050	-0.003	0.969	0.043	0.002	0.958	0.042	0.002	0.930	0.036	-0.011	1.008	0.046	-0.008	0.994	0.044	-0.031	1.019	0.056	-0.034	0.988	0.047
rough truth	Poisson	smooth	100	default	rich							0.001	1.032	0.059	0.004	2.029	0.326	-0.114	0.994	0.051	-0.214	2.024	0.333	-0.026	1.016	0.056	-0.069	2.017	0.326
rough truth	Poisson	indep.	100	default	rich							0.001	0.941	0.040	0.001	0.912	0.033	0.043	1.013	0.049	0.046	0.999	0.048	-0.042	1.025	0.056	-0.044	0.990	0.046
smooth effect	binary	smooth	100	default	rich							0.002	1.022	0.055	0.059	1.689	0.246							0.041	1.051	0.057	0.076	1.807	0.277
smooth effect	binary	indep.	100	default	rich							-0.059	0.815	0.018	-0.057	0.778	0.012							-0.017	1.080	0.070	-0.016	0.998	0.048
smooth effect	binary	rough	100	default	rich							-0.010	1.005	0.051	0.040	1.556	0.209							0.011	1.108	0.075	0.018	1.810	0.271
smooth effect	Gaussian	smooth	100	default	rich	-0.020	1.054	0.063	-0.041	2.133	0.350	-0.006	1.035	0.058	-0.013	2.123	0.350							0.007	1.008	0.050	0.006	2.115	0.352
smooth effect	Gaussian	indep.	100	default	rich	0.000	0.944	0.039	-0.001	0.915	0.032	-0.021	0.897	0.029	-0.022	0.878	0.024							-0.016	0.977	0.048	-0.019	0.951	0.038
smooth effect	Gaussian	rough	100	default	rich	-0.017	1.047	0.062	-0.036	2.031	0.332	0.000	1.016	0.055	0.000	2.000	0.323							-0.024	1.023	0.062	-0.057	2.062	0.339
smooth effect	Poisson	smooth	100	default	rich							0.009	1.031	0.057	0.027	2.027	0.329							0.015	1.017	0.054	0.025	2.035	0.336
smooth effect	Poisson	indep.	100	default	rich							-0.048	0.884	0.027	-0.047	0.863	0.022							-0.004	0.994	0.051	-0.008	0.963	0.041
smooth effect	Poisson	rough	100	default	rich							0.010	1.014	0.055	0.029	1.918	0.303							-0.012	1.024	0.062	-0.031	1.979	0.327
misreg.	Gaussian	misreg.	40	default	rich	0.009	1.059	0.064	-0.001	2.032	0.282	0.001	1.011	0.053	0.002	1.842	0.250	0.009	1.001	0.054	-0.005	1.847	0.240	-0.013	1.123	0.078	-0.043	2.211	0.302
misreg.	Gaussian	misreg.	100	default	rich	0.009	1.040	0.060	0.008	2.040	0.282	0.001	1.009	0.052	0.005	1.887	0.259	0.015	1.012	0.049	0.021	1.847	0.251	-0.021	1.074	0.070	-0.051	2.170	0.294
misreg. (AR(1))	Gaussian	misreg.	100	default	rich	0.007	1.051	0.063	0.006	2.122	0.294	0.001	1.026	0.056	0.004	1.961	0.271	0.012	1.024	0.051	0.017	1.943	0.268	-0.020	1.087	0.072	-0.048	2.264	0.308

S1.5 Estimation

Under dependent errors, the NCV/REML ratio of mean squared errors for $\beta(s,t)$ (geometric mean over the main-factorial settings of each error process) is 0.12–0.21 (independent errors: 0.69–1.05); for the mean 0.58–0.74 (same summary; per setting at $G = 100$, the mean's ratio is 0.59–0.80). NCV's advantage for β does not vanish under independent errors: for Gaussian responses, the per-setting ratio is 0.55–0.94 there. Part of the gap therefore reflects the smooth truths, which favor more smoothing, and not only the dependence. With the rough truth and smooth errors, the ratio for β is 0.21–0.43.

REML's excess error is variance: the squared-bias share of REML's β error in the dependent settings at $G = 100$ is 0.01, of NCV's 0.12–0.20; with the rough truth, NCV's share rises to 0.26–0.42. REML's surface EDF exceeds NCV's by a factor 1.92–2.31 under dependence (1.09–1.31 under independent errors).

Pointwise NCV, which leaves out single grid points, is no substitute for leaving out whole curves: its mean squared error for β is 2.86–4.86 times REML's under smooth and sign-changing errors (Gaussian responses, $G = 100$). With a low-rank functional covariate, the NCV/REML ratio for β is 0.09–0.12 under smooth errors ($G = 100$, default signal).

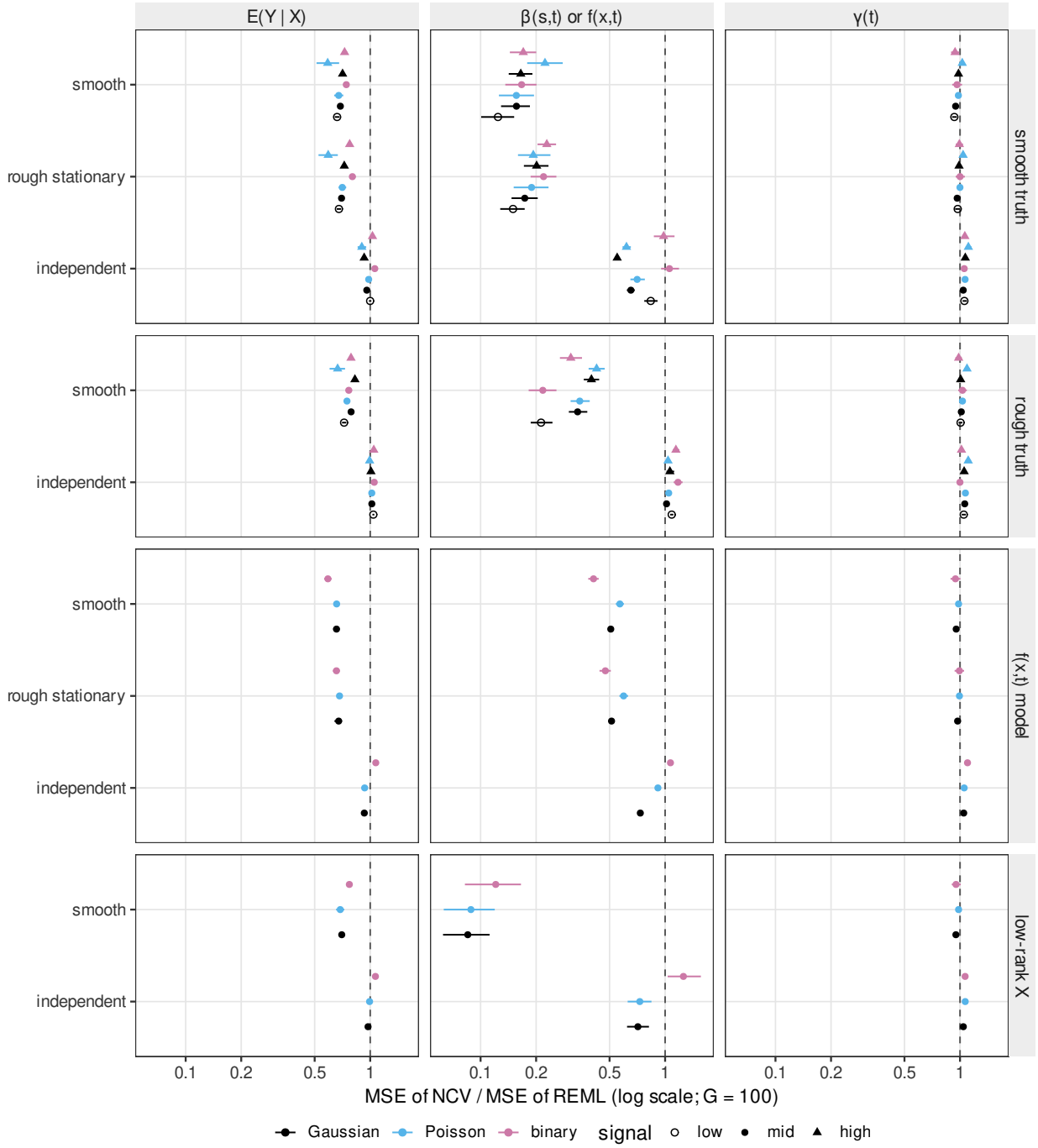


Figure 7: Mean squared error of the NCV fit relative to the REML fit (log scale) with 95% replicate-bootstrap intervals, $G = 100$, by error process, family and signal level: smooth truth, rough truth, smooth-effect model and low-rank covariate. The surface column shows $\beta(s,t)$ or $f(x,t)$. Dashed: equal error.

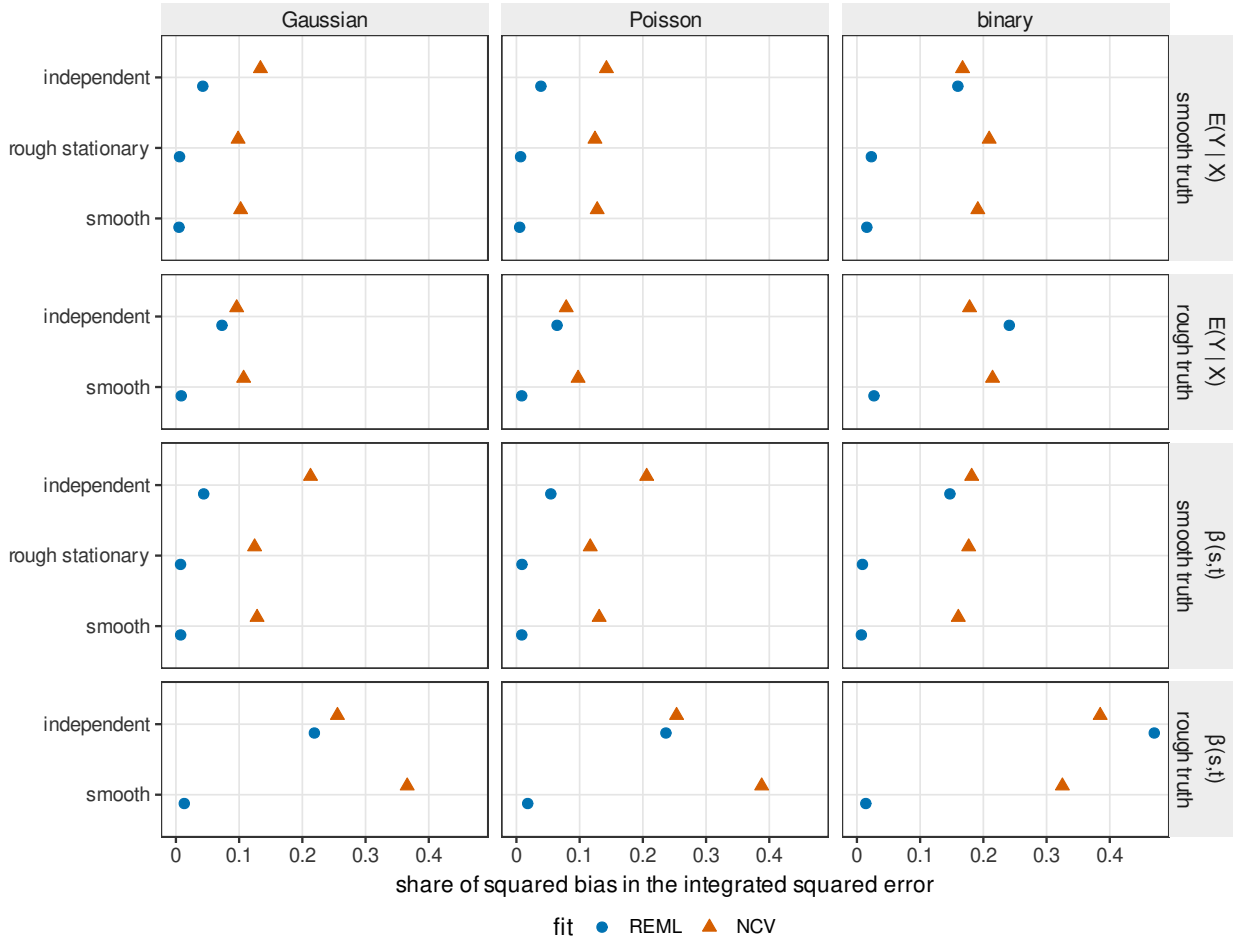


Figure 8: Share of squared bias in the integrated squared error of $\beta(s,t)$ and the mean (the remainder is variance), REML and NCV fits, $G = 100$, default signal, by family, truth and errors.

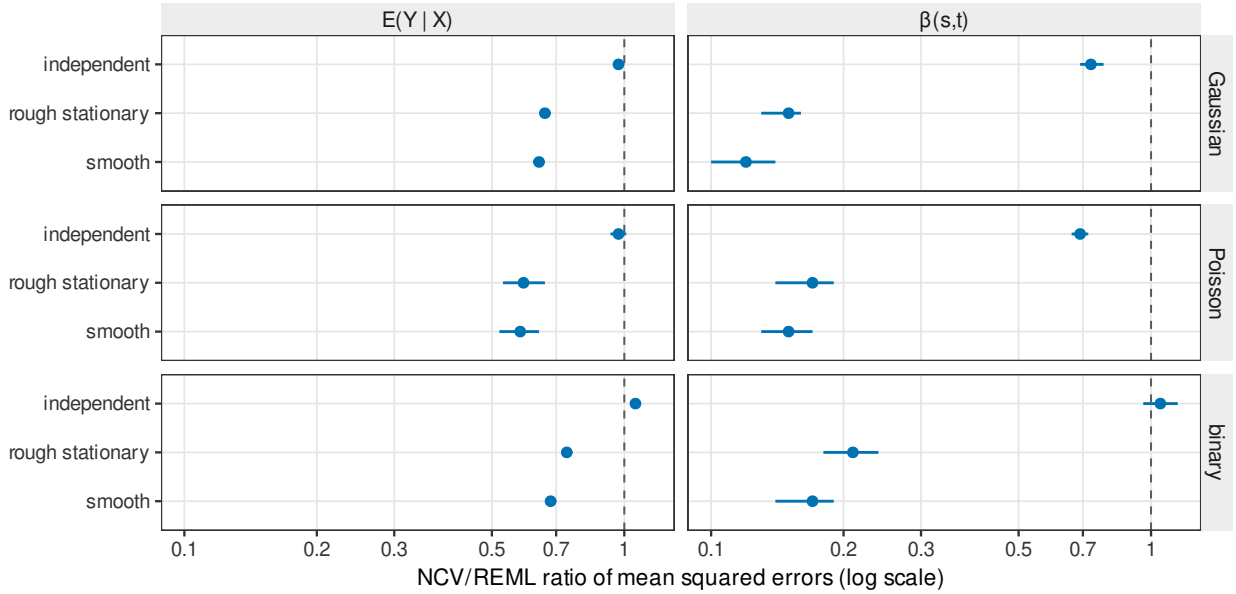


Figure 9: NCV/REML ratio of mean squared errors, geometric mean over the main-factorial settings of each error process (all G and signal levels), with 95% bootstrap intervals; log scale. Dashed: equal error. Values: S7.

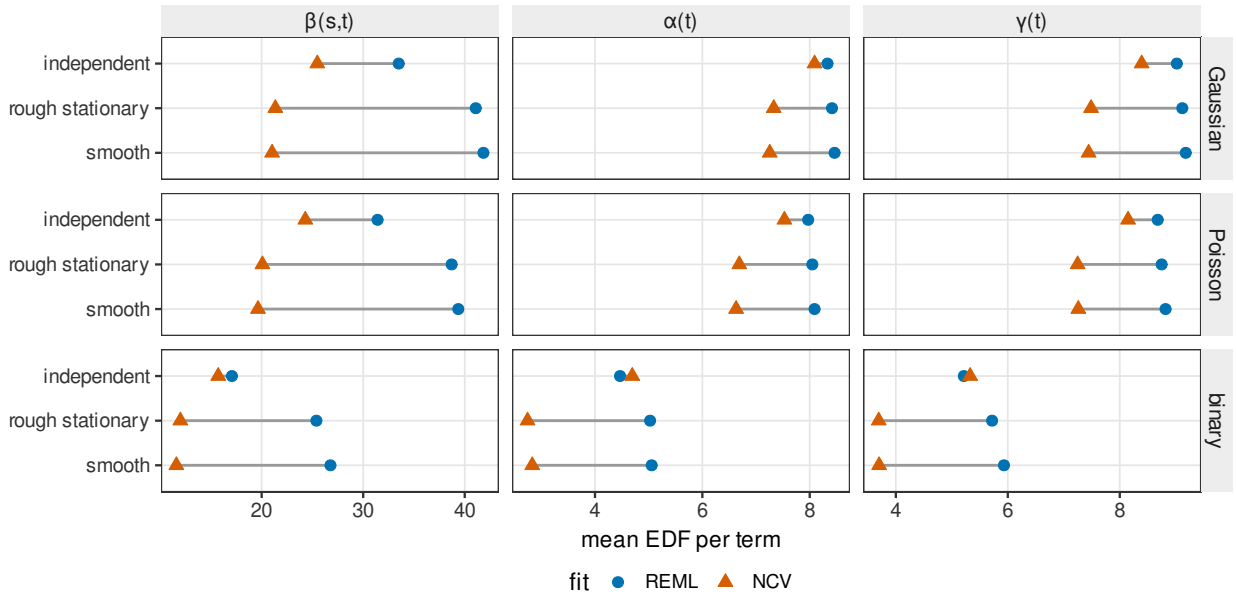


Figure 10: Mean EDF per term of the REML and NCV fits, main factorial at $G = 100$, default signal. Values and ratios: S7.

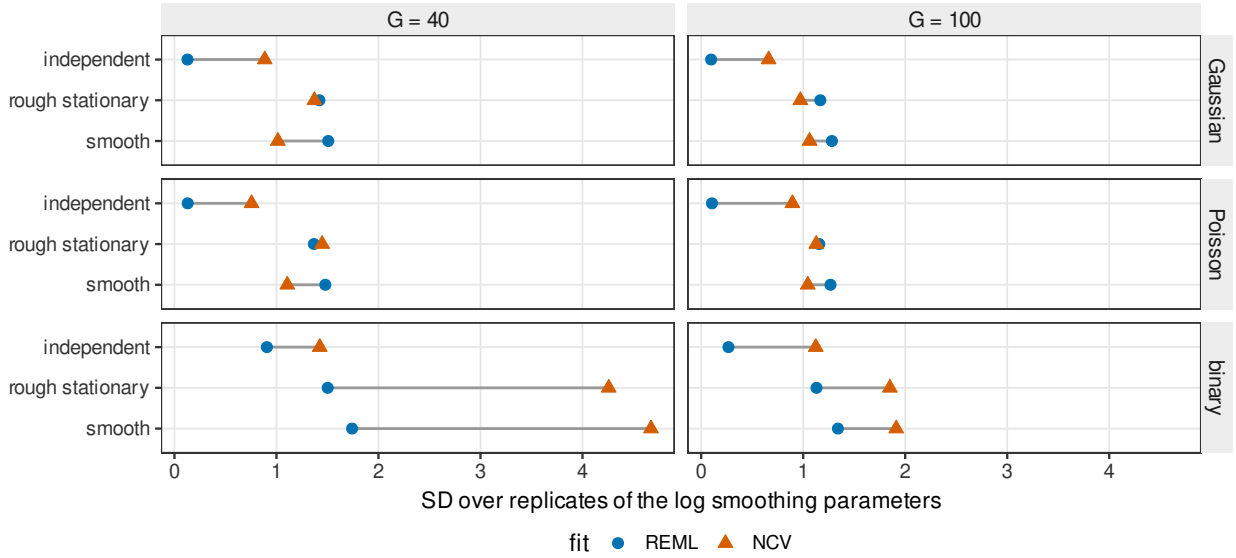


Figure 11: SD over replicates of the selected log smoothing parameters of the surface term (mean over its two penalties), main factorial at the default signal. Values: S7.

Table 7: Pointwise (non-blocked) NCV relative to REML: MSE ratio with 95% bootstrap interval, Gaussian, $G = 100$.

error	estimand	MSE pointwise NCV / REML
indep.	β(s,t)	0.65 [0.62, 0.68]
indep.	E(Y | X)	0.96 [0.95, 0.97]
smooth	β(s,t)	2.86 [2.57, 3.21]
smooth	E(Y | X)	1.34 [1.32, 1.37]
sign-chg.	β(s,t)	4.86 [4.35, 5.38]
sign-chg.	E(Y | X)	1.45 [1.42, 1.47]

Table 8: NCV/REML MSE ratio with the rich and the low-rank functional covariate, $G = 100$, default signal.

family	error	estimand	rich	low-rank
binary	smooth	α(t)	0.88 [0.84, 0.93]	0.87 [0.82, 0.91]
binary	smooth	β(s,t)	0.17 [0.14, 0.20]	0.12 [0.08, 0.17]
binary	smooth	γ(t)	0.96 [0.91, 1.03]	0.95 [0.90, 1.01]
binary	smooth	E(Y | X)	0.74 [0.71, 0.77]	0.77 [0.74, 0.80]
binary	indep.	α(t)	1.10 [1.05, 1.17]	1.06 [1.03, 1.11]
binary	indep.	β(s,t)	1.06 [0.95, 1.19]	1.26 [1.03, 1.56]
binary	indep.	γ(t)	1.05 [1.03, 1.09]	1.07 [1.04, 1.10]
binary	indep.	E(Y | X)	1.06 [1.04, 1.08]	1.07 [1.05, 1.09]
Gaussian	smooth	α(t)	0.97 [0.94, 0.99]	0.97 [0.94, 0.99]
Gaussian	smooth	β(s,t)	0.16 [0.13, 0.19]	0.09 [0.06, 0.11]
Gaussian	smooth	γ(t)	0.95 [0.92, 0.97]	0.95 [0.93, 0.97]
Gaussian	smooth	E(Y | X)	0.69 [0.67, 0.71]	0.70 [0.68, 0.72]
Gaussian	indep.	α(t)	1.04 [1.02, 1.07]	1.04 [1.02, 1.07]
Gaussian	indep.	β(s,t)	0.65 [0.62, 0.68]	0.71 [0.62, 0.82]
Gaussian	indep.	γ(t)	1.04 [1.02, 1.07]	1.04 [1.02, 1.07]
Gaussian	indep.	E(Y | X)	0.96 [0.95, 0.97]	0.97 [0.96, 0.99]
Poisson	smooth	α(t)	0.94 [0.90, 0.97]	0.93 [0.90, 0.97]
Poisson	smooth	β(s,t)	0.16 [0.13, 0.19]	0.09 [0.06, 0.12]
Poisson	smooth	γ(t)	0.98 [0.95, 1.01]	0.98 [0.95, 1.01]
Poisson	smooth	E(Y | X)	0.68 [0.64, 0.71]	0.69 [0.65, 0.72]
Poisson	indep.	α(t)	1.04 [1.01, 1.07]	1.03 [1.00, 1.06]
Poisson	indep.	β(s,t)	0.71 [0.65, 0.78]	0.73 [0.63, 0.84]
Poisson	indep.	γ(t)	1.07 [1.04, 1.10]	1.07 [1.04, 1.10]
Poisson	indep.	E(Y | X)	0.98 [0.96, 1.00]	0.99 [0.97, 1.01]

Table 9: Integrated MSE of the REML and NCV fits per setting and estimand, and their paired ratio NCV/REML (95% bootstrap interval); for the smooth-effect settings, the $\beta(s,t)$ columns hold $f(x,t)$.

							E(Y | X)			β(s,t)			α(t)			γ(t)		
block	family	error	G	signal	truth	basis	REML	NCV	ratio	REML	NCV	ratio	REML	NCV	ratio	REML	NCV	ratio
main factorial	Gaussian	indep.	40	low	smooth	default	0.0619	0.0636	1.03 [1.01, 1.04]	0.1677	0.1575	0.94 [0.86, 1.04]	0.0129	0.0137	1.06 [1.03, 1.09]	0.0126	0.0134	1.06 [1.04, 1.09]
main factorial	Gaussian	indep.	100	low	smooth	default	0.0271	0.0271	1.00 [0.98, 1.01]	0.0930	0.0777	0.84 [0.77, 0.91]	0.0056	0.0059	1.05 [1.02, 1.08]	0.0054	0.0057	1.06 [1.03, 1.09]
main factorial	Gaussian	rough	40	low	smooth	default	0.4799	0.2958	0.62 [0.59, 0.64]	8.1491	0.9605	0.12 [0.09, 0.15]	0.0657	0.0668	1.02 [0.95, 1.08]	0.0625	0.0614	0.98 [0.92, 1.04]
main factorial	Gaussian	rough	100	low	smooth	default	0.1718	0.1163	0.68 [0.66, 0.70]	2.3958	0.3591	0.15 [0.13, 0.17]	0.0244	0.0236	0.97 [0.94, 1.00]	0.0250	0.0243	0.97 [0.94, 1.00]
main factorial	Gaussian	smooth	40	low	smooth	default	0.5272	0.3064	0.58 [0.56, 0.61]	10.4813	0.9421	0.09 [0.07, 0.11]	0.0728	0.0694	0.95 [0.90, 1.01]	0.0669	0.0615	0.92 [0.87, 0.98]
main factorial	Gaussian	smooth	100	low	smooth	default	0.1827	0.1209	0.66 [0.64, 0.68]	2.8137	0.3496	0.12 [0.10, 0.15]	0.0260	0.0255	0.98 [0.94, 1.03]	0.0255	0.0237	0.93 [0.91, 0.96]
main factorial	Gaussian	indep.	40	default	smooth	default	0.0196	0.0192	0.98 [0.96, 1.00]	0.0742	0.0633	0.85 [0.70, 1.13]	0.0039	0.0040	1.04 [1.01, 1.06]	0.0038	0.0040	1.04 [1.02, 1.06]
main factorial	Gaussian	indep.	100	default	smooth	default	0.0084	0.0081	0.96 [0.95, 0.97]	0.0412	0.0269	0.65 [0.62, 0.68]	0.0016	0.0017	1.04 [1.02, 1.07]	0.0016	0.0017	1.04 [1.02, 1.07]
main factorial	Gaussian	rough	40	default	smooth	default	0.1298	0.0799	0.62 [0.59, 0.64]	2.1864	0.2788	0.13 [0.10, 0.16]	0.0179	0.0166	0.93 [0.90, 0.96]	0.0172	0.0161	0.93 [0.90, 0.97]
main factorial	Gaussian	rough	100	default	smooth	default	0.0464	0.0325	0.70 [0.68, 0.72]	0.6473	0.1124	0.17 [0.15, 0.20]	0.0066	0.0063	0.96 [0.95, 0.98]	0.0068	0.0065	0.97 [0.94, 0.99]
main factorial	Gaussian	smooth	40	default	smooth	default	0.1415	0.0832	0.59 [0.57, 0.61]	2.7504	0.2633	0.10 [0.08, 0.12]	0.0198	0.0180	0.91 [0.87, 0.95]	0.0183	0.0163	0.89 [0.85, 0.93]
main factorial	Gaussian	smooth	100	default	smooth	default	0.0494	0.0340	0.69 [0.67, 0.71]	0.7524	0.1177	0.16 [0.13, 0.19]	0.0070	0.0068	0.97 [0.94, 0.99]	0.0069	0.0066	0.95 [0.92, 0.97]
main factorial	Gaussian	indep.	40	high	smooth	default	0.0061	0.0057	0.94 [0.92, 0.95]	0.0330	0.0205	0.62 [0.58, 0.67]	0.0011	0.0011	1.03 [1.01, 1.05]	0.0012	0.0012	1.04 [1.01, 1.06]
main factorial	Gaussian	indep.	100	high	smooth	default	0.0026	0.0024	0.93 [0.92, 0.94]	0.0177	0.0097	0.55 [0.53, 0.57]	0.0004	0.0005	1.03 [1.01, 1.05]	0.0005	0.0005	1.07 [1.04, 1.10]
main factorial	Gaussian	rough	40	high	smooth	default	0.0351	0.0221	0.63 [0.61, 0.65]	0.6178	0.0735	0.12 [0.10, 0.14]	0.0048	0.0044	0.93 [0.90, 0.96]	0.0047	0.0043	0.93 [0.90, 0.96]
main factorial	Gaussian	rough	100	high	smooth	default	0.0126	0.0091	0.72 [0.71, 0.74]	0.1895	0.0381	0.20 [0.17, 0.23]	0.0017	0.0017	0.98 [0.96, 1.00]	0.0018	0.0018	0.99 [0.97, 1.01]
main factorial	Gaussian	smooth	40	high	smooth	default	0.0381	0.0233	0.61 [0.59, 0.63]	0.7526	0.0795	0.11 [0.09, 0.13]	0.0052	0.0048	0.91 [0.88, 0.95]	0.0050	0.0045	0.91 [0.87, 0.94]
main factorial	Gaussian	smooth	100	high	smooth	default	0.0134	0.0095	0.71 [0.69, 0.73]	0.2135	0.0353	0.17 [0.14, 0.19]	0.0019	0.0018	0.97 [0.95, 1.00]	0.0018	0.0018	0.98 [0.96, 1.00]
main factorial	Poisson	indep.	40	default	smooth	default	0.0816	0.0817	1.00 [0.98, 1.03]	0.0235	0.0170	0.72 [0.68, 0.76]	0.0014	0.0015	1.04 [1.02, 1.06]	0.0014	0.0015	1.04 [1.02, 1.06]
main factorial	Poisson	indep.	100	default	smooth	default	0.0337	0.0330	0.98 [0.96, 1.00]	0.0130	0.0092	0.71 [0.65, 0.78]	0.0006	0.0007	1.04 [1.01, 1.07]	0.0006	0.0006	1.07 [1.04, 1.10]
main factorial	Poisson	rough	40	default	smooth	default	0.5989	0.3617	0.60 [0.55, 0.65]	0.6181	0.0854	0.14 [0.11, 0.17]	0.0072	0.0061	0.86 [0.81, 0.90]	0.0062	0.0060	0.97 [0.92, 1.03]
main factorial	Poisson	rough	100	default	smooth	default	0.1949	0.1375	0.71 [0.67, 0.74]	0.1986	0.0375	0.19 [0.15, 0.23]	0.0026	0.0024	0.92 [0.88, 0.95]	0.0023	0.0023	1.00 [0.97, 1.02]
main factorial	Poisson	smooth	40	default	smooth	default	0.6682	0.3816	0.57 [0.53, 0.61]	0.8135	0.0794	0.10 [0.08, 0.12]	0.0078	0.0065	0.83 [0.78, 0.89]	0.0064	0.0062	0.96 [0.92, 1.01]
main factorial	Poisson	smooth	100	default	smooth	default	0.2194	0.1481	0.68 [0.64, 0.71]	0.2277	0.0356	0.16 [0.13, 0.19]	0.0027	0.0026	0.94 [0.90, 0.97]	0.0024	0.0023	0.98 [0.95, 1.01]
main factorial	Poisson	indep.	40	high	smooth	default	0.5135	0.5123	1.00 [0.92, 1.08]	0.0539	0.0385	0.71 [0.66, 0.79]	0.0031	0.0034	1.08 [1.05, 1.12]	0.0027	0.0030	1.09 [1.06, 1.14]
main factorial	Poisson	indep.	100	high	smooth	default	0.1768	0.1592	0.90 [0.85, 0.95]	0.0292	0.0180	0.62 [0.59, 0.65]	0.0013	0.0014	1.06 [1.03, 1.10]	0.0010	0.0011	1.11 [1.07, 1.15]
main factorial	Poisson	rough	40	high	smooth	default	4.3810	2.1726	0.50 [0.37, 0.64]	0.9085	0.1389	0.15 [0.13, 0.18]	0.0136	0.0128	0.93 [0.87, 1.01]	0.0107	0.0104	0.97 [0.91, 1.03]
main factorial	Poisson	rough	100	high	smooth	default	1.0830	0.6411	0.59 [0.52, 0.67]	0.3030	0.0585	0.19 [0.16, 0.24]	0.0051	0.0049	0.96 [0.92, 1.00]	0.0037	0.0039	1.04 [1.01, 1.07]
main factorial	Poisson	smooth	40	high	smooth	default	5.0480	2.4766	0.49 [0.36, 0.63]	1.1357	0.1547	0.14 [0.11, 0.17]	0.0152	0.0134	0.88 [0.81, 0.96]	0.0107	0.0106	0.99 [0.93, 1.04]
main factorial	Poisson	smooth	100	high	smooth	default	1.2444	0.7319	0.59 [0.51, 0.68]	0.3238	0.0723	0.22 [0.18, 0.28]	0.0053	0.0051	0.96 [0.92, 1.01]	0.0038	0.0039	1.03 [0.99, 1.06]
main factorial	binary	indep.	40	default	smooth	default	0.0019	0.0020	1.08 [1.05, 1.12]	0.1508	0.1733	1.15 [0.99, 1.36]	0.0174	0.0186	1.07 [1.01, 1.13]	0.0168	0.0196	1.17 [1.10, 1.25]
main factorial	binary	indep.	100	default	smooth	default	0.0008	0.0009	1.06 [1.04, 1.08]	0.0754	0.0796	1.06 [0.95, 1.19]	0.0073	0.0080	1.10 [1.05, 1.17]	0.0075	0.0079	1.05 [1.03, 1.09]
main factorial	binary	rough	40	default	smooth	default	0.0094	0.0061	0.66 [0.62, 0.69]	3.7624	0.7505	0.20 [0.15, 0.25]	0.0711	0.0482	0.68 [0.63, 0.73]	0.0646	0.0523	0.81 [0.75, 0.87]
main factorial	binary	rough	100	default	smooth	default	0.0035	0.0028	0.80 [0.77, 0.83]	1.1349	0.2490	0.22 [0.19, 0.26]	0.0264	0.0241	0.91 [0.87, 0.97]	0.0242	0.0242	1.00 [0.95, 1.06]
main factorial	binary	smooth	40	default	smooth	default	0.0116	0.0069	0.59 [0.56, 0.62]	6.5792	1.0127	0.15 [0.11, 0.20]	0.0959	0.0541	0.56 [0.52, 0.61]	0.0793	0.0543	0.69 [0.64, 0.74]
main factorial	binary	smooth	100	default	smooth	default	0.0041	0.0031	0.74 [0.71, 0.77]	1.6613	0.2772	0.17 [0.14, 0.20]	0.0309	0.0272	0.88 [0.84, 0.93]	0.0268	0.0258	0.96 [0.91, 1.03]
main factorial	binary	indep.	40	high	smooth	default	0.0022	0.0023	1.06 [1.03, 1.09]	0.2715	0.2799	1.03 [0.90, 1.21]	0.0239	0.0261	1.09 [1.05, 1.13]	0.0251	0.0267	1.06 [1.03, 1.11]
main factorial	binary	indep.	100	high	smooth	default	0.0010	0.0010	1.03 [1.01, 1.05]	0.1475	0.1447	0.98 [0.87, 1.12]	0.0105	0.0113	1.08 [1.05, 1.12]	0.0108	0.0115	1.06 [1.04, 1.09]
main factorial	binary	rough	40	high	smooth	default	0.0098	0.0072	0.74 [0.71, 0.77]	4.4148	0.8701	0.20 [0.16, 0.24]	0.1037	0.0761	0.73 [0.68, 0.79]	0.0863	0.0785	0.91 [0.84, 0.99]
main factorial	binary	rough	100	high	smooth	default	0.0037	0.0028	0.77 [0.75, 0.80]	1.4186	0.3241	0.23 [0.20, 0.26]	0.0359	0.0308	0.86 [0.82, 0.90]	0.0314	0.0311	0.99 [0.95, 1.04]
main factorial	binary	smooth	40	high	smooth	default	0.0121	0.0080	0.66 [0.63, 0.70]	7.7267	1.3164	0.17 [0.12, 0.23]	0.1431	0.0844	0.59 [0.54, 0.64]	0.1101	0.0892	0.81 [0.75, 0.88]
main factorial	binary	smooth	100	high	smooth	default	0.0043	0.0032	0.73 [0.70, 0.75]	2.1147	0.3603	0.17 [0.14, 0.20]	0.0437	0.0364	0.83 [0.79, 0.88]	0.0364	0.0342	0.94 [0.89, 0.99]
misreg.	Gaussian	misreg.	40	high	smooth	default	0.0351	0.0288	0.82 [0.79, 0.85]	0.3887	0.1711	0.44 [0.33, 0.56]	0.0042	0.0043	1.02 [0.97, 1.06]	0.0063	0.0067	1.06 [1.00, 1.13]
misreg.	Gaussian	misreg.	40	high	rough	default	0.0475	0.0431	0.91 [0.88, 0.93]	0.5156	0.3330	0.65 [0.54, 0.78]	0.0052	0.0052	1.01 [0.96, 1.06]	0.0090	0.0101	1.12 [1.06, 1.18]
misreg.	Gaussian	misreg.	100	high	smooth	default	0.0140	0.0117	0.83 [0.82, 0.85]	0.1520	0.0712	0.47 [0.39, 0.55]	0.0017	0.0016	0.98 [0.96, 1.01]	0.0025	0.0025	1.02 [1.00, 1.04]
misreg.	Gaussian	misreg.	100	high	rough	default	0.0194	0.0175	0.91 [0.89, 0.92]	0.2543	0.1733	0.68 [0.62, 0.74]	0.0022	0.0022	0.98 [0.95, 1.01]	0.0034	0.0035	1.02 [1.01, 1.03]
misreg.	Poisson	misreg.	40	high	smooth	default	12.9278	8.2514	0.64 [0.54, 0.74]									
misreg.	Poisson	misreg.	40	high	rough	default	12.8997	7.8017	0.60 [0.46, 0.80]									
misreg.	Poisson	misreg.	100	high	smooth	default	6.4191	4.6246	0.72 [0.64, 0.81]									
misreg.	Poisson	misreg.	100	high	rough	default	5.7093	4.1304	0.72 [0.64, 0.82]									
misreg.	binary	misreg.	40	high	smooth	default	0.0026	0.0028	1.09 [1.06, 1.11]									

Table 9: Integrated MSE of the REML and NCV fits per setting and estimand, and their paired ratio NCV/REML (95% bootstrap interval); for the smooth-effect settings, the $\beta(s,t)$ columns hold $f(x,t)$. *(continued)*

block	family	error	G	signal	truth	basis	REML	NCV	ratio	REML	NCV	ratio	REML	NCV	ratio	REML	NCV	ratio
misreg.	binary	misreg.	40	high	rough	default	0.0029	0.0032	1.08 [1.06, 1.11]									
misreg.	binary	misreg.	100	high	smooth	default	0.0012	0.0012	1.01 [1.00, 1.03]									
misreg.	binary	misreg.	100	high	rough	default	0.0014	0.0015	1.04 [1.03, 1.06]									
dense grid	Gaussian	indep.	100	default	smooth	default	0.0027	0.0026	0.94 [0.93, 0.96]	0.0185	0.0107	0.58 [0.54, 0.64]	0.0005	0.0005	1.04 [1.01, 1.07]	0.0005	0.0006	1.12 [1.09, 1.15]
dense grid	Gaussian	rough	100	default	smooth	default	0.0605	0.0317	0.52 [0.51, 0.54]	2.0279	0.1111	0.05 [0.05, 0.07]	0.0067	0.0061	0.91 [0.89, 0.94]	0.0070	0.0065	0.92 [0.89, 0.94]
dense grid	Gaussian	smooth	100	default	smooth	default	0.0687	0.0342	0.50 [0.48, 0.51]	2.3248	0.1150	0.05 [0.04, 0.06]	0.0078	0.0067	0.87 [0.84, 0.90]	0.0077	0.0067	0.87 [0.84, 0.90]
dense grid	Poisson	indep.	100	default	smooth	default	0.0107	0.0100	0.94 [0.92, 0.96]	0.0058	0.0033	0.57 [0.55, 0.59]	0.0002	0.0002	1.05 [1.02, 1.08]	0.0002	0.0002	1.13 [1.09, 1.17]
dense grid	Poisson	rough	100	default	smooth	default	0.2725	0.1346	0.49 [0.47, 0.52]	0.6655	0.0383	0.06 [0.05, 0.07]	0.0028	0.0023	0.83 [0.80, 0.87]	0.0024	0.0023	0.94 [0.91, 0.97]
dense grid	Poisson	smooth	100	default	smooth	default	0.3205	0.1468	0.46 [0.43, 0.49]	0.7644	0.0361	0.05 [0.04, 0.06]	0.0032	0.0025	0.80 [0.76, 0.85]	0.0026	0.0023	0.88 [0.85, 0.92]
dense grid	binary	indep.	100	default	smooth	default	0.0003	0.0003	1.01 [1.00, 1.03]	0.0323	0.0272	0.84 [0.78, 0.93]	0.0022	0.0024	1.07 [1.04, 1.11]	0.0023	0.0024	1.06 [1.04, 1.07]
dense grid	binary	rough	100	default	smooth	default	0.0055	0.0026	0.47 [0.45, 0.49]	6.4136	0.2320	0.04 [0.03, 0.04]	0.0342	0.0224	0.66 [0.61, 0.71]	0.0286	0.0226	0.79 [0.74, 0.85]
dense grid	binary	smooth	100	default	smooth	default	0.0073	0.0030	0.41 [0.39, 0.42]	9.5703	0.2991	0.03 [0.03, 0.04]	0.0472	0.0263	0.56 [0.52, 0.60]	0.0369	0.0251	0.68 [0.63, 0.73]
rough truth	Gaussian	indep.	100	low	rough	default	0.0330	0.0343	1.04 [1.03, 1.05]	0.2073	0.2251	1.09 [1.07, 1.11]	0.0057	0.0060	1.05 [1.03, 1.09]	0.0067	0.0070	1.05 [1.02, 1.08]
rough truth	Gaussian	smooth	100	low	rough	default	0.1931	0.1394	0.72 [0.70, 0.74]	3.1309	0.6660	0.21 [0.19, 0.25]	0.0262	0.0256	0.98 [0.94, 1.02]	0.0279	0.0281	1.01 [0.98, 1.03]
rough truth	Gaussian	indep.	100	default	rough	default	0.0100	0.0102	1.02 [1.01, 1.03]	0.0835	0.0850	1.02 [1.01, 1.03]	0.0016	0.0017	1.05 [1.02, 1.07]	0.0018	0.0019	1.06 [1.04, 1.09]
rough truth	Gaussian	smooth	100	default	rough	default	0.0531	0.0418	0.79 [0.77, 0.81]	0.9297	0.3120	0.34 [0.30, 0.38]	0.0071	0.0069	0.98 [0.96, 1.01]	0.0074	0.0075	1.02 [1.00, 1.03]
rough truth	Gaussian	indep.	100	high	rough	default	0.0030	0.0030	1.01 [1.00, 1.02]	0.0307	0.0325	1.06 [1.02, 1.12]	0.0005	0.0005	1.04 [1.02, 1.06]	0.0005	0.0005	1.05 [1.04, 1.07]
rough truth	Gaussian	smooth	100	high	rough	default	0.0146	0.0121	0.83 [0.81, 0.84]	0.2927	0.1167	0.40 [0.36, 0.44]	0.0019	0.0019	0.99 [0.97, 1.01]	0.0019	0.0019	1.01 [0.99, 1.02]
rough truth	Poisson	indep.	100	default	rough	default	0.0391	0.0398	1.02 [1.01, 1.03]	0.0267	0.0279	1.05 [1.02, 1.07]	0.0007	0.0007	1.02 [0.99, 1.05]	0.0007	0.0007	1.07 [1.04, 1.10]
rough truth	Poisson	smooth	100	default	rough	default	0.2270	0.1698	0.75 [0.72, 0.78]	0.2706	0.0933	0.34 [0.31, 0.39]	0.0027	0.0026	0.93 [0.90, 0.96]	0.0025	0.0026	1.03 [1.01, 1.05]
rough truth	Poisson	indep.	100	high	rough	default	0.1632	0.1620	0.99 [0.97, 1.02]	0.0547	0.0568	1.04 [1.01, 1.06]	0.0014	0.0014	1.05 [1.01, 1.09]	0.0011	0.0012	1.11 [1.07, 1.15]
rough truth	Poisson	smooth	100	high	rough	default	1.0961	0.7299	0.67 [0.60, 0.73]	0.4569	0.1944	0.43 [0.39, 0.47]	0.0054	0.0050	0.92 [0.89, 0.96]	0.0039	0.0042	1.09 [1.06, 1.13]
rough truth	binary	indep.	100	default	rough	default	0.0010	0.0010	1.05 [1.03, 1.07]	0.1416	0.1663	1.17 [1.11, 1.24]	0.0073	0.0079	1.08 [1.03, 1.14]	0.0101	0.0101	1.00 [0.97, 1.03]
rough truth	binary	smooth	100	default	rough	default	0.0043	0.0033	0.77 [0.74, 0.80]	1.7833	0.3877	0.22 [0.18, 0.26]	0.0309	0.0268	0.87 [0.82, 0.92]	0.0293	0.0302	1.03 [0.98, 1.09]
rough truth	binary	indep.	100	high	rough	default	0.0012	0.0012	1.05 [1.03, 1.06]	0.3097	0.3544	1.14 [1.10, 1.19]	0.0106	0.0111	1.05 [1.03, 1.08]	0.0140	0.0143	1.02 [1.00, 1.04]
rough truth	binary	smooth	100	high	rough	default	0.0046	0.0036	0.79 [0.76, 0.81]	2.3240	0.7165	0.31 [0.27, 0.35]	0.0435	0.0368	0.85 [0.80, 0.90]	0.0400	0.0393	0.98 [0.94, 1.02]
smooth effect	Gaussian	indep.	100	default	smooth	default	0.0103	0.0095	0.93 [0.91, 0.95]	0.0052	0.0039	0.73 [0.72, 0.75]				0.0016	0.0017	1.05 [1.02, 1.08]
smooth effect	Gaussian	rough	100	default	smooth	default	0.0565	0.0381	0.67 [0.64, 0.70]	0.0303	0.0155	0.51 [0.49, 0.53]				0.0068	0.0066	0.97 [0.95, 0.99]
smooth effect	Gaussian	smooth	100	default	smooth	default	0.0638	0.0419	0.66 [0.63, 0.68]	0.0346	0.0175	0.51 [0.49, 0.53]				0.0070	0.0066	0.95 [0.93, 0.98]
smooth effect	Poisson	indep.	100	default	smooth	default	0.0311	0.0291	0.93 [0.92, 0.95]	0.0019	0.0017	0.91 [0.89, 0.94]				0.0006	0.0006	1.05 [1.03, 1.08]
smooth effect	Poisson	rough	100	default	smooth	default	0.1623	0.1107	0.68 [0.66, 0.70]	0.0111	0.0066	0.60 [0.57, 0.63]				0.0022	0.0022	0.99 [0.97, 1.02]
smooth effect	Poisson	smooth	100	default	smooth	default	0.1818	0.1195	0.66 [0.64, 0.68]	0.0128	0.0073	0.57 [0.54, 0.60]				0.0023	0.0023	0.98 [0.96, 1.01]
smooth effect	binary	indep.	100	default	smooth	default	0.0007	0.0008	1.07 [1.05, 1.10]	0.0140	0.0150	1.07 [1.04, 1.11]				0.0069	0.0076	1.10 [1.06, 1.14]
smooth effect	binary	rough	100	default	smooth	default	0.0038	0.0025	0.66 [0.63, 0.68]	0.0931	0.0442	0.47 [0.44, 0.51]				0.0245	0.0243	0.99 [0.93, 1.05]
smooth effect	binary	smooth	100	default	smooth	default	0.0048	0.0028	0.59 [0.56, 0.62]	0.1257	0.0515	0.41 [0.38, 0.44]				0.0280	0.0265	0.94 [0.89, 1.01]
families	scaled t	indep.	100	default	smooth	default	0.0063	0.0060	0.95 [0.94, 0.97]	0.0336	0.0203	0.60 [0.58, 0.63]	0.0012	0.0012	1.04 [1.02, 1.07]	0.0012	0.0012	1.05 [1.02, 1.09]
families	scaled t	smooth	100	default	smooth	default	0.0336	0.0240	0.71 [0.70, 0.74]	0.4734	0.1031	0.22 [0.18, 0.26]	0.0048	0.0047	0.97 [0.95, 1.00]	0.0048	0.0046	0.95 [0.93, 0.98]
families	beta	indep.	100	default	smooth	default	0.0001	0.0001	0.98 [0.96, 0.99]	0.0100	0.0075	0.75 [0.65, 0.91]	0.0004	0.0004	1.06 [1.03, 1.09]	0.0004	0.0004	1.08 [1.05, 1.12]
families	beta	smooth	100	default	smooth	default	0.0005	0.0003	0.70 [0.68, 0.72]	0.1764	0.0322	0.18 [0.15, 0.22]	0.0017	0.0016	0.97 [0.94, 1.00]	0.0016	0.0016	0.96 [0.94, 0.99]
families	negative binomial	indep.	100	default	smooth	default	0.0570	0.0604	1.06 [1.03, 1.09]	0.0173	0.0153	0.89 [0.75, 1.05]	0.0009	0.0010	1.06 [1.03, 1.10]	0.0009	0.0009	1.10 [1.06, 1.14]
families	negative binomial	smooth	100	default	smooth	default	0.4070	0.2795	0.69 [0.64, 0.74]	0.3909	0.0685	0.18 [0.14, 0.21]	0.0042	0.0039	0.92 [0.88, 0.96]	0.0037	0.0036	0.97 [0.94, 1.00]
families	negative binomial	indep.	100	default	smooth	default	0.0683	0.0615	0.90 [0.87, 0.93]	0.0249	0.0133	0.53 [0.49, 0.59]	0.0010	0.0010	1.00 [0.97, 1.04]	0.0009	0.0010	1.04 [1.01, 1.08]
families	negative binomial	smooth	100	default	smooth	default	0.5344	0.2957	0.55 [0.52, 0.59]	0.7490	0.0647	0.09 [0.07, 0.11]	0.0045	0.0039	0.87 [0.82, 0.92]	0.0041	0.0040	0.96 [0.92, 0.99]
exact AR(1)	Gaussian	AR(1)	100	default	smooth	default	0.0516	0.0346	0.67 [0.66, 0.69]	0.7611	0.1115	0.15 [0.13, 0.17]	0.0072	0.0068	0.95 [0.93, 0.97]	0.0075	0.0071	0.95 [0.92, 0.97]
sign-changing	Gaussian	sign-chg.	100	default	smooth	default	0.0485	0.0294	0.61 [0.59, 0.62]	0.4781	0.0877	0.18 [0.17, 0.20]	0.0073	0.0065	0.89 [0.84, 0.93]	0.0073	0.0063	0.86 [0.82, 0.90]
sign-changing	Poisson	sign-chg.	100	default	smooth	default	0.2034	0.1195	0.59 [0.56, 0.62]	0.1488	0.0287	0.19 [0.18, 0.21]	0.0029	0.0023	0.79 [0.75, 0.82]	0.0025	0.0023	0.92 [0.88, 0.97]
sign-changing	binary	sign-chg.	100	default	smooth	default	0.0033	0.0020	0.61 [0.58, 0.63]	0.7819	0.2016	0.26 [0.21, 0.32]	0.0270	0.0173	0.64 [0.60, 0.68]	0.0245	0.0179	0.73 [0.69, 0.77]
misreg. (AR(1))	Gaussian	misreg.	100	high	rough	default	0.0194	0.0175	0.91 [0.89, 0.92]	0.2543	0.1733	0.68 [0.62, 0.74]	0.0022	0.0022	0.98 [0.95, 1.01]	0.0034	0.0035	1.02 [1.01, 1.03]
heteroskedastic	Gaussian	var(t)	40	default	smooth	default	0.1444	0.0865	0.60 [0.58, 0.62]	2.8124	0.3693	0.13 [0.10, 0.17]	0.0197	0.0180	0.91 [0.88, 0.95]	0.0185	0.0167	0.90 [0.86, 0.94]
heteroskedastic	Gaussian	var(t)	100	default	smooth	default	0.0504	0.0345	0.68 [0.66, 0.70]	0.7678	0.1270	0.17 [0.14, 0.20]	0.0070	0.0068	0.96 [0.94, 0.99]	0.0071	0.0066	0.94 [0.92, 0.96]
heteroskedastic	Gaussian	var(z)	40	default	smooth	default	0.1638	0.1036	0.63 [0.61, 0.66]	2.6201	0.2672	0.10 [0.08, 0.13]	0.0188	0.0176	0.93 [0.89, 0.97]	0.0372	0.0309	0.83 [0.79, 0.88]
heteroskedastic	Gaussian	var(z)	100	default	smooth	default	0.0587	0.0424	0.72 [0.70, 0.74]	0.7612	0.1238	0.16 [0.14, 0.19]	0.0070	0.0067	0.97 [0.94, 0.99]	0.0140	0.0126	0.90 [0.87, 0.92]
heteroskedastic	Gaussian	var(t,z)	40	default	smooth	default	0.1649	0.1044	0.63 [0.61, 0.66]	2.5654	0.2735	0.11 [0.09, 0.13]	0.0187	0.0174	0.93 [0.89, 0.97]	0.0376	0.0313	0.83 [0.79, 0.88]

Table 9: Integrated MSE of the REML and NCV fits per setting and estimand, and their paired ratio NCV/REML (95% bootstrap interval); for the smooth-effect settings, the $\beta(s,t)$ columns hold $f(x,t)$. *(continued)*

block	family	error	G	signal	truth	basis	REML	NCV	ratio	REML	NCV	ratio	REML	NCV	ratio	REML	NCV	ratio
heteroskedastic	Gaussian	var(t,z)	100	default	smooth	default	0.0597	0.0424	0.71 [0.69, 0.73]	0.7931	0.1203	0.15 [0.13, 0.18]	0.0069	0.0067	0.96 [0.93, 0.98]	0.0141	0.0125	0.89 [0.85, 0.92]
low-rank covariate	Gaussian	indep.	100	default	smooth	default	0.0081	0.0079	0.97 [0.96, 0.99]	0.1017	0.0724	0.71 [0.62, 0.82]	0.0016	0.0017	1.04 [1.02, 1.07]	0.0016	0.0017	1.04 [1.02, 1.07]
low-rank covariate	Gaussian	smooth	100	default	smooth	default	0.0476	0.0334	0.70 [0.68, 0.72]	3.7471	0.3194	0.09 [0.06, 0.11]	0.0070	0.0068	0.97 [0.94, 0.99]	0.0069	0.0066	0.95 [0.93, 0.97]
low-rank covariate	Poisson	indep.	100	default	smooth	default	0.0323	0.0320	0.99 [0.97, 1.01]	0.0312	0.0228	0.73 [0.63, 0.84]	0.0006	0.0007	1.03 [1.00, 1.06]	0.0006	0.0006	1.07 [1.04, 1.10]
low-rank covariate	Poisson	smooth	100	default	smooth	default	0.2097	0.1442	0.69 [0.65, 0.72]	1.1068	0.0981	0.09 [0.06, 0.12]	0.0027	0.0026	0.93 [0.90, 0.97]	0.0023	0.0023	0.98 [0.95, 1.01]
low-rank covariate	binary	indep.	100	default	smooth	default	0.0008	0.0009	1.07 [1.05, 1.09]	0.1574	0.1978	1.26 [1.03, 1.56]	0.0073	0.0077	1.06 [1.03, 1.11]	0.0075	0.0080	1.07 [1.04, 1.10]
low-rank covariate	binary	smooth	100	default	smooth	default	0.0040	0.0031	0.77 [0.74, 0.80]	7.0153	0.8475	0.12 [0.08, 0.17]	0.0305	0.0264	0.87 [0.82, 0.91]	0.0267	0.0254	0.95 [0.90, 1.01]
basis size	Gaussian	indep.	100	default	smooth	2× finer knots	0.0114	0.0102	0.89 [0.88, 0.90]	0.0678	0.0377	0.56 [0.53, 0.60]	0.0020	0.0019	0.94 [0.93, 0.96]	0.0021	0.0021	1.01 [0.98, 1.03]
basis size	Gaussian	indep.	100	default	smooth	4× finer knots	0.0138	0.0117	0.85 [0.83, 0.86]	0.0895	0.0438	0.49 [0.47, 0.51]	0.0024	0.0021	0.89 [0.87, 0.91]	0.0024	0.0023	0.96 [0.94, 0.98]
basis size	Gaussian	smooth	100	default	smooth	2× finer knots	0.0801	0.0376	0.47 [0.45, 0.49]	3.6193	0.1201	0.03 [0.03, 0.04]	0.0083	0.0071	0.85 [0.82, 0.88]	0.0083	0.0072	0.87 [0.83, 0.90]
basis size	Gaussian	smooth	100	default	smooth	4× finer knots	0.1209	0.0400	0.33 [0.32, 0.34]	11.7721	0.1482	0.01 [0.01, 0.02]	0.0090	0.0073	0.81 [0.77, 0.85]	0.0092	0.0075	0.82 [0.79, 0.85]
basis size	Poisson	indep.	100	default	smooth	2× finer knots	0.0458	0.0418	0.91 [0.89, 0.93]	0.0204	0.0113	0.55 [0.53, 0.57]	0.0008	0.0007	0.96 [0.94, 0.99]	0.0007	0.0007	1.05 [1.03, 1.07]
basis size	Poisson	indep.	100	default	smooth	4× finer knots	0.0558	0.0488	0.87 [0.86, 0.89]	0.0261	0.0134	0.52 [0.49, 0.54]	0.0009	0.0008	0.92 [0.90, 0.95]	0.0008	0.0008	1.03 [1.01, 1.06]
basis size	Poisson	smooth	100	default	smooth	2× finer knots	0.3825	0.1637	0.43 [0.40, 0.46]	1.0140	0.0349	0.03 [0.03, 0.04]	0.0032	0.0026	0.82 [0.77, 0.86]	0.0027	0.0025	0.92 [0.88, 0.95]
basis size	Poisson	smooth	100	default	smooth	4× finer knots	0.5536	0.1747	0.32 [0.29, 0.34]	3.2858	0.0437	0.01 [0.01, 0.02]	0.0035	0.0027	0.75 [0.71, 0.81]	0.0030	0.0026	0.89 [0.85, 0.93]
basis size	binary	indep.	100	default	smooth	2× finer knots	0.0009	0.0010	1.06 [1.04, 1.09]	0.0942	0.0985	1.05 [0.91, 1.22]	0.0073	0.0077	1.06 [1.02, 1.10]	0.0079	0.0086	1.10 [1.05, 1.16]
basis size	binary	indep.	100	default	smooth	4× finer knots	0.0010	0.0010	1.05 [1.03, 1.08]	0.1043	0.1073	1.03 [0.92, 1.16]	0.0075	0.0078	1.05 [1.00, 1.11]	0.0081	0.0089	1.10 [1.05, 1.18]
basis size	binary	smooth	100	default	smooth	2× finer knots	0.0065	0.0032	0.49 [0.46, 0.51]	7.3259	0.2938	0.04 [0.03, 0.05]	0.0384	0.0263	0.68 [0.64, 0.73]	0.0310	0.0257	0.83 [0.78, 0.88]
basis size	binary	smooth	100	default	smooth	4× finer knots	0.0102	0.0032	0.31 [0.30, 0.33]	32.6957	0.3265	0.01 [0.01, 0.01]	0.0504	0.0256	0.51 [0.47, 0.56]	0.0349	0.0256	0.74 [0.69, 0.78]

S1.6 Relative error, half-width and detection per setting

Under dependent errors, the median relative error of $\beta(s,t)$ over settings is 0.67 for REML and 0.21 for NCV; under independent errors, it is 0.16 and 0.12. In the main factorial, the REML/NCV ratio of relative errors is 2.1–3.7 under dependent and 1.1–1.4 under independent errors. Wrong-sign exclusions of the CL2 intervals stay rare (at most 0.023 of the grid for REML + CL2 and 0.040 for NCV + CL2); model-based REML intervals reach 0.20.

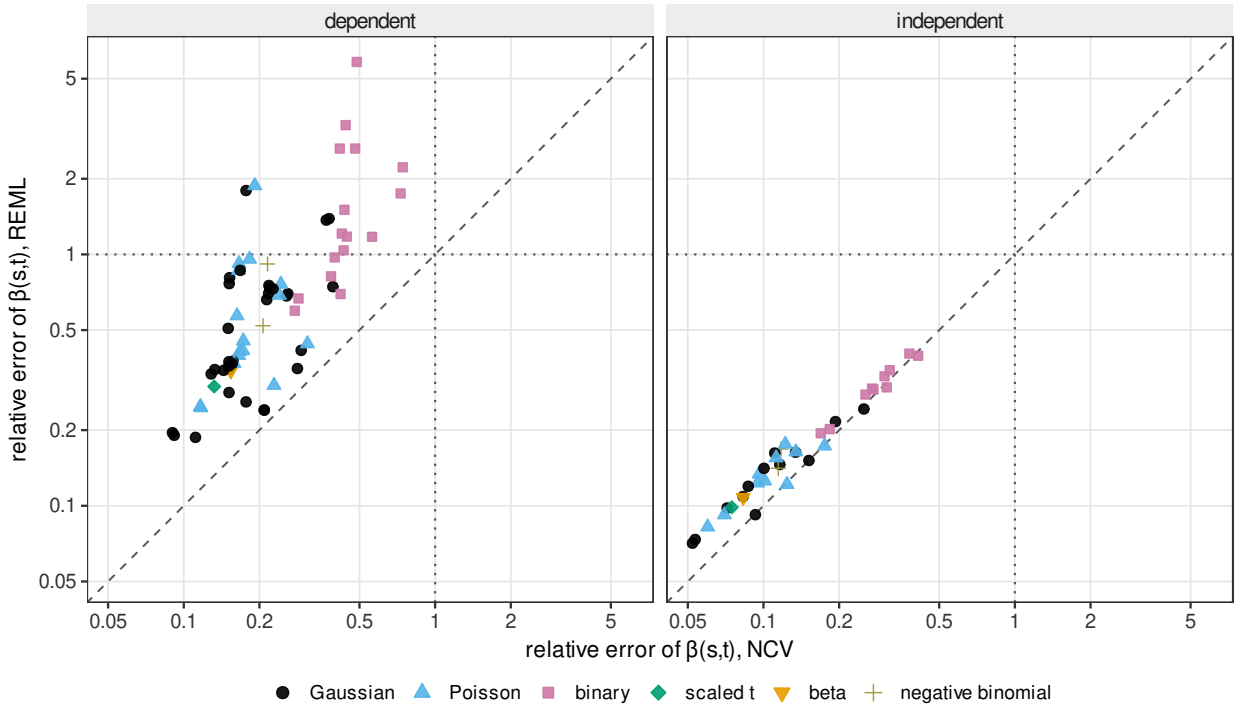


Figure 12: Relative error of the REML and NCV estimates of $\beta(s,t)$ per setting (median over replicates; log scales), by dependence. Dashed: equal error; dotted: error as large as the effect.

Table 10: Relative error and informativeness per estimand, dependence and interval (all synthetic settings): median coverage, median and maximum relative error, settings with relative error above 0.5, mean share of replicates with relative error above 1, median relative half-width (hw), median and minimum detection rate, maximum wrong-sign rate.

estimand	dependence	interval	settings	cov	rel	rel max	settings >0.5	reps >1	hw	det	det min	wrong max
E(Y | X)	dependent	REML, model-based	83	0.65	0.28	1.19	18	0.04	0.27			
E(Y | X)	dependent	REML + CL2	83	0.94	0.28	1.19	18	0.04	0.58			
E(Y | X)	dependent	NCV + CL2	83	0.92	0.23	0.91	17	0.01	0.42			
E(Y | X)	dependent	NCV + CL2, NCV–REML widening	83	0.95	0.23	0.91	17	0.01	0.53			
E(Y | X)	dependent	AR(1) working model	9	0.95	0.18	0.21	0	0.00	0.38			
E(Y | X)	independent	REML, model-based	40	0.97	0.11	0.49	0	0.00	0.24			
E(Y | X)	independent	REML + CL2	40	0.96	0.11	0.49	0	0.00	0.24			
E(Y | X)	independent	NCV + CL2	40	0.95	0.10	0.50	1	0.00	0.22			
E(Y | X)	independent	NCV + CL2, NCV–REML widening	40	0.95	0.10	0.50	1	0.00	0.23			
β(s,t)	dependent	REML, model-based	69	0.67	0.67	5.83	40	0.28	0.57	0.83	0.16	0.20
β(s,t)	dependent	REML + CL2	69	0.94	0.67	5.83	40	0.28	1.35	0.54	0.04	0.02
β(s,t)	dependent	NCV + CL2	69	0.92	0.21	0.74	3	0.02	0.37	0.95	0.40	0.04
β(s,t)	dependent	NCV + CL2, NCV–REML widening	69	0.98	0.21	0.74	3	0.02	1.14	0.65	0.04	0.01
β(s,t)	dependent	AR(1) working model	9	0.99	0.19	0.21	0	0.00	0.50	0.94	0.93	0.00
β(s,t)	independent	REML, model-based	37	0.99	0.16	0.40	0	0.00	0.40	0.96	0.43	0.00
β(s,t)	independent	REML + CL2	37	0.99	0.16	0.40	0	0.00	0.40	0.96	0.47	0.00
β(s,t)	independent	NCV + CL2	37	0.98	0.12	0.41	0	0.00	0.31	0.97	0.56	0.00
β(s,t)	independent	NCV + CL2, NCV–REML widening	37	0.99	0.12	0.41	0	0.00	0.32	0.97	0.55	0.00
α(t)	dependent	REML, model-based	69	0.66	0.31	1.60	18	0.11	0.31			
α(t)	dependent	REML + CL2	69	0.95	0.31	1.60	18	0.11	0.65			
α(t)	dependent	NCV + CL2	69	0.92	0.28	1.23	17	0.09	0.57			
α(t)	dependent	NCV + CL2, NCV–REML widening	69	0.94	0.28	1.23	17	0.09	0.64			
α(t)	dependent	AR(1) working model	9	0.95	0.23	0.24	0	0.00	0.49			
α(t)	independent	REML, model-based	37	0.96	0.16	0.74	1	0.00	0.34			
α(t)	independent	REML + CL2	37	0.95	0.16	0.74	1	0.00	0.35			
α(t)	independent	NCV + CL2	37	0.94	0.16	0.74	4	0.01	0.32			
α(t)	independent	NCV + CL2, NCV–REML widening	37	0.94	0.16	0.74	4	0.01	0.34			
γ(t)	dependent	REML, model-based	75	0.66	0.22	1.14	17	0.04	0.23	0.99	0.41	0.07
γ(t)	dependent	REML + CL2	75	0.94	0.22	1.14	17	0.04	0.46	0.94	0.15	0.01
γ(t)	dependent	NCV + CL2	75	0.92	0.22	1.00	17	0.03	0.42	0.96	0.12	0.01
γ(t)	dependent	NCV + CL2, NCV–REML widening	75	0.93	0.22	1.00	17	0.03	0.44	0.95	0.06	0.01
γ(t)	dependent	AR(1) working model	9	0.92	0.17	0.24	0	0.00	0.37	0.98	0.94	0.01
γ(t)	independent	REML, model-based	40	0.96	0.11	0.54	1	0.00	0.24	1.00	0.35	0.01
γ(t)	independent	REML + CL2	40	0.95	0.11	0.54	1	0.00	0.25	1.00	0.45	0.01
γ(t)	independent	NCV + CL2	40	0.94	0.12	0.58	1	0.00	0.23	1.00	0.44	0.01
γ(t)	independent	NCV + CL2, NCV–REML widening	40	0.94	0.12	0.58	1	0.00	0.23	1.00	0.42	0.01

Table 11: $\beta(s,t)$ per setting: relative error of the REML and NCV estimates, relative half-width (hw) and detection rate (det) of REML + CL2 and NCV + CL2.

block	family	signal	error	G	rel NCV	rel REML	hw NCV+CL2	hw REML+CL2	det NCV+CL2	det REML+CL2
main factorial	Gaussian	low	indep.	40	0.19	0.22	0.49	0.60	0.92	0.90
main factorial	Gaussian	low	indep.	100	0.13	0.16	0.34	0.44	0.96	0.95
main factorial	Gaussian	low	rough	40	0.37	1.37	0.66	2.40	0.77	0.27
main factorial	Gaussian	low	rough	100	0.26	0.68	0.47	1.36	0.89	0.49
main factorial	Gaussian	low	smooth	40	0.38	1.39	0.67	2.69	0.76	0.25
main factorial	Gaussian	low	smooth	100	0.26	0.70	0.47	1.44	0.90	0.47
main factorial	Gaussian	default	indep.	40	0.12	0.15	0.29	0.40	0.97	0.96
main factorial	Gaussian	default	indep.	100	0.08	0.11	0.20	0.29	0.99	0.98
main factorial	Gaussian	default	rough	40	0.21	0.66	0.42	1.24	0.92	0.54
main factorial	Gaussian	default	rough	100	0.15	0.35	0.29	0.71	0.97	0.80
main factorial	Gaussian	default	smooth	40	0.22	0.70	0.41	1.35	0.92	0.51
main factorial	Gaussian	default	smooth	100	0.15	0.36	0.29	0.75	0.97	0.78
main factorial	Gaussian	high	indep.	40	0.07	0.10	0.18	0.25	1.00	0.99
main factorial	Gaussian	high	indep.	100	0.05	0.07	0.12	0.18	1.00	1.00
main factorial	Gaussian	high	rough	40	0.13	0.33	0.25	0.64	0.99	0.83
main factorial	Gaussian	high	rough	100	0.09	0.19	0.17	0.39	1.00	0.95
main factorial	Gaussian	high	smooth	40	0.13	0.35	0.26	0.68	0.99	0.80
main factorial	Gaussian	high	smooth	100	0.09	0.20	0.17	0.40	1.00	0.94
main factorial	Poisson	default	indep.	40	0.13	0.16	0.34	0.45	0.96	0.95
main factorial	Poisson	default	indep.	100	0.09	0.12	0.23	0.32	0.98	0.97
main factorial	Poisson	default	rough	40	0.24	0.69	0.46	1.36	0.89	0.53
main factorial	Poisson	default	rough	100	0.17	0.40	0.32	0.80	0.96	0.77
main factorial	Poisson	default	smooth	40	0.24	0.76	0.46	1.48	0.90	0.49
main factorial	Poisson	default	smooth	100	0.17	0.41	0.32	0.84	0.96	0.75
main factorial	Poisson	high	indep.	40	0.10	0.13	0.24	0.33	0.98	0.97
main factorial	Poisson	high	indep.	100	0.07	0.09	0.16	0.23	1.00	0.99
main factorial	Poisson	high	rough	40	0.17	0.42	0.32	0.85	0.95	0.77
main factorial	Poisson	high	rough	100	0.12	0.25	0.22	0.51	0.99	0.91
main factorial	Poisson	high	smooth	40	0.17	0.45	0.33	0.89	0.95	0.73
main factorial	Poisson	high	smooth	100	0.12	0.25	0.22	0.52	0.98	0.91
main factorial	binary	default	indep.	40	0.38	0.40	0.93	0.99	0.66	0.66

Table 11: $\beta(s,t)$ per setting: relative error of the REML and NCV estimates, relative half-width (hw) and detection rate (det) of REML + CL2 and NCV + CL2. *(continued)*

block	family	signal	error	G	rel NCV	rel REML	hw NCV+CL2	hw REML+CL2	det NCV+CL2	det REML+CL2
main factorial	binary	default	indep.	100	0.27	0.29	0.66	0.78	0.84	0.81
main factorial	binary	default	rough	40	0.73	1.75	1.02	3.51	0.46	0.17
main factorial	binary	default	rough	100	0.43	1.04	0.77	2.16	0.68	0.31
main factorial	binary	default	smooth	40	0.74	2.22	1.04	4.16	0.40	0.15
main factorial	binary	default	smooth	100	0.45	1.18	0.79	2.41	0.66	0.27
main factorial	binary	high	indep.	40	0.25	0.28	0.61	0.74	0.86	0.83
main factorial	binary	high	indep.	100	0.18	0.20	0.44	0.55	0.93	0.92
main factorial	binary	high	rough	40	0.40	0.97	0.71	1.99	0.75	0.36
main factorial	binary	high	rough	100	0.28	0.60	0.51	1.25	0.88	0.57
main factorial	binary	high	smooth	40	0.43	1.21	0.74	2.37	0.72	0.30
main factorial	binary	high	smooth	100	0.29	0.67	0.52	1.38	0.88	0.52
misreg.	Gaussian	high	misreg.	40	0.15	0.28	0.30	0.59	0.96	0.84
misreg.	Gaussian	high	misreg.	40	0.28	0.35	0.42	0.72	0.90	0.77
misreg.	Gaussian	high	misreg.	100	0.11	0.19	0.21	0.40	0.99	0.95
misreg.	Gaussian	high	misreg.	100	0.21	0.24	0.34	0.49	0.97	0.91
dense grid	Gaussian	default	indep.	100	0.05	0.07	0.12	0.18	1.00	1.00
dense grid	Gaussian	default	rough	100	0.15	0.77	0.27	1.51	0.98	0.57
dense grid	Gaussian	default	smooth	100	0.15	0.81	0.27	1.65	0.98	0.52
dense grid	Poisson	default	indep.	100	0.06	0.08	0.14	0.20	1.00	0.99
dense grid	Poisson	default	rough	100	0.17	0.86	0.29	1.74	0.96	0.54
dense grid	Poisson	default	smooth	100	0.17	0.92	0.30	1.88	0.96	0.49
dense grid	binary	default	indep.	100	0.17	0.19	0.41	0.52	0.94	0.93
dense grid	binary	default	rough	100	0.42	2.64	0.68	5.32	0.75	0.11
dense grid	binary	default	smooth	100	0.44	3.27	0.72	6.64	0.71	0.09
rough truth	Gaussian	low	indep.	100	0.25	0.24	0.50	0.50	0.95	0.95
rough truth	Gaussian	low	smooth	100	0.39	0.74	0.54	1.47	0.81	0.42
rough truth	Gaussian	default	indep.	100	0.15	0.15	0.32	0.33	0.99	0.99
rough truth	Gaussian	default	smooth	100	0.29	0.42	0.42	0.80	0.94	0.75
rough truth	Gaussian	high	indep.	100	0.09	0.09	0.19	0.21	1.00	1.00
rough truth	Gaussian	high	smooth	100	0.18	0.26	0.31	0.48	0.99	0.95
rough truth	Poisson	default	indep.	100	0.18	0.17	0.37	0.37	0.99	0.99
rough truth	Poisson	default	smooth	100	0.31	0.44	0.43	0.90	0.91	0.72
rough truth	Poisson	high	indep.	100	0.12	0.12	0.26	0.27	0.99	0.99
rough truth	Poisson	high	smooth	100	0.23	0.30	0.36	0.61	0.97	0.90
rough truth	binary	default	indep.	100	0.41	0.40	0.76	0.80	0.73	0.74
rough truth	binary	default	smooth	100	0.56	1.17	0.80	2.50	0.58	0.24
rough truth	binary	high	indep.	100	0.31	0.30	0.61	0.60	0.88	0.89
rough truth	binary	high	smooth	100	0.42	0.70	0.57	1.46	0.78	0.47
families	scaled t	default	indep.	100	0.07	0.10	0.18	0.25	1.00	0.99
families	scaled t	default	smooth	100	0.13	0.30	0.26	0.62	0.98	0.86
families	beta	default	indep.	100	0.08	0.11	0.20	0.28	0.99	0.98
families	beta	default	smooth	100	0.15	0.34	0.28	0.72	0.97	0.80
families	negative binomial	default	indep.	100	0.11	0.14	0.28	0.38	0.97	0.96
families	negative binomial	default	smooth	100	0.21	0.52	0.39	1.08	0.93	0.62
families	negative binomial	default	indep.	100	0.12	0.17	0.25	0.40	0.98	0.96
families	negative binomial	default	smooth	100	0.22	0.92	0.37	1.81	0.93	0.46
exact AR(1)	Gaussian	default	AR(1)	100	0.16	0.38	0.29	0.76	0.97	0.77
sign-changing	Gaussian	default	sign-chg.	100	0.14	0.35	0.26	0.68	0.97	0.84
sign-changing	Poisson	default	sign-chg.	100	0.16	0.37	0.28	0.75	0.96	0.81
sign-changing	binary	default	sign-chg.	100	0.39	0.82	0.70	1.79	0.77	0.39
misreg. (AR(1))	Gaussian	high	misreg.	100	0.21	0.24	0.34	0.49	0.97	0.91
heteroskedastic	Gaussian	default	var(t)	40	0.23	0.73	0.41	1.39	0.91	0.50
heteroskedastic	Gaussian	default	var(t)	100	0.16	0.37	0.29	0.76	0.97	0.77
heteroskedastic	Gaussian	default	var(z)	40	0.22	0.75	0.41	1.46	0.92	0.52
heteroskedastic	Gaussian	default	var(z)	100	0.15	0.36	0.29	0.74	0.97	0.79
heteroskedastic	Gaussian	default	var(t,z)	40	0.22	0.74	0.41	1.39	0.92	0.52
heteroskedastic	Gaussian	default	var(t,z)	100	0.15	0.37	0.29	0.76	0.97	0.78
low-rank covariate	Gaussian	default	indep.	100	0.09	0.12	0.22	0.34	0.99	0.97
low-rank covariate	Gaussian	default	smooth	100	0.15	0.51	0.29	1.06	0.96	0.63
low-rank covariate	Poisson	default	indep.	100	0.10	0.13	0.25	0.38	0.98	0.96
low-rank covariate	Poisson	default	smooth	100	0.16	0.57	0.32	1.20	0.95	0.60
low-rank covariate	binary	default	indep.	100	0.27	0.29	0.70	0.84	0.81	0.78
low-rank covariate	binary	default	smooth	100	0.44	1.50	0.78	3.13	0.68	0.22
basis size	Gaussian	default	indep.	100	0.10	0.14	0.28	0.44	0.98	0.96
basis size	Gaussian	default	indep.	100	0.11	0.16	0.37	0.60	0.97	0.91
basis size	Gaussian	default	smooth	100	0.17	0.86	0.34	1.75	0.96	0.37
basis size	Gaussian	default	smooth	100	0.18	1.80	0.39	3.78	0.95	0.12
basis size	Poisson	default	indep.	100	0.11	0.16	0.31	0.48	0.97	0.94
basis size	Poisson	default	indep.	100	0.12	0.18	0.40	0.66	0.95	0.88
basis size	Poisson	default	smooth	100	0.18	0.96	0.37	1.94	0.95	0.35
basis size	Poisson	default	smooth	100	0.19	1.88	0.42	3.97	0.93	0.12
basis size	binary	default	indep.	100	0.30	0.33	0.85	1.04	0.71	0.64
basis size	binary	default	indep.	100	0.32	0.35	1.04	1.30	0.56	0.47
basis size	binary	default	smooth	100	0.48	2.64	0.87	5.69	0.59	0.08
basis size	binary	default	smooth	100	0.49	5.83	0.96	12.44	0.51	0.04

S1.7 Satterthwaite critical values beyond the main factorial

REML + CL2 intervals with the per-point Satterthwaite critical value of Appendix A.7 of the paper were computed from the stored REML fits for every synthetic setting outside the main factorial (110 settings, including the settings with 20 curves and those with a zero surface) and for the plasmode combinations with curve flips (one independent random sign per residual curve; S2.1). Paired with the normal critical value on the same replicates, the Satterthwaite interval never covered less: the smallest difference over all settings, plasmode combinations and estimands is 0.01 pp. Over the extension settings (those outside the main factorial, each block changing one aspect of it), it adds on average 2.0 pp of coverage with 40 curves and 0.7 pp with 100, at 1.10 and 1.04 times the width (figure below; values in S7). Coverage stays below 0.93 in 13 combinations of setting and estimand (first table below): the mean of misregistered count responses, and, for binary responses, the intercept $\alpha(t)$ or the scalar-covariate effect $\gamma(t)$ in the dense grid, rough truth, low-rank covariate, larger bases settings. For misregistered count and binary responses, only the mean is scored, here as in the main study: the targets of their coefficient functions on the observed clock are link-scale approximations, not exact (Appendix B.1 of the paper). The per-point degrees of freedom (second table below) are about a third of $G - 1$ for the mean, $\beta(s,t)$ and $\gamma(t)$ in every synthetic setting and larger for $\alpha(t)$; on the plasmode data, they spread more.

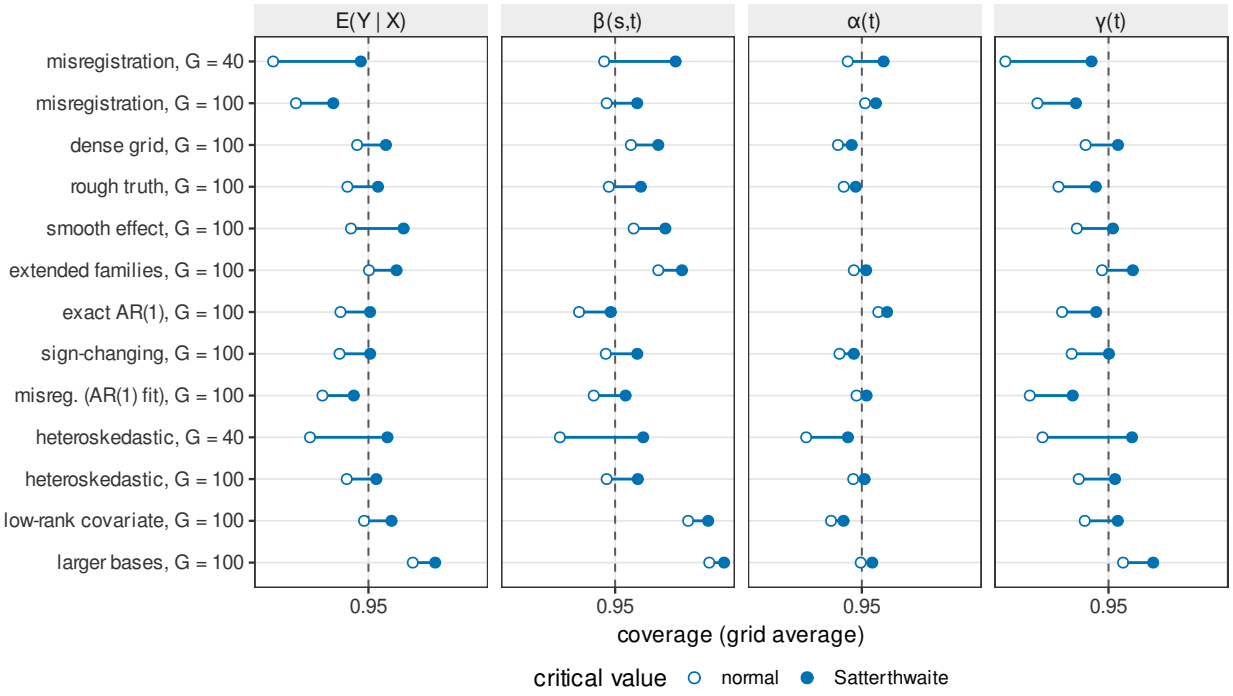


Figure 13: REML + CL2 with the normal and the Satterthwaite critical value in the extension settings (outside the main factorial): grid-averaged coverage per block, number of curves and estimand (mean over the block's settings; β includes $f(x,t)$ of the smooth-effect settings; misregistered count and binary settings score the mean only). Dashed: nominal 0.95. Values and width ratios: S7.

Table 12: Extension settings (outside the main factorial) and estimands in which REML + CL2 with the Satterthwaite critical value covers less than 0.93: coverage with z and Satterthwaite critical value, and the 5% quantile of pointwise coverage of the Satterthwaite interval.

block	family	errors	G	signal	truth	estimand	z	Satt.	q05 Satt.
misregistration	Poisson	misreg.	40	high	rough	E(Y | X)	0.899	0.930	0.865
misregistration	Poisson	misreg.	100	high	smooth	E(Y | X)	0.910	0.924	0.850
misregistration	Poisson	misreg.	100	high	rough	E(Y | X)	0.906	0.919	0.850
dense grid	binary	rough	100	default	smooth	α(t)	0.919	0.925	0.887
dense grid	binary	smooth	100	default	smooth	α(t)	0.920	0.925	0.890
rough truth	binary	indep.	100	default	rough	α(t)	0.921	0.924	0.775
rough truth	binary	smooth	100	default	rough	α(t)	0.920	0.925	0.875
rough truth	binary	indep.	100	default	rough	γ(t)	0.877	0.890	0.710
rough truth	binary	indep.	100	high	rough	γ(t)	0.916	0.928	0.868
low-rank covariate	binary	indep.	100	default	smooth	α(t)	0.923	0.927	0.785
low-rank covariate	binary	smooth	100	default	smooth	α(t)	0.922	0.926	0.885
larger bases	binary	smooth	100	default	smooth	α(t)	0.915	0.920	0.865
larger bases	binary	smooth	100	default	smooth	α(t)	0.899	0.904	0.853

Table 13: Per-point Satterthwaite degrees of freedom of REML + CL2: range over settings (plasmode: combinations) of the median over grid points and replicates, and that median relative to $G - 1$ (minimum, median and maximum over settings), by data source, number of curves (all: all subjects) and estimand; β includes $f(x, t)$.

data	G	estimand	settings	df min	df max	df/(G-1) min	df/(G-1) median	df/(G-1) max
plasmode	40	E(Y | X)	60	11.43	19.11	0.29	0.35	0.49
plasmode	40	β(s,t)	60	10.65	13.52	0.27	0.31	0.35
plasmode	40	α(t)	60	12.36	20.92	0.32	0.35	0.54
plasmode	40	γ(t)	60	7.03	25.35	0.18	0.41	0.65
plasmode	all	E(Y | X)	60	21.45	50.84	0.28	0.35	0.51
plasmode	all	β(s,t)	60	19.81	47.94	0.20	0.30	0.35
plasmode	all	α(t)	60	22.62	57.98	0.24	0.34	0.59
plasmode	all	γ(t)	60	13.56	96.92	0.18	0.46	0.83
synthetic	20	E(Y | X)	21	6.40	7.52	0.34	0.35	0.40
synthetic	20	β(s,t)	21	6.00	6.94	0.32	0.33	0.37
synthetic	20	α(t)	21	8.44	12.20	0.44	0.52	0.64
synthetic	20	γ(t)	21	6.10	6.58	0.32	0.33	0.35
synthetic	40	E(Y | X)	22	12.68	18.25	0.33	0.36	0.47
synthetic	40	β(s,t)	18	12.65	13.77	0.32	0.34	0.35
synthetic	40	α(t)	18	24.53	35.97	0.63	0.68	0.92
synthetic	40	γ(t)	18	12.36	13.45	0.32	0.33	0.34
synthetic	100	E(Y | X)	85	21.85	46.26	0.22	0.36	0.47
synthetic	100	β(s,t)	81	24.32	44.31	0.25	0.34	0.45
synthetic	100	α(t)	72	71.99	96.05	0.73	0.84	0.97
synthetic	100	γ(t)	81	23.97	34.01	0.24	0.33	0.34

S1.8 A zero coefficient surface

The zero-surface settings set $\beta(s, t) \equiv 0$ in the smooth-truth setting at the default signal (Appendix B.1 of the paper). The share of grid points at which a pointwise 95% interval excludes zero estimates the pointwise false-positive rate, nominally 0.05 (first table). Under smooth errors, REML + CL2 has a rate of 0.054–0.076 with the normal critical value, 0.051–0.068 with t_{G-1} and 0.046–0.052 with the Satterthwaite critical value. The model-based intervals have 0.23–0.40 (REML) and 0.46–0.57 (NCV), NCV + CL2 0.11–0.15. NCV + CL2 widened by the NCV–REML difference has the lowest rate under smooth errors (0.017–0.018), but not because it is calibrated: its variance adds the squared difference between the NCV and the REML estimate, and, under a zero surface, the REML estimate is far from zero (second table), so its intervals are wide.

Under independent errors, the Gaussian setting with 40 curves stands out: REML + CL2 has 0.095 (MC SE 0.018), with the Satterthwaite critical value 0.075, against 0.050 with 100 curves. The excluded points of REML + CL2 concentrate in a minority of replicates (0.18 have more than 5% of the grid excluding zero; the 90% quantile of the per-replicate rate is 0.24). In most replicates, REML shrinks the surface toward the null space of its penalty; intervals for terms shrunk this far are known to behave poorly (Marra and Wood 2012).

The second table is descriptive. The share of replicates with at least one grid point, or with more than 5% of the grid, excluding zero is not an error rate of pointwise intervals; whether β is zero everywhere needs a simultaneous band or a global test, which the study does not evaluate. Under smooth errors, REML's estimate of the zero surface has a root-mean-square size of 0.39–2.25 times that of the non-zero surface of the corresponding main-factorial setting, NCV's 0.06–0.45 times; under independent errors, both are small (0.01–0.16). The zero surface hardly changes the other estimands (third table): REML + CL2 coverage of $\alpha(t)$ and $\gamma(t)$ differs from that in the matching non-zero setting by -0.7 to 1.6 pp.

Table 14: Zero surface: pointwise false-positive rate (MC SE), the share of grid points at which the 95% interval excludes zero, mean over replicates; nominal 0.05. NCV+CL2 widened: NCV + CL2 with the squared NCV–REML difference added to the variance.

family	errors	G	REML model	REML+CL2	REML+CL2, t(G-1)	REML+CL2, Satt.	NCV model	NCV+CL2	NCV+CL2 widened
Gaussian	smooth	40	0.399 (0.011)	0.076 (0.005)	0.068 (0.004)	0.052 (0.004)	0.516 (0.026)	0.130 (0.015)	0.017 (0.003)
Gaussian	smooth	100	0.361 (0.010)	0.058 (0.004)	0.055 (0.004)	0.049 (0.003)	0.573 (0.026)	0.151 (0.019)	0.017 (0.003)
Gaussian	indep.	40	0.024 (0.009)	0.095 (0.018)	0.086 (0.017)	0.075 (0.017)	0.066 (0.014)	0.096 (0.017)	0.079 (0.017)
Gaussian	indep.	100	0.009 (0.004)	0.050 (0.013)	0.049 (0.013)	0.046 (0.013)	0.040 (0.009)	0.047 (0.010)	0.035 (0.010)
binary	smooth	40	0.250 (0.011)	0.071 (0.005)	0.063 (0.005)	0.048 (0.004)	0.462 (0.025)	0.111 (0.014)	0.018 (0.003)
binary	smooth	100	0.228 (0.011)	0.054 (0.004)	0.051 (0.004)	0.046 (0.004)	0.476 (0.027)	0.118 (0.016)	0.018 (0.004)
binary	indep.	40	0.009 (0.004)	0.056 (0.013)	0.048 (0.012)	0.029 (0.009)	0.040 (0.009)	0.060 (0.012)	0.035 (0.010)
binary	indep.	100	0.015 (0.007)	0.042 (0.012)	0.034 (0.011)	0.031 (0.010)	0.033 (0.009)	0.048 (0.011)	0.037 (0.011)

Table 15: Zero surface, descriptive: share of replicates with at least one grid point / with more than 5% of the grid excluding zero, per interval (not error rates of pointwise intervals); root-mean-square size of the REML and NCV estimates over the grid (mean over replicates), absolute and relative to the non-zero surface of the corresponding main-factorial setting.

family	errors	G	REML+CL2	REML+CL2, Satt.	NCV+CL2	RMS REML	RMS NCV	rel. REML	rel. NCV
Gaussian	smooth	40	0.92 / 0.62	0.83 / 0.41	0.41 / 0.41	1.398	0.205	0.771	0.113
Gaussian	smooth	100	0.86 / 0.48	0.81 / 0.41	0.41 / 0.40	0.702	0.117	0.387	0.064
Gaussian	indep.	40	0.20 / 0.18	0.15 / 0.14	0.29 / 0.22	0.036	0.059	0.020	0.032
Gaussian	indep.	100	0.14 / 0.11	0.12 / 0.10	0.24 / 0.17	0.021	0.034	0.012	0.019
binary	smooth	40	0.83 / 0.54	0.76 / 0.33	0.37 / 0.35	2.043	0.410	2.255	0.452
binary	smooth	100	0.77 / 0.42	0.72 / 0.34	0.34 / 0.33	1.005	0.208	1.109	0.229
binary	indep.	40	0.14 / 0.14	0.11 / 0.09	0.27 / 0.20	0.084	0.142	0.093	0.157
binary	indep.	100	0.13 / 0.10	0.10 / 0.08	0.27 / 0.16	0.053	0.083	0.059	0.091

Table 16: Zero surface: coverage of the mean, $\alpha(t)$ and $\gamma(t)$ per interval, and of REML + CL2 in the matching non-zero setting (same α and γ; the mean differs).

family	errors	G	estimand	REML model	REML+CL2	NCV+CL2	NCV+CL2 widened	REML+CL2, non-zero β
Gaussian	smooth	40	E(Y | X)	0.600	0.930	0.918	0.965	
Gaussian	smooth	40	α(t)	0.616	0.936	0.922	0.940	0.937
Gaussian	smooth	40	γ(t)	0.621	0.945	0.928	0.947	0.946
Gaussian	smooth	100	E(Y | X)	0.624	0.943	0.926	0.967	
Gaussian	smooth	100	α(t)	0.641	0.950	0.939	0.950	0.950
Gaussian	smooth	100	γ(t)	0.654	0.949	0.936	0.948	0.948
Gaussian	indep.	40	E(Y | X)	0.961	0.944	0.935	0.944	
Gaussian	indep.	40	α(t)	0.951	0.944	0.934	0.939	0.947
Gaussian	indep.	40	γ(t)	0.961	0.949	0.944	0.947	0.951
Gaussian	indep.	100	E(Y | X)	0.961	0.948	0.941	0.949	
Gaussian	indep.	100	α(t)	0.954	0.952	0.939	0.945	0.956
Gaussian	indep.	100	γ(t)	0.961	0.951	0.946	0.951	0.950
binary	smooth	40	E(Y | X)	0.705	0.923	0.872	0.953	
binary	smooth	40	α(t)	0.680	0.901	0.869	0.925	0.885
binary	smooth	40	γ(t)	0.699	0.930	0.845	0.912	0.922
binary	smooth	100	E(Y | X)	0.716	0.938	0.873	0.943	
binary	smooth	100	α(t)	0.714	0.925	0.860	0.917	0.920
binary	smooth	100	γ(t)	0.715	0.937	0.854	0.905	0.939
binary	indep.	40	E(Y | X)	0.941	0.893	0.898	0.918	
binary	indep.	40	α(t)	0.900	0.852	0.870	0.884	0.859
binary	indep.	40	γ(t)	0.938	0.888	0.878	0.897	0.890
binary	indep.	100	E(Y | X)	0.953	0.925	0.926	0.936	
binary	indep.	100	α(t)	0.943	0.923	0.923	0.930	0.924
binary	indep.	100	γ(t)	0.944	0.918	0.916	0.920	0.923

S1.9 A principal component term for a low-rank covariate

In the eight settings of the next table (all with smooth errors and REML fits), the functional term was also fitted with refund's `ffpc()` term, on the same simulated data as the `ff()` fits. `ffpc()` estimates the leading functional principal components of the covariate curves and regresses the response on their scores, with one coefficient function $b_k(t)$ per component in the basis and penalty used for $\alpha(t)$. The surface estimate is $\hat{\beta}(s,t) = \sum_{k=1}^{K} \psi_k(s)\, \hat{b}_k(t)$, where the ψ_k are the estimated eigenfunctions scaled so that the integrals of the centered covariate curves against them reproduce the scores. Its standard errors use the CL2 covariance of the b_k (with the leverage adjustment computed from the full block $\mathbf{R}_i$ of curve i, Section 2.3 of the paper, and with the smoothing-bias allowance $\mathbf{V}_p - \mathbf{V}_e$) and are conditional on the estimated eigenfunctions; the intervals use the normal critical value. The number of components K is fixed at 3 or 5, or is the smallest number that explains 95% or 90% of the covariate's variance. Accuracy is measured by the relative root-mean-square error (relative RMSE) $\{\sum_{r,j}(\hat{\beta}_{rj} - \beta_j)^2 / \sum_{r,j} \beta_j^2\}^{1/2}$, pooled over replicates r and grid points j (not the median over replicates of Section 2.4 of the paper); the `ff()` columns are the study's REML + CL2 and NCV + CL2 on the same replicates, with the same measures.

With the low-rank covariate, the components that explain 95% of its variance (median $K = 3$) estimate β with relative RMSE 0.20, against 0.76 for `ff()` with REML and 0.22 with NCV. Their CL2 intervals cover 0.915 (MC SE 0.005) at 0.32 times the half-width of `ff()` REML + CL2; with five components, they cover 0.95 at relative RMSE 0.42 and 0.73 times the half-width. Two components (90% of the variance) truncate the surface: they cover the true β at 0.79, its projection onto the span of the estimated components at 0.94.

With the default covariate, whose spectrum decays slowly, the variance-explained rules choose many components (median 11–11.5 for 95%, 6 for 90%), and the estimates are noisy: relative RMSE 1.0–9.7 at 95%, against 0.48–3.24 for `ff()` with REML. A fixed $K = 3$ has 0.69–1.14 times the relative RMSE of `ff()` with NCV, but it truncates the surface, and its intervals undercover: 0.84–0.93 for smooth truths and 0.64 for the rough truth. With $K = 5$, they cover 0.90–0.94 at 0.57–0.85 times the half-width of `ff()` REML + CL2, with relative RMSE between those of `ff()` with NCV and with REML in every setting. In a pilot with 20 replicates of three of the settings, the threshold of 0.99 (the default of current refund releases; at most 15 components) chose median K of 9–15 and gave relative RMSE 1.12–3.22 (`ff()` with REML: 0.44–0.95), also with the low-rank covariate. The model-based intervals of the `ffpc()` fits cover 0.34–0.68 in all settings and rules.

These results come from synthetic settings only. The smooth truth varies along s mainly like the first cosine functions, which are the covariate's leading principal components, so a small K suits it. No rule that chooses K from the response, such as cross-validation, was evaluated, and there is no plasmode check, in which truths fitted with `ff()` would not lie in the span of a few components.

Table 17: Principal component term (ffpc) and ff, smooth errors, REML fits, 200 replicates: relative RMSE of $\beta(s,t)$ pooled over replicates and grid (rel.) and grid-averaged coverage of nominal 95% CL2 intervals for the true $\beta(s,t)$ (cov.); ff with REML + CL2 and NCV + CL2; ffpc with a fixed number K of components or the number explaining 95% or 90% of the covariate's variance (K: median over replicates). Signal: default unless stated.

	ff, REML		ff, NCV		ffpc, K = 3		ffpc, K = 5		ffpc, 95%			ffpc, 90%		
setting	rel.	cov.	rel.	cov.	rel.	cov.	rel.	cov.	K	rel.	cov.	K	rel.	cov.
Gaussian, low-rank covariate, G = 100	0.76	0.945	0.22	0.936	0.20	0.915	0.42	0.948	3	0.20	0.915	2	0.15	0.788
Gaussian, G = 100	0.48	0.944	0.19	0.934	0.21	0.872	0.32	0.939	11.5	1.02	0.948	6	0.42	0.944
Gaussian, low signal, G = 40	1.79	0.923	0.54	0.902	0.52	0.913	0.93	0.932	11	3.70	0.929	6	1.37	0.932
Gaussian, G = 40	0.92	0.926	0.28	0.924	0.32	0.865	0.50	0.924	11	1.89	0.929	6	0.72	0.927
Gaussian, low signal, G = 20	3.24	0.909	1.07	0.839	0.74	0.892	1.42	0.914	11	9.65	0.904	6	2.23	0.913
Gaussian, G = 20	1.63	0.911	0.56	0.899	0.45	0.844	0.76	0.904	11	4.88	0.903	6	1.16	0.905
binary, G = 40	2.83	0.926	1.11	0.785	1.03	0.930	1.93	0.939	11	8.27	0.925	6	2.78	0.937
Gaussian, rough truth, G = 100	0.52	0.941	0.30	0.824	0.34	0.636	0.35	0.906	11.5	1.02	0.947	6	0.44	0.929

S1.10 Detection with the Satterthwaite critical value; the NCV estimate and the REML + CL2 interval

The detection rates of the paper and of S1.6 are for the normal critical value. For the 18 main-factorial settings at the default signal, the stored replicates were scored again with the per-point Satterthwaite critical value (next table; the scores with the normal critical value reproduce the study's detection rates and coverages exactly). Under dependent errors, the Satterthwaite critical value lowers REML + CL2's detection rate for $\beta(s,t)$ from 0.15–0.54 to 0.11–0.49 with 40 curves and from 0.27–0.80 to 0.25–0.78 with 100, and that for $\gamma(t)$ from 0.15–0.86 to 0.11–0.83 and from 0.30–0.98 to 0.27–0.98.

The same replicates show where the NCV estimate lies relative to the REML + CL2 interval, which matters when the two fits are reported side by side (Section 6 of the paper). Under dependent errors, the NCV estimate of $\beta(s,t)$ is outside the Satterthwaite interval at 1.2–2.4% of the grid points (normal critical value: 1.6–3.9%); under independent errors, at no more than 0.8%. The NCV surface is thus almost everywhere compatible with the pointwise REML + CL2 intervals. This share is not a coverage statement about the NCV estimate, and a display that combines the two fits was not evaluated as a procedure.

Table 18: Main factorial at the default signal, REML + CL2 with the normal (z) and the Satterthwaite (Satt.) critical value: detection rate for $\beta(s,t)$ and $\gamma(t)$, and share of grid points (%) at which the NCV estimate of $\beta(s,t)$ lies outside the REML + CL2 interval; means over 200 replicates.

family	errors	G	det. β, z	det. β, Satt.	det. γ, z	det. γ, Satt.	NCV outside, z (%)	NCV outside, Satt. (%)
Gaussian	independent	40	0.96	0.95	1.00	0.99	0.22	0.21
Gaussian	independent	100	0.98	0.98	1.00	1.00	0.00	0.00
Gaussian	rough stationary	40	0.54	0.49	0.86	0.83	3.39	2.11
Gaussian	rough stationary	100	0.80	0.78	0.98	0.98	1.62	1.30
Gaussian	smooth	40	0.51	0.46	0.84	0.80	3.67	2.27
Gaussian	smooth	100	0.78	0.77	0.97	0.97	1.75	1.42
Poisson	independent	40	0.95	0.94	0.99	0.98	0.04	0.01
Poisson	independent	100	0.97	0.97	1.00	1.00	0.07	0.06
Poisson	rough stationary	40	0.53	0.47	0.79	0.74	3.09	1.83
Poisson	rough stationary	100	0.77	0.75	0.96	0.95	1.75	1.38
Poisson	smooth	40	0.49	0.43	0.77	0.72	3.41	2.06
Poisson	smooth	100	0.75	0.73	0.95	0.94	1.93	1.53
binary	independent	40	0.66	0.60	0.45	0.37	0.95	0.77
binary	independent	100	0.81	0.79	0.77	0.75	0.38	0.35
binary	rough stationary	40	0.17	0.13	0.18	0.13	3.06	1.83
binary	rough stationary	100	0.31	0.28	0.34	0.31	1.56	1.25
binary	smooth	40	0.15	0.11	0.15	0.11	3.93	2.45
binary	smooth	100	0.27	0.25	0.30	0.27	1.65	1.28

S2 Plasmode study

The datasets, the fitted models and the construction of the plasmode data are described in Section 5.1 and Appendices B.2 and B.3 of the paper.

In a replicate, each subject's error is its fitted residual curve with a random sign, so it takes only two values: the construction is a Rademacher wild cluster bootstrap around the fitted residuals, which keeps each residual curve's shape and within-curve covariance but not the full error distribution of new real curves. Two limits follow. The contribution of a curve to the sandwich varies little between replicates, so the plasmode study tests whether CL2's standard error matches the spread of the estimate, but hardly the Satterthwaite reference distribution, which accounts for the variability of that standard error; the evidence for the reference distribution is the simulation (S1.2, S1.7). Fitted residuals are also somewhat smaller than the errors that produced them, more so for a more flexible fit; the heartbeat difference curves of S2.3 are an error source that no fit has shaped.

With curve flips (one independent random sign per residual curve; the main design), the simulated errors are independent across curves for every dataset, including the three whose observed curves are serially or spatially dependent (ocean, weather, electricity; below, "datasets with dependent source curves", and the other six "datasets with independent source curves"); only the block flips of S2.3 share one sign within a block of neighboring curves. Coverage ranges are therefore also given over all nine datasets.

All plasmode truths are fits of the model itself, so the plasmode study cannot evaluate misspecification of the mean, and the comparator results of S3 on plasmode data describe this generator, not real data in general.

S2.1 Coverage per dataset, truth source and residual source

On the datasets with independent source curves (means over each dataset's truth × residual combinations, all subjects), model-based intervals cover $\beta(s,t)$ at 0.51–0.67 and the mean at 0.54–0.75; REML + CL2 covers them at 0.92–0.94 and 0.93–0.95; NCV + CL2 at 0.73–0.91 and 0.90–0.92. On the datasets with dependent source curves (curve flips, so independent simulated errors), REML + CL2 covers β at 0.88–0.93. Over all nine datasets, it covers β at 0.88–0.94. The lowest is the weather dataset: 0.88 on average (combinations 0.84–0.91) with all 73 stations and 0.83 (0.77–0.87) with 40, on a covariate of rank 4.

With 40 subjects, REML + CL2 loses 0.4–1.7 pp on the datasets with independent source curves. On these datasets, the ratio of REML + CL2's root mean square standard error to the replicate SD of the estimate is 0.90–1.01 (median over grid points; S4.1(a)). For a centered normal estimator, a ratio r gives coverage of about $2\Phi(1.96\,r) - 1$, for example 0.92 at $r = 0.90$. The ratios are therefore consistent with a standard error a little short and with residual bias; the shortfall was not decomposed further.

The lower tail of REML + CL2 for β depends on the truth source: under the REML-fitted truth, its 5% quantile on these datasets is 0.85–0.90; under the NCV-fitted truth, it is 0.88–0.90.

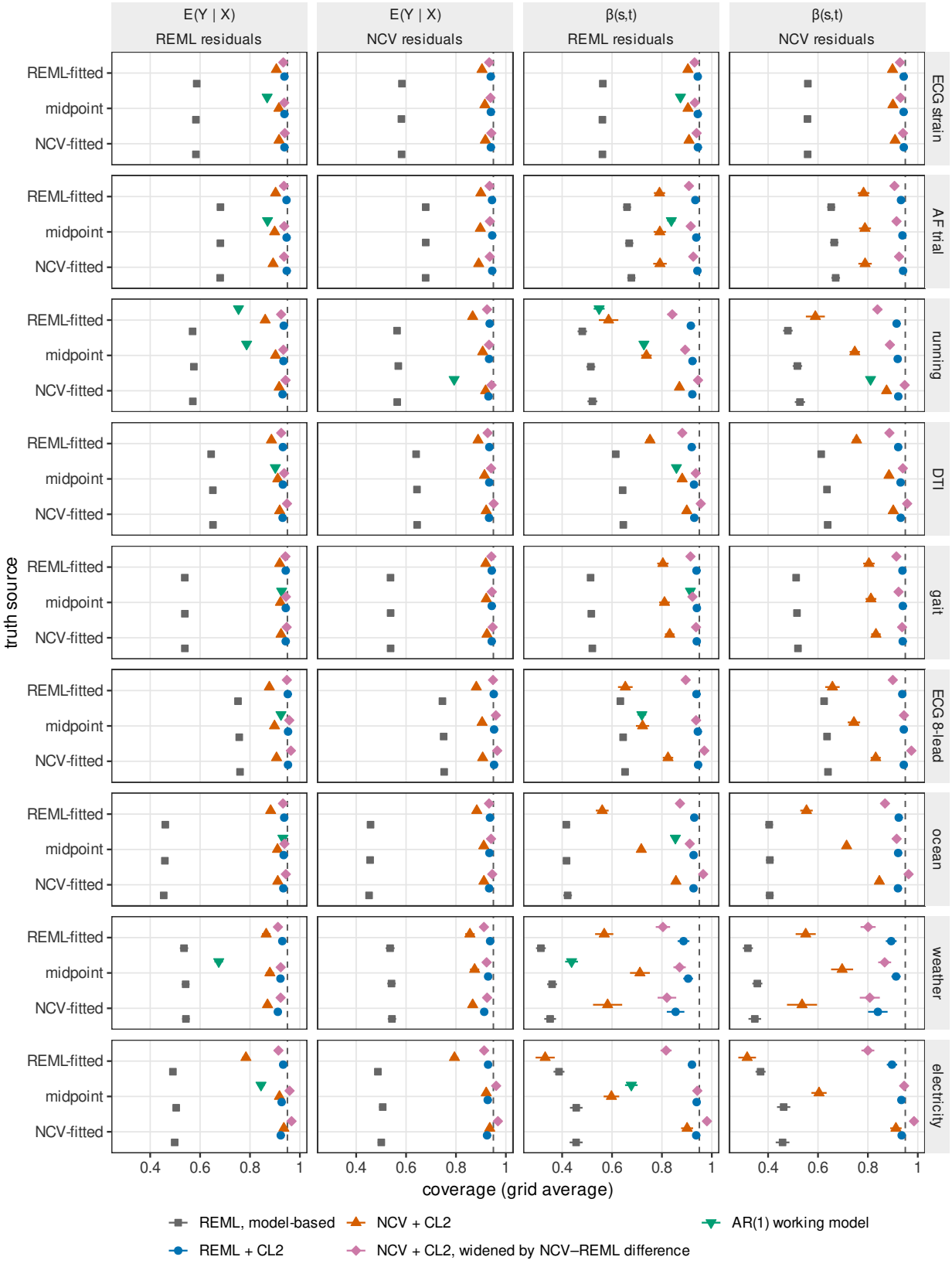


Figure 14: Plasmode coverage of the mean and of $\beta(s,t)$, all subjects, by dataset (rows), truth source (y axis) and interval (color and symbol). Columns: estimand and residual source. The AR(1) working model was fitted with the midpoint truth and REML residuals and in two further ECG strain and running combinations. Bars: ±2 MC SE; dashed: nominal 0.95.

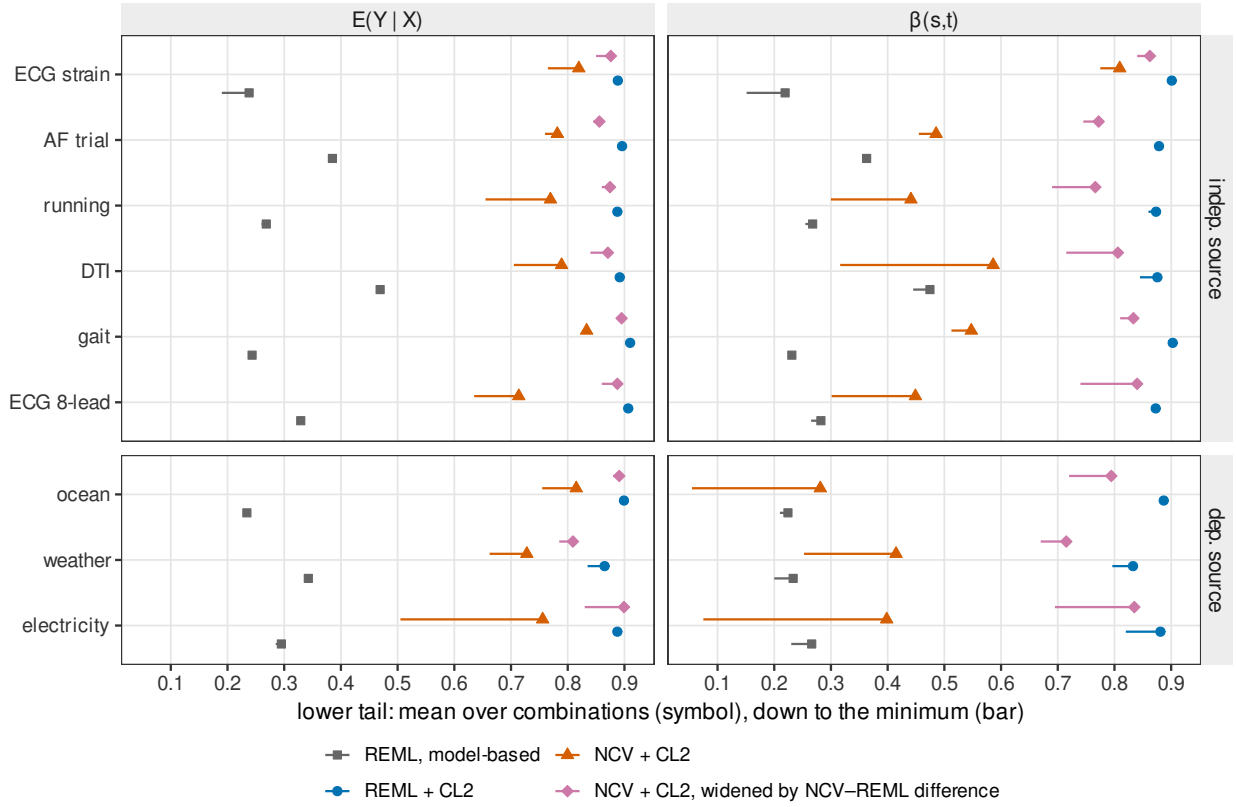


Figure 15: Lower tail (5% quantile of pointwise coverage) per dataset and interval, all subjects: mean over each dataset's combinations (symbol), down to their minimum (bar). Values: S7.

Table 19: REML + CL2 for $\beta(s,t)$, all subjects, by truth source (mean over the two fitted residual sources): coverage and 5% quantile of pointwise coverage.

dataset	role	cov REML truth	cov midpoint	cov NCV truth	q05 REML truth	q05 midpoint	q05 NCV truth
ECG strain	indep. curves	0.943	0.943	0.944	0.901	0.900	0.903
AF trial	indep. curves	0.933	0.938	0.942	0.873	0.878	0.885
running	indep. curves	0.915	0.921	0.922	0.860	0.880	0.880
DTI	indep. curves	0.920	0.930	0.931	0.847	0.889	0.891
gait	indep. curves	0.939	0.940	0.940	0.903	0.905	0.900
ECG 8-lead	indep. curves	0.938	0.944	0.944	0.870	0.873	0.876
ocean	dep. source	0.927	0.924	0.924	0.888	0.890	0.883
weather	dep. source	0.890	0.909	0.847	0.830	0.860	0.808
electricity	dep. source	0.909	0.937	0.936	0.845	0.897	0.901

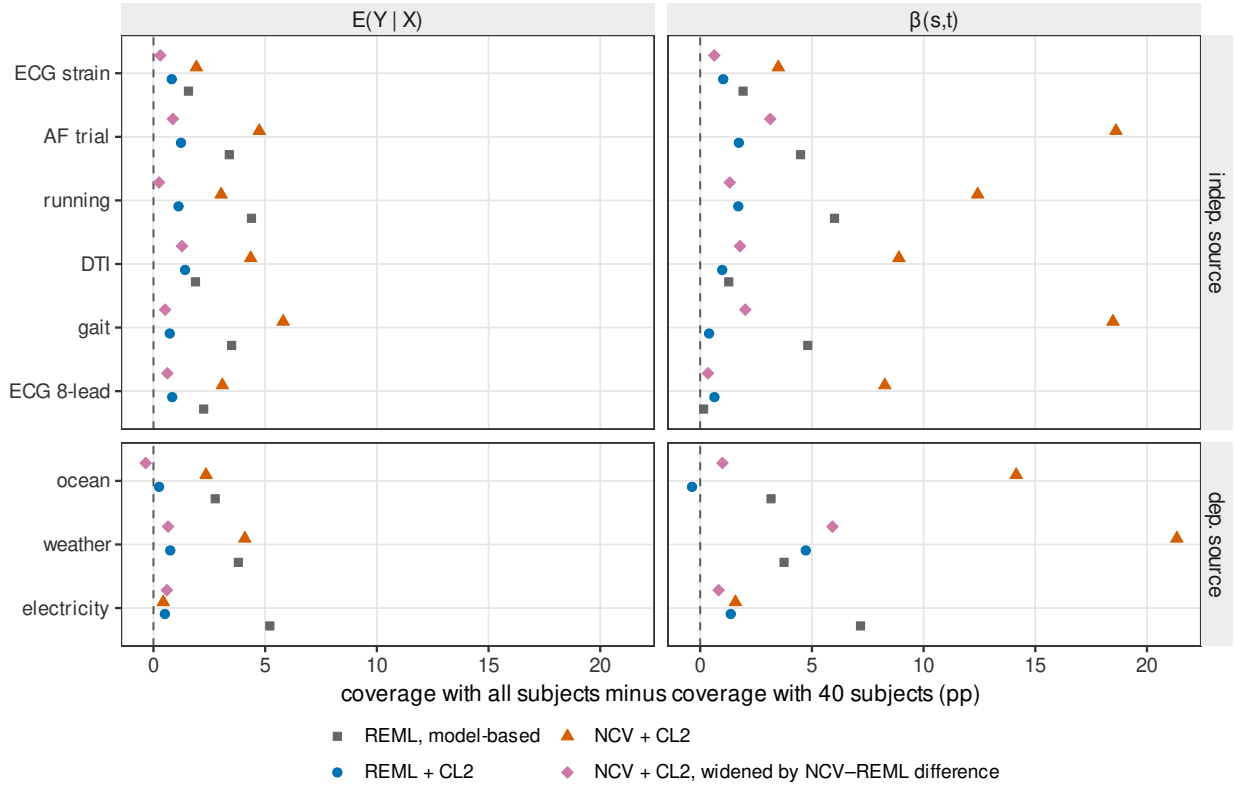


Figure 16: Coverage with all subjects minus coverage with 40 subjects (mean of paired differences over truth × residual combinations), in percentage points. Dashed: no difference. Values: S7.

Satterthwaite critical values. With the per-point Satterthwaite critical value, REML + CL2 covers β at 0.94 on average over the nine datasets with all subjects and 0.94 with 40 (normal critical value: 0.93 and 0.91), at 1.05 and 1.13 times the width (next figure; means over each dataset's combinations of truth and residual source). Coverage stays below 0.93 only for $\alpha(t)$ of the gait data (0.925 with 40 subjects); β of the weather data (0.863 with 40 subjects, 0.899 with all subjects); $\gamma(t)$ of the weather data (0.921 with 40 subjects, 0.927 with all subjects); $\alpha(t)$ of the electricity data (0.879 with 40 subjects, 0.912 with all subjects). The curve flips make the simulated errors independent across curves, so these shortfalls do not come from dependence between curves. The paired gain over the normal critical value is positive in every combination (smallest 0.13 pp). Coverage of β is below 0.95 in 120 of the 120 combinations with the normal critical value and in 63 with the Satterthwaite critical value.

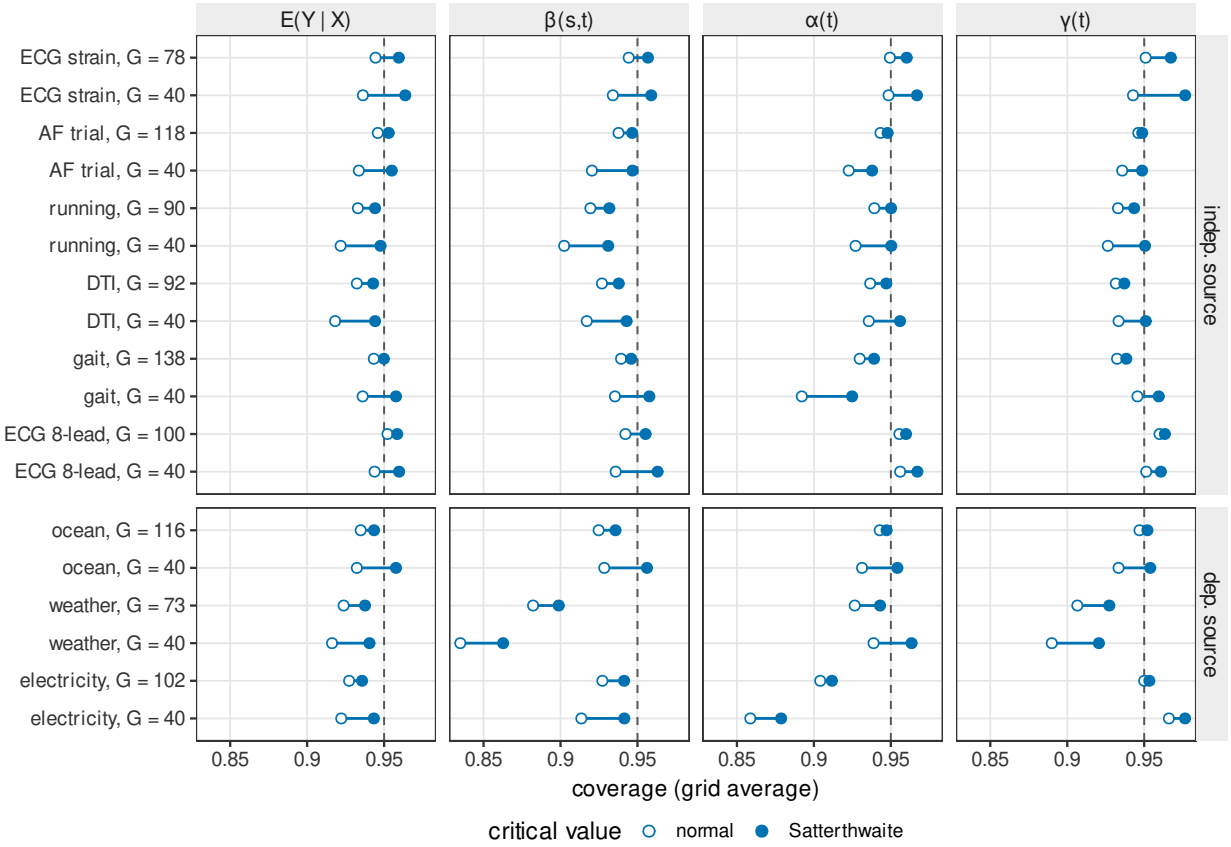


Figure 17: Plasmode study, curve flips: grid-averaged coverage of REML + CL2 with the normal and the Satterthwaite critical value per dataset and number of subjects G (mean over the dataset's combinations of truth and residual source). Dashed: nominal 0.95. Values and width ratios for $\beta(s,t)$: S7.

Table 20: Grid-averaged coverage (MC SE) per plasmode combination and interval, for $\beta(s,t)$ and the mean. REML model: model-based REML interval; NCV wid.: NCV + CL2 widened by the NCV–REML difference; AR(1): AR(1) working model.

						β(s,t)					E(Y | X)				
dataset	truth	residual	flip	error	G	REML model	REML+CL2	NCV+CL2	NCV wid.	AR(1)	REML model	REML+CL2	NCV+CL2	NCV wid.	AR(1)
ECG strain	midpoint	NCV residuals	curve	attached	78	0.558 (0.006)	0.943 (0.003)	0.900 (0.005)	0.931 (0.004)		0.581 (0.004)	0.940 (0.002)	0.916 (0.003)	0.938 (0.002)	
ECG strain	midpoint	REML residuals	curve	attached	78	0.561 (0.006)	0.944 (0.003)	0.904 (0.005)	0.932 (0.004)	0.874 (0.005)	0.583 (0.004)	0.938 (0.002)	0.916 (0.003)	0.937 (0.002)	0.869 (0.003)
ECG strain	midpoint	beat differences (own scale)	curve	attached	78	0.534 (0.005)	0.945 (0.004)	0.933 (0.004)	0.946 (0.003)		0.570 (0.003)	0.950 (0.002)	0.940 (0.002)	0.951 (0.002)	
ECG strain	midpoint	beat differences (variance-matched)	curve	attached	78	0.524 (0.005)	0.945 (0.004)	0.910 (0.005)	0.940 (0.003)	0.837 (0.004)	0.564 (0.003)	0.950 (0.002)	0.923 (0.002)	0.947 (0.002)	0.847 (0.003)
ECG strain	NCV-fitted	NCV residuals	curve	attached	78	0.559 (0.006)	0.944 (0.003)	0.909 (0.005)	0.941 (0.003)		0.582 (0.004)	0.940 (0.002)	0.918 (0.003)	0.942 (0.002)	
ECG strain	NCV-fitted	REML residuals	curve	attached	78	0.562 (0.006)	0.944 (0.003)	0.909 (0.005)	0.939 (0.003)		0.583 (0.004)	0.938 (0.003)	0.916 (0.003)	0.939 (0.002)	
ECG strain	NCV-fitted	beat differences (own scale)	curve	attached	78	0.532 (0.005)	0.945 (0.004)	0.928 (0.004)	0.946 (0.003)		0.569 (0.003)	0.950 (0.002)	0.935 (0.002)	0.950 (0.002)	
ECG strain	NCV-fitted	beat differences (variance-matched)	curve	attached	78	0.523 (0.005)	0.945 (0.004)	0.906 (0.005)	0.942 (0.003)		0.564 (0.003)	0.950 (0.002)	0.921 (0.002)	0.948 (0.002)	
ECG strain	REML-fitted	NCV residuals	curve	attached	78	0.559 (0.006)	0.943 (0.003)	0.899 (0.005)	0.929 (0.004)		0.583 (0.004)	0.939 (0.002)	0.904 (0.003)	0.933 (0.002)	
ECG strain	REML-fitted	REML residuals	curve	attached	78	0.563 (0.006)	0.943 (0.003)	0.904 (0.005)	0.931 (0.004)		0.586 (0.004)	0.938 (0.002)	0.906 (0.003)	0.933 (0.002)	
ECG strain	REML-fitted	beat differences (own scale)	curve	attached	78	0.536 (0.005)	0.945 (0.003)	0.932 (0.004)	0.944 (0.003)		0.579 (0.003)	0.950 (0.002)	0.940 (0.002)	0.948 (0.002)	
ECG strain	REML-fitted	beat differences (variance-matched)	curve	attached	78	0.526 (0.005)	0.944 (0.004)	0.910 (0.004)	0.939 (0.003)		0.571 (0.003)	0.949 (0.002)	0.921 (0.002)	0.945 (0.002)	
ECG strain	midpoint	NCV residuals	curve	attached	40	0.529 (0.006)	0.935 (0.004)	0.873 (0.006)	0.928 (0.004)		0.562 (0.004)	0.932 (0.002)	0.901 (0.003)	0.937 (0.002)	
ECG strain	midpoint	REML residuals	curve	attached	40	0.532 (0.006)	0.932 (0.004)	0.873 (0.006)	0.925 (0.004)		0.564 (0.004)	0.932 (0.003)	0.902 (0.003)	0.936 (0.003)	
ECG strain	midpoint	beat differences (own scale)	curve	attached	40	0.524 (0.006)	0.934 (0.004)	0.913 (0.005)	0.936 (0.004)		0.559 (0.004)	0.941 (0.002)	0.926 (0.003)	0.945 (0.002)	
ECG strain	midpoint	beat differences (variance-matched)	curve	attached	40	0.516 (0.006)	0.935 (0.004)	0.845 (0.011)	0.930 (0.004)		0.554 (0.004)	0.941 (0.003)	0.890 (0.006)	0.943 (0.002)	
ECG strain	NCV-fitted	NCV residuals	curve	attached	40	0.530 (0.006)	0.936 (0.004)	0.887 (0.006)	0.942 (0.004)		0.562 (0.004)	0.933 (0.002)	0.908 (0.003)	0.943 (0.002)	
ECG strain	NCV-fitted	REML residuals	curve	attached	40	0.533 (0.006)	0.933 (0.004)	0.887 (0.006)	0.941 (0.004)		0.564 (0.004)	0.932 (0.003)	0.908 (0.003)	0.942 (0.003)	
ECG strain	NCV-fitted	beat differences (own scale)	curve	attached	40	0.522 (0.006)	0.935 (0.004)	0.910 (0.005)	0.939 (0.004)		0.559 (0.004)	0.941 (0.002)	0.924 (0.003)	0.946 (0.002)	
ECG strain	NCV-fitted	beat differences (variance-matched)	curve	attached	40	0.515 (0.006)	0.935 (0.004)	0.854 (0.011)	0.938 (0.004)		0.554 (0.004)	0.941 (0.003)	0.896 (0.005)	0.947 (0.002)	
ECG strain	REML-fitted	NCV residuals	curve	attached	40	0.530 (0.006)	0.934 (0.004)	0.867 (0.006)	0.922 (0.004)		0.562 (0.004)	0.931 (0.002)	0.887 (0.003)	0.930 (0.002)	
ECG strain	REML-fitted	REML residuals	curve	attached	40	0.533 (0.006)	0.932 (0.004)	0.868 (0.006)	0.921 (0.004)		0.565 (0.004)	0.931 (0.003)	0.885 (0.003)	0.930 (0.003)	
ECG strain	REML-fitted	beat differences (own scale)	curve	attached	40	0.526 (0.006)	0.934 (0.004)	0.911 (0.004)	0.933 (0.004)		0.566 (0.004)	0.940 (0.002)	0.924 (0.003)	0.941 (0.002)	
ECG strain	REML-fitted	beat differences (variance-matched)	curve	attached	40	0.517 (0.006)	0.934 (0.004)	0.837 (0.012)	0.927 (0.004)		0.558 (0.004)	0.940 (0.002)	0.876 (0.006)	0.938 (0.002)	
AF trial	midpoint	NCV residuals	curve	attached	118	0.665 (0.008)	0.939 (0.005)	0.788 (0.012)	0.915 (0.006)		0.679 (0.004)	0.945 (0.002)	0.898 (0.004)	0.936 (0.003)	
AF trial	midpoint	REML residuals	curve	attached	118	0.669 (0.008)	0.938 (0.004)	0.792 (0.012)	0.915 (0.006)	0.838 (0.008)	0.681 (0.004)	0.947 (0.002)	0.899 (0.005)	0.937 (0.003)	0.870 (0.004)
AF trial	NCV-fitted	NCV residuals	curve	attached	118	0.671 (0.009)	0.941 (0.004)	0.789 (0.013)	0.925 (0.005)		0.678 (0.004)	0.945 (0.003)	0.891 (0.005)	0.936 (0.003)	
AF trial	NCV-fitted	REML residuals	curve	attached	118	0.678 (0.008)	0.942 (0.004)	0.792 (0.013)	0.926 (0.005)		0.681 (0.004)	0.947 (0.002)	0.893 (0.005)	0.937 (0.003)	
AF trial	REML-fitted	NCV residuals	curve	attached	118	0.653 (0.008)	0.933 (0.005)	0.783 (0.012)	0.907 (0.006)		0.679 (0.004)	0.945 (0.002)	0.900 (0.004)	0.935 (0.003)	
AF trial	REML-fitted	REML residuals	curve	attached	118	0.660 (0.008)	0.934 (0.004)	0.790 (0.011)	0.908 (0.006)		0.681 (0.004)	0.946 (0.002)	0.903 (0.004)	0.936 (0.003)	
AF trial	midpoint	NCV residuals	curve	attached	40	0.620 (0.010)	0.920 (0.006)	0.596 (0.018)	0.883 (0.007)		0.644 (0.006)	0.933 (0.004)	0.849 (0.006)	0.926 (0.004)	
AF trial	midpoint	REML residuals	curve	attached	40	0.625 (0.010)	0.922 (0.006)	0.589 (0.018)	0.883 (0.007)		0.648 (0.006)	0.934 (0.004)	0.849 (0.006)	0.927 (0.004)	
AF trial	NCV-fitted	NCV residuals	curve	attached	40	0.628 (0.011)	0.925 (0.006)	0.640 (0.016)	0.904 (0.006)		0.645 (0.006)	0.934 (0.004)	0.853 (0.006)	0.930 (0.004)	
AF trial	NCV-fitted	REML residuals	curve	attached	40	0.631 (0.011)	0.926 (0.006)	0.649 (0.016)	0.906 (0.006)		0.647 (0.006)	0.935 (0.004)	0.856 (0.005)	0.931 (0.003)	
AF trial	REML-fitted	NCV residuals	curve	attached	40	0.609 (0.010)	0.913 (0.006)	0.570 (0.019)	0.868 (0.008)		0.644 (0.006)	0.932 (0.004)	0.845 (0.006)	0.924 (0.004)	
AF trial	REML-fitted	REML residuals	curve	attached	40	0.614 (0.010)	0.916 (0.006)	0.572 (0.018)	0.865 (0.008)		0.648 (0.005)	0.934 (0.004)	0.847 (0.006)	0.925 (0.004)	
running	midpoint	NCV residuals	curve	attached	90	0.518 (0.009)	0.919 (0.006)	0.747 (0.010)	0.888 (0.006)		0.569 (0.005)	0.933 (0.003)	0.907 (0.003)	0.933 (0.003)	
running	midpoint	REML residuals	curve	attached	90	0.515 (0.009)	0.922 (0.005)	0.738 (0.011)	0.893 (0.006)	0.728 (0.006)	0.575 (0.004)	0.934 (0.003)	0.902 (0.003)	0.934 (0.003)	0.787 (0.004)
running	NCV-fitted	NCV residuals	curve	attached	90	0.528 (0.010)	0.922 (0.005)	0.875 (0.006)	0.948 (0.004)	0.811 (0.005)	0.565 (0.005)	0.930 (0.003)	0.919 (0.003)	0.943 (0.003)	0.793 (0.004)
running	NCV-fitted	NCV residuals	curve	detached	90	0.506 (0.011)	0.926 (0.006)	0.844 (0.010)	0.947 (0.005)		0.577 (0.005)	0.936 (0.003)	0.917 (0.004)	0.947 (0.003)	
running	NCV-fitted	REML residuals	curve	attached	90	0.521 (0.010)	0.922 (0.005)	0.870 (0.006)	0.945 (0.004)		0.571 (0.005)	0.931 (0.003)	0.917 (0.003)	0.943 (0.003)	
running	REML-fitted	NCV residuals	curve	attached	90	0.479 (0.010)	0.915 (0.007)	0.589 (0.019)	0.838 (0.009)		0.564 (0.005)	0.935 (0.003)	0.867 (0.005)	0.925 (0.003)	
running	REML-fitted	REML residuals	curve	attached	90	0.481 (0.009)	0.916 (0.007)	0.586 (0.020)	0.841 (0.010)	0.548 (0.011)	0.570 (0.004)	0.935 (0.003)	0.861 (0.005)	0.925 (0.003)	0.754 (0.004)
running	REML-fitted	REML residuals	curve	detached	90	0.468 (0.008)	0.919 (0.006)	0.505 (0.018)	0.850 (0.009)		0.582 (0.005)	0.939 (0.003)	0.841 (0.006)	0.932 (0.003)	
running	midpoint	NCV residuals	curve	attached	40	0.452 (0.010)	0.903 (0.007)	0.636 (0.015)	0.880 (0.007)		0.527 (0.006)	0.922 (0.004)	0.871 (0.005)	0.930 (0.003)	
running	midpoint	REML residuals	curve	attached	40	0.467 (0.010)	0.912 (0.007)	0.638 (0.014)	0.889 (0.006)		0.531 (0.006)	0.923 (0.004)	0.871 (0.005)	0.932 (0.003)	
running	NCV-fitted	NCV residuals	curve	attached	40	0.463 (0.011)	0.910 (0.007)	0.760 (0.014)	0.927 (0.006)		0.522 (0.006)	0.921 (0.004)	0.885 (0.005)	0.937 (0.003)	
running	NCV-fitted	NCV residuals	curve	detached	40	0.453 (0.011)	0.926 (0.006)	0.747 (0.013)	0.929 (0.005)		0.533 (0.006)	0.926 (0.004)	0.888 (0.005)	0.941 (0.004)	
running	NCV-fitted	REML residuals	curve	attached	40	0.463 (0.011)	0.914 (0.006)	0.767 (0.013)	0.928 (0.005)		0.527 (0.006)	0.921 (0.004)	0.886 (0.005)	0.938 (0.003)	
running	REML-fitted	NCV residuals	curve	attached	40	0.414 (0.010)	0.888 (0.009)	0.432 (0.015)	0.823 (0.010)		0.520 (0.006)	0.922 (0.004)	0.841 (0.005)	0.924 (0.004)	
running	REML-fitted	REML residuals	curve	attached	40	0.421 (0.010)	0.887 (0.009)	0.427 (0.015)	0.829 (0.011)		0.522 (0.005)	0.921 (0.004)	0.837 (0.006)	0.925 (0.004)	
running	REML-fitted	REML residuals	curve	detached	40	0.422 (0.009)	0.909 (0.007)	0.412 (0.014)	0.836 (0.009)		0.536 (0.005)	0.928 (0.004)	0.839 (0.005)	0.929 (0.004)	
DTI	midpoint	NCV residuals	curve	attached	92	0.636 (0.005)	0.931 (0.003)	0.885 (0.005)	0.940 (0.003)		0.644 (0.004)	0.933 (0.002)	0.914 (0.003)	0.941 (0.002)	
DTI	midpoint	REML residuals	curve	attached	92	0.643 (0.005)	0.929 (0.003)	0.881 (0.005)	0.936 (0.003)	0.858 (0.007)	0.651 (0.004)	0.932 (0.002)	0.911 (0.003)	0.937 (0.002)	0.902 (0.004)
DTI	NCV-fitted	NCV residuals	curve	attached	92	0.639 (0.006)	0.931 (0.003)	0.902 (0.005)	0.957 (0.003)		0.644 (0.004)	0.933 (0.002)	0.921 (0.003)	0.951 (0.002)	
DTI	NCV-fitted	REML residuals	curve	attached	92	0.645 (0.006)	0.930 (0.003)	0.900 (0.005)	0.956 (0.003)		0.652 (0.004)	0.931 (0.002)	0.919 (0.003)	0.948 (0.002)	
DTI	REML-fitted	NCV residuals	curve	attached	92	0.613 (0.005)	0.922 (0.004)	0.754 (0.008)	0.886 (0.005)		0.640 (0.004)	0.934 (0.002)	0.889 (0.003)	0.927 (0.002)	

Table 20: Grid-averaged coverage (MC SE) per plasmode combination and interval, for $\beta(s,t)$ and the mean. REML model: model-based REML interval; NCV wid.: NCV + CL2 widened by the NCV–REML difference; AR(1): AR(1) working model. *(continued)*

dataset	truth	residual	flip	error	G	REML model	REML+CL2	NCV+CL2	NCV wid.	AR(1)	REML model	REML+CL2	NCV+CL2	NCV wid.	AR(1)
DTI	REML-fitted	REML residuals	curve	attached	92	0.615 (0.005)	0.919 (0.004)	0.752 (0.008)	0.882 (0.005)		0.644 (0.004)	0.932 (0.002)	0.886 (0.003)	0.925 (0.002)	
DTI	midpoint	NCV residuals	curve	attached	40	0.620 (0.006)	0.919 (0.003)	0.787 (0.010)	0.919 (0.004)		0.623 (0.004)	0.918 (0.003)	0.867 (0.004)	0.926 (0.003)	
DTI	midpoint	REML residuals	curve	attached	40	0.631 (0.006)	0.922 (0.003)	0.799 (0.009)	0.921 (0.004)		0.633 (0.004)	0.919 (0.003)	0.870 (0.004)	0.925 (0.003)	
DTI	NCV-fitted	NCV residuals	curve	attached	40	0.626 (0.007)	0.921 (0.003)	0.803 (0.012)	0.933 (0.005)		0.625 (0.004)	0.918 (0.003)	0.877 (0.005)	0.934 (0.003)	
DTI	NCV-fitted	REML residuals	curve	attached	40	0.635 (0.007)	0.922 (0.004)	0.801 (0.013)	0.929 (0.005)		0.634 (0.004)	0.917 (0.003)	0.874 (0.005)	0.931 (0.003)	
DTI	REML-fitted	NCV residuals	curve	attached	40	0.599 (0.006)	0.909 (0.004)	0.671 (0.011)	0.874 (0.006)		0.620 (0.004)	0.919 (0.003)	0.844 (0.005)	0.919 (0.003)	
DTI	REML-fitted	REML residuals	curve	attached	40	0.604 (0.006)	0.909 (0.004)	0.679 (0.010)	0.873 (0.005)		0.628 (0.004)	0.919 (0.003)	0.847 (0.004)	0.918 (0.003)	
gait	midpoint	NCV residuals	curve	attached	138	0.516 (0.007)	0.940 (0.004)	0.812 (0.011)	0.923 (0.004)		0.538 (0.004)	0.943 (0.003)	0.922 (0.003)	0.944 (0.002)	
gait	midpoint	REML residuals	curve	attached	138	0.517 (0.007)	0.940 (0.004)	0.810 (0.011)	0.922 (0.004)	0.914 (0.004)	0.539 (0.004)	0.943 (0.003)	0.922 (0.003)	0.944 (0.002)	0.927 (0.003)
gait	NCV-fitted	NCV residuals	curve	attached	138	0.520 (0.007)	0.940 (0.004)	0.832 (0.010)	0.937 (0.004)		0.538 (0.004)	0.943 (0.003)	0.924 (0.003)	0.947 (0.002)	
gait	NCV-fitted	REML residuals	curve	attached	138	0.521 (0.007)	0.940 (0.005)	0.831 (0.010)	0.937 (0.004)		0.539 (0.004)	0.943 (0.003)	0.924 (0.003)	0.947 (0.002)	
gait	REML-fitted	NCV residuals	curve	attached	138	0.512 (0.006)	0.939 (0.004)	0.804 (0.011)	0.914 (0.004)		0.538 (0.004)	0.943 (0.003)	0.919 (0.003)	0.942 (0.002)	
gait	REML-fitted	REML residuals	curve	attached	138	0.514 (0.006)	0.938 (0.004)	0.804 (0.011)	0.914 (0.004)		0.539 (0.004)	0.943 (0.003)	0.919 (0.003)	0.942 (0.002)	
gait	midpoint	NCV residuals	curve	attached	40	0.469 (0.006)	0.936 (0.004)	0.626 (0.014)	0.904 (0.004)		0.503 (0.004)	0.936 (0.003)	0.863 (0.006)	0.939 (0.003)	
gait	midpoint	REML residuals	curve	attached	40	0.470 (0.006)	0.936 (0.004)	0.625 (0.013)	0.905 (0.004)		0.504 (0.004)	0.936 (0.003)	0.864 (0.006)	0.940 (0.003)	
gait	NCV-fitted	NCV residuals	curve	attached	40	0.470 (0.007)	0.935 (0.004)	0.672 (0.014)	0.915 (0.004)		0.503 (0.004)	0.936 (0.003)	0.868 (0.006)	0.940 (0.003)	
gait	NCV-fitted	REML residuals	curve	attached	40	0.471 (0.007)	0.936 (0.004)	0.673 (0.014)	0.917 (0.004)		0.503 (0.004)	0.936 (0.003)	0.869 (0.006)	0.941 (0.003)	
gait	REML-fitted	NCV residuals	curve	attached	40	0.465 (0.006)	0.934 (0.004)	0.595 (0.013)	0.893 (0.004)		0.503 (0.004)	0.936 (0.003)	0.860 (0.006)	0.937 (0.003)	
gait	REML-fitted	REML residuals	curve	attached	40	0.466 (0.006)	0.935 (0.004)	0.595 (0.013)	0.893 (0.004)		0.504 (0.004)	0.936 (0.003)	0.860 (0.006)	0.938 (0.003)	
ECG 8-lead	midpoint	NCV residuals	curve	attached	100	0.636 (0.006)	0.944 (0.003)	0.744 (0.012)	0.945 (0.003)		0.751 (0.004)	0.953 (0.002)	0.905 (0.004)	0.960 (0.002)	
ECG 8-lead	midpoint	REML residuals	curve	attached	100	0.644 (0.006)	0.944 (0.003)	0.722 (0.013)	0.937 (0.004)	0.720 (0.009)	0.757 (0.004)	0.953 (0.002)	0.898 (0.005)	0.957 (0.002)	0.924 (0.003)
ECG 8-lead	NCV-fitted	NCV residuals	curve	attached	100	0.640 (0.006)	0.944 (0.003)	0.831 (0.010)	0.974 (0.002)		0.753 (0.004)	0.953 (0.002)	0.907 (0.005)	0.965 (0.002)	
ECG 8-lead	NCV-fitted	REML residuals	curve	attached	100	0.652 (0.006)	0.945 (0.003)	0.824 (0.010)	0.970 (0.003)		0.760 (0.004)	0.952 (0.002)	0.906 (0.005)	0.964 (0.002)	
ECG 8-lead	REML-fitted	NCV residuals	curve	attached	100	0.624 (0.005)	0.938 (0.003)	0.658 (0.014)	0.900 (0.006)		0.746 (0.004)	0.951 (0.002)	0.882 (0.005)	0.948 (0.002)	
ECG 8-lead	REML-fitted	REML residuals	curve	attached	100	0.633 (0.005)	0.938 (0.004)	0.653 (0.015)	0.895 (0.006)		0.752 (0.004)	0.951 (0.002)	0.878 (0.005)	0.948 (0.002)	
ECG 8-lead	midpoint	NCV residuals	curve	attached	40	0.639 (0.006)	0.937 (0.004)	0.637 (0.014)	0.940 (0.003)		0.729 (0.004)	0.944 (0.002)	0.863 (0.005)	0.951 (0.002)	
ECG 8-lead	midpoint	REML residuals	curve	attached	40	0.641 (0.006)	0.937 (0.004)	0.646 (0.014)	0.936 (0.004)		0.734 (0.005)	0.944 (0.003)	0.867 (0.005)	0.950 (0.003)	
ECG 8-lead	NCV-fitted	NCV residuals	curve	attached	40	0.640 (0.007)	0.936 (0.004)	0.770 (0.012)	0.965 (0.003)		0.730 (0.005)	0.944 (0.003)	0.881 (0.005)	0.959 (0.002)	
ECG 8-lead	NCV-fitted	REML residuals	curve	attached	40	0.645 (0.007)	0.937 (0.004)	0.774 (0.012)	0.965 (0.003)		0.736 (0.005)	0.944 (0.003)	0.884 (0.005)	0.957 (0.002)	
ECG 8-lead	REML-fitted	NCV residuals	curve	attached	40	0.625 (0.005)	0.934 (0.004)	0.550 (0.015)	0.900 (0.004)		0.724 (0.004)	0.944 (0.003)	0.846 (0.006)	0.945 (0.002)	
ECG 8-lead	REML-fitted	REML residuals	curve	attached	40	0.631 (0.005)	0.934 (0.004)	0.562 (0.016)	0.895 (0.005)		0.730 (0.004)	0.944 (0.003)	0.849 (0.006)	0.943 (0.003)	
ocean	midpoint	NCV residuals	curve	attached	116	0.407 (0.008)	0.921 (0.006)	0.715 (0.006)	0.915 (0.004)		0.456 (0.004)	0.934 (0.003)	0.911 (0.004)	0.941 (0.003)	
ocean	midpoint	REML residuals	curve	attached	116	0.418 (0.008)	0.927 (0.006)	0.717 (0.005)	0.912 (0.004)	0.854 (0.003)	0.459 (0.004)	0.935 (0.003)	0.910 (0.004)	0.938 (0.003)	0.932 (0.003)
ocean	NCV-fitted	NCV residuals	curve	attached	116	0.406 (0.008)	0.920 (0.006)	0.846 (0.008)	0.963 (0.003)		0.452 (0.004)	0.933 (0.003)	0.912 (0.004)	0.945 (0.003)	
ocean	NCV-fitted	NCV residuals	block	attached	116	0.420 (0.009)	0.934 (0.006)	0.855 (0.007)	0.966 (0.002)		0.448 (0.004)	0.931 (0.003)	0.902 (0.004)	0.937 (0.003)	
ocean	NCV-fitted	REML residuals	curve	attached	116	0.422 (0.009)	0.927 (0.006)	0.856 (0.007)	0.965 (0.002)		0.454 (0.004)	0.934 (0.003)	0.911 (0.004)	0.944 (0.003)	
ocean	REML-fitted	NCV residuals	curve	attached	116	0.404 (0.008)	0.924 (0.006)	0.554 (0.012)	0.869 (0.007)		0.458 (0.004)	0.936 (0.003)	0.883 (0.004)	0.933 (0.003)	
ocean	REML-fitted	REML residuals	curve	attached	116	0.417 (0.008)	0.930 (0.006)	0.561 (0.013)	0.872 (0.006)		0.461 (0.004)	0.937 (0.003)	0.883 (0.004)	0.932 (0.003)	
ocean	REML-fitted	REML residuals	block	attached	116	0.417 (0.007)	0.935 (0.006)	0.564 (0.014)	0.883 (0.006)		0.456 (0.004)	0.933 (0.003)	0.875 (0.004)	0.929 (0.003)	
ocean	midpoint	NCV residuals	curve	attached	40	0.376 (0.007)	0.927 (0.005)	0.585 (0.014)	0.914 (0.004)		0.427 (0.004)	0.931 (0.003)	0.882 (0.004)	0.942 (0.002)	
ocean	midpoint	REML residuals	curve	attached	40	0.387 (0.007)	0.933 (0.005)	0.589 (0.014)	0.914 (0.004)		0.432 (0.004)	0.933 (0.003)	0.882 (0.004)	0.942 (0.002)	
ocean	NCV-fitted	NCV residuals	curve	attached	40	0.375 (0.008)	0.929 (0.005)	0.689 (0.012)	0.944 (0.003)		0.425 (0.004)	0.931 (0.003)	0.885 (0.004)	0.946 (0.003)	
ocean	NCV-fitted	REML residuals	curve	attached	40	0.388 (0.008)	0.934 (0.005)	0.700 (0.012)	0.946 (0.003)		0.430 (0.004)	0.933 (0.003)	0.886 (0.004)	0.945 (0.003)	
ocean	REML-fitted	NCV residuals	curve	attached	40	0.375 (0.007)	0.922 (0.005)	0.416 (0.012)	0.860 (0.005)		0.428 (0.004)	0.932 (0.003)	0.867 (0.004)	0.940 (0.002)	
ocean	REML-fitted	REML residuals	curve	attached	40	0.383 (0.007)	0.926 (0.005)	0.420 (0.012)	0.858 (0.005)		0.432 (0.004)	0.934 (0.003)	0.868 (0.004)	0.939 (0.002)	
weather	midpoint	NCV residuals	curve	attached	73	0.357 (0.010)	0.913 (0.009)	0.697 (0.022)	0.867 (0.013)		0.542 (0.008)	0.929 (0.005)	0.875 (0.009)	0.923 (0.006)	
weather	midpoint	REML residuals	curve	attached	73	0.360 (0.010)	0.905 (0.010)	0.712 (0.020)	0.871 (0.012)	0.438 (0.013)	0.542 (0.008)	0.922 (0.005)	0.880 (0.008)	0.923 (0.006)	0.675 (0.009)
weather	NCV-fitted	NCV residuals	curve	attached	73	0.347 (0.013)	0.840 (0.020)	0.536 (0.030)	0.808 (0.020)		0.543 (0.008)	0.913 (0.006)	0.867 (0.007)	0.925 (0.005)	
weather	NCV-fitted	NCV residuals	block	attached	73	0.369 (0.011)	0.896 (0.016)	0.540 (0.031)	0.831 (0.017)		0.489 (0.006)	0.897 (0.004)	0.820 (0.008)	0.899 (0.005)	
weather	NCV-fitted	REML residuals	curve	attached	73	0.352 (0.012)	0.855 (0.018)	0.582 (0.029)	0.820 (0.019)		0.543 (0.008)	0.912 (0.005)	0.870 (0.007)	0.923 (0.005)	
weather	REML-fitted	NCV residuals	curve	attached	73	0.319 (0.010)	0.893 (0.011)	0.551 (0.020)	0.801 (0.015)		0.536 (0.009)	0.937 (0.005)	0.856 (0.010)	0.912 (0.007)	
weather	REML-fitted	REML residuals	curve	attached	73	0.315 (0.010)	0.887 (0.012)	0.569 (0.018)	0.804 (0.014)		0.536 (0.008)	0.929 (0.006)	0.865 (0.009)	0.912 (0.006)	
weather	REML-fitted	REML residuals	block	attached	73	0.350 (0.013)	0.911 (0.009)	0.605 (0.018)	0.814 (0.014)		0.522 (0.007)	0.914 (0.004)	0.842 (0.008)	0.898 (0.005)	
weather	midpoint	NCV residuals	curve	attached	40	0.310 (0.012)	0.866 (0.018)	0.454 (0.030)	0.795 (0.019)		0.496 (0.008)	0.918 (0.006)	0.818 (0.009)	0.909 (0.006)	
weather	midpoint	REML residuals	curve	attached	40	0.327 (0.012)	0.863 (0.017)	0.453 (0.030)	0.806 (0.018)		0.504 (0.008)	0.912 (0.006)	0.820 (0.009)	0.909 (0.007)	
weather	NCV-fitted	NCV residuals	curve	attached	40	0.298 (0.013)	0.798 (0.022)	0.357 (0.028)	0.741 (0.023)		0.505 (0.008)	0.915 (0.005)	0.859 (0.007)	0.923 (0.005)	
weather	NCV-fitted	REML residuals	curve	attached	40	0.305 (0.014)	0.769 (0.023)	0.351 (0.028)	0.730 (0.023)		0.511 (0.008)	0.907 (0.006)	0.855 (0.007)	0.919 (0.006)	
weather	REML-fitted	NCV residuals	curve	attached	40	0.288 (0.011)	0.861 (0.018)	0.386 (0.026)	0.769 (0.019)		0.496 (0.008)	0.925 (0.006)	0.809 (0.009)	0.908 (0.007)	
weather	REML-fitted	REML residuals	curve	attached	40	0.295 (0.011)	0.853 (0.017)	0.366 (0.025)	0.776 (0.018)		0.502 (0.008)	0.921 (0.006)	0.806 (0.009)	0.910 (0.006)	

Table 20: Grid-averaged coverage (MC SE) per plasmode combination and interval, for $\beta(s,t)$ and the mean. REML model: model-based REML interval; NCV wid.: NCV + CL2 widened by the NCV–REML difference; AR(1): AR(1) working model. *(continued)*

dataset	truth	residual	flip	error	G	REML model	REML+CL2	NCV+CL2	NCV wid.	AR(1)	REML model	REML+CL2	NCV+CL2	NCV wid.	AR(1)
electricity	midpoint	NCV residuals	curve	attached	102	0.462 (0.013)	0.934 (0.006)	0.604 (0.015)	0.945 (0.006)		0.506 (0.007)	0.927 (0.005)	0.921 (0.006)	0.961 (0.004)	
electricity	midpoint	REML residuals	curve	attached	102	0.457 (0.013)	0.939 (0.005)	0.598 (0.016)	0.942 (0.005)	0.677 (0.013)	0.504 (0.006)	0.927 (0.005)	0.918 (0.006)	0.959 (0.004)	0.844 (0.007)
electricity	NCV-fitted	NCV residuals	curve	attached	102	0.458 (0.014)	0.935 (0.007)	0.912 (0.011)	0.985 (0.004)		0.501 (0.007)	0.925 (0.005)	0.935 (0.006)	0.968 (0.004)	
electricity	NCV-fitted	NCV residuals	block	attached	102	0.452 (0.013)	0.935 (0.006)	0.915 (0.011)	0.991 (0.002)		0.490 (0.007)	0.926 (0.004)	0.925 (0.007)	0.966 (0.004)	
electricity	NCV-fitted	REML residuals	curve	attached	102	0.456 (0.013)	0.937 (0.006)	0.901 (0.012)	0.980 (0.004)		0.498 (0.006)	0.923 (0.005)	0.936 (0.006)	0.967 (0.004)	
electricity	REML-fitted	NCV residuals	curve	attached	102	0.370 (0.010)	0.897 (0.010)	0.316 (0.018)	0.800 (0.013)		0.487 (0.007)	0.929 (0.005)	0.794 (0.008)	0.913 (0.006)	
electricity	REML-fitted	REML residuals	curve	attached	102	0.387 (0.011)	0.920 (0.009)	0.332 (0.019)	0.817 (0.011)		0.491 (0.006)	0.933 (0.005)	0.784 (0.007)	0.914 (0.005)	
electricity	REML-fitted	REML residuals	block	attached	102	0.390 (0.010)	0.905 (0.011)	0.308 (0.020)	0.799 (0.014)		0.493 (0.007)	0.935 (0.005)	0.773 (0.008)	0.911 (0.005)	
electricity	midpoint	NCV residuals	curve	attached	40	0.385 (0.012)	0.936 (0.007)	0.672 (0.011)	0.967 (0.004)		0.451 (0.007)	0.924 (0.005)	0.905 (0.007)	0.953 (0.005)	
electricity	midpoint	REML residuals	curve	attached	40	0.378 (0.011)	0.932 (0.007)	0.656 (0.012)	0.959 (0.005)		0.455 (0.008)	0.923 (0.005)	0.908 (0.007)	0.953 (0.005)	
electricity	NCV-fitted	NCV residuals	curve	attached	40	0.381 (0.013)	0.936 (0.006)	0.869 (0.016)	0.982 (0.004)		0.446 (0.007)	0.923 (0.005)	0.910 (0.008)	0.957 (0.005)	
electricity	NCV-fitted	REML residuals	curve	attached	40	0.386 (0.012)	0.938 (0.006)	0.880 (0.014)	0.982 (0.004)		0.452 (0.008)	0.923 (0.005)	0.915 (0.007)	0.957 (0.005)	
electricity	REML-fitted	NCV residuals	curve	attached	40	0.312 (0.009)	0.861 (0.012)	0.252 (0.016)	0.752 (0.014)		0.433 (0.006)	0.917 (0.005)	0.816 (0.007)	0.911 (0.006)	
electricity	REML-fitted	REML residuals	curve	attached	40	0.319 (0.009)	0.878 (0.010)	0.239 (0.015)	0.778 (0.012)		0.438 (0.007)	0.924 (0.005)	0.808 (0.007)	0.915 (0.005)	

S2.2 Estimation by truth source; relative error and detection

The NCV/REML ratio of mean squared errors for $\beta(s,t)$ (geometric mean over residual sources, all subjects) is 0.04–0.78 under the NCV-fitted truth, 0.05–0.90 under the midpoint truth (smoothing parameters at the geometric mean of the REML and NCV values) and 0.32–2.24 under the REML-fitted truth; the ratio exceeds 1 under the REML-fitted truth on 4 of 9 datasets (electricity, DTI, ocean, running). A truth taken from one fit has that fit's smoothness, so each truth source favors its own selector. The midpoint truth does not remove this effect: its fit lies between the REML and NCV fits in EDF for every term on only 3 of 9 datasets, and its surface EDF exceeds both fits' on running, ocean, weather, where it is the roughest of the three truths. The relative error of β is large for both fits on several datasets (next table), possibly because β is weakly identified on these covariates.

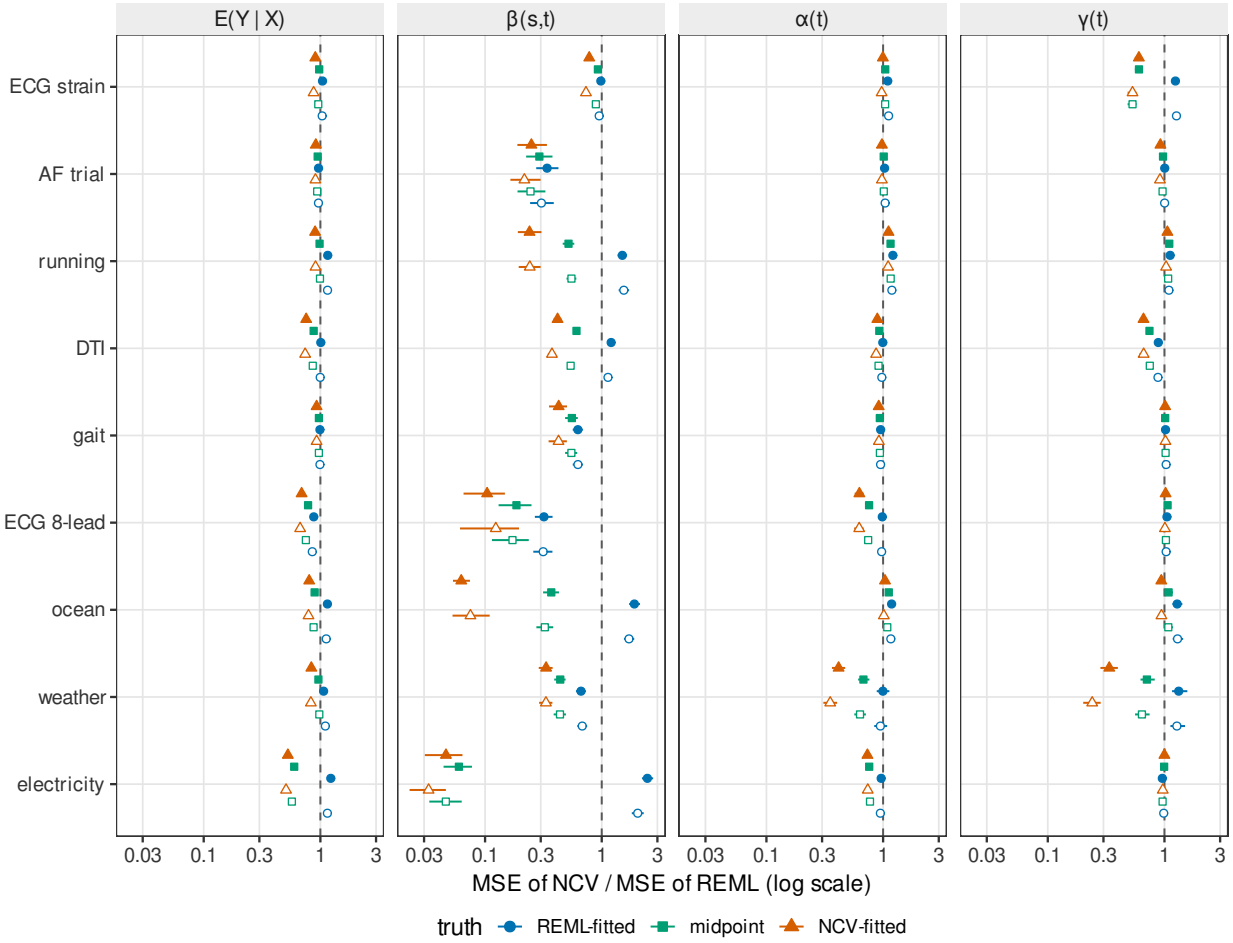


Figure 18: NCV/REML ratio of mean squared errors per dataset and truth source (color and symbol), all subjects, with 95% paired bootstrap intervals; log scale. Filled symbols: REML residuals; open symbols: NCV residuals. Dashed: equal error.

Table 21: NCV/REML MSE ratio per dataset and truth source (geometric mean over residual sources, all subjects).

role	dataset	estimand	REML	midpoint	NCV
indep. curves	ECG strain	α(t)	1.071	1.029	0.980
indep. curves	AF trial	α(t)	1.035	1.013	0.977
indep. curves	running	α(t)	1.201	1.160	1.108
indep. curves	DTI	α(t)	0.984	0.921	0.881
indep. curves	gait	α(t)	0.952	0.938	0.919
indep. curves	ECG 8-lead	α(t)	0.977	0.751	0.625
dep. source	ocean	α(t)	1.174	1.102	1.026
dep. source	weather	α(t)	0.971	0.656	0.383
dep. source	electricity	α(t)	0.955	0.766	0.736
indep. curves	ECG strain	β(s,t)	0.952	0.904	0.780
indep. curves	AF trial	β(s,t)	0.321	0.268	0.233
indep. curves	running	β(s,t)	1.523	0.534	0.241
indep. curves	DTI	β(s,t)	1.167	0.573	0.396
indep. curves	gait	β(s,t)	0.625	0.552	0.427
indep. curves	ECG 8-lead	β(s,t)	0.317	0.179	0.113
dep. source	ocean	β(s,t)	1.806	0.347	0.068
dep. source	weather	β(s,t)	0.672	0.440	0.333
dep. source	electricity	β(s,t)	2.237	0.052	0.039
indep. curves	ECG strain	γ(t)	1.184	0.634	0.628
indep. curves	AF trial	γ(t)	1.004	0.968	0.919
indep. curves	running	γ(t)	1.106	1.088	1.046
indep. curves	DTI	γ(t)	0.883	0.746	0.662
indep. curves	gait	γ(t)	1.028	1.018	1.017
indep. curves	ECG 8-lead	γ(t)	1.042	1.046	1.016
dep. source	ocean	γ(t)	1.290	1.082	0.941
dep. source	weather	γ(t)	1.301	0.673	0.284
dep. source	electricity	γ(t)	0.972	0.979	0.983
indep. curves	ECG strain	E(Y | X)	1.016	0.966	0.907
indep. curves	AF trial	E(Y | X)	0.964	0.943	0.910
indep. curves	running	E(Y | X)	1.154	0.988	0.904
indep. curves	DTI	E(Y | X)	1.002	0.868	0.748
indep. curves	gait	E(Y | X)	0.991	0.971	0.928
indep. curves	ECG 8-lead	E(Y | X)	0.865	0.767	0.681
dep. source	ocean	E(Y | X)	1.136	0.884	0.796
dep. source	weather	E(Y | X)	1.084	0.970	0.833
dep. source	electricity	E(Y | X)	1.188	0.583	0.517

Table 22: $\beta(s,t)$, all subjects: coverage, median relative error of the estimate, median relative half-width (hw) and median detection rate (det) per interval and dataset (over truth × residual combinations); NCV–REML widening: NCV + CL2 widened by the NCV–REML difference (S4.1).

role	dataset	cov NCV+CL2	cov NCV–REML widening	cov REML+CL2	rel NCV+CL2	rel REML+CL2	hw NCV+CL2	hw NCV–REML widening	hw REML+CL2	det NCV+CL2	det NCV–REML widening	det REML+CL2
indep. curves	ECG strain	0.912	0.938	0.944	0.676	0.698	1.177	1.30	1.42	0.461	0.404	0.442
indep. curves	AF trial	0.789	0.916	0.938	0.714	0.865	0.963	1.53	1.82	0.533	0.303	0.372
indep. curves	running	0.734	0.892	0.919	0.661	0.825	0.678	1.52	1.85	0.735	0.578	0.637
indep. curves	DTI	0.846	0.926	0.927	0.469	0.633	0.760	1.08	1.17	0.772	0.562	0.638
indep. curves	gait	0.816	0.925	0.939	0.765	1.007	1.024	1.93	2.39	0.650	0.486	0.546
indep. curves	ECG 8-lead	0.739	0.937	0.942	0.806	2.427	0.851	4.37	4.88	0.568	0.210	0.216
dep. source	ocean	0.708	0.916	0.925	0.667	1.035	0.527	2.11	2.25	0.528	0.178	0.450
dep. source	weather	0.608	0.828	0.882	0.678	1.007	0.771	1.79	1.98	0.538	0.228	0.321
dep. source	electricity	0.611	0.911	0.927	0.815	3.418	0.564	5.62	5.72	0.512	0.094	0.076

Table 23: NCV/REML MSE ratio per plasmode combination and estimand (95% paired bootstrap intervals).

dataset	truth	residual	flip	error	G	alpha	beta	gamma	mean
ECG strain	midpoint	NCV residuals	curve	attached	78	1.04 [1.00, 1.09]	0.89 [0.86, 0.92]	0.53 [0.48, 0.58]	0.96 [0.94, 0.98]
ECG strain	midpoint	REML residuals	curve	attached	78	1.04 [1.00, 1.09]	0.93 [0.90, 0.96]	0.60 [0.56, 0.65]	0.98 [0.96, 1.00]
ECG strain	midpoint	beat differences (own scale)	curve	attached	78	1.01 [0.99, 1.04]	0.94 [0.93, 0.96]	0.72 [0.68, 0.75]	0.97 [0.97, 0.98]
ECG strain	midpoint	beat differences (variance-matched)	curve	attached	78	1.02 [0.98, 1.07]	0.86 [0.83, 0.88]	0.70 [0.66, 0.73]	0.95 [0.94, 0.97]
ECG strain	NCV-fitted	NCV residuals	curve	attached	78	0.97 [0.93, 1.02]	0.73 [0.71, 0.76]	0.53 [0.48, 0.58]	0.87 [0.86, 0.89]
ECG strain	NCV-fitted	REML residuals	curve	attached	78	1.00 [0.95, 1.04]	0.78 [0.76, 0.80]	0.60 [0.56, 0.64]	0.90 [0.89, 0.92]
ECG strain	NCV-fitted	beat differences (own scale)	curve	attached	78	1.00 [0.97, 1.04]	0.89 [0.87, 0.91]	0.70 [0.67, 0.73]	0.96 [0.95, 0.98]
ECG strain	NCV-fitted	beat differences (variance-matched)	curve	attached	78	0.95 [0.90, 1.00]	0.73 [0.71, 0.75]	0.69 [0.65, 0.72]	0.89 [0.87, 0.90]
ECG strain	REML-fitted	NCV residuals	curve	attached	78	1.11 [1.07, 1.16]	0.95 [0.92, 0.98]	1.27 [1.19, 1.36]	1.04 [1.01, 1.06]
ECG strain	REML-fitted	REML residuals	curve	attached	78	1.10 [1.06, 1.14]	0.98 [0.95, 1.01]	1.24 [1.17, 1.33]	1.04 [1.02, 1.06]
ECG strain	REML-fitted	beat differences (own scale)	curve	attached	78	1.01 [0.99, 1.03]	0.97 [0.96, 0.98]	1.05 [1.04, 1.06]	0.99 [0.99, 1.00]
ECG strain	REML-fitted	beat differences (variance-matched)	curve	attached	78	1.07 [1.02, 1.12]	0.91 [0.88, 0.93]	1.20 [1.13, 1.27]	0.99 [0.98, 1.01]
ECG strain	midpoint	NCV residuals	curve	attached	40	0.95 [0.90, 1.01]	0.74 [0.71, 0.77]	0.54 [0.49, 0.59]	0.87 [0.85, 0.90]
ECG strain	midpoint	REML residuals	curve	attached	40	0.96 [0.91, 1.01]	0.78 [0.75, 0.81]	0.58 [0.53, 0.64]	0.89 [0.87, 0.92]
ECG strain	midpoint	beat differences (own scale)	curve	attached	40	0.99 [0.94, 1.05]	0.89 [0.86, 0.91]	0.69 [0.64, 0.73]	0.95 [0.93, 0.97]
ECG strain	midpoint	beat differences (variance-matched)	curve	attached	40	0.94 [0.86, 1.03]	0.69 [0.66, 0.73]	0.68 [0.63, 0.73]	0.93 [0.88, 0.97]
ECG strain	NCV-fitted	NCV residuals	curve	attached	40	0.83 [0.79, 0.89]	0.59 [0.56, 0.62]	0.53 [0.47, 0.58]	0.77 [0.75, 0.80]
ECG strain	NCV-fitted	REML residuals	curve	attached	40	0.85 [0.80, 0.91]	0.62 [0.59, 0.65]	0.59 [0.53, 0.64]	0.80 [0.77, 0.82]
ECG strain	NCV-fitted	beat differences (own scale)	curve	attached	40	0.93 [0.87, 0.99]	0.77 [0.75, 0.80]	0.68 [0.64, 0.72]	0.89 [0.87, 0.91]
ECG strain	NCV-fitted	beat differences (variance-matched)	curve	attached	40	0.82 [0.75, 0.90]	0.56 [0.53, 0.59]	0.67 [0.62, 0.71]	0.81 [0.77, 0.85]
ECG strain	REML-fitted	NCV residuals	curve	attached	40	1.05 [1.00, 1.11]	0.81 [0.78, 0.84]	1.05 [0.99, 1.11]	0.95 [0.92, 0.97]
ECG strain	REML-fitted	REML residuals	curve	attached	40	1.07 [1.02, 1.14]	0.85 [0.82, 0.88]	1.13 [1.07, 1.20]	0.97 [0.95, 1.00]
ECG strain	REML-fitted	beat differences (own scale)	curve	attached	40	1.03 [0.98, 1.09]	0.94 [0.91, 0.97]	1.14 [1.09, 1.21]	1.00 [0.97, 1.02]
ECG strain	REML-fitted	beat differences (variance-matched)	curve	attached	40	1.07 [0.98, 1.17]	0.75 [0.71, 0.79]	1.27 [1.20, 1.36]	1.01 [0.97, 1.06]
AF trial	midpoint	NCV residuals	curve	attached	118	1.01 [0.97, 1.06]	0.25 [0.19, 0.33]	0.96 [0.93, 1.00]	0.94 [0.91, 0.97]
AF trial	midpoint	REML residuals	curve	attached	118	1.01 [0.97, 1.06]	0.29 [0.22, 0.38]	0.97 [0.93, 1.01]	0.95 [0.92, 0.98]
AF trial	NCV-fitted	NCV residuals	curve	attached	118	0.98 [0.94, 1.02]	0.22 [0.16, 0.30]	0.92 [0.88, 0.95]	0.91 [0.88, 0.94]
AF trial	NCV-fitted	REML residuals	curve	attached	118	0.98 [0.94, 1.02]	0.25 [0.19, 0.34]	0.92 [0.89, 0.96]	0.91 [0.88, 0.94]
AF trial	REML-fitted	NCV residuals	curve	attached	118	1.04 [1.00, 1.09]	0.30 [0.24, 0.39]	1.00 [0.97, 1.04]	0.96 [0.93, 0.99]
AF trial	REML-fitted	REML residuals	curve	attached	118	1.03 [0.99, 1.07]	0.34 [0.27, 0.43]	1.01 [0.97, 1.05]	0.96 [0.94, 0.99]
AF trial	midpoint	NCV residuals	curve	attached	40	0.97 [0.90, 1.04]	0.17 [0.10, 0.28]	0.84 [0.78, 0.89]	0.78 [0.74, 0.82]
AF trial	midpoint	REML residuals	curve	attached	40	0.97 [0.91, 1.05]	0.19 [0.11, 0.31]	0.84 [0.79, 0.90]	0.79 [0.75, 0.83]
AF trial	NCV-fitted	NCV residuals	curve	attached	40	0.93 [0.87, 1.01]	0.12 [0.07, 0.21]	0.81 [0.76, 0.86]	0.72 [0.68, 0.76]
AF trial	NCV-fitted	REML residuals	curve	attached	40	0.93 [0.88, 1.00]	0.13 [0.08, 0.23]	0.81 [0.76, 0.87]	0.73 [0.70, 0.77]
AF trial	REML-fitted	NCV residuals	curve	attached	40	0.99 [0.92, 1.06]	0.19 [0.12, 0.29]	0.86 [0.81, 0.92]	0.81 [0.77, 0.85]
AF trial	REML-fitted	REML residuals	curve	attached	40	1.01 [0.95, 1.10]	0.20 [0.13, 0.31]	0.87 [0.82, 0.93]	0.82 [0.79, 0.86]
running	midpoint	NCV residuals	curve	attached	90	1.16 [1.10, 1.23]	0.55 [0.50, 0.61]	1.08 [1.04, 1.12]	0.99 [0.98, 1.01]
running	midpoint	REML residuals	curve	attached	90	1.16 [1.09, 1.23]	0.52 [0.46, 0.58]	1.10 [1.06, 1.15]	0.98 [0.97, 1.00]
running	NCV-fitted	NCV residuals	curve	attached	90	1.10 [1.04, 1.17]	0.24 [0.19, 0.30]	1.03 [0.99, 1.08]	0.91 [0.89, 0.92]
running	NCV-fitted	NCV residuals	curve	detached	90	1.06 [1.00, 1.13]	0.15 [0.12, 0.20]	1.09 [1.03, 1.17]	0.87 [0.85, 0.90]
running	NCV-fitted	REML residuals	curve	attached	90	1.11 [1.05, 1.18]	0.24 [0.19, 0.30]	1.06 [1.02, 1.12]	0.90 [0.88, 0.92]
running	REML-fitted	NCV residuals	curve	attached	90	1.19 [1.12, 1.26]	1.55 [1.40, 1.71]	1.09 [1.05, 1.13]	1.15 [1.12, 1.18]
running	REML-fitted	REML residuals	curve	attached	90	1.21 [1.13, 1.29]	1.50 [1.37, 1.65]	1.12 [1.07, 1.17]	1.16 [1.13, 1.19]
running	REML-fitted	REML residuals	curve	detached	90	1.14 [1.07, 1.23]	1.35 [1.22, 1.51]	1.17 [1.09, 1.25]	1.20 [1.17, 1.24]
running	midpoint	NCV residuals	curve	attached	40	1.09 [1.02, 1.18]	0.31 [0.25, 0.38]	0.99 [0.92, 1.05]	0.92 [0.88, 0.96]
running	midpoint	REML residuals	curve	attached	40	1.11 [1.03, 1.21]	0.29 [0.24, 0.36]	0.99 [0.92, 1.06]	0.91 [0.87, 0.95]
running	NCV-fitted	NCV residuals	curve	attached	40	1.00 [0.93, 1.09]	0.15 [0.11, 0.20]	0.92 [0.87, 0.99]	0.83 [0.79, 0.86]
running	NCV-fitted	NCV residuals	curve	detached	40	1.00 [0.92, 1.08]	0.14 [0.10, 0.19]	0.82 [0.77, 0.89]	0.77 [0.74, 0.81]
running	NCV-fitted	REML residuals	curve	attached	40	1.02 [0.95, 1.10]	0.15 [0.11, 0.20]	0.93 [0.87, 0.99]	0.82 [0.79, 0.86]
running	REML-fitted	NCV residuals	curve	attached	40	1.13 [1.05, 1.24]	0.82 [0.72, 0.93]	0.99 [0.93, 1.06]	1.01 [0.97, 1.06]
running	REML-fitted	REML residuals	curve	attached	40	1.15 [1.06, 1.27]	0.80 [0.71, 0.91]	1.00 [0.93, 1.08]	1.01 [0.97, 1.05]
running	REML-fitted	REML residuals	curve	detached	40	1.10 [1.02, 1.20]	0.73 [0.65, 0.83]	0.97 [0.90, 1.06]	0.97 [0.94, 1.01]
DTI	midpoint	NCV residuals	curve	attached	92	0.91 [0.88, 0.95]	0.54 [0.51, 0.57]	0.75 [0.70, 0.79]	0.86 [0.85, 0.87]
DTI	midpoint	REML residuals	curve	attached	92	0.93 [0.89, 0.97]	0.61 [0.57, 0.64]	0.74 [0.70, 0.78]	0.88 [0.86, 0.89]

Table 23: NCV/REML MSE ratio per plasmode combination and estimand (95% paired bootstrap intervals). *(continued)*

dataset	truth	residual	flip	error	G	alpha	beta	gamma	mean
DTI	NCV-fitted	NCV residuals	curve	attached	92	0.87 [0.83, 0.91]	0.37 [0.35, 0.41]	0.66 [0.61, 0.71]	0.74 [0.72, 0.76]
DTI	NCV-fitted	REML residuals	curve	attached	92	0.89 [0.85, 0.93]	0.42 [0.39, 0.45]	0.66 [0.61, 0.70]	0.76 [0.74, 0.77]
DTI	REML-fitted	NCV residuals	curve	attached	92	0.98 [0.94, 1.01]	1.13 [1.08, 1.18]	0.88 [0.83, 0.93]	1.00 [0.98, 1.01]
DTI	REML-fitted	REML residuals	curve	attached	92	0.99 [0.96, 1.03]	1.20 [1.16, 1.25]	0.89 [0.84, 0.93]	1.01 [0.99, 1.02]
DTI	midpoint	NCV residuals	curve	attached	40	0.91 [0.86, 0.97]	0.49 [0.46, 0.53]	0.68 [0.62, 0.74]	0.81 [0.78, 0.84]
DTI	midpoint	REML residuals	curve	attached	40	0.91 [0.86, 0.95]	0.53 [0.49, 0.57]	0.68 [0.63, 0.73]	0.82 [0.80, 0.85]
DTI	NCV-fitted	NCV residuals	curve	attached	40	0.92 [0.86, 0.98]	0.37 [0.33, 0.40]	0.63 [0.57, 0.68]	0.71 [0.68, 0.73]
DTI	NCV-fitted	REML residuals	curve	attached	40	0.91 [0.85, 0.97]	0.40 [0.37, 0.44]	0.64 [0.59, 0.70]	0.73 [0.71, 0.76]
DTI	REML-fitted	NCV residuals	curve	attached	40	0.91 [0.86, 0.97]	0.84 [0.80, 0.89]	0.75 [0.70, 0.81]	0.87 [0.85, 0.90]
DTI	REML-fitted	REML residuals	curve	attached	40	0.92 [0.86, 0.97]	0.90 [0.85, 0.94]	0.75 [0.70, 0.81]	0.88 [0.86, 0.91]
gait	midpoint	NCV residuals	curve	attached	138	0.94 [0.91, 0.97]	0.55 [0.49, 0.62]	1.02 [0.99, 1.06]	0.97 [0.95, 0.99]
gait	midpoint	REML residuals	curve	attached	138	0.94 [0.91, 0.97]	0.55 [0.49, 0.62]	1.01 [0.98, 1.05]	0.97 [0.95, 0.99]
gait	NCV-fitted	NCV residuals	curve	attached	138	0.92 [0.88, 0.96]	0.43 [0.35, 0.50]	1.02 [0.98, 1.06]	0.93 [0.91, 0.95]
gait	NCV-fitted	REML residuals	curve	attached	138	0.92 [0.88, 0.95]	0.43 [0.35, 0.50]	1.01 [0.98, 1.05]	0.93 [0.91, 0.95]
gait	REML-fitted	NCV residuals	curve	attached	138	0.95 [0.92, 0.98]	0.63 [0.56, 0.69]	1.03 [1.00, 1.07]	0.99 [0.97, 1.01]
gait	REML-fitted	REML residuals	curve	attached	138	0.95 [0.93, 0.98]	0.62 [0.56, 0.69]	1.02 [0.99, 1.05]	0.99 [0.97, 1.01]
gait	midpoint	NCV residuals	curve	attached	40	0.86 [0.80, 0.92]	0.26 [0.20, 0.33]	0.88 [0.83, 0.95]	0.91 [0.87, 0.94]
gait	midpoint	REML residuals	curve	attached	40	0.86 [0.80, 0.92]	0.25 [0.19, 0.32]	0.89 [0.83, 0.96]	0.91 [0.87, 0.94]
gait	NCV-fitted	NCV residuals	curve	attached	40	0.85 [0.79, 0.91]	0.22 [0.16, 0.29]	0.87 [0.81, 0.93]	0.88 [0.85, 0.92]
gait	NCV-fitted	REML residuals	curve	attached	40	0.85 [0.79, 0.91]	0.22 [0.15, 0.29]	0.87 [0.81, 0.94]	0.88 [0.85, 0.92]
gait	REML-fitted	NCV residuals	curve	attached	40	0.86 [0.80, 0.92]	0.29 [0.23, 0.36]	0.89 [0.83, 0.96]	0.92 [0.89, 0.95]
gait	REML-fitted	REML residuals	curve	attached	40	0.87 [0.81, 0.93]	0.29 [0.23, 0.36]	0.89 [0.84, 0.97]	0.92 [0.89, 0.96]
ECG 8-lead	midpoint	NCV residuals	curve	attached	100	0.74 [0.70, 0.80]	0.17 [0.11, 0.24]	1.03 [0.98, 1.08]	0.75 [0.73, 0.78]
ECG 8-lead	midpoint	REML residuals	curve	attached	100	0.76 [0.71, 0.81]	0.19 [0.13, 0.25]	1.06 [1.02, 1.12]	0.78 [0.75, 0.81]
ECG 8-lead	NCV-fitted	NCV residuals	curve	attached	100	0.62 [0.56, 0.68]	0.12 [0.06, 0.20]	1.01 [0.95, 1.07]	0.67 [0.64, 0.70]
ECG 8-lead	NCV-fitted	REML residuals	curve	attached	100	0.63 [0.57, 0.68]	0.10 [0.07, 0.15]	1.02 [0.97, 1.09]	0.69 [0.66, 0.72]
ECG 8-lead	REML-fitted	NCV residuals	curve	attached	100	0.97 [0.91, 1.04]	0.31 [0.26, 0.38]	1.03 [0.99, 1.08]	0.85 [0.83, 0.88]
ECG 8-lead	REML-fitted	REML residuals	curve	attached	100	0.98 [0.92, 1.06]	0.32 [0.27, 0.38]	1.05 [1.01, 1.10]	0.88 [0.85, 0.91]
ECG 8-lead	midpoint	NCV residuals	curve	attached	40	0.67 [0.61, 0.74]	0.10 [0.07, 0.13]	1.00 [0.93, 1.07]	0.66 [0.62, 0.69]
ECG 8-lead	midpoint	REML residuals	curve	attached	40	0.68 [0.62, 0.75]	0.12 [0.08, 0.16]	0.98 [0.91, 1.05]	0.68 [0.64, 0.71]
ECG 8-lead	NCV-fitted	NCV residuals	curve	attached	40	0.57 [0.51, 0.64]	0.08 [0.05, 0.12]	0.93 [0.87, 1.01]	0.58 [0.55, 0.61]
ECG 8-lead	NCV-fitted	REML residuals	curve	attached	40	0.58 [0.52, 0.64]	0.09 [0.06, 0.13]	0.93 [0.86, 1.00]	0.60 [0.56, 0.63]
ECG 8-lead	REML-fitted	NCV residuals	curve	attached	40	0.83 [0.76, 0.92]	0.14 [0.11, 0.18]	0.99 [0.92, 1.05]	0.71 [0.68, 0.74]
ECG 8-lead	REML-fitted	REML residuals	curve	attached	40	0.83 [0.76, 0.91]	0.18 [0.14, 0.23]	1.00 [0.93, 1.07]	0.74 [0.71, 0.77]
ocean	midpoint	NCV residuals	curve	attached	116	1.09 [1.04, 1.13]	0.33 [0.28, 0.38]	1.08 [1.00, 1.18]	0.87 [0.86, 0.89]
ocean	midpoint	REML residuals	curve	attached	116	1.12 [1.07, 1.17]	0.37 [0.31, 0.43]	1.08 [1.00, 1.18]	0.89 [0.88, 0.91]
ocean	NCV-fitted	NCV residuals	curve	attached	116	1.01 [0.95, 1.07]	0.07 [0.05, 0.11]	0.94 [0.88, 1.02]	0.79 [0.77, 0.81]
ocean	NCV-fitted	NCV residuals	block	attached	116	1.00 [0.95, 1.05]	0.05 [0.04, 0.06]	0.84 [0.79, 0.89]	0.81 [0.79, 0.83]
ocean	NCV-fitted	REML residuals	curve	attached	116	1.04 [0.98, 1.09]	0.06 [0.05, 0.07]	0.94 [0.88, 1.02]	0.80 [0.78, 0.82]
ocean	REML-fitted	NCV residuals	curve	attached	116	1.16 [1.10, 1.23]	1.71 [1.55, 1.90]	1.29 [1.18, 1.45]	1.12 [1.10, 1.15]
ocean	REML-fitted	REML residuals	curve	attached	116	1.18 [1.12, 1.25]	1.90 [1.72, 2.13]	1.29 [1.17, 1.42]	1.15 [1.12, 1.18]
ocean	REML-fitted	REML residuals	block	attached	116	1.16 [1.09, 1.24]	1.82 [1.62, 2.07]	1.22 [1.11, 1.33]	1.15 [1.12, 1.18]
ocean	midpoint	NCV residuals	curve	attached	40	0.90 [0.83, 0.97]	0.13 [0.11, 0.15]	0.76 [0.70, 0.83]	0.76 [0.74, 0.79]
ocean	midpoint	REML residuals	curve	attached	40	0.93 [0.86, 1.00]	0.13 [0.11, 0.15]	0.76 [0.70, 0.82]	0.77 [0.75, 0.80]
ocean	NCV-fitted	NCV residuals	curve	attached	40	0.85 [0.79, 0.92]	0.04 [0.03, 0.06]	0.64 [0.59, 0.69]	0.70 [0.67, 0.73]
ocean	NCV-fitted	REML residuals	curve	attached	40	0.88 [0.81, 0.95]	0.04 [0.03, 0.05]	0.65 [0.60, 0.69]	0.71 [0.68, 0.74]
ocean	REML-fitted	NCV residuals	curve	attached	40	0.94 [0.87, 1.02]	0.64 [0.57, 0.72]	0.86 [0.78, 0.95]	0.85 [0.82, 0.88]
ocean	REML-fitted	REML residuals	curve	attached	40	0.97 [0.90, 1.05]	0.69 [0.61, 0.78]	0.86 [0.79, 0.95]	0.87 [0.83, 0.90]
weather	midpoint	NCV residuals	curve	attached	73	0.63 [0.57, 0.71]	0.44 [0.39, 0.49]	0.64 [0.56, 0.75]	0.98 [0.93, 1.03]
weather	midpoint	REML residuals	curve	attached	73	0.68 [0.61, 0.76]	0.44 [0.39, 0.49]	0.71 [0.62, 0.83]	0.96 [0.92, 1.01]
weather	NCV-fitted	NCV residuals	curve	attached	73	0.35 [0.31, 0.40]	0.33 [0.29, 0.38]	0.24 [0.20, 0.28]	0.83 [0.80, 0.87]
weather	NCV-fitted	NCV residuals	block	attached	73	0.32 [0.28, 0.37]	0.41 [0.36, 0.48]	0.20 [0.17, 0.24]	0.86 [0.83, 0.90]
weather	NCV-fitted	REML residuals	curve	attached	73	0.42 [0.36, 0.47]	0.33 [0.29, 0.38]	0.34 [0.28, 0.40]	0.84 [0.80, 0.87]
weather	REML-fitted	NCV residuals	curve	attached	73	0.95 [0.84, 1.08]	0.68 [0.62, 0.75]	1.28 [1.13, 1.50]	1.10 [1.05, 1.16]

Table 23: NCV/REML MSE ratio per plasmode combination and estimand (95% paired bootstrap intervals). *(continued)*

dataset	truth	residual	flip	error	G	alpha	beta	gamma	mean
weather	REML-fitted	REML residuals	curve	attached	73	1.00 [0.88, 1.13]	0.66 [0.60, 0.73]	1.32 [1.16, 1.57]	1.07 [1.02, 1.11]
weather	REML-fitted	REML residuals	block	attached	73	0.82 [0.71, 0.94]	0.83 [0.75, 0.90]	1.16 [1.02, 1.31]	1.10 [1.06, 1.14]
weather	midpoint	NCV residuals	curve	attached	40	0.50 [0.41, 0.61]	0.32 [0.28, 0.38]	0.61 [0.49, 0.76]	0.90 [0.85, 0.95]
weather	midpoint	REML residuals	curve	attached	40	0.59 [0.49, 0.69]	0.35 [0.31, 0.41]	0.75 [0.61, 0.94]	0.93 [0.88, 0.98]
weather	NCV-fitted	NCV residuals	curve	attached	40	0.30 [0.24, 0.38]	0.21 [0.17, 0.25]	0.32 [0.25, 0.40]	0.71 [0.67, 0.75]
weather	NCV-fitted	REML residuals	curve	attached	40	0.38 [0.31, 0.46]	0.22 [0.19, 0.27]	0.39 [0.31, 0.49]	0.74 [0.70, 0.78]
weather	REML-fitted	NCV residuals	curve	attached	40	0.64 [0.52, 0.78]	0.44 [0.39, 0.51]	1.00 [0.80, 1.23]	0.97 [0.91, 1.02]
weather	REML-fitted	REML residuals	curve	attached	40	0.77 [0.63, 0.91]	0.45 [0.39, 0.51]	1.18 [0.97, 1.44]	0.99 [0.94, 1.05]
electricity	midpoint	NCV residuals	curve	attached	102	0.77 [0.73, 0.82]	0.05 [0.03, 0.06]	0.96 [0.92, 1.01]	0.57 [0.54, 0.60]
electricity	midpoint	REML residuals	curve	attached	102	0.76 [0.72, 0.80]	0.06 [0.04, 0.08]	0.99 [0.94, 1.04]	0.60 [0.57, 0.62]
electricity	NCV-fitted	NCV residuals	curve	attached	102	0.74 [0.70, 0.78]	0.03 [0.02, 0.05]	0.97 [0.92, 1.02]	0.51 [0.48, 0.54]
electricity	NCV-fitted	NCV residuals	block	attached	102	0.74 [0.70, 0.78]	0.03 [0.02, 0.04]	0.95 [0.90, 0.99]	0.52 [0.49, 0.55]
electricity	NCV-fitted	REML residuals	curve	attached	102	0.73 [0.69, 0.77]	0.05 [0.03, 0.06]	1.00 [0.95, 1.05]	0.53 [0.50, 0.56]
electricity	REML-fitted	NCV residuals	curve	attached	102	0.95 [0.88, 1.01]	2.03 [1.81, 2.29]	0.99 [0.95, 1.02]	1.15 [1.10, 1.19]
electricity	REML-fitted	REML residuals	curve	attached	102	0.96 [0.90, 1.02]	2.46 [2.21, 2.76]	0.96 [0.93, 0.99]	1.23 [1.18, 1.28]
electricity	REML-fitted	REML residuals	block	attached	102	0.94 [0.89, 1.00]	2.38 [2.12, 2.71]	0.97 [0.94, 1.01]	1.26 [1.21, 1.32]
electricity	midpoint	NCV residuals	curve	attached	40	0.76 [0.71, 0.82]	0.03 [0.02, 0.04]	0.88 [0.81, 0.95]	0.49 [0.45, 0.53]
electricity	midpoint	REML residuals	curve	attached	40	0.78 [0.73, 0.85]	0.04 [0.02, 0.06]	0.89 [0.82, 0.96]	0.49 [0.45, 0.53]
electricity	NCV-fitted	NCV residuals	curve	attached	40	0.75 [0.69, 0.81]	0.02 [0.02, 0.04]	0.89 [0.82, 0.96]	0.46 [0.43, 0.50]
electricity	NCV-fitted	REML residuals	curve	attached	40	0.76 [0.71, 0.82]	0.04 [0.02, 0.06]	0.89 [0.82, 0.96]	0.46 [0.43, 0.50]
electricity	REML-fitted	NCV residuals	curve	attached	40	0.87 [0.81, 0.94]	0.84 [0.75, 0.94]	0.87 [0.81, 0.94]	0.76 [0.72, 0.80]
electricity	REML-fitted	REML residuals	curve	attached	40	0.82 [0.76, 0.89]	0.98 [0.88, 1.10]	0.86 [0.79, 0.93]	0.78 [0.74, 0.81]

S2.3 Truth source against residual source; heartbeat errors; block flips

On the 2×2 design of REML/NCV truths and REML/NCV residuals (all subjects), the truth source shifts the log NCV/REML ratio of mean squared errors for $\beta(s,t)$ by -4.05 to -0.24 (NCV-fitted minus REML-fitted truth), the residual source by -0.27 to 0.08: on these datasets, the truth source dominates.

With heartbeat difference curves of the ECG strain data as errors, REML + CL2's coverage changes by 0.1–1.2 pp (variance-matched) and 0.1–1.2 pp (own scale) relative to the REML residuals.

For the three datasets with dependent source curves, block flips share one sign within each block of neighboring curves, so the simulated errors are independent only across blocks (ocean 39 blocks, weather 16 blocks, electricity 51 blocks). Block flips change REML + curve-clustered CL2 coverage by -1.8 to 2.1 pp relative to curve flips; block-wise NCV, which leaves out these blocks instead of single curves, changes the mean squared error by a factor 0.99–1.11.

Table 24: Truth-source and residual-source main effects and their interaction (paired, 95% bootstrap intervals) on the log NCV/REML MSE ratio and on coverage. Residual verdict: small = interval inside ±2 points or log[0.95, 1.05].

dataset	estimand	quantity	truth	residual	interaction	residual verdict
ECG strain	β(s,t)	coverage NCV + CL2, NCV–REML widening	0.010 [0.007, 0.013]	-0.000 [-0.003, 0.002]	0.004 [0.002, 0.006]	small
AF trial	β(s,t)	coverage NCV + CL2, NCV–REML widening	0.018 [0.009, 0.026]	-0.001 [-0.004, 0.002]	0.001 [-0.004, 0.005]	small
running	β(s,t)	coverage NCV + CL2, NCV–REML widening	0.106 [0.090, 0.123]	-0.000 [-0.006, 0.005]	0.006 [-0.006, 0.017]	small
DTI	β(s,t)	coverage NCV + CL2, NCV–REML widening	0.073 [0.063, 0.082]	0.003 [0.000, 0.006]	-0.003 [-0.008, 0.002]	small
gait	β(s,t)	coverage NCV + CL2, NCV–REML widening	0.023 [0.017, 0.028]	0.000 [-0.002, 0.002]	0.001 [-0.002, 0.003]	small
ECG 8-lead	β(s,t)	coverage NCV + CL2, NCV–REML widening	0.075 [0.064, 0.085]	0.005 [0.001, 0.008]	-0.000 [-0.006, 0.006]	small
ocean	β(s,t)	coverage NCV + CL2, NCV–REML widening	0.094 [0.081, 0.106]	-0.003 [-0.007, 0.001]	0.002 [-0.007, 0.011]	small
weather	β(s,t)	coverage NCV + CL2, NCV–REML widening	0.011 [-0.015, 0.036]	-0.008 [-0.019, 0.004]	-0.009 [-0.029, 0.010]	small
electricity	β(s,t)	coverage NCV + CL2, NCV–REML widening	0.174 [0.153, 0.197]	-0.006 [-0.014, 0.002]	0.021 [0.005, 0.037]	small
ECG strain	E(Y | X)	coverage NCV + CL2, NCV–REML widening	0.007 [0.006, 0.009]	0.001 [-0.001, 0.003]	0.003 [0.001, 0.004]	small
AF trial	E(Y | X)	coverage NCV + CL2, NCV–REML widening	0.001 [-0.002, 0.003]	-0.001 [-0.002, 0.000]	0.000 [-0.001, 0.002]	small
running	E(Y | X)	coverage NCV + CL2, NCV–REML widening	0.018 [0.014, 0.023]	0.000 [-0.002, 0.002]	0.000 [-0.003, 0.003]	small
DTI	E(Y | X)	coverage NCV + CL2, NCV–REML widening	0.024 [0.021, 0.027]	0.002 [0.001, 0.004]	0.001 [-0.001, 0.003]	small
gait	E(Y | X)	coverage NCV + CL2, NCV–REML widening	0.005 [0.004, 0.006]	0.000 [-0.000, 0.001]	0.000 [-0.001, 0.001]	small
ECG 8-lead	E(Y | X)	coverage NCV + CL2, NCV–REML widening	0.017 [0.013, 0.020]	0.001 [-0.000, 0.003]	0.001 [-0.001, 0.003]	small
ocean	E(Y | X)	coverage NCV + CL2, NCV–REML widening	0.012 [0.008, 0.016]	0.001 [-0.001, 0.003]	0.002 [-0.001, 0.004]	small
weather	E(Y | X)	coverage NCV + CL2, NCV–REML widening	0.012 [0.006, 0.017]	0.001 [-0.002, 0.005]	0.002 [-0.003, 0.006]	small
electricity	E(Y | X)	coverage NCV + CL2, NCV–REML widening	0.054 [0.046, 0.062]	0.000 [-0.004, 0.004]	0.003 [-0.003, 0.009]	small
ECG strain	β(s,t)	coverage REML + CL2	0.000 [-0.000, 0.001]	-0.000 [-0.002, 0.002]	-0.000 [-0.001, 0.001]	small
AF trial	β(s,t)	coverage REML + CL2	0.008 [0.005, 0.012]	-0.001 [-0.004, 0.001]	0.000 [-0.002, 0.002]	small
running	β(s,t)	coverage REML + CL2	0.006 [-0.002, 0.014]	-0.000 [-0.007, 0.006]	0.001 [-0.007, 0.008]	small
DTI	β(s,t)	coverage REML + CL2	0.010 [0.005, 0.015]	0.002 [-0.001, 0.005]	-0.000 [-0.003, 0.003]	small
gait	β(s,t)	coverage REML + CL2	0.001 [-0.000, 0.002]	0.000 [-0.001, 0.001]	0.000 [-0.001, 0.001]	small
ECG 8-lead	β(s,t)	coverage REML + CL2	0.006 [0.003, 0.010]	-0.001 [-0.004, 0.002]	-0.001 [-0.004, 0.003]	small
ocean	β(s,t)	coverage REML + CL2	-0.003 [-0.006, -0.000]	-0.006 [-0.011, -0.002]	-0.000 [-0.003, 0.003]	small
weather	β(s,t)	coverage REML + CL2	-0.043 [-0.071, -0.017]	-0.004 [-0.016, 0.007]	-0.022 [-0.043, -0.003]	small
electricity	β(s,t)	coverage REML + CL2	0.028 [0.013, 0.044]	-0.013 [-0.023, -0.003]	0.022 [0.005, 0.038]	undetermined
ECG strain	E(Y | X)	coverage REML + CL2	0.000 [-0.000, 0.001]	0.002 [-0.000, 0.003]	0.000 [-0.001, 0.001]	small
AF trial	E(Y | X)	coverage REML + CL2	0.001 [-0.000, 0.002]	-0.002 [-0.003, -0.001]	-0.000 [-0.001, 0.001]	small
running	E(Y | X)	coverage REML + CL2	-0.005 [-0.006, -0.003]	-0.000 [-0.003, 0.002]	0.000 [-0.001, 0.001]	small
DTI	E(Y | X)	coverage REML + CL2	-0.001 [-0.002, 0.000]	0.002 [0.000, 0.003]	-0.000 [-0.001, 0.001]	small
gait	E(Y | X)	coverage REML + CL2	-0.000 [-0.001, 0.000]	0.000 [-0.000, 0.001]	-0.000 [-0.000, 0.000]	small
ECG 8-lead	E(Y | X)	coverage REML + CL2	0.001 [0.000, 0.002]	0.000 [-0.001, 0.002]	0.000 [-0.001, 0.001]	small
ocean	E(Y | X)	coverage REML + CL2	-0.003 [-0.004, -0.003]	-0.001 [-0.002, 0.001]	0.000 [-0.001, 0.001]	small
weather	E(Y | X)	coverage REML + CL2	-0.021 [-0.027, -0.015]	0.005 [0.001, 0.009]	-0.006 [-0.011, -0.002]	small
electricity	E(Y | X)	coverage REML + CL2	-0.007 [-0.011, -0.002]	-0.002 [-0.006, 0.003]	0.006 [0.001, 0.010]	small
ECG strain	β(s,t)	coverage REML, model-based	-0.001 [-0.002, 0.000]	-0.003 [-0.007, 0.000]	0.001 [-0.001, 0.002]	small
AF trial	β(s,t)	coverage REML, model-based	0.018 [0.010, 0.026]	-0.007 [-0.012, -0.002]	0.001 [-0.007, 0.008]	small
running	β(s,t)	coverage REML, model-based	0.044 [0.036, 0.053]	0.002 [-0.007, 0.013]	0.008 [-0.001, 0.017]	small
DTI	β(s,t)	coverage REML, model-based	0.028 [0.022, 0.034]	-0.004 [-0.009, 0.001]	-0.004 [-0.010, 0.001]	small
gait	β(s,t)	coverage REML, model-based	0.007 [0.005, 0.010]	-0.002 [-0.003, 0.000]	0.000 [-0.001, 0.001]	small
ECG 8-lead	β(s,t)	coverage REML, model-based	0.017 [0.012, 0.023]	-0.011 [-0.015, -0.006]	-0.003 [-0.010, 0.003]	small
ocean	β(s,t)	coverage REML, model-based	0.004 [-0.002, 0.009]	-0.014 [-0.021, -0.007]	-0.003 [-0.010, 0.003]	undetermined
weather	β(s,t)	coverage REML, model-based	0.032 [0.019, 0.047]	-0.001 [-0.013, 0.014]	-0.009 [-0.024, 0.006]	small
electricity	β(s,t)	coverage REML, model-based	0.078 [0.060, 0.099]	-0.008 [-0.022, 0.006]	0.020 [-0.001, 0.041]	undetermined
ECG strain	E(Y | X)	coverage REML, model-based	-0.002 [-0.003, -0.001]	-0.002 [-0.004, 0.000]	0.002 [0.000, 0.003]	small
AF trial	E(Y | X)	coverage REML, model-based	-0.001 [-0.002, 0.001]	-0.002 [-0.004, -0.000]	0.001 [-0.001, 0.002]	small
running	E(Y | X)	coverage REML, model-based	0.001 [-0.001, 0.003]	-0.006 [-0.010, -0.002]	-0.000 [-0.002, 0.001]	small
DTI	E(Y | X)	coverage REML, model-based	0.006 [0.003, 0.008]	-0.006 [-0.009, -0.002]	-0.003 [-0.006, -0.001]	small
gait	E(Y | X)	coverage REML, model-based	0.000 [-0.000, 0.001]	-0.001 [-0.002, -0.000]	0.000 [-0.000, 0.001]	small
ECG 8-lead	E(Y | X)	coverage REML, model-based	0.008 [0.006, 0.009]	-0.007 [-0.009, -0.005]	-0.001 [-0.002, -0.000]	small
ocean	E(Y | X)	coverage REML, model-based	-0.006 [-0.008, -0.005]	-0.003 [-0.005, -0.000]	0.001 [-0.001, 0.002]	small
weather	E(Y | X)	coverage REML, model-based	0.007 [0.003, 0.012]	0.000 [-0.005, 0.006]	-0.000 [-0.004, 0.004]	small
electricity	E(Y | X)	coverage REML, model-based	0.010 [0.005, 0.015]	-0.001 [-0.008, 0.007]	0.006 [-0.000, 0.013]	small
ECG strain	β(s,t)	log MSE NCV/REML	-0.245 [-0.267, -0.225]	-0.049 [-0.059, -0.038]	-0.032 [-0.043, -0.022]	undetermined
AF trial	β(s,t)	log MSE NCV/REML	-0.323 [-0.566, -0.089]	-0.125 [-0.208, -0.046]	-0.027 [-0.153, 0.105]	undetermined
running	β(s,t)	log MSE NCV/REML	-1.844 [-2.042, -1.623]	0.017 [-0.060, 0.101]	-0.024 [-0.159, 0.113]	undetermined

Table 24: Truth-source and residual-source main effects and their interaction (paired, 95% bootstrap intervals) on the log NCV/REML MSE ratio and on coverage. Residual verdict: small = interval inside ±2 points or log[0.95, 1.05]. *(continued)*

dataset	estimand	quantity	truth	residual	interaction	residual verdict
DTI	β(s,t)	log MSE NCV/REML	-1.082 [-1.142, -1.017]	-0.086 [-0.118, -0.059]	-0.048 [-0.092, -0.009]	not small
gait	β(s,t)	log MSE NCV/REML	-0.381 [-0.503, -0.287]	-0.001 [-0.021, 0.018]	-0.004 [-0.041, 0.033]	small
ECG 8-lead	β(s,t)	log MSE NCV/REML	-1.029 [-1.469, -0.781]	0.078 [-0.113, 0.259]	0.186 [-0.173, 0.559]	undetermined
ocean	β(s,t)	log MSE NCV/REML	-3.273 [-3.480, -3.060]	0.038 [-0.135, 0.240]	0.287 [-0.013, 0.678]	undetermined
weather	β(s,t)	log MSE NCV/REML	-0.703 [-0.786, -0.625]	0.011 [-0.050, 0.072]	-0.027 [-0.104, 0.048]	undetermined
electricity	β(s,t)	log MSE NCV/REML	-4.051 [-4.404, -3.753]	-0.267 [-0.403, -0.125]	-0.151 [-0.386, 0.066]	not small
ECG strain	E(Y | X)	log MSE NCV/REML	-0.155 [-0.168, -0.143]	-0.020 [-0.029, -0.012]	-0.027 [-0.038, -0.017]	small
AF trial	E(Y | X)	log MSE NCV/REML	-0.057 [-0.077, -0.035]	-0.005 [-0.013, 0.004]	-0.006 [-0.019, 0.007]	small
running	E(Y | X)	log MSE NCV/REML	-0.243 [-0.263, -0.223]	0.002 [-0.008, 0.013]	0.012 [-0.003, 0.025]	small
DTI	E(Y | X)	log MSE NCV/REML	-0.293 [-0.312, -0.273]	-0.016 [-0.025, -0.007]	-0.008 [-0.020, 0.003]	small
gait	E(Y | X)	log MSE NCV/REML	-0.066 [-0.075, -0.056]	0.000 [-0.002, 0.003]	-0.000 [-0.005, 0.004]	small
ECG 8-lead	E(Y | X)	log MSE NCV/REML	-0.240 [-0.267, -0.212]	-0.029 [-0.044, -0.015]	-0.001 [-0.021, 0.020]	small
ocean	E(Y | X)	log MSE NCV/REML	-0.355 [-0.375, -0.333]	-0.018 [-0.029, -0.008]	0.009 [-0.003, 0.022]	small
weather	E(Y | X)	log MSE NCV/REML	-0.264 [-0.301, -0.229]	0.013 [-0.007, 0.033]	-0.043 [-0.073, -0.013]	small
electricity	E(Y | X)	log MSE NCV/REML	-0.832 [-0.893, -0.767]	-0.051 [-0.078, -0.024]	0.030 [-0.015, 0.077]	undetermined

Table 25: ECG strain: coverage and NCV/REML MSE ratio with fitted residuals and with heartbeat differences (variance-matched and own scale).

truth	residual	estimand	NCV + CL2	NCV + CL2, NCV–REML widening	REML + CL2	REML, model-based	AR(1)	MSE NCV/REML
REML	NCV residuals	β(s,t)	0.899	0.929	0.943	0.559		0.951
REML	REML residuals	β(s,t)	0.904	0.931	0.943	0.563		0.982
REML	beat differences (own scale)	β(s,t)	0.932	0.944	0.945	0.536		0.970
REML	beat differences (variance-matched)	β(s,t)	0.910	0.939	0.944	0.526		0.907
midpoint	NCV residuals	β(s,t)	0.900	0.931	0.943	0.558		0.890
midpoint	REML residuals	β(s,t)	0.904	0.932	0.944	0.561	0.874	0.927
midpoint	beat differences (own scale)	β(s,t)	0.933	0.946	0.945	0.534		0.943
midpoint	beat differences (variance-matched)	β(s,t)	0.910	0.940	0.945	0.524	0.837	0.857
NCV	NCV residuals	β(s,t)	0.909	0.941	0.944	0.559		0.732
NCV	REML residuals	β(s,t)	0.909	0.939	0.944	0.562		0.782
NCV	beat differences (own scale)	β(s,t)	0.928	0.946	0.945	0.532		0.887
NCV	beat differences (variance-matched)	β(s,t)	0.906	0.942	0.945	0.523		0.728
REML	NCV residuals	E(Y | X)	0.904	0.933	0.939	0.583		1.036
REML	REML residuals	E(Y | X)	0.906	0.933	0.938	0.586		1.043
REML	beat differences (own scale)	E(Y | X)	0.940	0.948	0.950	0.579		0.995
REML	beat differences (variance-matched)	E(Y | X)	0.921	0.945	0.949	0.571		0.992
midpoint	NCV residuals	E(Y | X)	0.916	0.938	0.940	0.581		0.959
midpoint	REML residuals	E(Y | X)	0.916	0.937	0.938	0.583	0.869	0.977
midpoint	beat differences (own scale)	E(Y | X)	0.940	0.951	0.950	0.570		0.974
midpoint	beat differences (variance-matched)	E(Y | X)	0.923	0.947	0.950	0.564	0.847	0.953
NCV	NCV residuals	E(Y | X)	0.918	0.942	0.940	0.582		0.875
NCV	REML residuals	E(Y | X)	0.916	0.939	0.938	0.583		0.905
NCV	beat differences (own scale)	E(Y | X)	0.935	0.950	0.950	0.569		0.962
NCV	beat differences (variance-matched)	E(Y | X)	0.921	0.948	0.950	0.564		0.888

Table 26: Datasets with dependent source curves: block-NCV/NCV MSE ratio, mismatched-partition ratio and coverage changes (95% bootstrap intervals), under block flips and curve flips.

dataset	estimand	verdict	flips	MSE block-NCV/NCV	coverage loss	coverage loss (method)	MSE mismatched blocks/NCV
ocean	β(s,t)	inconclusive	block flips	1.105 [0.979, 1.278]	0.006 [0.005, 0.007]	0.006 [0.003, 0.009]	
ocean	β(s,t)	inconclusive	curve flips	1.168 [1.029, 1.290]	0.001 [-0.000, 0.002]	0.004 [0.001, 0.007]	
ocean	E(Y | X)	cost only	block flips	0.995 [0.989, 1.001]	-0.006 [-0.007, -0.005]	-0.005 [-0.007, -0.004]	
ocean	E(Y | X)	cost only	curve flips	1.003 [0.997, 1.010]	0.003 [0.001, 0.004]	0.003 [0.002, 0.005]	
weather	β(s,t)	inconclusive	block flips	1.017 [0.955, 1.090]	0.011 [0.009, 0.014]	0.014 [0.008, 0.019]	1.039 [0.957, 1.128]
weather	β(s,t)	inconclusive	curve flips	1.061 [1.002, 1.132]	0.010 [0.007, 0.014]	-0.007 [-0.014, -0.001]	
weather	E(Y | X)	hurts	block flips	1.014 [0.995, 1.032]	-0.006 [-0.010, -0.002]	-0.003 [-0.007, 0.002]	0.999 [0.982, 1.015]
weather	E(Y | X)	hurts	curve flips	1.009 [0.986, 1.030]	0.029 [0.023, 0.033]	0.026 [0.021, 0.031]	
electricity	β(s,t)	inconclusive	block flips	1.021 [1.003, 1.045]	0.000 [-0.000, 0.001]	-0.002 [-0.003, -0.000]	
electricity	β(s,t)	inconclusive	curve flips	1.055 [0.977, 1.161]	0.001 [-0.000, 0.001]	0.000 [-0.001, 0.002]	
electricity	E(Y | X)	cost only	block flips	1.001 [0.995, 1.008]	-0.008 [-0.009, -0.006]	-0.008 [-0.009, -0.007]	
electricity	E(Y | X)	cost only	curve flips	1.004 [0.995, 1.013]	0.001 [-0.000, 0.003]	0.001 [-0.001, 0.002]	

Table 27: Coverage of curve- and block-clustered intervals under curve flips and block flips. Widened: NCV + CL2 widened by the NCV–REML difference; alt. blocks: the alternative partition (weather).

dataset	estimand	flips	NCV + curve CL2, widened	block NCV + block CL2, widened	NCV + block CL2, widened	block NCV + curve CL2, widened	REML + curve CL2	REML + block CL2	mismatched blocks, widened
ocean	β(s,t)	block flips	0.925	0.919	0.920	0.925	0.934	0.929	
ocean	β(s,t)	curve flips	0.916	0.912	0.916	0.913	0.925	0.922	
ocean	E(Y | X)	block flips	0.933	0.939	0.940	0.933	0.932	0.936	
ocean	E(Y | X)	curve flips	0.939	0.936	0.937	0.938	0.935	0.930	
weather	β(s,t)	block flips	0.823	0.809	0.813	0.821	0.903	0.889	0.806
weather	β(s,t)	curve flips	0.828	0.836	0.824	0.846	0.882	0.867	
weather	E(Y | X)	block flips	0.898	0.901	0.908	0.895	0.906	0.902	0.900
weather	E(Y | X)	curve flips	0.920	0.894	0.898	0.922	0.924	0.892	
electricity	β(s,t)	block flips	0.895	0.896	0.894	0.897	0.920	0.921	
electricity	β(s,t)	curve flips	0.911	0.911	0.911	0.912	0.927	0.925	
electricity	E(Y | X)	block flips	0.938	0.946	0.946	0.939	0.931	0.938	
electricity	E(Y | X)	curve flips	0.947	0.946	0.946	0.947	0.927	0.926	

S2.4 Studentized errors

On the datasets with independent source curves, the pooled SD of the studentized REML error for $\beta(s,t)$ with the CL2 standard error is 1.01–1.14 (mean -0.00 to 0.03). On the datasets with dependent source curves, it is 1.07–2.86. With the model-based standard error, it is 2.20–3.95 on the datasets with independent source curves.

Table 28: Pooled studentized errors of the REML fit (all subjects, fitted residuals; means over residual sources).

dataset	role	truth	estimand	mean z (CL2)	SD z (CL2)	share |z|>1.96 (CL2)	mean z (model)	SD z (model)	share |z|>1.96 (model)
ECG strain	indep. curves	REML	β(s,t)	0.000	1.024	0.057	-0.001	3.24	0.439
ECG strain	indep. curves	midpoint	β(s,t)	-0.003	1.025	0.057	0.000	3.25	0.440
ECG strain	indep. curves	NCV	β(s,t)	-0.003	1.022	0.056	0.001	3.25	0.440
AF trial	indep. curves	REML	β(s,t)	0.032	1.067	0.067	0.077	2.46	0.343
AF trial	indep. curves	midpoint	β(s,t)	0.022	1.044	0.062	0.060	2.42	0.333
AF trial	indep. curves	NCV	β(s,t)	0.021	1.028	0.058	0.063	2.37	0.326
running	indep. curves	REML	β(s,t)	0.003	1.143	0.085	-0.003	3.65	0.520
running	indep. curves	midpoint	β(s,t)	0.002	1.115	0.079	-0.004	3.40	0.484
running	indep. curves	NCV	β(s,t)	0.001	1.115	0.078	-0.001	3.38	0.475
DTI	indep. curves	REML	β(s,t)	0.006	1.118	0.080	0.014	2.36	0.386
DTI	indep. curves	midpoint	β(s,t)	0.009	1.079	0.070	0.018	2.22	0.361
DTI	indep. curves	NCV	β(s,t)	0.012	1.078	0.069	0.024	2.20	0.358
gait	indep. curves	REML	β(s,t)	0.002	1.048	0.061	-0.006	3.95	0.487
gait	indep. curves	midpoint	β(s,t)	0.001	1.044	0.060	-0.006	3.94	0.484
gait	indep. curves	NCV	β(s,t)	0.000	1.045	0.060	-0.007	3.93	0.480
ECG 8-lead	indep. curves	REML	β(s,t)	0.011	1.046	0.062	0.025	2.87	0.371
ECG 8-lead	indep. curves	midpoint	β(s,t)	0.008	1.015	0.056	0.017	2.79	0.360
ECG 8-lead	indep. curves	NCV	β(s,t)	0.009	1.011	0.056	0.018	2.75	0.354
ocean	dep. source	REML	β(s,t)	0.001	1.095	0.073	0.004	4.46	0.589
ocean	dep. source	midpoint	β(s,t)	0.002	1.108	0.076	0.006	4.35	0.588
ocean	dep. source	NCV	β(s,t)	0.001	1.111	0.076	0.005	4.37	0.586
weather	dep. source	REML	β(s,t)	-0.040	2.809	0.110	-0.047	5.83	0.683
weather	dep. source	midpoint	β(s,t)	-0.018	1.866	0.091	-0.016	4.60	0.642
weather	dep. source	NCV	β(s,t)	-0.087	2.861	0.153	-0.083	5.47	0.651
electricity	dep. source	REML	β(s,t)	0.001	1.392	0.091	-0.016	4.70	0.621
electricity	dep. source	midpoint	β(s,t)	-0.005	1.077	0.063	-0.023	3.74	0.541
electricity	dep. source	NCV	β(s,t)	-0.002	1.068	0.064	-0.002	3.77	0.543
ECG strain	indep. curves	REML	E(Y | X)	-0.006	1.040	0.061	-0.016	3.07	0.415
ECG strain	indep. curves	midpoint	E(Y | X)	-0.006	1.040	0.061	-0.018	3.09	0.418
ECG strain	indep. curves	NCV	E(Y | X)	-0.006	1.039	0.061	-0.017	3.08	0.417
AF trial	indep. curves	REML	E(Y | X)	-0.027	1.015	0.055	-0.097	2.34	0.320
AF trial	indep. curves	midpoint	E(Y | X)	-0.028	1.012	0.054	-0.098	2.34	0.320
AF trial	indep. curves	NCV	E(Y | X)	-0.028	1.011	0.054	-0.094	2.34	0.320
running	indep. curves	REML	E(Y | X)	-0.011	1.062	0.065	-0.028	3.30	0.433
running	indep. curves	midpoint	E(Y | X)	-0.010	1.068	0.067	-0.026	3.27	0.428
running	indep. curves	NCV	E(Y | X)	-0.011	1.079	0.070	-0.028	3.29	0.432
DTI	indep. curves	REML	E(Y | X)	0.010	1.067	0.067	0.013	2.20	0.358
DTI	indep. curves	midpoint	E(Y | X)	0.012	1.069	0.067	0.019	2.17	0.353
DTI	indep. curves	NCV	E(Y | X)	0.012	1.074	0.068	0.019	2.17	0.352
gait	indep. curves	REML	E(Y | X)	-0.015	1.028	0.057	-0.069	3.62	0.462
gait	indep. curves	midpoint	E(Y | X)	-0.014	1.027	0.057	-0.068	3.62	0.462
gait	indep. curves	NCV	E(Y | X)	-0.014	1.029	0.057	-0.066	3.62	0.462
ECG 8-lead	indep. curves	REML	E(Y | X)	-0.007	0.986	0.049	-0.006	2.21	0.251
ECG 8-lead	indep. curves	midpoint	E(Y | X)	-0.009	0.979	0.047	-0.006	2.19	0.246
ECG 8-lead	indep. curves	NCV	E(Y | X)	-0.011	0.980	0.047	-0.007	2.17	0.244
ocean	dep. source	REML	E(Y | X)	0.002	1.059	0.063	0.003	3.86	0.541
ocean	dep. source	midpoint	E(Y | X)	0.003	1.066	0.065	0.004	3.86	0.543
ocean	dep. source	NCV	E(Y | X)	0.003	1.071	0.067	0.002	3.88	0.547
weather	dep. source	REML	E(Y | X)	0.021	1.082	0.067	0.061	2.86	0.464
weather	dep. source	midpoint	E(Y | X)	0.022	1.102	0.075	0.056	2.81	0.458
weather	dep. source	NCV	E(Y | X)	0.006	1.148	0.088	0.050	2.80	0.457
electricity	dep. source	REML	E(Y | X)	0.016	1.073	0.069	0.032	3.43	0.511
electricity	dep. source	midpoint	E(Y | X)	0.004	1.089	0.073	-0.004	3.32	0.495
electricity	dep. source	NCV	E(Y | X)	0.003	1.101	0.076	-0.001	3.34	0.501

S3 Comparators

S3.1 AR(1) working model

The AR(1) working model fits the REML model with an AR(1) residual correlation along t, ρ profiled (mgcv `bam`), and uses its model-based intervals. On stationary homoskedastic synthetic errors, it covers 0.93–0.99 (Gaussian, $G = 100$; next table). It fails under misregistration (mean 0.84, $\beta(s,t)$ 0.86–0.88), for $\gamma(t)$ under covariate-dependent variance, and in the plasmode study: with the midpoint truth (smoothing parameters at the geometric mean of

the REML and NCV values) and REML residuals, it covers the mean at 0.67–0.93 and β at 0.44–0.91 across the nine datasets. The median fit time is 39 s against 2.0 s for REML (Appendix C of the paper).

Table 29: AR(1) working model (Gaussian, $G = 100$): coverage next to REML intervals, interval score relative to REML + CL2, MSE relative to NCV (95% bootstrap interval).

error	G	truth	estimand	AR(1) working model	NCV + CL2, widened by NCV–REML difference	REML + CL2	REML, model-based	MSE AR(1) / NCV	IS AR(1) / REML+CL2
smooth	100	smooth	β(s,t)	0.993	0.983	0.944	0.657	0.98 [0.82, 1.18]	0.532
smooth	100	smooth	E(Y | X)	0.972	0.959	0.944	0.634	1.04 [1.02, 1.05]	0.846
var(z)	100	smooth	β(s,t)	0.992	0.984	0.949	0.660	0.92 [0.76, 1.09]	0.550
var(z)	100	smooth	E(Y | X)	0.949	0.955	0.944	0.606	1.05 [1.03, 1.07]	0.925
var(t)	100	smooth	β(s,t)	0.991	0.983	0.944	0.662	1.00 [0.80, 1.21]	0.558
var(t)	100	smooth	E(Y | X)	0.972	0.959	0.944	0.642	1.10 [1.08, 1.12]	0.902
var(t,z)	100	smooth	β(s,t)	0.990	0.984	0.950	0.666	1.06 [0.88, 1.25]	0.580
var(t,z)	100	smooth	E(Y | X)	0.951	0.955	0.944	0.615	1.12 [1.10, 1.14]	0.980
sign-chg.	100	smooth	β(s,t)	0.992	0.985	0.942	0.700	1.13 [1.01, 1.27]	0.551
sign-chg.	100	smooth	E(Y | X)	0.963	0.957	0.942	0.646	1.03 [1.01, 1.06]	0.778
rough	100	smooth	β(s,t)	0.972	0.983	0.940	0.672	1.30 [1.11, 1.49]	0.521
rough	100	smooth	E(Y | X)	0.934	0.958	0.942	0.646	1.04 [1.03, 1.06]	0.834
AR(1)	100	smooth	β(s,t)	0.989	0.984	0.940	0.651	1.12 [0.96, 1.27]	0.493
AR(1)	100	smooth	E(Y | X)	0.961	0.959	0.942	0.625	1.02 [1.01, 1.04]	0.805
misreg.	100	smooth	β(s,t)	0.875	0.976	0.951	0.753	1.29 [1.00, 1.62]	1.059
misreg.	100	smooth	E(Y | X)	0.844	0.952	0.942	0.730	1.09 [1.07, 1.11]	1.347
misreg.	100	rough	β(s,t)	0.860	0.934	0.944	0.729	0.84 [0.70, 0.98]	1.102
misreg.	100	rough	E(Y | X)	0.838	0.938	0.937	0.706	1.00 [0.99, 1.01]	1.391

Table 30: AR(1) working model in the plasmode study: coverage, 5% quantile of pointwise coverage, REML + CL2 coverage on the same data, profiled ρ and fit time.

dataset	role	truth	residual	estimand	AR(1)	AR(1) q05	REML + CL2	rho	fit time s
ECG strain	indep. curves	midpoint	REML	β(s,t)	0.874	0.650	0.944	0.976	44.5
ECG strain	indep. curves	midpoint	REML	γ(t)	0.926	0.725	0.948	0.976	44.5
ECG strain	indep. curves	midpoint	REML	E(Y | X)	0.869	0.620	0.938	0.976	44.5
ECG strain	indep. curves	midpoint	beat differences (variance-matched)	β(s,t)	0.837	0.475	0.945	0.979	39.5
ECG strain	indep. curves	midpoint	beat differences (variance-matched)	γ(t)	0.943	0.782	0.957	0.979	39.5
ECG strain	indep. curves	midpoint	beat differences (variance-matched)	E(Y | X)	0.847	0.497	0.950	0.979	39.5
AF trial	indep. curves	midpoint	REML	β(s,t)	0.838	0.520	0.938	0.940	27.6
AF trial	indep. curves	midpoint	REML	γ(t)	0.877	0.685	0.946	0.940	27.6
AF trial	indep. curves	midpoint	REML	E(Y | X)	0.870	0.630	0.947	0.940	27.6
running	indep. curves	midpoint	REML	β(s,t)	0.728	0.445	0.922	0.990	52.8
running	indep. curves	midpoint	REML	γ(t)	0.806	0.540	0.934	0.990	52.8
running	indep. curves	midpoint	REML	E(Y | X)	0.787	0.500	0.934	0.990	52.8
running	indep. curves	REML-fitted	REML	β(s,t)	0.548	0.165	0.916	0.990	52.5
running	indep. curves	NCV-fitted	NCV	β(s,t)	0.811	0.545	0.922	0.990	52.7
running	indep. curves	REML-fitted	REML	γ(t)	0.805	0.538	0.937	0.990	52.5
running	indep. curves	NCV-fitted	NCV	γ(t)	0.798	0.545	0.929	0.990	52.7
running	indep. curves	REML-fitted	REML	E(Y | X)	0.754	0.470	0.935	0.990	52.5
running	indep. curves	NCV-fitted	NCV	E(Y | X)	0.793	0.495	0.930	0.990	52.7
DTI	indep. curves	midpoint	REML	β(s,t)	0.858	0.470	0.929	0.817	42.7
DTI	indep. curves	midpoint	REML	γ(t)	0.900	0.855	0.934	0.817	42.7
DTI	indep. curves	midpoint	REML	E(Y | X)	0.902	0.667	0.932	0.817	42.7
gait	indep. curves	midpoint	REML	β(s,t)	0.914	0.730	0.940	0.987	161.6
gait	indep. curves	midpoint	REML	γ(t)	0.919	0.770	0.933	0.987	161.6
gait	indep. curves	midpoint	REML	E(Y | X)	0.927	0.760	0.943	0.987	161.6
ECG 8-lead	indep. curves	midpoint	REML	β(s,t)	0.720	0.170	0.944	0.910	120.4
ECG 8-lead	indep. curves	midpoint	REML	γ(t)	0.940	0.620	0.961	0.910	120.4
ECG 8-lead	indep. curves	midpoint	REML	E(Y | X)	0.924	0.555	0.953	0.910	120.4
ocean	dep. source	midpoint	REML	β(s,t)	0.854	0.299	0.927	0.990	157.0
ocean	dep. source	midpoint	REML	γ(t)	0.934	0.815	0.945	0.990	157.0
ocean	dep. source	midpoint	REML	E(Y | X)	0.932	0.775	0.935	0.990	157.0
weather	dep. source	midpoint	REML	β(s,t)	0.438	0.320	0.905	0.445	44.3
weather	dep. source	midpoint	REML	γ(t)	0.567	0.500	0.909	0.445	44.3
weather	dep. source	midpoint	REML	E(Y | X)	0.675	0.440	0.922	0.445	44.3
electricity	dep. source	midpoint	REML	β(s,t)	0.677	0.220	0.939	0.979	40.7
electricity	dep. source	midpoint	REML	γ(t)	0.908	0.823	0.956	0.979	40.7
electricity	dep. source	midpoint	REML	E(Y | X)	0.844	0.640	0.927	0.979	40.7

S3.2 Random curve effect, GLS and curve bootstrap

Three further methods address dependent residual curves. All are fitted to the same data as the study's REML fit, with common random numbers, in the 4 Gaussian main-factorial settings with dependent errors at the default signal (rough stationary and smooth errors, $G = 40$ and 100; below: the *four synthetic Gaussian settings with dependent errors*), in the corresponding settings with independent errors, and in the six plasmode datasets with

independent source curves (observed curves independent across subjects; midpoint truth, i.e. smoothing parameters at the geometric mean of the REML and NCV values; REML residuals; all subjects and 40: 12 plasmode combinations). The evidence on these methods is limited to these four Gaussian settings with dependent errors and these plasmode combinations.

Functional random effect per curve (`pcre()` in refund): the REML model plus a curve-level random function expanded in the leading eigenfunctions of the REML residual curves (95% of their variance), with model-based intervals.

Generalized least squares (GLS). Estimate the within-curve residual covariance from the REML residual curves, either as their raw covariance or as an FPCA estimate plus white noise; prewhiten every curve with the inverse square root of that covariance; refit by REML; use model-based intervals. This procedure is the algorithm of refund's former `pffr_gls()`, rebuilt from refund's internal functions; a singular estimated covariance (fewer curves than grid points) is repaired by the nearest positive-definite matrix.

Curve bootstrap: resample whole curves, refit by REML (199 resamples), and form percentile intervals or Wald intervals with the bootstrap standard error around the REML estimate.

Results. In the four synthetic Gaussian settings with dependent errors, the random curve effect and GLS with the FPCA covariance cover $\beta(s,t)$ at 0.93–0.97, at 0.51–0.66 times REML + CL2's width and 0.52–0.57 times its interval score. The random curve effect undercovers the mean and $\alpha(t)$ there: it covers the mean at 0.88–0.91 and $\alpha(t)$ at 0.51–0.59. On the plasmode data, both methods fall to 0.59–0.86 for β, while REML + CL2 covers 0.91–0.94. A common working covariance that fits the synthetic error processes is only an approximation to the subject-specific two-point error law of the plasmode generator (each subject's error is plus or minus its own residual curve; S2), and the model-based intervals of these fits do not hold their level there; this finding concerns that generator and is not a general statement about real data.

GLS with the raw covariance undercovers everywhere (0.59–0.89 even under independent errors), because the raw covariance of fewer curves than grid points is a poor, singular estimate. The percentile bootstrap covers β at 0.93–0.97 on the plasmode data with intervals 1.42 times as wide as REML + CL2's, at the cost of 199 refits (Appendix C of the paper).

As estimators, the random curve effect and GLS (FPCA) remove most of REML's excess error. Their mean squared error for β relative to the NCV fit's (geometric mean over settings) is 1.75 and 1.79 in the four synthetic Gaussian settings with dependent errors (REML 7.4), and 0.72 and 0.68 on the plasmode datasets (REML 2.73; per setting 0.23–1.25 and 0.21–1.20). Both thus improve much on REML, but, in the synthetic settings, they are worse than NCV. Their plasmode advantage over NCV rests on the midpoint truth with REML residuals only. The GLS covariance is estimated from the residual curves of a REML pilot fit to each simulated dataset; because these errors are sign-flipped REML residuals of the original fit, those pilot residuals closely resemble the curves that generate them. The random curve effect is expensive: median 21 s per fit including the REML pilot fit, up to 1356 s, against 2.4 s for REML.

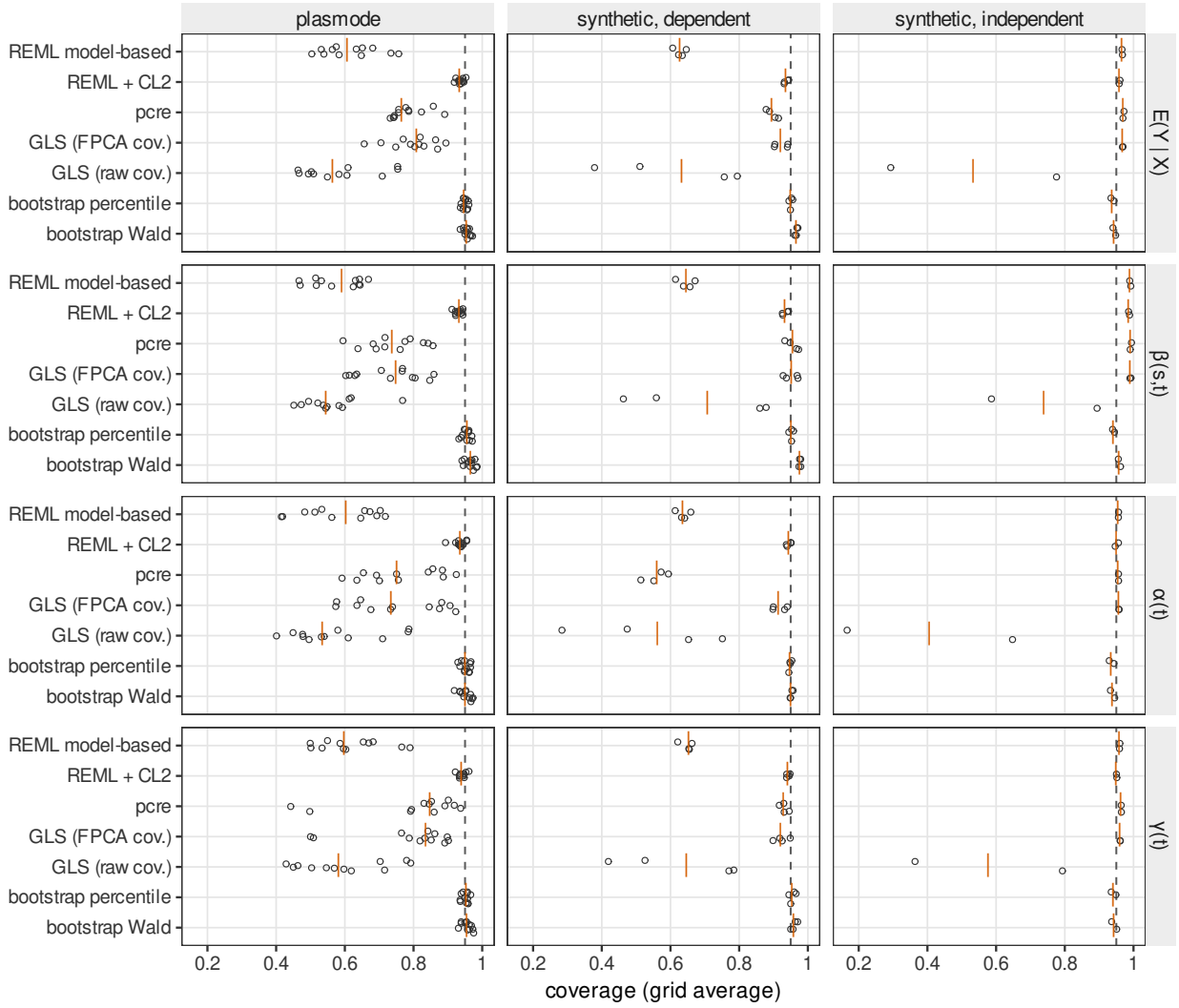


Figure 19: Random curve effect (pcre), GLS and curve bootstrap: grid-averaged coverage per setting or plasmode combination (one circle each, jittered vertically; bar: median), by estimand and data source. Dashed: nominal 0.95.

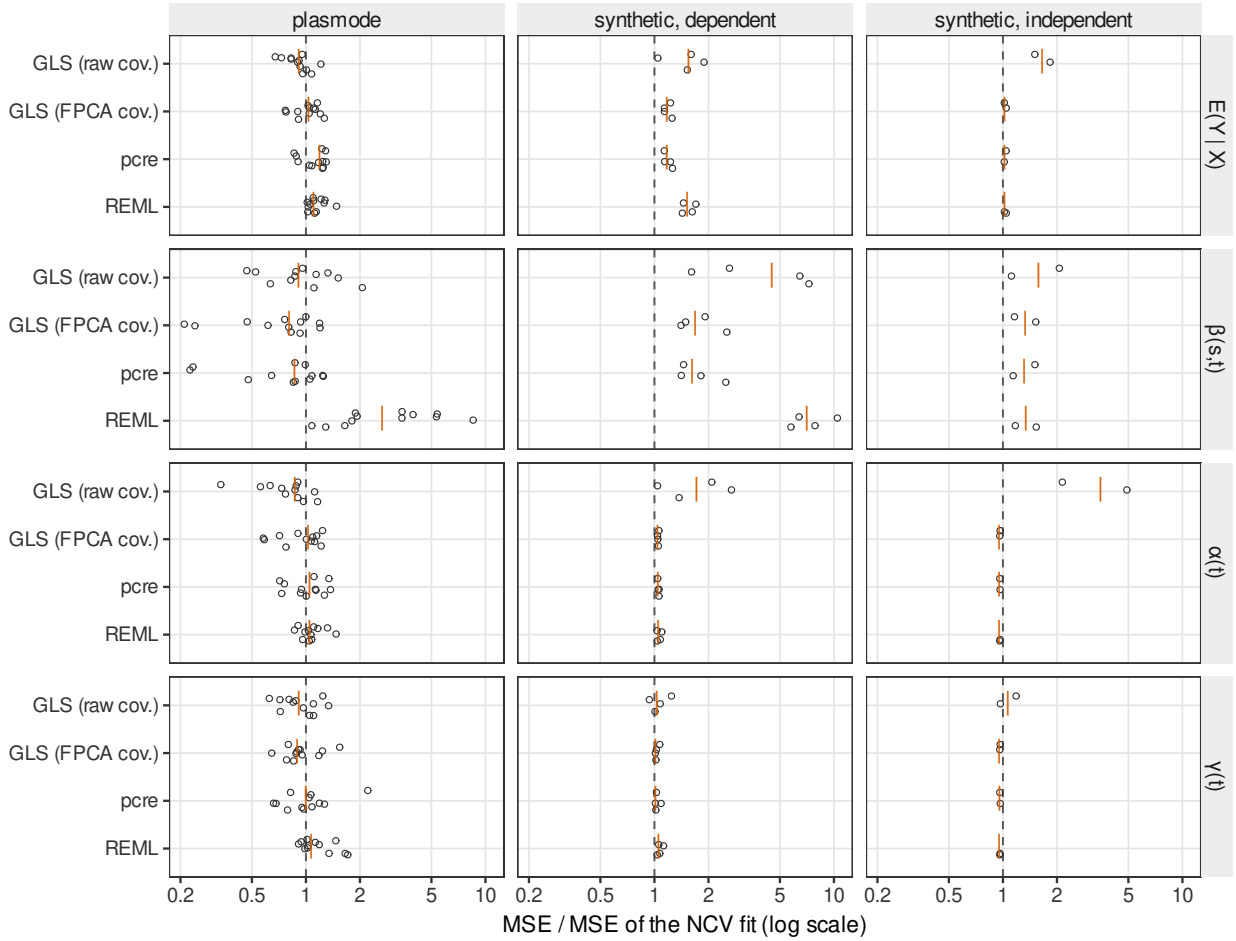


Figure 20: Mean squared error of each estimator divided by that of the NCV fit, per setting or plasmode combination (one circle each, jittered vertically; bar: median; log scale; below 1 = better than NCV). The bootstrap intervals use the REML estimate.

Table 31: Comparators: mean and minimum coverage, mean 5% quantile of pointwise coverage, width and interval score relative to REML + CL2 (geometric means of per-setting ratios).

setting	estimand	interval	settings	coverage	min coverage	q05	width rel.	IS rel.
plasmode	E(Y | X)	REML model-based	12	0.62	0.50	0.32	0.42	2.70
plasmode	E(Y | X)	REML + CL2	12	0.94	0.92	0.88	1.00	1.00
plasmode	E(Y | X)	pcre	12	0.78	0.73	0.54	0.63	1.46
plasmode	E(Y | X)	GLS (FPCA cov.)	12	0.80	0.66	0.58	0.63	1.35
plasmode	E(Y | X)	GLS (raw cov.)	12	0.58	0.46	0.43	0.38	1.99
plasmode	E(Y | X)	bootstrap percentile	12	0.95	0.94	0.90	1.14	1.05
plasmode	E(Y | X)	bootstrap Wald	12	0.96	0.94	0.91	1.12	1.00
plasmode	β(s,t)	REML model-based	12	0.58	0.47	0.32	0.39	2.90
plasmode	β(s,t)	REML + CL2	12	0.93	0.91	0.88	1.00	1.00
plasmode	β(s,t)	pcre	12	0.74	0.59	0.42	0.40	1.05
plasmode	β(s,t)	GLS (FPCA cov.)	12	0.73	0.60	0.45	0.38	1.03
plasmode	β(s,t)	GLS (raw cov.)	12	0.56	0.45	0.38	0.30	1.62
plasmode	β(s,t)	bootstrap percentile	12	0.95	0.93	0.91	1.42	1.22
plasmode	β(s,t)	bootstrap Wald	12	0.97	0.94	0.93	1.39	1.15
plasmode	α(t)	REML model-based	12	0.58	0.42	0.33	0.39	2.83
plasmode	α(t)	REML + CL2	12	0.93	0.89	0.89	1.00	1.00
plasmode	α(t)	pcre	12	0.76	0.59	0.63	0.62	1.52
plasmode	α(t)	GLS (FPCA cov.)	12	0.75	0.57	0.63	0.58	1.50
plasmode	α(t)	GLS (raw cov.)	12	0.57	0.40	0.50	0.35	1.95
plasmode	α(t)	bootstrap percentile	12	0.95	0.93	0.92	1.07	0.97
plasmode	α(t)	bootstrap Wald	12	0.95	0.92	0.91	1.06	0.96
plasmode	γ(t)	REML model-based	12	0.62	0.50	0.37	0.43	2.67
plasmode	γ(t)	REML + CL2	12	0.94	0.92	0.90	1.00	1.00
plasmode	γ(t)	pcre	12	0.80	0.44	0.64	0.71	1.33

Table 31: Comparators: mean and minimum coverage, mean 5% quantile of pointwise coverage, width and interval score relative to REML + CL2 (geometric means of per-setting ratios). *(continued)*

setting	estimand	interval	settings	coverage	min coverage	q05	width rel.	IS rel.
plasmode	γ(t)	GLS (FPCA cov.)	12	0.79	0.50	0.64	0.65	1.33
plasmode	γ(t)	GLS (raw cov.)	12	0.60	0.43	0.48	0.40	1.89
plasmode	γ(t)	bootstrap percentile	12	0.95	0.94	0.92	1.13	1.05
plasmode	γ(t)	bootstrap Wald	12	0.96	0.93	0.91	1.11	1.02
syn. dep.	E(Y | X)	REML model-based	4	0.63	0.61	0.54	0.47	2.05
syn. dep.	E(Y | X)	REML + CL2	4	0.94	0.93	0.91	1.00	1.00
syn. dep.	E(Y | X)	pcre	4	0.90	0.88	0.84	0.76	0.91
syn. dep.	E(Y | X)	GLS (FPCA cov.)	4	0.92	0.90	0.88	0.81	0.88
syn. dep.	E(Y | X)	GLS (raw cov.)	4	0.61	0.38	0.55	0.43	2.03
syn. dep.	E(Y | X)	bootstrap percentile	4	0.95	0.94	0.91	1.17	1.10
syn. dep.	E(Y | X)	bootstrap Wald	4	0.97	0.96	0.93	1.16	1.02
syn. dep.	β(s,t)	REML model-based	4	0.65	0.61	0.56	0.49	2.06
syn. dep.	β(s,t)	REML + CL2	4	0.93	0.93	0.91	1.00	1.00
syn. dep.	β(s,t)	pcre	4	0.96	0.93	0.92	0.59	0.55
syn. dep.	β(s,t)	GLS (FPCA cov.)	4	0.95	0.93	0.91	0.58	0.55
syn. dep.	β(s,t)	GLS (raw cov.)	4	0.69	0.46	0.63	0.40	1.33
syn. dep.	β(s,t)	bootstrap percentile	4	0.95	0.94	0.91	1.60	1.41
syn. dep.	β(s,t)	bootstrap Wald	4	0.98	0.97	0.94	1.57	1.28
syn. dep.	α(t)	REML model-based	4	0.64	0.61	0.56	0.47	2.04
syn. dep.	α(t)	REML + CL2	4	0.94	0.94	0.92	1.00	1.00
syn. dep.	α(t)	pcre	4	0.56	0.51	0.50	0.40	2.41
syn. dep.	α(t)	GLS (FPCA cov.)	4	0.92	0.90	0.87	0.90	1.02
syn. dep.	α(t)	GLS (raw cov.)	4	0.54	0.28	0.50	0.46	3.07
syn. dep.	α(t)	bootstrap percentile	4	0.95	0.94	0.92	1.05	1.02
syn. dep.	α(t)	bootstrap Wald	4	0.95	0.95	0.92	1.04	1.00
syn. dep.	γ(t)	REML model-based	4	0.65	0.62	0.58	0.47	1.96
syn. dep.	γ(t)	REML + CL2	4	0.94	0.94	0.92	1.00	1.00
syn. dep.	γ(t)	pcre	4	0.93	0.92	0.90	0.92	0.97
syn. dep.	γ(t)	GLS (FPCA cov.)	4	0.92	0.90	0.89	0.90	0.98
syn. dep.	γ(t)	GLS (raw cov.)	4	0.62	0.42	0.58	0.45	2.00
syn. dep.	γ(t)	bootstrap percentile	4	0.96	0.94	0.92	1.07	1.01
syn. dep.	γ(t)	bootstrap Wald	4	0.96	0.95	0.93	1.05	0.97
syn. indep.	E(Y | X)	REML model-based	2	0.97	0.97	0.94	1.01	0.98
syn. indep.	E(Y | X)	REML + CL2	2	0.96	0.96	0.93	1.00	1.00
syn. indep.	E(Y | X)	pcre	2	0.97	0.97	0.94	1.03	0.99
syn. indep.	E(Y | X)	GLS (FPCA cov.)	2	0.97	0.97	0.94	1.02	0.98
syn. indep.	E(Y | X)	GLS (raw cov.)	2	0.53	0.29	0.47	0.42	3.05
syn. indep.	E(Y | X)	bootstrap percentile	2	0.94	0.93	0.90	0.93	1.01
syn. indep.	E(Y | X)	bootstrap Wald	2	0.94	0.94	0.90	0.92	0.98
syn. indep.	β(s,t)	REML model-based	2	0.99	0.99	0.97	1.02	1.01
syn. indep.	β(s,t)	REML + CL2	2	0.99	0.99	0.97	1.00	1.00
syn. indep.	β(s,t)	pcre	2	0.99	0.99	0.98	1.04	1.02
syn. indep.	β(s,t)	GLS (FPCA cov.)	2	0.99	0.99	0.98	1.02	1.01
syn. indep.	β(s,t)	GLS (raw cov.)	2	0.74	0.59	0.65	0.49	1.50
syn. indep.	β(s,t)	bootstrap percentile	2	0.94	0.94	0.90	0.82	0.98
syn. indep.	β(s,t)	bootstrap Wald	2	0.96	0.96	0.92	0.81	0.90
syn. indep.	α(t)	REML model-based	2	0.96	0.96	0.93	1.01	0.98
syn. indep.	α(t)	REML + CL2	2	0.95	0.95	0.92	1.00	1.00
syn. indep.	α(t)	pcre	2	0.96	0.96	0.93	1.00	0.98
syn. indep.	α(t)	GLS (FPCA cov.)	2	0.96	0.96	0.94	1.01	0.98
syn. indep.	α(t)	GLS (raw cov.)	2	0.41	0.17	0.36	0.42	5.72
syn. indep.	α(t)	bootstrap percentile	2	0.94	0.93	0.90	0.95	1.01
syn. indep.	α(t)	bootstrap Wald	2	0.94	0.93	0.91	0.95	0.99
syn. indep.	γ(t)	REML model-based	2	0.96	0.96	0.93	1.01	0.97
syn. indep.	γ(t)	REML + CL2	2	0.95	0.95	0.92	1.00	1.00
syn. indep.	γ(t)	pcre	2	0.96	0.96	0.94	1.04	0.98
syn. indep.	γ(t)	GLS (FPCA cov.)	2	0.96	0.96	0.94	1.02	0.97
syn. indep.	γ(t)	GLS (raw cov.)	2	0.58	0.36	0.51	0.41	2.40
syn. indep.	γ(t)	bootstrap percentile	2	0.94	0.94	0.89	0.95	1.00
syn. indep.	γ(t)	bootstrap Wald	2	0.94	0.94	0.90	0.94	0.97

Table 32: Comparators as estimators: median relative error, MSE relative to REML and to NCV (geometric means of per-setting ratios; range for NCV).

setting	estimand	fit	settings	rel. error	MSE/REML	MSE/NCV min	MSE/NCV max	MSE/NCV
plasmode	E(Y | X)	REML	12	0.21	1.00	1.02	1.48	1.14
plasmode	E(Y | X)	NCV	12	0.19	0.87	1.00	1.00	1.00
plasmode	E(Y | X)	pcre	12	0.21	0.97	0.86	1.30	1.11
plasmode	E(Y | X)	GLS (FPCA cov.)	12	0.19	0.89	0.77	1.27	1.01
plasmode	E(Y | X)	GLS (raw cov.)	12	0.18	0.79	0.67	1.21	0.91
plasmode	β(s,t)	REML	12	1.49	1.00	1.08	8.54	2.73
plasmode	β(s,t)	NCV	12	0.96	0.37	1.00	1.00	1.00
plasmode	β(s,t)	pcre	12	0.78	0.26	0.23	1.25	0.72
plasmode	β(s,t)	GLS (FPCA cov.)	12	0.76	0.25	0.21	1.20	0.68

Table 32: Comparators as estimators: median relative error, MSE relative to REML and to NCV (geometric means of per-setting ratios; range for NCV). *(continued)*

setting	estimand	fit	settings	rel. error	MSE/REML	MSE/NCV min	MSE/NCV max	MSE/NCV
plasmode	β(s,t)	GLS (raw cov.)	12	0.89	0.35	0.47	2.06	0.94
plasmode	α(t)	REML	12	0.64	1.00	0.86	1.47	1.07
plasmode	α(t)	NCV	12	0.63	0.93	1.00	1.00	1.00
plasmode	α(t)	pcre	12	0.65	0.95	0.71	1.37	1.01
plasmode	α(t)	GLS (FPCA cov.)	12	0.66	0.86	0.58	1.24	0.92
plasmode	α(t)	GLS (raw cov.)	12	0.61	0.73	0.34	1.16	0.78
plasmode	γ(t)	REML	12	0.85	1.00	0.91	1.71	1.17
plasmode	γ(t)	NCV	12	0.82	0.85	1.00	1.00	1.00
plasmode	γ(t)	pcre	12	0.89	0.86	0.66	2.21	1.01
plasmode	γ(t)	GLS (FPCA cov.)	12	0.87	0.80	0.64	1.54	0.94
plasmode	γ(t)	GLS (raw cov.)	12	0.75	0.79	0.63	1.34	0.93
syn. dep.	E(Y | X)	REML	4	0.29	1.00	1.43	1.70	1.55
syn. dep.	E(Y | X)	NCV	4	0.23	0.65	1.00	1.00	1.00
syn. dep.	E(Y | X)	pcre	4	0.25	0.77	1.13	1.26	1.19
syn. dep.	E(Y | X)	GLS (FPCA cov.)	4	0.25	0.77	1.14	1.25	1.19
syn. dep.	E(Y | X)	GLS (raw cov.)	4	0.29	0.96	1.04	1.89	1.48
syn. dep.	β(s,t)	REML	4	0.65	1.00	5.76	10.45	7.41
syn. dep.	β(s,t)	NCV	4	0.24	0.13	1.00	1.00	1.00
syn. dep.	β(s,t)	pcre	4	0.31	0.24	1.41	2.50	1.75
syn. dep.	β(s,t)	GLS (FPCA cov.)	4	0.31	0.24	1.41	2.53	1.79
syn. dep.	β(s,t)	GLS (raw cov.)	4	0.52	0.51	1.61	7.25	3.75
syn. dep.	α(t)	REML	4	0.32	1.00	1.03	1.10	1.06
syn. dep.	α(t)	NCV	4	0.31	0.94	1.00	1.00	1.00
syn. dep.	α(t)	pcre	4	0.32	0.99	1.04	1.06	1.05
syn. dep.	α(t)	GLS (FPCA cov.)	4	0.32	0.99	1.04	1.06	1.05
syn. dep.	α(t)	GLS (raw cov.)	4	0.41	1.58	1.04	2.69	1.68
syn. dep.	γ(t)	REML	4	0.24	1.00	1.04	1.12	1.07
syn. dep.	γ(t)	NCV	4	0.24	0.93	1.00	1.00	1.00
syn. dep.	γ(t)	pcre	4	0.24	0.97	1.01	1.09	1.04
syn. dep.	γ(t)	GLS (FPCA cov.)	4	0.24	0.96	1.01	1.07	1.03
syn. dep.	γ(t)	GLS (raw cov.)	4	0.25	0.99	0.94	1.24	1.06
syn. indep.	E(Y | X)	REML	2	0.11	1.00	1.02	1.04	1.03
syn. indep.	E(Y | X)	NCV	2	0.11	0.97	1.00	1.00	1.00
syn. indep.	E(Y | X)	pcre	2	0.11	1.00	1.02	1.04	1.03
syn. indep.	E(Y | X)	GLS (FPCA cov.)	2	0.11	1.00	1.02	1.04	1.03
syn. indep.	E(Y | X)	GLS (raw cov.)	2	0.15	1.61	1.51	1.83	1.66
syn. indep.	β(s,t)	REML	2	0.13	1.00	1.17	1.53	1.34
syn. indep.	β(s,t)	NCV	2	0.11	0.75	1.00	1.00	1.00
syn. indep.	β(s,t)	pcre	2	0.13	0.98	1.14	1.51	1.31
syn. indep.	β(s,t)	GLS (FPCA cov.)	2	0.13	0.99	1.16	1.52	1.33
syn. indep.	β(s,t)	GLS (raw cov.)	2	0.14	1.13	1.12	2.06	1.52
syn. indep.	α(t)	REML	2	0.15	1.00	0.96	0.96	0.96
syn. indep.	α(t)	NCV	2	0.16	1.04	1.00	1.00	1.00
syn. indep.	α(t)	pcre	2	0.15	1.00	0.96	0.96	0.96
syn. indep.	α(t)	GLS (FPCA cov.)	2	0.15	1.00	0.96	0.96	0.96
syn. indep.	α(t)	GLS (raw cov.)	2	0.30	3.37	2.14	4.90	3.24
syn. indep.	γ(t)	REML	2	0.12	1.00	0.96	0.96	0.96
syn. indep.	γ(t)	NCV	2	0.12	1.04	1.00	1.00	1.00
syn. indep.	γ(t)	pcre	2	0.12	1.00	0.96	0.97	0.96
syn. indep.	γ(t)	GLS (FPCA cov.)	2	0.12	1.00	0.96	0.96	0.96
syn. indep.	γ(t)	GLS (raw cov.)	2	0.12	1.11	0.97	1.18	1.07

CL2 on the GLS and random-curve-effect fits. GLS whitens each curve with $\hat{\Sigma}^{-1/2}$. The transform acts within curves, so curves stay independent clusters, and CL2 (Bayesian form, curves as clusters) applies to the whitened fit; it does not need $\hat{\Sigma}$ to be correct. GLS-FPCA + CL2 is CL2 on the GLS fit with the FPCA covariance, pcre + CL2 is CL2 on the fit with the random curve effect (`pcre()`). The comparison uses replicates 1–100, where all constructions exist; one pcre + CL2 replicate failed in a singular value decomposition and is left out for that construction only. The eigenvalue floor of the leverage adjustment (Appendix A.2 of the paper) bound in no GLS fit and, among the pcre fits, only on the running data, there in 70 of 100 replicates with all subjects and in 60 of 100 with 40, so pcre + CL2 on that dataset depends on the floor. In the four synthetic Gaussian settings with dependent errors, GLS-FPCA + CL2 covers $\beta(s,t)$ at 0.96, at 0.61 times REML + CL2's width (interval score 0.55 times). On the plasmode data, it covers β at 0.82 (by dataset 0.65–0.90; GLS-FPCA without CL2 0.73, REML + CL2 0.93), at 0.46 times the width. The standard error of GLS-FPCA + CL2 is close to the spread of the GLS estimate there (geometric mean SE/SD 0.98). This agreement is consistent with residual smoothing bias of the GLS estimate under the plasmode generator, but an aggregate ratio near one does not exclude local or tail errors of the standard error, and no single cause is established. pcre + CL2 behaves alike (β 0.83) and repairs the model-based intervals

of the random curve effect for $\alpha(t)$ on synthetic data (0.56 to 0.95). GLS with the raw residual covariance fails with CL2 as well (0.65 under dependent synthetic errors): with more grid points than curves, the raw covariance absorbs the residuals themselves.

Table 33: CL2 on the GLS and pcre fits (replicates 1–100): mean and minimum coverage over settings or plasmode combinations, mean 5% quantile of pointwise coverage, width and interval score relative to REML + CL2, MSE relative to the NCV fit (geometric means), and detection rate, by data source, estimand and construction.

setting	estimand	interval	coverage	min cov.	q05	width rel.	IS rel.	MSE/NCV	detection
plasmode	E(Y | X)	REML + CL2	0.94	0.92	0.87	1.00	1.00	1.15	0.97
plasmode	E(Y | X)	NCV + CL2	0.89	0.85	0.74	0.81	1.04	1.00	0.99
plasmode	E(Y | X)	GLS (FPCA cov.)	0.80	0.66	0.57	0.63	1.34	1.01	0.99
plasmode	E(Y | X)	GLS (FPCA cov.) + CL2	0.89	0.83	0.74	0.79	1.02	1.01	0.99
plasmode	E(Y | X)	GLS (raw cov.) + CL2	0.84	0.51	0.73	0.66	1.10	0.90	0.99
plasmode	E(Y | X)	pcre	0.78	0.73	0.54	0.63	1.45	1.10	0.99
plasmode	E(Y | X)	pcre + CL2	0.91	0.86	0.74	0.86	1.04	1.10	0.99
plasmode	β(s,t)	REML + CL2	0.93	0.91	0.87	1.00	1.00	2.75	0.38
plasmode	β(s,t)	NCV + CL2	0.76	0.60	0.46	0.46	1.13	1.00	0.54
plasmode	β(s,t)	GLS (FPCA cov.)	0.73	0.60	0.45	0.38	1.02	0.67	0.55
plasmode	β(s,t)	GLS (FPCA cov.) + CL2	0.82	0.63	0.56	0.46	0.83	0.67	0.48
plasmode	β(s,t)	GLS (raw cov.) + CL2	0.79	0.52	0.63	0.48	0.95	0.90	0.57
plasmode	β(s,t)	pcre	0.74	0.59	0.42	0.40	1.03	0.71	0.49
plasmode	β(s,t)	pcre + CL2	0.83	0.66	0.53	0.49	0.85	0.71	0.42
plasmode	α(t)	REML + CL2	0.94	0.91	0.88	1.00	1.00	1.06	0.76
plasmode	α(t)	NCV + CL2	0.89	0.79	0.77	0.85	1.12	1.00	0.79
plasmode	α(t)	GLS (FPCA cov.)	0.75	0.57	0.62	0.58	1.52	0.93	0.86
plasmode	α(t)	GLS (FPCA cov.) + CL2	0.87	0.77	0.77	0.74	1.15	0.93	0.80
plasmode	α(t)	GLS (raw cov.) + CL2	0.83	0.56	0.74	0.61	1.10	0.77	0.86
plasmode	α(t)	pcre	0.77	0.59	0.63	0.62	1.54	1.02	0.85
plasmode	α(t)	pcre + CL2	0.89	0.80	0.77	0.83	1.15	1.02	0.78
plasmode	γ(t)	REML + CL2	0.94	0.92	0.89	1.00	1.00	1.19	0.41
plasmode	γ(t)	NCV + CL2	0.90	0.86	0.79	0.84	1.01	1.00	0.45
plasmode	γ(t)	GLS (FPCA cov.)	0.79	0.50	0.63	0.65	1.34	0.96	0.52
plasmode	γ(t)	GLS (FPCA cov.) + CL2	0.85	0.65	0.74	0.73	1.10	0.96	0.46
plasmode	γ(t)	GLS (raw cov.) + CL2	0.82	0.52	0.73	0.64	1.16	0.94	0.55
plasmode	γ(t)	pcre	0.80	0.45	0.65	0.71	1.33	1.02	0.49
plasmode	γ(t)	pcre + CL2	0.87	0.54	0.74	0.83	1.13	1.02	0.40
syn. dep.	E(Y | X)	REML + CL2	0.94	0.93	0.90	1.00	1.00	1.59	0.98
syn. dep.	E(Y | X)	NCV + CL2	0.93	0.92	0.87	0.76	0.79	1.00	0.99
syn. dep.	E(Y | X)	GLS (FPCA cov.)	0.93	0.91	0.87	0.81	0.87	1.19	0.99
syn. dep.	E(Y | X)	GLS (FPCA cov.) + CL2	0.94	0.93	0.90	0.86	0.87	1.19	0.99
syn. dep.	E(Y | X)	GLS (raw cov.) + CL2	0.59	0.28	0.51	0.39	2.11	1.50	1.00
syn. dep.	E(Y | X)	pcre	0.90	0.88	0.84	0.76	0.89	1.19	0.99
syn. dep.	E(Y | X)	pcre + CL2	0.98	0.97	0.95	1.05	0.91	1.19	0.98
syn. dep.	β(s,t)	REML + CL2	0.94	0.93	0.89	1.00	1.00	7.72	0.66
syn. dep.	β(s,t)	NCV + CL2	0.93	0.92	0.87	0.36	0.37	1.00	0.95
syn. dep.	β(s,t)	GLS (FPCA cov.)	0.95	0.93	0.91	0.58	0.55	1.84	0.87
syn. dep.	β(s,t)	GLS (FPCA cov.) + CL2	0.96	0.94	0.92	0.61	0.55	1.84	0.86
syn. dep.	β(s,t)	GLS (raw cov.) + CL2	0.65	0.36	0.57	0.36	1.43	3.87	0.91
syn. dep.	β(s,t)	pcre	0.96	0.93	0.92	0.59	0.55	1.81	0.87
syn. dep.	β(s,t)	pcre + CL2	0.98	0.96	0.96	0.71	0.58	1.81	0.82
syn. dep.	α(t)	REML + CL2	0.95	0.94	0.91	1.00	1.00	1.07	0.98
syn. dep.	α(t)	NCV + CL2	0.93	0.93	0.86	0.93	0.98	1.00	0.99
syn. dep.	α(t)	GLS (FPCA cov.)	0.92	0.89	0.84	0.90	1.03	1.06	1.00
syn. dep.	α(t)	GLS (FPCA cov.) + CL2	0.93	0.92	0.84	0.95	1.02	1.06	0.99
syn. dep.	α(t)	GLS (raw cov.) + CL2	0.52	0.21	0.46	0.42	3.08	1.65	1.00
syn. dep.	α(t)	pcre	0.56	0.51	0.47	0.40	2.37	1.05	1.00
syn. dep.	α(t)	pcre + CL2	0.95	0.94	0.92	1.00	0.99	1.05	0.99
syn. dep.	γ(t)	REML + CL2	0.95	0.95	0.92	1.00	1.00	1.05	0.92
syn. dep.	γ(t)	NCV + CL2	0.93	0.92	0.87	0.91	0.99	1.00	0.94
syn. dep.	γ(t)	GLS (FPCA cov.)	0.93	0.91	0.89	0.90	0.98	1.02	0.94
syn. dep.	γ(t)	GLS (FPCA cov.) + CL2	0.94	0.93	0.90	0.95	0.99	1.02	0.93
syn. dep.	γ(t)	GLS (raw cov.) + CL2	0.61	0.33	0.54	0.41	2.06	1.04	0.99
syn. dep.	γ(t)	pcre	0.94	0.93	0.89	0.92	0.97	1.03	0.93
syn. dep.	γ(t)	pcre + CL2	0.98	0.98	0.96	1.22	1.08	1.03	0.87
syn. indep.	E(Y | X)	REML + CL2	0.96	0.96	0.92	1.00	1.00	1.03	1.00
syn. indep.	E(Y | X)	NCV + CL2	0.95	0.95	0.90	0.92	0.97	1.00	1.00
syn. indep.	E(Y | X)	GLS (FPCA cov.)	0.97	0.97	0.94	1.02	0.98	1.03	1.00
syn. indep.	E(Y | X)	GLS (FPCA cov.) + CL2	0.96	0.96	0.92	1.00	1.00	1.03	1.00
syn. indep.	E(Y | X)	GLS (raw cov.) + CL2	0.46	0.14	0.38	0.28	3.45	1.66	1.00
syn. indep.	E(Y | X)	pcre	0.97	0.97	0.94	1.03	0.99	1.03	1.00
syn. indep.	E(Y | X)	pcre + CL2	0.97	0.97	0.94	1.02	1.01	1.03	1.00
syn. indep.	β(s,t)	REML + CL2	0.99	0.99	0.97	1.00	1.00	1.27	0.97
syn. indep.	β(s,t)	NCV + CL2	0.98	0.98	0.94	0.75	0.78	1.00	0.98
syn. indep.	β(s,t)	GLS (FPCA cov.)	0.99	0.99	0.97	1.02	1.01	1.26	0.97
syn. indep.	β(s,t)	GLS (FPCA cov.) + CL2	0.99	0.99	0.97	1.00	1.00	1.26	0.97
syn. indep.	β(s,t)	GLS (raw cov.) + CL2	0.64	0.38	0.54	0.37	1.84	1.44	0.99
syn. indep.	β(s,t)	pcre	0.99	0.99	0.98	1.03	1.02	1.25	0.97

Table 33: CL2 on the GLS and pcre fits (replicates 1–100): mean and minimum coverage over settings or plasmode combinations, mean 5% quantile of pointwise coverage, width and interval score relative to REML + CL2, MSE relative to the NCV fit (geometric means), and detection rate, by data source, estimand and construction. *(continued)*

setting	estimand	interval	coverage	min cov.	q05	width rel.	IS rel.	MSE/NCV	detection
syn. indep.	β(s,t)	pcre + CL2	0.99	0.99	0.97	1.01	1.01	1.25	0.97
syn. indep.	α(t)	REML + CL2	0.95	0.95	0.92	1.00	1.00	0.98	1.00
syn. indep.	α(t)	NCV + CL2	0.95	0.94	0.90	0.98	1.01	1.00	1.00
syn. indep.	α(t)	GLS (FPCA cov.)	0.96	0.96	0.92	1.01	0.98	0.98	1.00
syn. indep.	α(t)	GLS (FPCA cov.) + CL2	0.95	0.95	0.92	1.00	1.00	0.98	1.00
syn. indep.	α(t)	GLS (raw cov.) + CL2	0.36	0.06	0.30	0.25	6.22	3.36	1.00
syn. indep.	α(t)	pcre	0.96	0.96	0.92	1.00	0.98	0.98	1.00
syn. indep.	α(t)	pcre + CL2	0.96	0.95	0.92	1.00	1.00	0.98	1.00
syn. indep.	γ(t)	REML + CL2	0.96	0.95	0.92	1.00	1.00	0.96	1.00
syn. indep.	γ(t)	NCV + CL2	0.95	0.95	0.88	0.96	1.01	1.00	1.00
syn. indep.	γ(t)	GLS (FPCA cov.)	0.97	0.96	0.93	1.02	0.98	0.96	1.00
syn. indep.	γ(t)	GLS (FPCA cov.) + CL2	0.96	0.96	0.92	1.00	1.00	0.96	1.00
syn. indep.	γ(t)	GLS (raw cov.) + CL2	0.48	0.16	0.42	0.26	2.87	1.06	1.00
syn. indep.	γ(t)	pcre	0.97	0.97	0.94	1.04	0.99	0.96	1.00
syn. indep.	γ(t)	pcre + CL2	0.96	0.96	0.92	1.03	1.01	0.96	1.00

S3.3 Functional GEE

Simulation design of the functional GEE paper

We reran the function-on-scalar simulation of Loewinger et al. (2026, sec. 4) with their data generator unchanged: $Y_{ij}(s) = \beta_0(s) + X_{1i}\beta_1(s) + X_{2i}\beta_2(s) + \varepsilon_{ij}(s)$ (in their notation, j indexes the curves of subject i and s the grid) on 50 grid points, with Gaussian errors (variance 10) or Poisson and binary responses generated through a Gaussian copula whose latent correlation is AR(1) (0.75) along s and exchangeable (0.75) between the curves of a subject. The design crosses the three families with 25, 100, or 500 subjects and 5 or 1 curves per subject (5 as in their study, 1 as in ours), with 200 replicates per setting; 1 one-step fit failed (singular system). The initial fit is their pffr call (bam with fREML, discretization, P-spline bases with 16 functions); all intervals labeled "initial fit" in Table 34 are computed on that fit, the REML row on a refit of the same model and bases by gam with REML, and all clusters are subjects.

Every interval is scored through one evaluation matrix per coefficient function; these matrices reproduce fast-FGEE's fitted curves and standard errors, so centering and basis differences between methods do not enter. As in their Table 3, coverage is the pointwise 95% coverage averaged over the grid, over replicates and over β_0, β_1 and β_2; its Monte Carlo standard error is at most 0.010.

Table 34: Simulation design of the functional GEE paper: ranges over the nine settings (three families, three numbers of subjects) of the coverage of nominal 95% pointwise intervals, averaged over the grid and the three coefficient functions, by number of curves per subject. Initial fit: the pffr call of that paper (bam). fGEE: one-step update with the stated functional and longitudinal working correlations. Calibrated crit.: fastFGEE's wild-bootstrap critical value with its t adjustment; z: normal critical value.

Intervals	5 curves per subject	1 curve per subject
Initial fit, model-based	0.43–0.57	0.65–0.79
Initial-fit sandwich of fastFGEE, z	0.60–0.87	0.61–0.80
Initial-fit sandwich of fastFGEE, calibrated crit.	0.94–1.00	0.95–1.00
Initial-fit sandwich of fastFGEE, penalty rescaled, z	0.90–0.95	0.87–0.94
Initial fit, cluster sandwich, frequentist, centered scores	0.90–0.95	0.87–0.94
Initial fit, CR1	0.90–0.95	0.89–0.95
Initial fit, CL2	0.92–0.95	0.92–0.95
REML (gam) fit, CL2	0.92–0.95	0.92–0.95
fGEE, independence × independence, z	0.74–0.94	0.75–0.94
fGEE, independence × independence, calibrated crit.	0.93–0.96	0.90–0.95
fGEE, AR(1) × exchangeable, z	0.75–0.94	0.71–0.94
fGEE, AR(1) × exchangeable, calibrated crit.	0.94–0.96	0.88–0.95

With 5 curves per subject, CL2 clustered by subject on the initial fit covers 0.94–0.95 with 100 or 500 subjects and 0.92–0.93 with 25. Refitting by gam and REML changes the pooled CL2 coverage by at most 0.013. The intervals of fastFGEE's functional GEE (fGEE), a one-step update of the initial fit under the stated working correlations, with its calibrated critical values cover 0.93–0.96 in this design; with a normal critical value, they cover 0.74–0.92 with at most 100 subjects and 0.92–0.94 with 500. The one-step estimate of β_1 has a root-mean-square error of 0.85–1.01 (independence) and 0.81–1.04 (AR(1) × exchangeable) times that of the initial fit.

The initial-fit sandwich and penalty-scale conventions

mgcv's smoothing parameters are on the scale of the summed log-likelihood: the penalized estimate maximizes $\sum_{i=1}^{n} \ell_i(\theta) - \theta^\top \mathrm{S}_\lambda \theta/(2\phi)$, with $\mathrm{S}_\lambda = \sum_r \lambda_r \mathrm{S}_r$, n clusters (subjects) and the sum over all observations of cluster i in ℓ_i (for the Gaussian, the residual sum of squares). A sandwich written with averaged quantities therefore needs the bread

$$\bar{\mathbf{A}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{X}_i^\top \mathbf{W}_i \mathbf{X}_i + \frac{\mathrm{S}_\lambda}{n\phi} = (n\,\mathbf{V}_p)^{-1},$$

with $\mathbf{W}_i$ the Fisher weights of the main text, and $\bar{\mathbf{A}}^{-1}\bar{\mathbf{M}}\bar{\mathbf{A}}^{-1}/n$ with the averaged meat $\bar{\mathbf{M}}$ is then the cluster sandwich $\mathbf{V}_p(\sum_i \mathbf{U}_i\mathbf{U}_i^\top)\mathbf{V}_p$ up to degrees-of-freedom factors.

The initial-fit sandwich of fastFGEE 0.2.2 (`gee.fit = FALSE`), which the simulation code of Loewinger et al. (2026) calls for their pffr baseline, differs in the penalty scale: it adds S_λ to the averaged bread, a penalty larger by the factor $n\phi$ (for the Gaussian, approximately $n\sigma^2$). The estimate is the unchanged pffr estimate, but the inverse bread, and with it the standard errors, shrink. In their design, the ratio of mean standard error to the empirical standard deviation of $\hat{\beta}_1$ is 0.44–0.76 for this sandwich and does not improve with the number of subjects (0.45–0.69 with 25, 0.44–0.76 with 500), because the penalty-scale difference does not vanish.

With 5 curves per subject, its pooled coverage is 0.60–0.64 for Gaussian, 0.84–0.87 for Poisson and 0.72–0.76 for binary responses (Table 34); its coverage here is in line with their Table 3 for the pffr baseline. With its wild-bootstrap critical value, which widens the intervals instead, coverage is 0.97–1.00 with 100 or more subjects.

Dividing the penalty by $n\phi$ and leaving everything else unchanged gives pooled coverage of 0.87–0.95 over all 18 settings. The rescaled sandwich then coincides with refund's frequentist cluster sandwich (no $\mathbf{V}_p - \mathbf{V}_e$ term) with centered cluster scores: pooled coverages agree to three decimals in every Poisson and binary setting and differ by at most 0.008 in the Gaussian settings, where fastFGEE estimates the working variance per grid point instead of using one σ^2.

CL2 on the same fit covers 0.94–0.95 with 100 or 500 subjects and 0.92–0.93 with 25; the lowest single coefficient function is 0.90 (β_0, binary, 25 subjects, 1 curve per subject). Its gain over the rescaled sandwich at 25 subjects, 0.025–0.048, combines uncentered scores, the $\mathbf{V}_p - \mathbf{V}_e$ term and the leverage adjustment.

The main factorial settings of our simulation

fastFGEE was started from the REML fit of the main study in its 18 main factorial settings at the default signal level (Gaussian, Poisson and binary responses; independent, rough stationary and smooth errors; $G = 40$ and 100; model $\mathrm{ff}(X) + z\gamma(t)$; one curve per subject; 200 replicates, the main study's datasets), with curves as clusters, independent longitudinal correlation, functional working correlation independence, AR(1) or FPCA, and the package defaults otherwise (cluster cross-validation of the smoothing parameters, wild bootstrap).

We score Wald intervals from its covariance with the normal critical value and intervals with its calibrated critical value (studentized wild cluster bootstrap with t adjustment), both with the same estimate and standard error. The bootstrap gives one critical value per dataset, a quantile of the absolute studentized statistics pooled over the evaluation points, so it targets coverage averaged over these points and not coverage at each point.

The interval interface of fastFGEE covers the functional intercept and scalar-covariate effects but not ff surfaces or fitted means. For $\alpha(t)$ and $\gamma(t)$, we transfer its critical value for the term to our evaluation points; the calibration on the package's own term grid need not carry over to other points. For $\beta(s,t)$ and the mean, we call the package's internal bootstrap routine with our evaluation matrix as the linear map and all other inputs as `fgee()` sets them; with the package's own term maps, this call reproduced its stored critical values exactly in the settings we checked.

The t adjustment multiplies the critical value by the ratio of t and normal quantiles, with degrees of freedom equal to the number of curves minus the effective degrees of freedom of the coefficients the map involves, floored at 1. The surface and the mean involve many more coefficients than the package's term maps, so, with few curves, the floor can be reached, where the factor is 6.48. We therefore report these intervals with and without the adjustment. Monte Carlo standard errors of coverage are at most 0.018.

Table 35: Main factorial settings of our simulation: ranges of grid-averaged coverage of nominal 95% pointwise intervals over the 12 settings with dependent errors (rough stationary, smooth) and the 6 settings with independent errors, and the ratio SE/SD (grid average of mean standard error over the empirical SD of the estimate) under dependence. fGEE intervals: Wald intervals from its covariance, or with its calibrated critical value (for α and γ transferred from its term grid; for β and the mean from its internal routine, with and without its t adjustment). Mean on the link scale.

Estimand	Intervals	Cov. dependent	Cov. independent	SE/SD dependent
β(s,t)	REML, model-based	0.61–0.82	0.98–0.99	0.41–0.62
	REML + CL2	0.93–0.95	0.97–0.99	0.83–0.94
	fGEE, independence	0.73–0.89	0.83–0.90	0.46–0.76
	fGEE, AR(1)	0.72–0.89	0.83–0.90	0.45–0.76
	fGEE, FPCA	0.67–0.86	0.82–0.89	0.41–0.68
	fGEE, independence, calibrated crit.	0.97–0.99	0.90–0.97	
	fGEE, independence, calibrated crit. without t	0.96–0.99	0.89–0.97	
	fGEE, AR(1), calibrated crit.	0.97–0.99	0.90–0.97	
	fGEE, AR(1), calibrated crit. without t	0.95–0.99	0.89–0.97	
	fGEE, FPCA, calibrated crit.	0.91–0.99	0.90–0.97	
	fGEE, FPCA, calibrated crit. without t	0.89–0.99	0.89–0.97	
mean (link)	REML, model-based	0.61–0.75	0.95–0.97	0.44–0.59
	REML + CL2	0.92–0.95	0.93–0.96	0.92–1.00
	fGEE, independence	0.81–0.91	0.83–0.90	0.70–0.91
	fGEE, AR(1)	0.81–0.90	0.83–0.90	0.70–0.91
	fGEE, FPCA	0.71–0.88	0.83–0.90	0.55–0.84
	fGEE, independence, calibrated crit.	0.96–0.99	0.89–0.99	
	fGEE, independence, calibrated crit. without t	0.94–0.96	0.86–0.93	
	fGEE, AR(1), calibrated crit.	0.95–0.99	0.89–0.99	
	fGEE, AR(1), calibrated crit. without t	0.91–0.96	0.86–0.93	
	fGEE, FPCA, calibrated crit.	0.92–0.99	0.90–0.99	
	fGEE, FPCA, calibrated crit. without t	0.86–0.95	0.87–0.93	
α(t)	REML, model-based	0.61–0.75	0.90–0.96	0.45–0.59
	REML + CL2	0.88–0.95	0.86–0.96	0.89–1.02
	fGEE, independence	0.79–0.92	0.79–0.91	0.72–0.95
	fGEE, AR(1)	0.79–0.92	0.79–0.91	0.72–0.94
	fGEE, FPCA	0.66–0.89	0.79–0.91	0.55–0.88
	fGEE, independence, calibrated crit.	0.91–0.95	0.83–0.94	
	fGEE, AR(1), calibrated crit.	0.90–0.95	0.83–0.94	
	fGEE, FPCA, calibrated crit.	0.82–0.94	0.84–0.94	
γ(t)	REML, model-based	0.62–0.74	0.93–0.96	0.47–0.59
	REML + CL2	0.92–0.95	0.89–0.96	0.93–1.03
	fGEE, independence	0.77–0.91	0.78–0.91	0.70–0.94
	fGEE, AR(1)	0.76–0.90	0.78–0.91	0.70–0.94
	fGEE, FPCA	0.70–0.87	0.78–0.90	0.56–0.86
	fGEE, independence, calibrated crit.	0.89–0.93	0.83–0.93	
	fGEE, AR(1), calibrated crit.	0.89–0.93	0.83–0.93	
	fGEE, FPCA, calibrated crit.	0.82–0.91	0.84–0.93	

The Wald intervals for β and the mean use the sandwich covariance fastFGEE 0.2.2 returns for its one-step estimate under the stated working correlation. Under dependent errors, they cover $\beta(s,t)$ at 0.67–0.89 against 0.93–0.95

for REML + CL2 (Table 35). Smoothing bias alone does not explain the shortfall: the root mean squared bias of the fGEE surface estimate is 0.11–0.35 of its empirical standard deviation (REML 0.09–0.13), while SE/SD is 0.41–0.76.

Table 36: Main factorial settings, β(s,t) and the mean (link scale): ranges over settings of grid-averaged coverage of nominal 95% pointwise intervals, of the 5% quantile of pointwise coverage over the grid (q05), and of width and interval score relative to REML + CL2 with the normal critical value in the same setting. fGEE with independence working correlation and its calibrated critical value, with and without its t adjustment.

Estimand	Errors	G	Intervals	Coverage	q05	Width	Interval score
β(s,t)	dependent	40	REML + CL2	0.93–0.94	0.90–0.91		
	dependent	40	fGEE, calibrated	0.97–0.98	0.94–0.97	0.97–1.10	0.76–0.94
	dependent	40	fGEE, calibrated without t	0.96–0.98	0.94–0.95	0.89–0.94	0.74–0.87
	dependent	100	REML + CL2	0.94–0.95	0.92–0.93		
	dependent	100	fGEE, calibrated	0.99	0.97–0.98	0.91–0.94	0.75–0.80
	dependent	100	fGEE, calibrated without t	0.99	0.96–0.98	0.90–0.93	0.74–0.80
	independent	40	REML + CL2	0.97–0.99	0.92–0.97		
	independent	40	fGEE, calibrated	0.90–0.95	0.82–0.91	0.76–0.88	0.88–1.24
	independent	40	fGEE, calibrated without t	0.89–0.93	0.80–0.88	0.66–0.83	0.89–1.29
	independent	100	REML + CL2	0.99	0.96–0.97		
	independent	100	fGEE, calibrated	0.94–0.97	0.87–0.93	0.70–0.75	0.77–0.97
	independent	100	fGEE, calibrated without t	0.93–0.97	0.86–0.93	0.68–0.74	0.77–0.97
mean (link)	dependent	40	REML + CL2	0.92–0.93	0.89–0.91		
	dependent	40	fGEE, calibrated	0.96–0.99	0.92–0.97	1.14–2.19	0.94–1.73
	dependent	40	fGEE, calibrated without t	0.94–0.95	0.89–0.91	0.86–0.96	0.83–0.88
	dependent	100	REML + CL2	0.94–0.95	0.91–0.92		
	dependent	100	fGEE, calibrated	0.96–0.97	0.89–0.94	0.93–0.97	0.86–0.89
	dependent	100	fGEE, calibrated without t	0.95–0.96	0.89–0.93	0.92–0.95	0.86–0.89
	independent	40	REML + CL2	0.93–0.96	0.82–0.93		
	independent	40	fGEE, calibrated	0.89–0.99	0.79–0.98	1.04–3.65	1.18–3.18
	independent	40	fGEE, calibrated without t	0.86–0.90	0.75–0.84	0.82–0.89	1.08–1.20
	independent	100	REML + CL2	0.95–0.96	0.89–0.93		
	independent	100	fGEE, calibrated	0.91–0.94	0.81–0.89	0.89–0.92	0.99–1.09
	independent	100	fGEE, calibrated without t	0.91–0.93	0.81–0.88	0.87–0.90	1.00–1.10

Under dependent errors, with independence working correlation, the calibrated intervals cover β at 0.97–0.99 (5% quantile of pointwise coverage 0.94–0.98), at 0.91–1.10 times the width and 0.75–0.94 times the interval score of REML + CL2 with the normal critical value, which covers 0.93–0.95 (0.90–0.93; Table 36). REML + CL2 with Satterthwaite degrees of freedom covers 0.95–0.96 (0.92–0.94), and the calibrated intervals are 0.88–0.99 times its width.

For fitted means, REML + CL2 with Satterthwaite degrees of freedom covers 0.94–0.95; without the t adjustment, the calibrated intervals cover 0.94–0.96 at 0.78–0.92 times its width, and, with the adjustment, they are up to 1.99 times its width with 40 curves.

The median critical value without the t adjustment is 4.32–5.37, whereas the normal quantile divided by SE/SD is 2.59–4.29. Over the three working correlations, the coverage is 0.91–0.99 for β and 0.92–0.99 for the mean, the lowest values for FPCA with 40 curves. Under independent errors, the intervals with independence working

correlation cover β at 0.90–0.97, the lowest values for binary responses, at 0.70–0.88 times the width and 0.77–1.24 times the interval score of REML + CL2, which covers 0.97–0.99.

With independence working correlation, the t adjustment changes the coverage of β by at most 0.021 (0.052 with FPCA), but the adjustment strongly affects the mean with 40 curves, where it sits at the floor on up to 51% of the replicates of a setting. There, the intervals for the mean are up to 3.65 times as wide as REML + CL2's, with an interval score up to 3.18 times that of REML + CL2, and cover 0.89–0.99. Without the adjustment, they cover 0.94–0.95 under dependent errors at 0.86–0.96 times REML + CL2's width and 0.86–0.90 under independent errors at 0.82–0.89 times that width. With 100 curves, the two versions differ by at most 0.006 in coverage. For α and γ, the calibrated intervals cover 0.82–0.95 under dependence. These results concern a use of fastFGEE's bootstrap routine that its interface does not offer, for targets other than those of its own function-on-scalar setting studied above.

Table 37: Main factorial settings: paired MSE ratios on identical datasets (ratio of mean squared errors over replicates, grid-averaged), ranges over settings. fGEE: independence working correlation. In brackets: number of settings whose 95% replicate-bootstrap interval lies entirely below one / entirely above one.

Estimand	Errors	Settings	fGEE / REML	NCV / REML	fGEE / NCV
β(s,t)	dependent	12	0.22–0.53 [12/0]	0.10–0.22 [12/0]	1.32–3.70 [0/12]
	independent	6	0.71–2.21 [2/3]	0.65–1.15 [3/0]	1.01–1.93 [0/5]
α(t)	dependent	12	0.65–0.99 [10/0]	0.56–0.97 [12/0]	0.97–1.16 [0/9]
	independent	6	1.07–1.10 [0/6]	1.04–1.10 [0/6]	0.99–1.05 [0/4]
γ(t)	dependent	12	0.75–1.01 [5/0]	0.69–1.00 [6/0]	0.98–1.10 [0/6]
	independent	6	1.10–1.20 [0/6]	1.04–1.17 [0/6]	1.03–1.08 [0/5]

Under dependent errors, the one-step update reduces the mean squared error of β relative to REML (0.22–0.53), by less than curve-blocked NCV (0.10–0.22): the ratio fGEE/NCV is 1.32–3.70, with every bootstrap interval above one (lowest lower bound 1.09). Choosing, after the fact, the working correlation with the lowest mean squared error in each setting gives a ratio of 1.27–3.70. Under independent errors, the ratio fGEE/REML for β is 0.71–2.21, above one in the settings with $G = 40$ (1.06–2.21). For α and γ, the ratio fGEE/NCV is 0.97–1.16.

Plasmode datasets

The plasmode comparison uses the six datasets with independent source curves (observed curves independent across subjects), the midpoint truth (refit at the geometric mean of the REML and NCV smoothing parameters), the REML residual curves attached to their own subjects with signs flipped per curve, all subjects and 40 subjects, 200 replicates per combination, and the plasmode study's own datasets; methods and interval definitions are those of the main factorial settings (Gaussian responses). For the two cardiology datasets, only aggregate results are reported. Monte Carlo standard errors of coverage are at most 0.025.

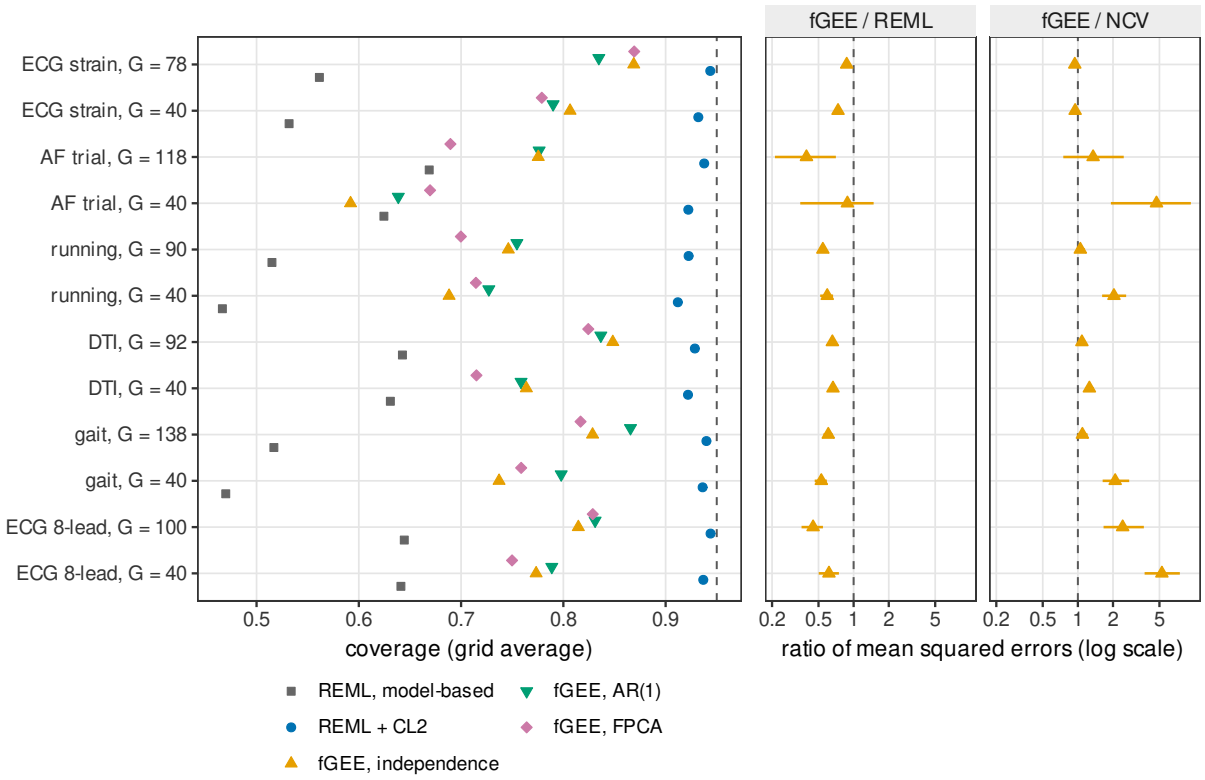


Figure 21: Plasmode datasets (midpoint truth, REML residuals), β(s,t), per dataset and number of subjects G. Left: grid-averaged coverage of nominal 95% pointwise intervals (REML model-based and REML + CL2; fGEE Wald intervals from its covariance under the independence, AR(1) and FPCA working correlations; dashed: 0.95). Right: paired ratios of mean squared errors with 95% replicate-bootstrap intervals, fGEE with independence working correlation (log scale; dashed: equal error). Values: S7.

Over the 12 combinations of dataset and number of subjects G, REML + CL2 covers $\beta(s,t)$ at 0.91–0.94, the mean at 0.92–0.95, $\alpha(t)$ at 0.89–0.95 and $\gamma(t)$ at 0.92–0.96; model-based REML intervals cover β at 0.47–0.67 (Figure 21). The fastFGEE 0.2.2 Wald intervals cover β at 0.59–0.87 with SE/SD 0.43–0.87 (REML + CL2: 0.80–1.05), the mean at 0.74–0.93, α at 0.64–0.93 and γ at 0.49–0.92, the lowest values for the randomized treatment effect in the AF trial (0.49–0.67). With calibrated critical values, the intervals for α and γ cover 0.80–1.00.

Table 38: Plasmode datasets (midpoint truth, REML residuals), β(s,t) and the mean: grid-averaged coverage (cov.) of nominal 95% pointwise intervals and 5% quantile of pointwise coverage (q05) for REML + CL2 with the normal critical value (CL2) and for fGEE (independence working correlation) with its calibrated critical value with its t adjustment (t) and without it (no t); width and interval score of the fGEE intervals relative to REML + CL2.

	Dataset	G	CL2 cov.	CL2 q05	t cov.	t q05	t width	t score	no t cov.	no t q05	no t width	no t score
β(s,t)	ECG strain	78	0.94	0.90	0.93	0.82	0.86	0.97	0.91	0.80	0.79	0.99
		40	0.93	0.87	1.00	1.00	4.68	3.51	0.89	0.77	0.72	0.92
	AF trial	118	0.94	0.88	0.93	0.75	0.73	0.79	0.93	0.74	0.72	0.79
		40	0.92	0.84	0.95	0.79	1.70	1.23	0.92	0.73	0.77	0.78
	running	90	0.92	0.89	0.93	0.80	0.99	0.99	0.92	0.77	0.95	1.00
		40	0.91	0.87	0.99	0.96	5.52	3.88	0.89	0.77	0.92	1.09
	DTI	92	0.93	0.89	0.94	0.85	0.91	0.87	0.94	0.84	0.89	0.87
		40	0.92	0.88	0.96	0.88	2.04	1.62	0.92	0.80	0.86	0.91
	gait	138	0.94	0.91	0.94	0.84	0.83	0.90	0.94	0.83	0.82	0.91
		40	0.94	0.90	1.00	0.99	5.27	4.10	0.94	0.84	0.83	0.93
	ECG 8-lead	100	0.94	0.88	0.98	0.94	0.99	0.80	0.98	0.93	0.97	0.80
		40	0.94	0.85	0.99	0.97	4.98	3.78	0.98	0.94	1.09	0.87
mean (link)	ECG strain	78	0.94	0.88	0.96	0.89	1.42	1.24	0.91	0.80	0.86	1.01
		40	0.93	0.86	1.00	1.00	5.12	3.82	0.89	0.78	0.79	0.98
	AF trial	118	0.95	0.90	0.93	0.82	0.88	1.00	0.92	0.82	0.87	1.00
		40	0.93	0.88	0.96	0.82	2.23	1.80	0.91	0.75	0.84	0.96
	running	90	0.93	0.89	0.93	0.85	1.00	1.04	0.91	0.83	0.91	1.06
		40	0.92	0.86	1.00	1.00	5.31	3.93	0.87	0.78	0.82	1.09
	DTI	92	0.93	0.89	0.93	0.86	0.94	0.95	0.92	0.85	0.91	0.96
		40	0.92	0.87	0.99	0.96	4.71	3.48	0.90	0.81	0.84	0.97
	gait	138	0.94	0.91	0.93	0.88	0.94	0.99	0.93	0.87	0.92	0.99
		40	0.94	0.90	1.00	1.00	5.45	4.28	0.91	0.84	0.84	0.97
	ECG 8-lead	100	0.95	0.91	1.00	0.98	4.74	3.85	0.96	0.91	0.89	0.84
		40	0.94	0.88	1.00	1.00	5.90	4.47	0.95	0.90	0.91	0.83

Table 39: The plasmode datasets of the previous table against REML + CL2 with the Satterthwaite critical value: its grid-averaged coverage (cov.) and 5% quantile of pointwise coverage (q05) for β(s,t), its coverage for the mean, and the width of the calibrated intervals for β(s,t) with and without the t adjustment as a multiple of its width.

Dataset	G	cov. beta	q05 beta	cov. mean	t width	no t width
ECG strain	78	0.96	0.92	0.95	0.81	0.74
	40	0.96	0.91	0.96	4.16	0.64
AF trial	118	0.95	0.89	0.95	0.70	0.69
	40	0.95	0.88	0.96	1.51	0.69
running	90	0.94	0.90	0.95	0.94	0.91
	40	0.94	0.91	0.95	4.93	0.82
DTI	92	0.94	0.90	0.94	0.87	0.85
	40	0.95	0.91	0.94	1.84	0.78
gait	138	0.95	0.91	0.95	0.81	0.80
	40	0.96	0.93	0.96	4.76	0.75
ECG 8-lead	100	0.96	0.91	0.96	0.93	0.91
	40	0.96	0.91	0.96	4.33	0.94

Plasmode errors are residual curves with independent random signs, the same distribution as the multipliers of the wild bootstrap, so this design resembles the bootstrap's own resampling scheme. With all subjects and independence working correlation, the calibrated intervals cover β at 0.93–0.98 (5% quantile of pointwise coverage 0.75–0.94; REML + CL2 with the normal critical value 0.92–0.94 and 0.88–0.91), at 0.73–0.99 times the width and 0.79–0.99 times the interval score of REML + CL2 (Table 38).

With 40 subjects, the t adjustment sits at the floor on 8–100% of the replicates for β and 24–100% for the mean. The intervals for β then cover 0.95–1.00 at 1.70–5.52 times REML + CL2's width, and, without the adjustment, 0.89–0.98 at 0.72–1.09 times that width (0.83–0.98 over the three working correlations).

For the mean, the intervals without the adjustment cover 0.91–0.96 with all subjects and 0.87–0.95 with 40, at 0.79–0.92 times REML + CL2's width. With the adjustment, they are up to 5.90 times as wide with 40 subjects and up to 4.74 times with all subjects: the floor also applies to the 100 subjects of the ECG 8-lead data, whose bases have more coefficients (76% of the replicates).

For β, the ratio fGEE/REML of mean squared errors is 0.39–0.88, below one in 12 of 12 combinations, with the interval below one in 11. Against curve-blocked NCV, fGEE is worse in 7 combinations, better in 2 (ECG strain) and not resolved in the remaining 3; the best working correlation per combination, chosen after the fact, gives a ratio fGEE/NCV of 0.89–4.06. For γ, the ratio fGEE/NCV is 0.59–1.10, with the interval below one in 7 combinations (gait, ECG 8-lead, ECG strain, and AF trial) and above one in 2; for α, it is 0.72–1.14.

S4 Intervals centered at the NCV fit and smoothing-parameter terms

This section collects approaches that did not work: intervals centered at the curve-blocked NCV estimate, a corrected version of mgcv's neighborhood covariance for NCV fits, and a smoothing-parameter uncertainty term for REML + CL2. Each approach is given as method, derivation, result and reason. The notation is that of §2 and Appendix A of the paper: $\mathbf{W}$ holds the working weights including $1/\phi$, $\mathbf{S} = \sum_k (\lambda_k/\phi)\mathbf{S}_k$, $\mathbf{V}_p = (\mathbf{X}^\top\mathbf{W}\mathbf{X} + \mathbf{S})^{-1}$ and $\mathbf{V}_e = \mathbf{V}_p\mathbf{X}^\top\mathbf{W}\mathbf{X}\mathbf{V}_p$. In addition, $\mathbf{F} = \mathbf{X}^\top\mathbf{W}\mathbf{X} + \mathbf{S} = \mathbf{V}_p^{-1}$ is the penalized Hessian, $\rho = \log\lambda$ and $\dot{\mathbf{S}}_k = \partial\mathbf{S}/\partial\rho_k = (\lambda_k/\phi)\mathbf{S}_k$. For a canonical link, the working weights are $w_{ij} = (d\mu/d\eta)_{ij}/\phi$, and the score of curve i is $\mathbf{u}_i = \mathbf{X}_i^\top(\mathbf{y}_i - \mu_i)/\phi$. The CL2 influence of curve i is $\mathbf{k}_i = \mathbf{V}_p\mathbf{U}_i$ with $\mathbf{U}_i$ as in Appendix A, so that $\hat{\mathbf{V}}_{\mathrm{CL2}} = \mathbf{V}_f + (\mathbf{V}_p - \mathbf{V}_e)$ with the frequentist form $\mathbf{V}_f = \frac{G}{G-1}\sum_i \mathbf{k}_i\mathbf{k}_i^\top$. $\mathbf{L}$ maps the coefficients to an estimand's values on its grid.

S4.1 Intervals centered at the NCV fit

Under dependent errors, curve-blocked NCV estimates $\beta(s,t)$ with a fraction of REML's error (S1.6), but NCV + CL2 undercovers. This section gives diagnostics for why it undercovers and evaluates six constructions that try to repair it. None of the constructions as implemented here gives intervals around the NCV estimate that hold their level across the plasmode datasets; the conclusions refer to these constructions, not to every possible correction.

(a) Why NCV + CL2 undercovers

Method. We use two diagnostics. *Bias-eliminated coverage* replaces each interval's center by the estimate minus its pointwise bias (estimated from the replicates) and shows the coverage without bias. A refit with every smoothing parameter fixed at the median NCV value of its plasmode combination removes the variability of the selected smoothing parameters. It also moves every replicate to the typical amount of smoothing, so it changes the bias as well as the variability; it uses the replicates themselves and is a diagnostic, not a decomposition of the shortfall and not a method.

Result. In the synthetic settings with dependent errors at $G = 100$, NCV + CL2 covers β at 0.93 (smooth truth) and 0.83 (rough truth); without bias, it would cover 0.95 and 0.94. On the six plasmode datasets with independent source curves (observed curves independent across subjects), the ratio of the root mean square standard error to the replicate SD of the estimate (SE/SD, median over grid points) is 0.75–0.98 for NCV + CL2; with fixed smoothing parameters, it is 1.03–1.15, but coverage stays at 0.69–0.92.

Reason. On synthetic data, the shortfall is mainly smoothing bias at the amount of smoothing NCV selects. On the plasmode data, the diagnostics suggest two contributions: the standard error is short, consistent with the variability of the selected smoothing parameters that it ignores, and coverage stays short at a fixed typical amount

of smoothing, consistent with smoothing bias. Fixing the smoothing parameters changes both the center and its sampling distribution, so the two cannot be separated exactly.

Table 40: Plasmode, $\beta(s,t)$, all subjects: coverage (5% quantile of pointwise coverage) and SE/SD for NCV + CL2, its bias-eliminated coverage, NCV + CL2 after a refit at the median NCV smoothing parameters, and REML + CL2 (100 replicates per combination).

dataset	NCV + CL2	bias removed	NCV, λ fixed	REML + CL2
ECG strain	0.91 (0.80) 0.98	0.93	0.92 (0.85) 1.03	0.94 (0.89) 1.01
AF trial	0.81 (0.52) 0.84	0.87	0.87 (0.57) 1.07	0.94 (0.86) 0.96
running	0.71 (0.40) 0.75	0.82	0.69 (0.29) 1.04	0.91 (0.84) 0.90
DTI	0.85 (0.58) 0.95	0.92	0.86 (0.57) 1.07	0.93 (0.86) 0.99
gait	0.81 (0.53) 0.90	0.88	0.88 (0.62) 1.07	0.94 (0.89) 1.00
ECG 8-lead	0.76 (0.48) 0.79	0.85	0.85 (0.46) 1.15	0.94 (0.88) 0.95
ocean	0.71 (0.29) 0.89	0.88	0.71 (0.24) 1.03	0.93 (0.87) 0.93
weather	0.59 (0.39) 0.68	0.70	0.69 (0.29) 1.14	0.88 (0.82) 0.93
electricity	0.62 (0.40) 0.69	0.76	0.64 (0.30) 1.06	0.93 (0.88) 0.89

(b) Widening by the NCV–REML difference

Method. Widen the NCV + CL2 interval in quadrature by the difference of the two estimates, $\mathbf{L}\hat{\theta}_N \pm z\{\mathrm{se}^2_{\mathrm{CL2},N} + (\mathbf{L}\hat{\theta}_N - \mathbf{L}\hat{\theta}_R)^2\}^{1/2}$, with the difference standing in for the bias of the NCV estimate. If the bias b were known, an interval $\pm z(\sigma^2 + b^2)^{1/2}$ would never undercover at the 95% level:

with $r = b/\sigma$, its coverage is $c(r) = \Phi\{z(1+r^2)^{1/2} - r\} - \Phi\{-z(1+r^2)^{1/2} - r\}$, and $c(r) - c(0) = z(z^2-3)\varphi(z)r^4/6 + O(r^6)$. Numerically, $c(r) \geq c(0)$ for every r exactly when $z \geq \sqrt{3}$, that is for nominal levels of at least 0.917.

Derivation. The interval covers when $|Z + r| \leq z(1 + r^2)^{1/2}$ with $Z \sim N(0,1)$. Write $u = z(1 + r^2)^{1/2} = z + zr^2/2 - zr^4/8 + O(r^6)$. Since $\Phi(-u - r) = 1 - \Phi(u + r)$,

$$c(r) = \Phi(u - r) + \Phi(u + r) - 1 = 2\Phi(u) - 1 + \varphi'(u)\, r^2 + \varphi'''(u)\, r^4/12 + O(r^6),$$

an even function of r. With $\varphi'(z) = -z\varphi(z)$, $\varphi''(z) = (z^2 - 1)\varphi(z)$ and $\varphi'''(z) = (3z - z^3)\varphi(z)$, expanding u around z gives $2\Phi(u) = 2\Phi(z) + z\varphi(z)r^2 - (z + z^3)\varphi(z)r^4/4 + O(r^6)$ and $\varphi'(u)r^2 = -z\varphi(z)r^2 + z(z^2 - 1)\varphi(z)r^4/2 + O(r^6)$. The terms in r^2 cancel, and the terms in r^4 add up to $\varphi(z)\{-(z+z^3)/4 + z(z^2-1)/2 + (3z - z^3)/12\} = z(z^2-3)\varphi(z)/6$. For small biases, the widened interval therefore overcovers if $z > \sqrt{3}$ and undercovers if $z < \sqrt{3}$; the global statement for all r is the numerical check above.

The plug-in replaces b by the estimated difference. For a contrast $\mathbf{a}$, let $\hat{\delta} = \mathbf{a}^\top(\hat{\theta}_N - \hat{\theta}_R)$, with biases b_N, b_R and variances σ^2_N, σ^2_R, covariance σ_{NR} of the two contrasts. Then

$$\mathrm{E}\hat{\delta}^2 = (b_N - b_R)^2 + \sigma^2_N + \sigma^2_R - 2\sigma_{NR},$$

so $\mathrm{se}^2_{\mathrm{CL2},N} + \hat{\delta}^2$ estimates $\sigma^2_N + b^2_N$ only if $b_R \approx 0$ and $\sigma^2_R \approx 2\sigma_{NR}$. Under within-curve dependence, σ^2_R is large (REML undersmooths), which inflates the added term $\hat{\delta}^2$, while a bias shared by both fits cancels in $b_N - b_R$.

Result. In the synthetic settings with dependent errors and a smooth truth, the widened interval overcovers β (0.98); under the REML-fitted plasmode truth, it covers 0.87 with a lower tail of 0.74 (next table and figure).

Reason. The estimated difference is not the bias. It carries REML's sampling variance, which is large under dependence, so the interval is too wide where REML is noisy; and it cannot see bias that both fits share, which concentrates where the truth is most curved.

(c) NCV estimate with the REML + CL2 width

Method. Center REML + CL2's pointwise half-width at the NCV estimate.

Result. The detection rate of this interval for β is lower than REML + CL2's on 9 of 9 plasmode datasets (0.05–0.51 against 0.22–0.62). Under the REML-fitted plasmode truth, it covers 0.91 with a lower tail of 0.67 (REML + CL2: 0.92, 0.87). Elsewhere it overcovers: 1.00 and 0.99 in the synthetic settings with dependent errors (smooth and rough truth), and 0.97 and 0.96 under the NCV-fitted and the midpoint plasmode truth (next table).

Reason. Its width equals REML + CL2's by construction, so a lower interval score only says that the NCV estimate is closer to the truth, and the wide band around a smoother center excludes zero less often. Its coverage relies on REML's standard error absorbing NCV's bias: the interval is then wider than needed where the NCV estimate is accurate, and it covers unevenly under the REML-fitted truth, the truth least favorable to the NCV fit.

Table 41: $\beta(s,t)$: coverage (5% quantile of pointwise coverage) and width / interval score relative to REML + CL2 (geometric means of per-setting ratios). Synthetic: main factorial at $G = 100$ and rough-truth settings with smooth errors; plasmode: all subjects, fitted residuals, by truth source. NCV–REML widening: NCV + CL2 widened in quadrature by the NCV–REML difference.

setting	combinations	REML + CL2	NCV + CL2	NCV–REML widening	NCV estimate, REML + CL2 width
dependent, rough truth	7	0.943 (0.92) 1.00 / 1.00	0.828 (0.54) 0.47 / 0.78	0.938 (0.83) 0.86 / 0.82	0.988 (0.96) 1.00 / 0.83
dependent, smooth truth	14	0.945 (0.92) 1.00 / 1.00	0.927 (0.85) 0.40 / 0.45	0.981 (0.94) 0.82 / 0.69	0.997 (0.99) 1.00 / 0.81
independent	7	0.985 (0.96) 1.00 / 1.00	0.981 (0.94) 0.77 / 0.79	0.988 (0.95) 0.85 / 0.84	0.991 (0.97) 1.00 / 1.00
plasmode, NCV-fitted truth	18	0.925 (0.88) 1.00 / 1.00	0.828 (0.66) 0.37 / 0.68	0.938 (0.86) 0.91 / 0.83	0.970 (0.92) 1.00 / 0.82
plasmode, REML-fitted truth	18	0.924 (0.87) 1.00 / 1.00	0.659 (0.36) 0.51 / 2.17	0.873 (0.74) 1.02 / 1.23	0.906 (0.67) 1.00 / 1.04
plasmode, midpoint truth	18	0.932 (0.89) 1.00 / 1.00	0.765 (0.45) 0.44 / 1.09	0.918 (0.80) 0.91 / 0.93	0.962 (0.88) 1.00 / 0.86

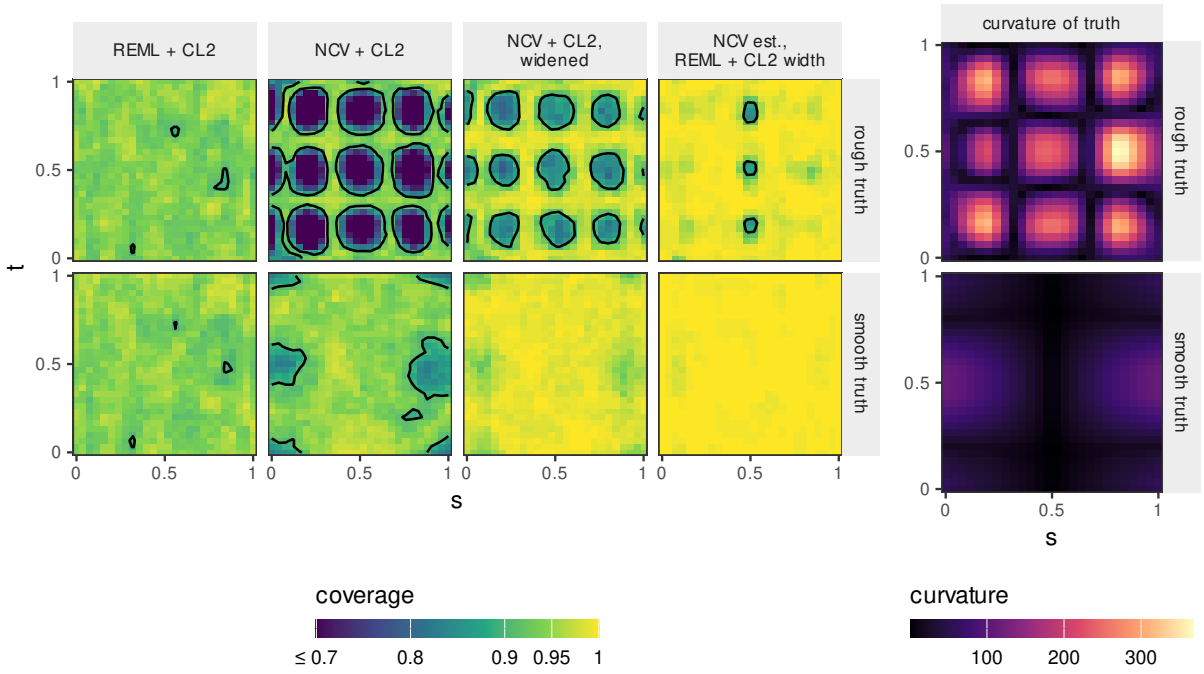


Figure 22: Pointwise coverage of $\beta(s,t)$ for REML + CL2 and three intervals centered at the NCV estimate, Gaussian, default signal, $G = 100$, smooth errors, rough truth (top) and smooth truth (bottom); black contour at 0.90; the color scale ends at 0.7 and values at or below it share its darkest color. Widened: NCV + CL2 widened by the NCV–REML difference; NCV est.: NCV estimate with REML + CL2's half-width. Right: curvature of the truth.

(d) One-step bias correction

Method. For a penalized fit at fixed λ and Gaussian responses, $\mathrm{E}\hat{\theta} = \mathbf{A}\theta$ with $\mathbf{A} = \mathbf{V}_e\mathbf{V}_p^{-1} = \mathbf{V}_p\mathbf{X}^\top\mathbf{W}\mathbf{X}$. Directions in which NCV shrinks the estimate almost completely ($\mathbf{A} \approx \mathbf{0}$) carry no information from the data, so no correction of this kind can restore them. The corrected estimate $\tilde{\theta} = (2\mathbf{I} - \mathbf{A})\hat{\theta}$ ("twicing", Tukey (1977)) has bias $-(\mathbf{I} - \mathbf{A})^2\theta$ instead of $-(\mathbf{I} - \mathbf{A})\theta$. Its covariance propagates the frequentist CL2 covariance $\mathbf{V}_f$ through the same map and keeps the Bayesian smoothing-bias term, $\mathbf{M}\mathbf{V}_f\mathbf{M}^\top + (\mathbf{I} - \mathbf{A})(\mathbf{V}_p - \mathbf{V}_e)(\mathbf{I} - \mathbf{A})^\top$ with $\mathbf{M} = 2\mathbf{I} - \mathbf{A}$, as in robust bias correction (Calonico, Cattaneo, and Farrell 2018). It needs no refit.

Derivation: mean and bias. For Gaussian responses with identity link and fixed λ, $\hat{\theta} = \mathbf{F}^{-1}\mathbf{X}^\top\mathbf{W}\mathbf{y}$, so $\mathrm{E}\hat{\theta} = \mathbf{F}^{-1}\mathbf{X}^\top\mathbf{W}\mathbf{X}\theta = \mathbf{A}\theta$ with

$$\mathbf{A} = \mathbf{V}_p\mathbf{X}^\top\mathbf{W}\mathbf{X} = (\mathbf{V}_p\mathbf{X}^\top\mathbf{W}\mathbf{X}\mathbf{V}_p)\mathbf{V}_p^{-1} = \mathbf{V}_e\mathbf{V}_p^{-1}, \qquad \mathbf{I} - \mathbf{A} = \mathbf{V}_p(\mathbf{F} - \mathbf{X}^\top\mathbf{W}\mathbf{X}) = \mathbf{V}_p\mathbf{S},$$

and the bias is $-(\mathbf{I} - \mathbf{A})\theta = -\mathbf{V}_p\mathbf{S}\theta$ (Appendix A.3 of the paper). Twicing applies the estimated bias map to the estimate, $\tilde{\theta} = \hat{\theta} + (\mathbf{I} - \mathbf{A})\hat{\theta} = (2\mathbf{I} - \mathbf{A})\hat{\theta}$, with $\mathrm{E}\tilde{\theta} = (2\mathbf{I} - \mathbf{A})\mathbf{A}\theta = \{\mathbf{I} - (\mathbf{I} - \mathbf{A})^2\}\theta$. Repeating the step m times gives $\mathbf{M}_m = \sum_{\ell=0}^{m}(\mathbf{I}-\mathbf{A})^\ell$ and $\mathrm{E}\,\mathbf{M}_m\hat{\theta} = \{\mathbf{I}-(\mathbf{I}-\mathbf{A})^{m+1}\}\theta$; two steps give $\mathbf{M}_2 = 3\mathbf{I}-3\mathbf{A}+\mathbf{A}^2$ with bias $-(\mathbf{I}-\mathbf{A})^3\theta$. If $\mathbf{X}^\top\mathbf{W}\mathbf{X}$ is invertible, $\mathbf{M}_m \to \mathbf{A}^{-1}$ as $m \to \infty$, and $\mathbf{A}^{-1}\hat{\theta}$ is the unpenalized estimate. For GLMs, the same algebra holds for the working linear model at the converged weights, to first order; $\mathbf{A}$ is computed as $\mathbf{V}_e\mathbf{V}_p^{-1}$ from the fit.

Derivation: covariance. At fixed λ, $\mathbf{M}$ is fixed, so $\mathrm{Var}(\tilde{\theta}) = \mathbf{M}\,\mathrm{Var}(\hat{\theta})\mathbf{M}^\top$, estimated by $\mathbf{M}\mathbf{V}_f\mathbf{M}^\top$ with the frequentist CL2. The smoothing-bias allowance follows from the Bayesian reading of Appendix A.3: under $\theta \sim N(\mathbf{0}, \mathbf{S}^-)$, the bias $-(\mathbf{I} - \mathbf{A})\theta$ of $\hat{\theta}$ has covariance $\mathbf{V}_p\mathbf{S}\mathbf{S}^-\mathbf{S}\mathbf{V}_p = \mathbf{V}_p\mathbf{S}\mathbf{V}_p = \mathbf{V}_p - \mathbf{V}_e$, which is CL2's allowance. The bias of $\tilde{\theta}$ is

$-(\mathbf{I}-\mathbf{A})\{(\mathbf{I}-\mathbf{A})\theta\}$, with prior covariance $(\mathbf{I}-\mathbf{A})(\mathbf{V}_p-\mathbf{V}_e)(\mathbf{I}-\mathbf{A})^\top$; after two steps, it is $(\mathbf{I}-\mathbf{A})^2(\mathbf{V}_p-\mathbf{V}_e)\{(\mathbf{I}-\mathbf{A})^2\}^\top$. The interval uses

$$\mathbf{M}\mathbf{V}_f\mathbf{M}^\top+(\mathbf{I}-\mathbf{A})(\mathbf{V}_p-\mathbf{V}_e)(\mathbf{I}-\mathbf{A})^\top,$$

not $\mathbf{M}\hat{\mathbf{V}}_{\mathrm{CL2}}\mathbf{M}^\top$, which propagates the allowance through the correction as if it were sampling variance.

Derivation: directions. Let $\mathbf{F}^{-1/2}\mathbf{X}^\top\mathbf{W}\mathbf{X}\mathbf{F}^{-1/2}=\mathbf{Q}\Lambda\mathbf{Q}^\top$ with $\Lambda=\mathrm{diag}(a_k)$, $0\le a_k\le 1$ ($a_k=1$ in the null space of the penalty), and $\mathbf{B}=\mathbf{F}^{-1/2}\mathbf{Q}$. Then $\mathbf{A}\mathbf{B}=\mathbf{B}\Lambda$, and

$$\mathbf{V}_e=\mathbf{B}\Lambda\mathbf{B}^\top,\quad \mathbf{V}_p-\mathbf{V}_e=\mathbf{B}(\mathbf{I}-\Lambda)\mathbf{B}^\top,\quad (\mathbf{I}-\mathbf{A})(\mathbf{V}_p-\mathbf{V}_e)(\mathbf{I}-\mathbf{A})^\top=\mathbf{B}(\mathbf{I}-\Lambda)^3\mathbf{B}^\top,\quad \mathbf{M}(\mathbf{V}_p-\mathbf{V}_e)\mathbf{M}^\top=\mathbf{B}(2\mathbf{I}-\Lambda)^2(\mathbf{I}-\Lambda)\mathbf{B}^\top.$$

In the coordinates $\vartheta=\mathbf{B}^{-1}\theta$, component k of the estimate has mean $a_k\vartheta_k$, working-model variance a_k and bias $-(1-a_k)\vartheta_k$; after twicing, the mean is $(2-a_k)a_k\vartheta_k$, the variance $(2-a_k)^2a_k$ and the bias $-(1-a_k)^2\vartheta_k$. The correctly propagated allowance is $(1-a_k)^3$, smaller than the original $1-a_k$; propagating the Bayesian CL2 through $\mathbf{M}$ gives $(2-a_k)^2(1-a_k)$, up to four times the original. In directions with $a_k\approx 0$, which NCV shrinks almost completely, the estimate carries almost no information about ϑ_k: any linear correction $p(\mathbf{A})\hat{\theta}$ has mean $p(a_k)a_k\vartheta_k$ and variance $p(a_k)^2a_k$, so recovering a fraction κ of ϑ_k needs $p(a_k)=\kappa/a_k$ and a variance κ^2/a_k that diverges as $a_k\to 0$. For $\mathbf{M}_m$, $p(a)=\{1-(1-a)^{m+1}\}/a\le m+1$, and the bias $-(1-a_k)^{m+1}\vartheta_k$ stays close to $-\vartheta_k$.

Result. On 200 fresh synthetic replicates, the corrected interval covers β at 0.93 (smooth truth, Gaussian and Poisson) and 0.90 (rough truth; NCV + CL2 0.85, REML + CL2 0.94), and binary β at 0.87. On the plasmode datasets, it adds 1.1–7.0 pp to the coverage of NCV + CL2 (figure in (f)).

Reason. One step removes only part of the bias, none of it in the directions NCV shrinks completely, and leaves the smoothing-parameter variability of (a) untouched.

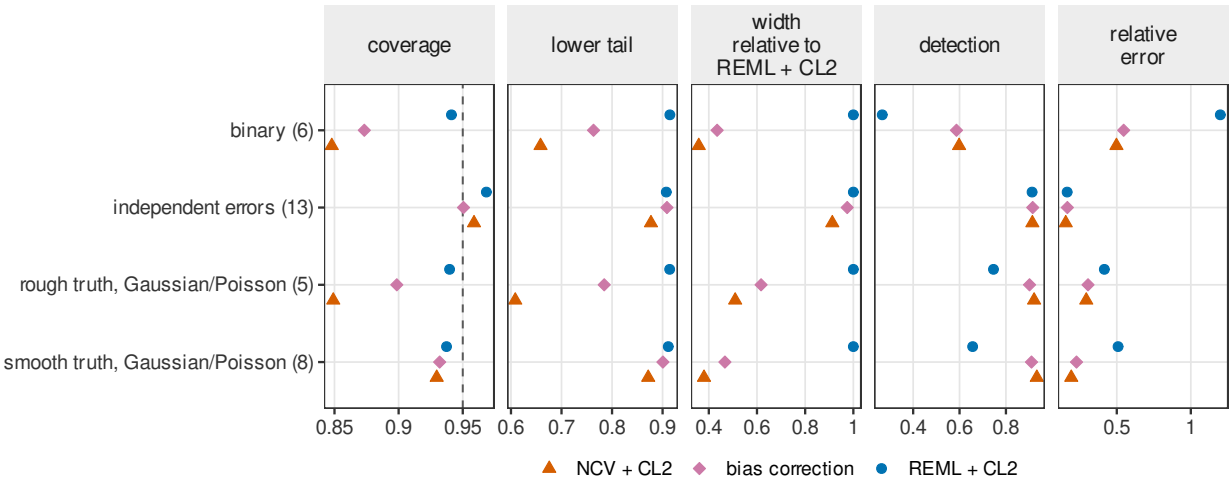


Figure 23: One-step bias correction, $\beta(s,t)$, 200 fresh replicates per setting: coverage, lower tail (5% quantile of pointwise coverage), width relative to REML + CL2 (geometric mean), detection rate and median relative error of the center, by group of settings (their number in brackets). Dashed: nominal 0.95 for coverage. Values: S7.

(e) Smoothing-parameter uncertainty term

Method. The NCV criterion with curve blocks has analytic derivatives with respect to a weight w_i on each curve. With the curve scores $\mathbf{s}_i=\partial^2\mathrm{NCV}/\partial\rho\,\partial w_i$ ($\rho=\log\lambda$) and the Hessian $\mathbf{H}$ of the criterion in ρ, the infinitesimal jackknife over curves gives the influence $\mathbf{d}_i=-\mathbf{H}^{-1}(\mathbf{s}_i-\bar{\mathbf{s}})$ of curve i on $\hat{\rho}$ and the curve-robust variance $\hat{\mathbf{V}}_\rho=$

$\sum_i \mathbf{d}_i\mathbf{d}_i^\top$. It enters the coefficient covariance as $\mathbf{J}\hat{\mathbf{V}}_\rho\mathbf{J}^\top$ with $\mathbf{J} = \partial\hat\theta/\partial\rho$ ("λ term"), optionally with the first-order cross term between the coefficient and the smoothing-parameter influences.

Derivation: the first-order term. Let $\hat\theta(\rho)$ be the penalized fit at fixed ρ and $\hat\rho$ the selected value with limit ρ_0. Implicit differentiation of the estimating equation $\sum_i \mathbf{u}_i(\hat\theta) = \mathrm{S}\hat\theta$ (Appendix A.1 of the paper) in ρ_k gives $-\mathbf{F}\,\partial\hat\theta/\partial\rho_k - \dot{\mathrm{S}}_k\hat\theta = \mathbf{0}$, so the columns of $\mathbf{J} = \partial\hat\theta/\partial\rho^\top$ are $\mathbf{J}_k = -\mathbf{V}_p\dot{\mathrm{S}}_k\hat\theta$ (mgcv's `db.drho`). To first order,

$$\hat\theta(\hat\rho) - \hat\theta(\rho_0) \approx \mathbf{J}(\hat\rho - \rho_0), \qquad \operatorname{Var}\hat\theta(\hat\rho) \approx \operatorname{Var}\hat\theta(\rho_0) + \mathbf{J}\operatorname{Var}(\hat\rho)\mathbf{J}^\top + \mathbf{C} + \mathbf{C}^\top,$$

with the cross covariance $\mathbf{C} = \operatorname{Cov}\{\hat\theta(\rho_0), \hat\rho\}\mathbf{J}^\top$. The terms left out (the dependence of $\mathbf{V}_p$ and $\mathbf{J}$ on ρ, the curvature of $\hat\theta(\rho)$) are of order G^{-1} relative to the first-order term when $\hat\rho$ is $\sqrt{G}$-regular. For the surface penalty under dependence, it is not: the sampling distribution of $\log\hat\lambda$ is flat or bimodal at $G = 40$, and its SD shrinks by a factor of only 0.84 from $G = 40$ to $G = 100$ (REML; $\sqrt{40/100} = 0.63$).

Derivation: mgcv's $\mathbf{V}_c$. Wood, Pya, and Säfken (2016) argue in the Bayesian model: given ρ, $\theta \mid \mathbf{y}, \rho \sim N\{\hat\theta(\rho), \mathbf{V}_p(\rho)\}$, and, with a flat prior, the Laplace approximation gives $\rho \mid \mathbf{y} \approx N(\hat\rho, \mathbf{H}^{-1})$, with $\mathbf{H}$ the Hessian of the negative log restricted likelihood in the outer parameters (block $\rho\rho$ of its inverse, if the scale is estimated). The law of total variance and the linearization $\hat\theta(\rho) \approx \hat\theta(\hat\rho) + \mathbf{J}(\rho - \hat\rho)$ give $\mathbf{V}_c = \mathbf{V}_p + \mathbf{J}\mathbf{H}^{-1}\mathbf{J}^\top$, to which mgcv adds a second-order term. Two working assumptions are built in. First, $\mathbf{H}^{-1}$ is the variance of $\hat\rho$ only if the information identity holds, i.e. if the variance of the gradient of the criterion equals its expected Hessian, which is the case when the N rows are independent with variance $\phi V(\mu)$. For Gaussian responses, the data enter the ρ_k-gradient through $\frac{1}{2}\hat\theta^\top\dot{\mathrm{S}}_k\hat\theta = \frac{1}{2}\mathbf{y}^\top\mathbf{Q}_k\mathbf{y}$ with $\mathbf{Q}_k = \mathbf{W}\mathbf{X}\mathbf{V}_p\dot{\mathrm{S}}_k\mathbf{V}_p\mathbf{X}^\top\mathbf{W}$; for $\mathrm{E}\mathbf{y} = \mathbf{m}$ and $\operatorname{Var}\mathbf{y} = \Sigma$, its variance is $\frac{1}{2}\operatorname{tr}(\mathbf{Q}_k\Sigma\mathbf{Q}_k\Sigma) + \mathbf{m}^\top\mathbf{Q}_k\Sigma\mathbf{Q}_k\mathbf{m}$, whereas $\mathbf{H}$ corresponds to $\Sigma = \phi\mathbf{I}$. In the main-factorial settings with dependent errors, mgcv's $\mathbf{H}^{-1}$ gives standard deviations of the surface penalty's $\log\hat\lambda$ of 0.26–0.52 times its Monte Carlo SD (median over replicates; REML). Second, the law of total variance has no cross term: in the Bayesian model, θ given ρ is the posterior. Over repeated samples with a fixed truth, $\hat\theta$ and the gradient are correlated: for symmetric errors $\operatorname{Cov}(\mathbf{y}^\top\mathbf{Q}_k\mathbf{y}, \mathbf{a}^\top\mathbf{y}) = 2\mathbf{m}^\top\mathbf{Q}_k\Sigma\mathbf{a}$, so $\operatorname{Cov}(\hat\theta^\top\dot{\mathrm{S}}_k\hat\theta, \hat\theta) = 2\mathbf{V}_p\mathbf{X}^\top\mathbf{W}\Sigma\mathbf{W}\mathbf{X}\mathbf{V}_p\dot{\mathrm{S}}_k\,\mathrm{E}\hat\theta$, which vanishes only if $\dot{\mathrm{S}}_k\mathrm{E}\hat\theta = \mathbf{0}$.

Derivation: curve contributions. Give curve i a weight c_i in the selection criterion $\mathscr{V}(\omega; \mathbf{c})$ (in its log-likelihood for REML; in its loss term and in every fit for NCV), so that $\mathbf{c} = \mathbf{1}$ is the data and $c_i = 0$ deletes the curve; ω collects the outer parameters (ρ, and $\tau = \log\phi$ for a Gaussian REML fit). The selected value solves $\nabla_\omega\mathscr{V}(\hat\omega; \mathbf{c}) = \mathbf{0}$, and the implicit-function theorem gives

$$\left.\frac{d\hat\omega}{dc_i}\right|_{\mathbf{c}=\mathbf{1}} = -\mathbf{H}_\omega^{-1}\mathbf{s}_i, \qquad \mathbf{s}_i = \left.\frac{\partial\nabla_\omega\mathscr{V}(\hat\omega; \mathbf{c})}{\partial c_i}\right|_{\mathbf{c}=\mathbf{1}}, \qquad \mathbf{H}_\omega = \nabla^2_\omega\mathscr{V},$$

where $\mathbf{s}_i$ is a total derivative that includes the dependence of $\hat\theta$, $\mathbf{W}$ and the leave-out fits on $\mathbf{c}$. Neither criterion is a sum of curve terms (log determinants; leave-out fits that use all other curves), so the decomposition $\hat\omega - \omega_0 \approx \sum_i \mathbf{d}_i$ holds only to first order: it is the influence-function expansion of $\hat\omega$ in the curves. The penalty does not scale with $\mathbf{c}$, so $\sum_i \mathbf{s}_i \neq \mathbf{0}$. Perturbing the empirical distribution of the curves at fixed total weight, $\mathbf{c} = \mathbf{1} + \varepsilon(\mathbf{e}_i - \mathbf{1}/G)$, gives the curve influence $\mathbf{d}_i = -\mathbf{H}_\omega^{-1}(\mathbf{s}_i - \bar{\mathbf{s}})$.

Derivation: estimators of $\operatorname{Var}(\hat\omega)$. (i) The infinitesimal jackknife (IJ),

$$\hat{\mathbf{V}}_\omega = \frac{G}{G-1}\sum_i \mathbf{d}_i\mathbf{d}_i^\top = \mathbf{H}_\omega^{-1}\hat{\mathbf{M}}\mathbf{H}_\omega^{-1}, \qquad \hat{\mathbf{M}} = \frac{G}{G-1}\sum_i (\mathbf{s}_i - \bar{\mathbf{s}})(\mathbf{s}_i - \bar{\mathbf{s}})^\top,$$

a sandwich for $\hat\omega$ with the curves as clusters. It differs from mgcv's $\mathbf{H}^{-1}$ in the meat and agrees with it when $\hat{\mathbf{M}} = \mathbf{H}_\omega$. (ii) The one-step leave-curve-out (OS) estimator: at $\hat\rho$, $\hat\theta_{(-i)} = \hat\theta - (\mathbf{F} - \tilde{\mathbf{X}}_i^\top\tilde{\mathbf{X}}_i)^{-1}\mathbf{u}_i$ (exact for Gaussian responses; for GLMs, the start of penalized IRLS at fixed $\hat\rho$, iterated to convergence), then one Newton step in the outer parameters, $\Delta_i = -\mathbf{H}_\omega^{-1}\{\nabla\mathscr{V}_{(-i)}(\hat\omega) - \nabla\mathscr{V}(\hat\omega)\}$, with the leave-out gradient evaluated exactly (re-evaluated

weights, $\mathbf{F}$ and number of observations), and $\hat{\mathbf{V}}_\omega = \frac{G-1}{G}\sum_i(\Delta_i-\bar{\Delta})(\Delta_i-\bar{\Delta})^\top$. Since $\nabla\mathscr{V}_{(-i)}-\nabla\mathscr{V} \approx -\mathbf{s}_i$, $\Delta_i-\bar{\Delta} \approx -\mathbf{d}_i$: IJ and OS agree to first order, and OS carries the leave-out inflation of the jackknife, as CR3 does relative to CR0. (iii) The exact curve jackknife, the reference: $\Delta_i = \hat{\omega}_{(-i)} - \hat{\omega}$ from a refit without curve i (same basis, constraints and penalties, warm start), with the same formula; it needs G refits. (iv) Regularized variants replace $\mathbf{H}_\omega^{-1}$ by $(\mathbf{H}_{\omega,+}+\mathbf{I}/10)^{-1}$, where $\mathbf{H}_{\omega,+}$ has its non-positive eigenvalues set to zero and $1/10$ is the prior precision of mgcv's second-order term. When a smoothing parameter drifts toward a boundary, $\mathbf{H}_\omega$ has a small eigenvalue along it and $\mathbf{H}_\omega^{-1}\mathbf{s}_i$ explodes while $\mathbf{J}$ is nearly zero there; the product is numerically uncontrolled and produced heavy upper tails of $\hat{\mathbf{V}}_\rho$ over replicates.

Derivation: adding the term to CL2. The influence of curve i on $\hat{\theta}(\hat{\rho})$ is $\psi_i = \mathbf{k}_i + \mathbf{J}\mathbf{d}_{i,\rho}$, where $\mathbf{d}_{i,\rho}$ is the ρ-block of $\mathbf{d}_i$ ($\hat{\theta}$ does not depend on τ). CL2 replaces the coefficient part by its leverage-adjusted version $\mathbf{k}_i$; the ρ-part is not adjusted. Without the cross term, $\hat{\mathbf{V}} = \hat{\mathbf{V}}_{\mathrm{CL2}} + \mathbf{J}\hat{\mathbf{V}}_\rho\mathbf{J}^\top$ with $\hat{\mathbf{V}}_\rho$ the $\rho\rho$-block of $\hat{\mathbf{V}}_\omega$. With the cross term, the sandwich of the stacked influences,

$$\hat{\mathbf{V}}_\times = \frac{G}{G-1}\sum_i(\mathbf{k}_i+\mathbf{J}\mathbf{d}_{i,\rho})(\mathbf{k}_i+\mathbf{J}\mathbf{d}_{i,\rho})^\top + (\mathbf{V}_p-\mathbf{V}_e) = \hat{\mathbf{V}}_{\mathrm{CL2}} + \mathbf{J}\hat{\mathbf{V}}_\rho\mathbf{J}^\top + \frac{G}{G-1}\sum_i(\mathbf{k}_i\mathbf{d}_{i,\rho}^\top\mathbf{J}^\top + \mathbf{J}\mathbf{d}_{i,\rho}\mathbf{k}_i^\top),$$

the plug-in for $\mathbf{C}+\mathbf{C}^\top$ above. For leave-out changes, $\mathbf{d}_{i,\rho}$ is replaced by $-(\Delta_{i,\rho}-\bar{\Delta}_\rho)$ rescaled by $(G-1)/G$, so that the $\rho\rho$-part equals the jackknife variance in both forms. The result is a CL2-based hybrid, not a sandwich derived from one estimating equation. Assumptions: independent curves, a correct mean model, a canonical link with unit prior weights and no offsets, smoothing parameters that are neither linked nor fixed, a converged fit at an interior optimum with positive definite $\mathbf{F}$, and, for the first-order term to be accurate, $\sqrt{G}$-regular $\hat{\rho}$.

Derivation: curve-blocked NCV, Gaussian responses. With curve weights, $\mathbf{F}_\mathbf{c} = \sum_h c_h\tilde{\mathbf{X}}_h^\top\tilde{\mathbf{X}}_h + \mathbf{S}$, the fit without curve i is

$$\mathbf{t}_i = \mathbf{F}_{(-i)}^{-1}\sum_{h\neq i} c_h\mathbf{X}_h^\top\mathbf{W}_h\mathbf{y}_h, \qquad \mathbf{F}_{(-i)} = \mathbf{F}_\mathbf{c} - c_i\tilde{\mathbf{X}}_i^\top\tilde{\mathbf{X}}_i,$$

and the curve-blocked NCV criterion of §2 of the paper, divided by ϕ, is $\mathscr{V}(\rho;\mathbf{c}) = \sum_i c_i\mathbf{r}_i^\top\mathbf{W}_i\mathbf{r}_i$ with $\mathbf{r}_i = \mathbf{y}_i - \mathbf{X}_i\mathbf{t}_i$. For Gaussian responses, mgcv's Newton-step leave-out fit is exact, so the criterion above is mgcv's criterion; the factor $1/\phi$ cancels in $\mathbf{H}_\rho^{-1}\mathbf{s}_i$, and there is no scale parameter among the outer parameters. A zero weight removes curve i from the loss and from every other leave-out fit. With $\mathbf{v}_{ik} = \mathbf{F}_{(-i)}^{-1}\dot{\mathbf{S}}_k\mathbf{t}_i$, $\partial\mathbf{t}_i/\partial\rho_k = -\mathbf{v}_{ik}$, and with $\mathbf{u}_i^{\mathrm{cv}} = \mathbf{F}_{(-i)}^{-1}\mathbf{X}_i^\top\mathbf{W}_i\mathbf{r}_i$ and $q_{ik} = \mathbf{u}_i^{\mathrm{cv}\top}\dot{\mathbf{S}}_k\mathbf{t}_i$,

$$\frac{\partial\mathscr{V}}{\partial\rho_k} = 2\sum_i c_i\,q_{ik}, \qquad H_{\rho,kl} = 2\sum_i c_i\big[(\mathbf{X}_i\mathbf{v}_{il})^\top\mathbf{W}_i\mathbf{X}_i\mathbf{v}_{ik} + \mathbb{1}_{k=l}\,q_{ik} - \mathbf{u}_i^{\mathrm{cv}\top}\dot{\mathbf{S}}_l\mathbf{v}_{ik} - \mathbf{u}_i^{\mathrm{cv}\top}\dot{\mathbf{S}}_k\mathbf{v}_{il}\big],$$

using $\partial\mathbf{u}_i^{\mathrm{cv}}/\partial\rho_l = \mathbf{F}_{(-i)}^{-1}(\mathbf{X}_i^\top\mathbf{W}_i\mathbf{X}_i\mathbf{v}_{il} - \dot{\mathbf{S}}_l\mathbf{u}_i^{\mathrm{cv}})$. For the curve scores, $\mathbf{t}_h$ does not depend on c_h, and for $i \neq h$ $\partial\mathbf{F}_{(-i)}/\partial c_h = \tilde{\mathbf{X}}_h^\top\tilde{\mathbf{X}}_h$ and $\partial\mathbf{t}_i/\partial c_h = \mathbf{F}_{(-i)}^{-1}\mathbf{X}_h^\top\mathbf{W}_h\mathbf{e}_{h|i}$, with $\mathbf{e}_{h|i} = \mathbf{y}_h - \mathbf{X}_h\mathbf{t}_i$ the residual of curve h under the fit without i. Differentiating q_{ik} gives, with $\mathbf{m}_{1,ik} = \mathbf{F}_{(-i)}^{-1}\mathbf{X}_i^\top\mathbf{W}_i\mathbf{X}_i\mathbf{v}_{ik}$ and $\mathbf{m}_{2,ik} = \mathbf{F}_{(-i)}^{-1}\dot{\mathbf{S}}_k\mathbf{u}_i^{\mathrm{cv}}$,

$$s_{hk} = \frac{\partial^2\mathscr{V}}{\partial\rho_k\,\partial c_h}\Big|_{\mathbf{c}=\mathbf{1}} = 2q_{hk} + 2\sum_{i\neq h}\big[\mathbf{e}_{h|i}^\top\mathbf{W}_h\mathbf{X}_h(\mathbf{m}_{2,ik}-\mathbf{m}_{1,ik}) - (\mathbf{X}_h\mathbf{u}_i^{\mathrm{cv}})^\top\mathbf{W}_h\mathbf{X}_h\mathbf{v}_{ik}\big].$$

The first term is curve h's own loss, the sum its effect on every other leave-out fit. The cost is $O(GKNp + Gp^3)$. Since $\mathscr{V}(\rho;c\mathbf{1}) = c\,\mathscr{V}(\rho-\log c;\mathbf{1})$, $\sum_h\mathbf{s}_h = \nabla_\rho\mathscr{V} - \mathbf{H}_\rho\mathbf{1}$, a check of signs and centering that the implementation meets to 10^{-12}; scores and Hessian also agree with central differences. NCV is a prediction loss, not a log-likelihood, so $\mathbf{H}_\rho^{-1}$ alone is not a covariance of $\hat{\rho}$; the IJ sandwich $\mathbf{H}_\rho^{-1}\hat{\mathbf{M}}\mathbf{H}_\rho^{-1}$ is, to first order, and it is invariant to rescaling the criterion. The regularized variants are not (the prior $1/10$ does not rescale), so they serve only as a sensitivity check. $\mathbf{J}$ is the same as for REML, because the penalized fit at given ρ is the same.

Derivation: curve-blocked NCV, Poisson and binary responses. With a canonical link and $\phi = 1$, $\mathbf{W} = \mathrm{diag}(d\mu/d\eta)$ depends on $\hat\theta$, and mgcv approximates the fit without curve i by one Newton step from $\hat\theta$: $\mathbf{F}_{(-i)} = \mathbf{F} - c_i\mathbf{X}_i^\top\mathbf{W}_i\mathbf{X}_i$, $\delta_i = \mathbf{F}_{(-i)}^{-1}\mathbf{X}_i^\top c_i(\mathbf{y}_i - \mu_i)$, $\eta_{(-i)} = \mathbf{X}_i(\hat\theta - \delta_i)$, and $\mathscr{V} = \sum_i c_i \sum_j \mathrm{dev}\{y_{ij}, \mu(\eta_{(-i),j})\}$. The derivatives in a direction $(\dot\rho, \dot{\mathbf{c}})$ follow by implicit differentiation of the score equation ($d\{c(y-\mu)\}/d\eta = -cw$ for a canonical link); with $\dot{\mathbf{S}} = \sum_k \dot\rho_k \dot{\mathbf{S}}_k$, $w' = dw/d\eta$ and $\circ$ the elementwise product,

$$\dot\theta = \mathbf{F}^{-1}\{\mathbf{X}^\top(\dot{\mathbf{c}} \circ (\mathbf{y} - \mu)) - \dot{\mathbf{S}}\hat\theta\}, \quad \dot\eta = \mathbf{X}\dot\theta, \quad (\mathbf{cw})^{\cdot} = \dot{\mathbf{c}} \circ \mathbf{w} + \mathbf{c} \circ \mathbf{w}' \circ \dot\eta, \quad \{\mathbf{c}(\mathbf{y}-\mu)\}^{\cdot} = \dot{\mathbf{c}} \circ (\mathbf{y} - \mu) - \mathbf{c} \circ \mathbf{w} \circ \dot\eta,$$

$$\dot{\mathbf{F}} = \mathbf{X}^\top \mathrm{diag}\{(\mathbf{cw})^{\cdot}\}\mathbf{X} + \dot{\mathbf{S}}, \quad \dot\delta_i = \mathbf{F}_{(-i)}^{-1}\big[\mathbf{X}_i^\top\{\mathbf{c}(\mathbf{y}-\mu)\}_i^{\cdot} - \{\dot{\mathbf{F}} - \mathbf{X}_i^\top \mathrm{diag}\{(\mathbf{cw})_i^{\cdot}\}\mathbf{X}_i\}\delta_i\big], \quad \dot\eta_{(-i)} = \mathbf{X}_i(\dot\theta - \dot\delta_i),$$

and, since $\partial\,\mathrm{dev}/\partial\eta = -2(y-\mu)$, $\dot{\mathscr{V}} = \sum_i\big[\dot c_i \sum_j \mathrm{dev}_{ij} - 2c_i\sum_j(y_{ij} - \mu_{(-i),j})\dot\eta_{(-i),j}\big]$. The K directions in ρ give the gradient (equal to mgcv's), the G directions in $\mathbf{c}$ give $\partial\mathscr{V}/\partial c_h$; $\mathbf{H}_\rho$ and the scores are central differences in ρ (step 10^{-3}) of these exact first derivatives. Compared with the Gaussian case, the weights, the Newton step and the leave-out predictions now move with ρ and $\mathbf{c}$ through $\hat\eta$, there is no scale parameter, and $\mathbf{J}_k = -\mathbf{V}_p\dot{\mathbf{S}}_k\hat\theta$ at the converged weights. A smoothing parameter whose curvature is below 10^{-3} times the largest diagonal entry of $\mathbf{H}_\rho$ lies on a flat stretch of the criterion ($\lambda_k \to \infty$, the fit no longer depends on it); as in mgcv's Newton iteration, it is treated as fixed ($d_{i,k} = 0$) and the IJ uses the sub-matrices of the free components. This rule cannot represent jumps of $\hat\rho$ between interior and boundary solutions.

Result. In the main factorial (200 replicates, dependent errors), the term without the cross term raises NCV + CL2's coverage of β from 0.92–0.93 to 0.95 for Gaussian responses, at 1.10–1.13 times the width; for binary responses with 40 curves, it reaches 0.86. On the plasmode datasets with independent source curves, it raises the mean coverage of β from 0.82 to 0.86 (range over combinations 0.58–0.94). Without the cross term, the variance is closer to the replicate variance (SE/SD 0.86–1.29 per plasmode dataset, against 0.99–1.38 with it) and coverage is lower by 0.5–2.6 pp per dataset (second figure in (f)).

Reason. The term addresses only the variance part of (a), not the smoothing bias. For binary responses, the selected smoothing parameters jump between interior and boundary solutions, which a first-order term cannot represent.

Table 42: Smoothing-parameter uncertainty term for curve-blocked NCV, $\beta(s,t)$, main-factorial settings with dependent errors at the default signal (200 replicates): coverage (5% quantile of pointwise coverage) and width relative to NCV + CL2. λ fixed: refit at the geometric-mean NCV smoothing parameters of the setting, computed from the same replicates (a diagnostic).

family	G	NCV + CL2	+ λ term	λ fixed (diagnostic)
binary	40	0.787 (0.61) 1.00	0.859 (0.72) 1.27	0.704 (0.23) 0.70
binary	100	0.899 (0.78) 1.00	0.933 (0.85) 1.16	0.949 (0.88) 0.96
Gaussian	40	0.923 (0.86) 1.00	0.948 (0.90) 1.13	0.949 (0.91) 1.00
Gaussian	100	0.934 (0.86) 1.00	0.951 (0.89) 1.10	0.950 (0.90) 1.00
Poisson	40	0.924 (0.88) 1.00	0.951 (0.91) 1.21	0.952 (0.92) 1.01
Poisson	100	0.935 (0.88) 1.00	0.953 (0.90) 1.09	0.953 (0.91) 1.00

(f) Bias correction and smoothing-parameter term combined

Method. Propagate the smoothing-parameter term of (e) through the bias-correction map of (d) (Gaussian responses only), with or without the cross term. The map $\mathbf{M} = 2\mathbf{I} - \mathbf{A}$ is held fixed at $\hat\rho$: the construction is a *frozen-map approximation* that ignores $\partial\mathbf{A}/\partial\rho$ (derivation below).

Derivation. The corrected estimate is $\tilde\theta(\hat\rho) = \mathbf{M}(\hat\rho)\hat\theta(\hat\rho)$. For Gaussian responses, $\partial\mathbf{V}_p/\partial\rho_k = -\mathbf{V}_p\dot{\mathbf{S}}_k\mathbf{V}_p$, so $\partial\mathbf{A}/\partial\rho_k = \partial(\mathbf{I} - \mathbf{V}_p\mathbf{S})/\partial\rho_k = -\mathbf{V}_p\dot{\mathbf{S}}_k\mathbf{A}$ and

$$\frac{\partial\tilde\theta}{\partial\rho_k} = \mathbf{M}\mathbf{J}_k - \frac{\partial\mathbf{A}}{\partial\rho_k}\hat\theta = \mathbf{M}\mathbf{J}_k + \mathbf{V}_p\dot{\mathbf{S}}_k\mathbf{A}\hat\theta.$$

The second term, $-(\partial \mathbf{A}/\partial \rho_k)\hat{\theta}$, is first order in the same smoothing-parameter perturbation as the first. The frozen-map construction keeps the first term only: the curve influences on $\tilde{\theta}$ are taken as $\mathbf{M}(\mathbf{k}_i + \mathbf{J}\mathbf{d}_{i,\rho})$, so the coefficient–smoothing-parameter cross term also uses $\mathbf{J} = \partial\hat{\theta}/\partial\rho$ and not $\partial\tilde{\theta}/\partial\rho$. The covariance is $\mathbf{M}\mathbf{V}_{f,\rho}\mathbf{M}^\top + (\mathbf{I}-\mathbf{A})(\mathbf{V}_p - \mathbf{V}_e)(\mathbf{I}-\mathbf{A})^\top$, where $\mathbf{V}_{f,\rho}$ is the frequentist CL2 plus the term of (e), with or without the cross term, and the allowance is propagated as in (d). With two steps, $\mathbf{M}_2$ and $(\mathbf{I}-\mathbf{A})^2$ replace $\mathbf{M}$ and $\mathbf{I}-\mathbf{A}$.

Result. On 100 fresh synthetic replicates per setting, the version with the cross term covers β at 0.97 on smooth truths and 0.94 on rough truths, at 0.59 and 0.76 times REML + CL2's width, with SE/SD 1.21–1.27; without the cross term, it covers 0.96 and 0.93.

On the plasmode data, the combined construction was run on replicates 1–50 of each plasmode combination; REML + CL2 comes from replicates 1–100 of the bias-correction run on the same combinations. The comparison is therefore only partly paired, and small differences are descriptive, not evidence of equivalence or of a precise ranking. The combined construction covers β at 0.83–0.94 on the datasets with independent source curves (REML + CL2 0.91–0.94) at 0.73–1.09 times the width, with lower tails 0.55–0.87 (REML + CL2 0.84–0.89), and 0.66–0.84 on the datasets with dependent source curves. By truth source, coverage is 0.81 (REML-fitted), 0.88 (midpoint truth, smoothing parameters at the geometric mean of the REML and NCV values), 0.91 (NCV-fitted). Without the cross term, it covers 0.81–0.94 on the datasets with independent source curves and stays below 0.85 on running, ocean, weather, electricity. Of the 3000 plasmode replicate fits, 8 failed the internal consistency check of the smoothing-parameter derivatives; they are reported as missing, not replaced.

Reason. On synthetic data, the excess variance of the term offsets the bias the correction leaves (SE/SD above 1). On the plasmode data, the remaining bias is larger, and this construction offers no width advantage over REML + CL2 while its lower tails are worse. The conclusion is about the frozen-map approximation evaluated here; whether the full derivative, including $-(\partial \mathbf{A}/\partial \rho)\hat{\theta}$, would change it was not studied.

Table 43: Synthetic, Gaussian, dependent errors, $\beta(s,t)$, 100 fresh replicates: coverage (5% quantile of pointwise coverage) and width relative to REML + CL2 for NCV + CL2, with the smoothing-parameter term, with the bias correction, and with both.

group	settings	NCV + CL2	+ λ term	bias correction	bias correction + λ term	REML + CL2
rough truth, Gaussian	3	0.846 (0.59) 0.50	0.889 (0.68) 0.59	0.894 (0.77) 0.61	0.941 (0.84) 0.76	0.940 (0.92)
smooth truth, Gaussian	4	0.925 (0.87) 0.38	0.957 (0.90) 0.46	0.928 (0.89) 0.47	0.971 (0.94) 0.59	0.938 (0.91)

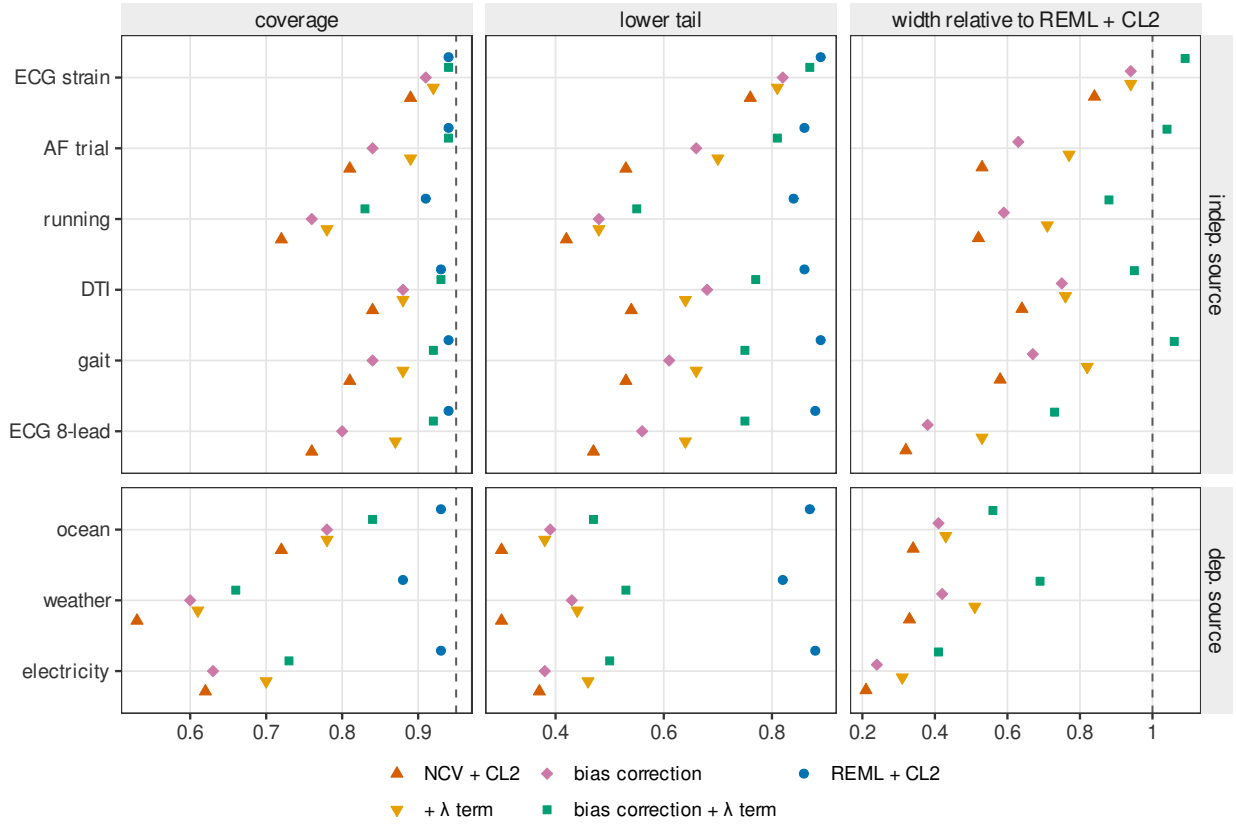


Figure 24: Plasmode, $\beta(s,t)$, all subjects, mean over truth × residual combinations: coverage, lower tail (5% quantile of pointwise coverage) and width relative to REML + CL2 for the four NCV-centered intervals (50 replicates), and REML + CL2 (100 replicates). Dashed: nominal 0.95 for coverage, 1 for the ratio. Values: S7.

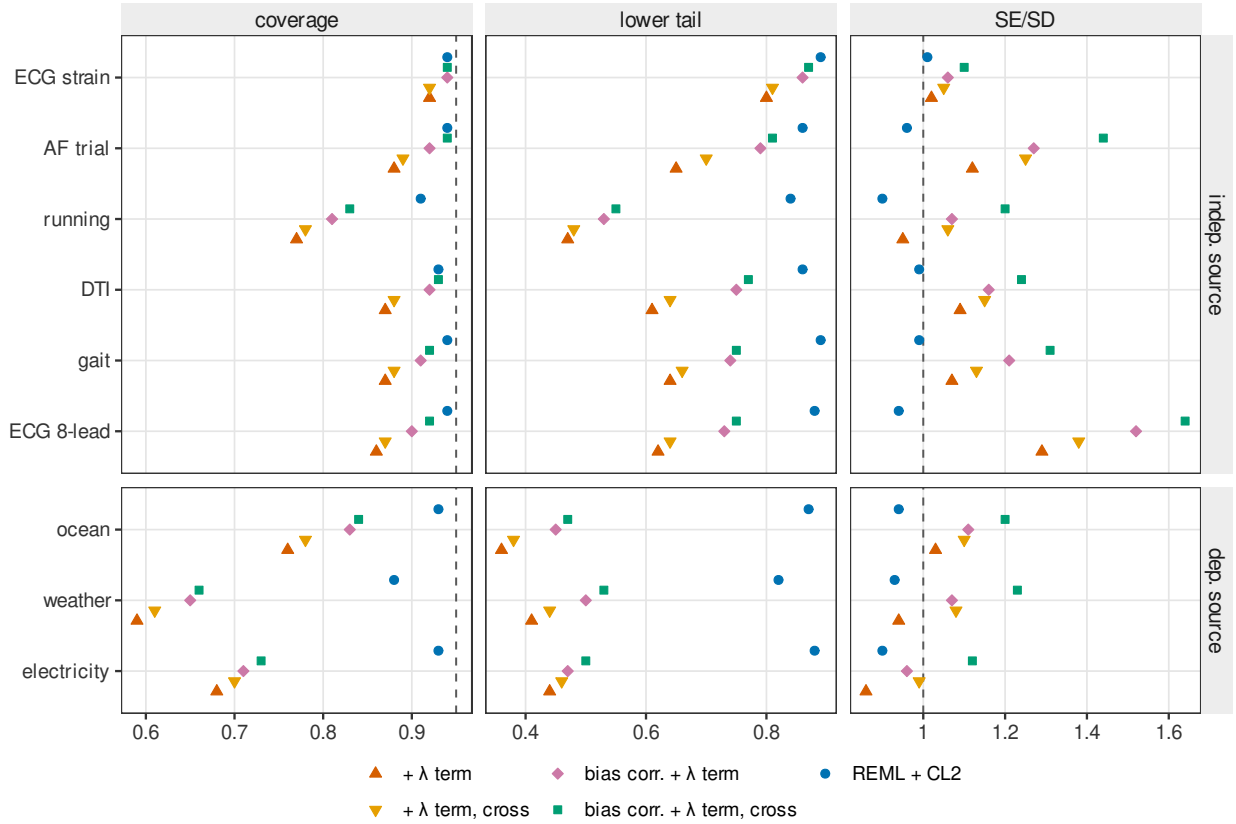


Figure 25: Plasmode, $\beta(s,t)$, all subjects, mean over truth × residual combinations: coverage, lower tail (5% quantile of pointwise coverage) and SE/SD for the smoothing-parameter term without and with the cross term, alone and combined with the one-step bias correction (50 replicates), and REML + CL2 (100 replicates). Dashed: nominal 0.95 for coverage, 1 for the ratio. Values: S7.

(g) Truths outside the spline space

Method. We evaluate NCV + CL2 and the corrections of (d)–(f), including their two-step versions, on the misspecified truths of S5.1 (default basis, 100 replicates).

Result. For T2 (the smooth truth plus $\sum_{k=12}^{16} \sqrt{2}\cos(k\pi s)\sin(\pi t)$; S5.1), whose missing part loads on directions the covariates barely excite, every NCV-centered interval covers $\beta(s,t)$ at 0.28–0.45. For T1 (the smooth truth plus a local Gaussian bump) under independent errors, they cover 0.85–0.90 with lower tails 0.11–0.25.

Reason. A bias that the spline space cannot represent is invisible to every estimate the corrections are built from, including the frozen-map combination of (f).

S4.2 mgcv's neighborhood covariance for curve-blocked NCV

Method. mgcv offers a covariance for NCV fits that accounts for the neighborhood structure (help page ?NCV, Wood (2024)). Computed correctly for curve neighborhoods, it was evaluated on the same NCV fits as NCV + CL2: in three main-factorial settings (Gaussian, Poisson, binary; smooth errors, $G = 100$, default signal; replicates 1–100) and in the 42 plasmode combinations of the six datasets with independent source curves (all subjects, curve flips, every truth × residual pair; replicates 1–100).

Derivation: what mgcv computes. The description follows `?NCV`, Wood (2024) and the source of mgcv 1.9-5 (functions `gam.outer` and `neicov`). mgcv specifies NCV per observation: each observation n is predicted from the fit with a set $\alpha(n)$ of observations dropped. For each n, mgcv computes the change of the coefficients when n alone is left out, $\mathbf{d}_n = \hat{\theta}_{-n} - \hat{\theta}$, by one Newton step at the selected λ, multiplied by $\{(N-1)/N\}^{0.4}$. To first order, $\mathbf{d}_n \approx -\mathbf{V}_p \tilde{\mathbf{x}}_n z_n$ (Appendix A.1 of the paper). The neighborhood-aware estimator is

$$\hat{\mathbf{V}}_{\text{nei}} = \frac{N}{N-\tau} \sum_{n=1}^{N} \tilde{\mathbf{d}}_n \Big(\sum_{m \in \alpha(n)} \tilde{\mathbf{d}}_m \Big)^{\top} \quad \text{(symmetrized)}, \qquad \tilde{\mathbf{d}}_n = \mathbf{d}_n \frac{r_n^{\text{cv}}}{r_n},$$

with τ the effective degrees of freedom and r_n^{cv}/r_n the ratio of the cross-validated to the ordinary deviance residual. mgcv returns $\hat{\mathbf{V}}_{\text{nei}} + a(\mathbf{V}_p - \mathbf{V}_e)$, projected onto the positive semidefinite matrices, where $a = \max\{\operatorname{tr} \hat{\mathbf{V}}^{(0)}_{\text{nei}} / \operatorname{tr} \mathbf{V}_e, 1\}$ and $\hat{\mathbf{V}}^{(0)}_{\text{nei}}$ is the same sum with $\mathbf{d}_n$ in place of $\tilde{\mathbf{d}}_n$. If $\alpha(n)$ is the curve of observation n, the sum equals $\frac{N}{N-\tau} \sum_i \mathbf{D}_i \mathbf{D}_i^{\top}$ with $\mathbf{D}_i = \sum_{n \in i} \tilde{\mathbf{d}}_n \approx -\mathbf{V}_p \tilde{\mathbf{X}}_i^{\top} \mathbf{z}_i$: a curve-cluster sandwich of CR0 type with the penalized bread and without CL2's leverage adjustment, plus a rescaled smoothing-bias allowance $a(\mathbf{V}_p - \mathbf{V}_e)$. With `nei$jackknife = TRUE`, mgcv instead returns the jackknife over the folds, $\sum_i \{(N - D_i)/N\}^{0.8} \mathbf{d}_i \mathbf{d}_i^{\top} + a'(\mathbf{V}_p - \mathbf{V}_e)$, with $\mathbf{d}_i$ the one-step change when curve i is left out and $a' = \operatorname{tr}(\cdot)/\operatorname{tr} \mathbf{V}_e$ of the first sum.

The block specification. refund specifies the same NCV criterion in block form: G folds, each dropping and predicting a whole curve. Both forms select the same smoothing parameters and give the same coefficients, but mgcv 1.9-5 computes its covariance only for the per-observation form. For the block form, it returns the Bayesian $\mathbf{V}_p$ when `jackknife` is unset. With `jackknife = TRUE`, its routine `nei_cov()` reads the $N \times p$ matrix of the changes $\mathbf{d}_n$ with row stride G, so it uses the wrong entries: the sum has a trace of about 10^{-3} times $\operatorname{tr} \mathbf{V}_e$ or less, $a = 1$, and the returned covariance is $\mathbf{V}_p - \mathbf{V}_e$, the smoothing-bias allowance alone. Its standard errors relative to the Bayesian ones are $\{\mathbf{a}^{\top}(\mathbf{V}_p - \mathbf{V}_e)\mathbf{a} / \mathbf{a}^{\top}\mathbf{V}_p\mathbf{a}\}^{1/2} < 1$ for every contrast $\mathbf{a}$ with $\mathbf{a}^{\top}\mathbf{V}_e\mathbf{a} > 0$. This returned covariance explains the result of `jackknife = TRUE` in the three synthetic settings (20 replicates): standard errors 0.49–0.77 times the Bayesian ones and coverage of the mean and $\beta(s,t)$ of 0.32–0.58. The correction will be part of a forthcoming mgcv release.

Corrected computation. The function `jackfix_vcov()` (code archive, `ncv-jack-recheck/jackfix-lib.R`) computes from the block fit what mgcv returns for the per-observation form with $\alpha(n)$ = the curve of n: the single-observation Newton changes, the residual ratios at the leave-curve-out fit, both sums, a and a'. It needs no refit and takes about 1 s per study fit (median), about as long as CL2. In small examples, it reproduces mgcv's covariance for the per-observation form to a relative difference of 10^{-14} (Gaussian) and 10^{-12} (Poisson); its leave-curve-out steps agree with exact leave-one-curve-out refits at fixed smoothing parameters to 10^{-13} for Gaussian responses and to within a few percent for Poisson responses (one Newton step). At study scale (study setting 12: Gaussian responses, smooth errors, $G = 100$, default signal; replicate 1), a pffr fit with per-observation neighborhoods selects the same smoothing parameters (largest difference in $\log \lambda$: 0), and `jackfix_vcov()` of the block fit reproduces its $\mathbf{V}_p$ and $\mathbf{V}_c$ to a relative difference of 1.3e-13. The per-observation fit took 801 s against 32 s for the block fit (one fit each, timed in a separate run, not comparable with the benchmark times of Appendix C of the paper). The NCV and REML estimates and CL2 standard errors of the re-evaluation reproduce the study's records (largest absolute difference 2e-06). No replicate fit failed (0 failures).

Result. On synthetic data (next table), the corrected covariance is conservative: it covers β at 0.96–0.98 (NCV + CL2 0.90–0.94, REML + CL2 0.95) with SE/SD 1.13–1.24, at 1.30–1.32 times NCV + CL2's width and 0.46–0.51 times REML + CL2's. Its sampling part alone behaves like CL2 (SE/SD for β 0.78–0.94). The median of a is 4.1–5.1 per setting, so the bias allowance $a(\mathbf{V}_p - \mathbf{V}_e)$ adds a large term. For binary responses, the corrected covariance covers the mean, $\alpha(t)$ and $\gamma(t)$ at 0.89–0.93, below REML + CL2 (0.93–0.94). The variant with the $\hat{\lambda}$ correction and mgcv's `jackknife = TRUE` estimate differ from it by at most 0.013 in coverage.

On the plasmode data, over the 42 combinations, the corrected covariance covers β at 0.87 on average (5% quantile 0.65, lowest combination 0.62), against 0.82 (0.59) for NCV + CL2 and 0.93 (0.87) for REML + CL2, at 0.68 times

REML + CL2's width (geometric mean). By truth source, averaged over the 14 combinations of each, it covers β at 0.82 under the REML-fitted truth, 0.87 under the midpoint truth and 0.91 under the NCV-fitted truth. Per dataset, it raises the coverage of NCV + CL2 for β by 0.01–0.08 and stays below REML + CL2 on every dataset (by 0.004–0.164).

Reason. Computed correctly, mgcv's neighborhood covariance is a curve-cluster sandwich plus a smoothing-bias allowance that is several times $\mathbf{V}_p - \mathbf{V}_e$. On synthetic data, this allowance makes the intervals conservative. On the plasmode data, they still undercover. Residual smoothing bias of the NCV estimate is a plausible contributor (S4.1(a)), which a covariance alone does not remove; variance and tail-calibration errors are not excluded (SE/SD 0.85 on ECG 8-lead and 0.93 on running; second table below). No unique cause of the failure to reach nominal coverage on these data was established.

Table 44: Intervals for the curve-blocked NCV fit (synthetic, smooth errors, $G = 100$, 100 replicates): coverage (5% quantile of pointwise coverage), SE/SD [width relative to NCV + CL2 / REML + CL2]. Bayesian Vp, NCV + CL2 and the corrected neighborhood covariance use the same NCV fit; REML + CL2 the REML fit to the same data.

estimand	interval	Gaussian	Poisson	binary
E(Y | X)	Bayesian Vp	0.618 (0.49) 0.46 [0.47/0.36]	0.620 (0.49) 0.47 [0.48/0.36]	0.612 (0.44) 0.49 [0.53/0.38]
E(Y | X)	NCV + CL2	0.936 (0.87) 1.02 [1.00/0.78]	0.929 (0.86) 1.00 [1.00/0.77]	0.886 (0.76) 0.93 [1.00/0.71]
E(Y | X)	corrected neighborhood covariance	0.963 (0.92) 1.14 [1.12/0.87]	0.960 (0.92) 1.13 [1.14/0.87]	0.929 (0.83) 1.05 [1.14/0.81]
E(Y | X)	... with $\bar{\chi}$ correction	0.964 (0.92) 1.15 [1.13/0.88]	0.961 (0.92) 1.14 [1.15/0.88]	0.936 (0.84) 1.08 [1.16/0.83]
E(Y | X)	its sampling part	0.936 (0.87) 1.03 [1.01/0.78]	0.930 (0.86) 1.02 [1.02/0.78]	0.874 (0.74) 0.90 [0.97/0.69]
E(Y | X)	mgcv per-point jackknife = TRUE	0.963 (0.92) 1.14 [1.12/0.87]	0.960 (0.91) 1.13 [1.13/0.87]	0.933 (0.84) 1.06 [1.15/0.82]
E(Y | X)	REML + CL2	0.949 (0.90) 1.02 [1.29/1.00]	0.948 (0.91) 1.02 [1.31/1.00]	0.943 (0.90) 1.00 [1.40/1.00]
β(s,t)	Bayesian Vp	0.703 (0.52) 0.50 [0.53/0.21]	0.710 (0.57) 0.51 [0.55/0.21]	0.715 (0.54) 0.53 [0.63/0.22]
β(s,t)	NCV + CL2	0.940 (0.86) 0.97 [1.00/0.39]	0.936 (0.87) 0.96 [1.00/0.38]	0.900 (0.79) 0.87 [1.00/0.35]
β(s,t)	corrected neighborhood covariance	0.981 (0.95) 1.24 [1.30/0.51]	0.981 (0.95) 1.23 [1.30/0.49]	0.960 (0.89) 1.13 [1.32/0.46]
β(s,t)	... with $\bar{\chi}$ correction	0.982 (0.95) 1.25 [1.30/0.51]	0.982 (0.95) 1.24 [1.31/0.50]	0.964 (0.90) 1.15 [1.34/0.47]
β(s,t)	its sampling part	0.925 (0.84) 0.93 [0.95/0.37]	0.921 (0.85) 0.94 [0.95/0.36]	0.858 (0.73) 0.78 [0.89/0.31]
β(s,t)	mgcv per-point jackknife = TRUE	0.983 (0.95) 1.26 [1.32/0.51]	0.983 (0.95) 1.25 [1.33/0.51]	0.965 (0.89) 1.16 [1.35/0.47]
β(s,t)	REML + CL2	0.947 (0.90) 0.97 [2.56/1.00]	0.946 (0.91) 0.96 [2.63/1.00]	0.952 (0.91) 0.96 [2.86/1.00]
α(t)	Bayesian Vp	0.617 (0.48) 0.46 [0.46/0.44]	0.607 (0.46) 0.45 [0.48/0.43]	0.557 (0.46) 0.47 [0.51/0.41]
α(t)	NCV + CL2	0.939 (0.85) 0.99 [1.00/0.94]	0.928 (0.85) 0.96 [1.00/0.91]	0.866 (0.69) 0.92 [1.00/0.80]
α(t)	corrected neighborhood covariance	0.955 (0.87) 1.07 [1.09/1.03]	0.950 (0.89) 1.05 [1.09/1.00]	0.894 (0.72) 1.02 [1.11/0.88]
α(t)	... with $\bar{\chi}$ correction	0.958 (0.89) 1.08 [1.10/1.03]	0.956 (0.90) 1.06 [1.11/1.01]	0.902 (0.74) 1.05 [1.14/0.91]
α(t)	its sampling part	0.941 (0.86) 0.99 [1.01/0.95]	0.928 (0.85) 0.97 [1.00/0.91]	0.860 (0.69) 0.90 [0.98/0.78]
α(t)	mgcv per-point jackknife = TRUE	0.954 (0.87) 1.07 [1.08/1.02]	0.950 (0.89) 1.05 [1.09/1.00]	0.896 (0.72) 1.03 [1.12/0.89]
α(t)	REML + CL2	0.950 (0.91) 1.04 [1.06/1.00]	0.949 (0.92) 1.00 [1.10/1.00]	0.935 (0.89) 0.99 [1.25/1.00]
γ(t)	Bayesian Vp	0.639 (0.58) 0.48 [0.46/0.43]	0.629 (0.56) 0.49 [0.48/0.44]	0.568 (0.47) 0.47 [0.51/0.41]
γ(t)	NCV + CL2	0.936 (0.87) 1.09 [1.00/0.92]	0.932 (0.84) 1.05 [1.00/0.92]	0.849 (0.74) 0.93 [1.00/0.79]
γ(t)	corrected neighborhood covariance	0.962 (0.94) 1.19 [1.11/1.02]	0.961 (0.94) 1.19 [1.12/1.03]	0.889 (0.79) 1.05 [1.12/0.89]
γ(t)	... with $\bar{\chi}$ correction	0.963 (0.94) 1.20 [1.12/1.03]	0.962 (0.94) 1.22 [1.13/1.04]	0.902 (0.81) 1.09 [1.16/0.92]
γ(t)	its sampling part	0.939 (0.88) 1.10 [1.02/0.94]	0.936 (0.85) 1.07 [1.02/0.94]	0.840 (0.73) 0.91 [0.98/0.78]
γ(t)	mgcv per-point jackknife = TRUE	0.962 (0.94) 1.18 [1.10/1.02]	0.960 (0.93) 1.18 [1.12/1.03]	0.894 (0.80) 1.07 [1.14/0.90]
γ(t)	REML + CL2	0.954 (0.93) 1.09 [1.08/1.00]	0.951 (0.90) 1.09 [1.09/1.00]	0.940 (0.91) 1.02 [1.26/1.00]

Table 45: Plasmode, six datasets with independent source curves, all subjects, 100 replicates per combination: coverage (5% quantile of pointwise coverage) SE/SD [width relative to NCV + CL2 / REML + CL2], means over each dataset's truth × residual combinations (SE/SD: median; widths: geometric mean).

estimand	dataset	combinations	Bayesian Vp	NCV + CL2	corrected neighborhood covariance	mgcv per-point jackknife = TRUE	REML + CL2
β(s,t)	AF trial	6	0.54 (0.21) 0.39	0.81 (0.52) 0.84	0.891 (0.63) 1.11 [1.38/0.72]	0.893 (0.64) 1.12 [1.41/0.74]	0.94 (0.86) 0.96
β(s,t)	DTI	6	0.56 (0.29) 0.46	0.85 (0.58) 0.95	0.904 (0.68) 1.12 [1.21/0.79]	0.905 (0.68) 1.12 [1.22/0.79]	0.93 (0.86) 0.99
β(s,t)	ECG strain	12	0.49 (0.20) 0.29	0.91 (0.80) 0.98	0.937 (0.84) 1.10 [1.12/0.95]	0.933 (0.83) 1.08 [1.10/0.93]	0.94 (0.89) 1.01
β(s,t)	ECG 8-lead	6	0.50 (0.18) 0.33	0.76 (0.48) 0.79	0.777 (0.49) 0.85 [1.05/0.32]	0.760 (0.48) 0.82 [1.01/0.31]	0.94 (0.88) 0.95
β(s,t)	gait	6	0.44 (0.15) 0.30	0.81 (0.53) 0.90	0.845 (0.60) 0.98 [1.10/0.63]	0.846 (0.60) 0.98 [1.10/0.63]	0.94 (0.89) 1.00
β(s,t)	running	6	0.44 (0.14) 0.32	0.71 (0.40) 0.75	0.788 (0.46) 0.93 [1.25/0.65]	0.792 (0.47) 0.94 [1.28/0.66]	0.91 (0.84) 0.90
E(Y | X)	AF trial	6	0.61 (0.31) 0.41	0.90 (0.79) 0.94	0.944 (0.84) 1.10 [1.22/1.04]	0.944 (0.84) 1.10 [1.23/1.05]	0.95 (0.89) 1.01
E(Y | X)	DTI	6	0.61 (0.39) 0.47	0.91 (0.79) 0.97	0.938 (0.84) 1.08 [1.12/0.95]	0.937 (0.84) 1.08 [1.12/0.95]	0.93 (0.88) 0.99
E(Y | X)	ECG strain	12	0.53 (0.22) 0.34	0.92 (0.81) 1.01	0.947 (0.86) 1.12 [1.12/1.02]	0.942 (0.85) 1.09 [1.09/1.00]	0.94 (0.88) 1.03
E(Y | X)	ECG 8-lead	6	0.77 (0.30) 0.78	0.90 (0.73) 0.97	0.910 (0.75) 1.01 [1.04/0.78]	0.899 (0.73) 0.98 [1.01/0.76]	0.95 (0.89) 1.04
E(Y | X)	gait	6	0.53 (0.23) 0.33	0.93 (0.82) 1.01	0.943 (0.85) 1.08 [1.06/0.97]	0.941 (0.85) 1.07 [1.05/0.96]	0.95 (0.91) 1.03
E(Y | X)	running	6	0.54 (0.23) 0.37	0.89 (0.75) 0.96	0.932 (0.82) 1.07 [1.13/1.03]	0.930 (0.81) 1.06 [1.12/1.02]	0.93 (0.88) 0.99

S4.3 Smoothing-parameter uncertainty term for REML

Method. The analogous term for REML uses curve weights in mgcv's REML criterion: the implicit-function theorem gives each curve's influence on $(\log\hat\lambda, \log\hat\phi)$ through analytic curve scores, and the infinitesimal jackknife over curves gives a curve-robust variance that enters as $\mathbf{J}\hat{\mathbf{V}}_\rho\mathbf{J}^\top$ on top of CL2.

Derivation: criterion and gradient. With curve weights c_i on the log-likelihood contributions (including their normalizing constants), $\hat\theta(\rho, \mathbf{c})$ maximizes $\ell_{\mathbf{c}}(\theta) - \frac{1}{2}\theta^\top \mathrm{S}\theta$ with $\ell_{\mathbf{c}} = \sum_i c_i\ell_i$, and mgcv's Laplace-approximate REML criterion (Wood 2011) is, up to a constant that depends on neither ω nor $\mathbf{c}$,

$$\mathscr{V}(\omega; \mathbf{c}) = -\ell_{\mathbf{c}}(\hat\theta) + \tfrac{1}{2}\hat\theta^\top \mathrm{S}\hat\theta + \tfrac{1}{2}\log|\mathbf{X}^\top\mathbf{W}_{\mathbf{c}}\mathbf{X} + \mathrm{S}| - \tfrac{1}{2}\log|\mathrm{S}|_+,$$

with $\mathbf{W}_{\mathbf{c}} = \mathrm{diag}(c_{i(n)}w_n)$, M_p the dimension of the null space of the penalty, $\omega = (\rho, \tau)$ with $\tau = \log\phi$ for Gaussian responses and $\omega = \rho$, $\phi = 1$ for Poisson and binary responses. For Gaussian responses, $-\ell_{\mathbf{c}} = \mathscr{D}_{\mathbf{c}}/(2\phi) + (N_{\mathbf{c}}/2)\log(2\pi\phi)$ with the weighted residual sum of squares $\mathscr{D}_{\mathbf{c}}$ and $N_{\mathbf{c}} = \sum_i c_i D_i$. Write $\mathbf{F}_{\mathbf{c}} = \mathbf{X}^\top\mathbf{W}_{\mathbf{c}}\mathbf{X} + \mathrm{S}$. Because $\hat\theta$ maximizes the penalized log-likelihood, its derivative does not enter the first two terms (envelope theorem), and

$$\frac{\partial\mathscr{V}}{\partial\rho_k} = \tfrac{1}{2}\hat\theta^\top\dot{\mathrm{S}}_k\hat\theta + \tfrac{1}{2}\,\mathrm{tr}(\mathbf{F}_{\mathbf{c}}^{-1}\dot{\mathbf{F}}_k) - \tfrac{1}{2}\,\mathrm{tr}(\mathrm{S}^+\dot{\mathrm{S}}_k), \qquad \dot{\mathbf{F}}_k = \dot{\mathrm{S}}_k + \mathbf{X}^\top\mathrm{diag}(c_{i(n)}w_n'\,\mathbf{x}_n^\top\mathbf{J}_k)\mathbf{X},$$

$$\frac{\partial\mathscr{V}}{\partial\tau} = -\frac{1}{2}\Big(\frac{\mathscr{D}_{\mathbf{c}}}{\phi} + \hat\theta^\top\mathrm{S}\hat\theta\Big) + \frac{N_{\mathbf{c}} - M_p}{2} \quad \text{(Gaussian)},$$

where $w_n' = dw_n/d\eta_n$ (zero for Gaussian responses), $\mathbf{x}_n^\top$ is row n of $\mathbf{X}$ and $\mathbf{J}_k = -\mathbf{F}_{\mathbf{c}}^{-1}\dot{\mathrm{S}}_k\hat\theta$. In the τ-derivative, $\mathrm{S} = \sum_k(\lambda_k/\phi)\mathrm{S}_k$ moves with ϕ at fixed λ, and the two log-determinants contribute $-p/2$ and $(p - M_p)/2$. $\mathbf{H}_\omega$ is mgcv's Hessian of this criterion.

Derivation: curve scores. At $\mathbf{c} = \mathbf{1}$, the fixed-λ influence of curve i on $\hat\theta$ is $\delta_i = \partial\hat\theta/\partial c_i = \mathbf{V}_p\mathbf{u}_i$ (implicit differentiation of $\sum_h c_h\mathbf{u}_h = \mathrm{S}\hat\theta$), and $\partial\mathbf{F}/\partial c_i = \mathbf{X}^\top\mathrm{diag}(\mathbf{b}_i)\mathbf{X}$ with $b_{i,n} = \mathbb{1}[n \in i]\,w_n + w_n'\,\mathbf{x}_n^\top\delta_i$. With $h_n = \mathbf{x}_n^\top\mathbf{V}_p\mathbf{x}_n$, $q_{kn} = \mathbf{x}_n^\top\mathbf{V}_p\dot{\mathbf{F}}_k\mathbf{V}_p\mathbf{x}_n$ and $\tilde{\mathbf{a}} = \mathbf{V}_p\mathbf{X}^\top(\mathbf{w}' \circ \mathbf{h})$, differentiating the three parts of $\partial\mathscr{V}/\partial\rho_k$ in c_i gives

$$s_{i,\rho_k} = \hat\theta^\top\dot{\mathrm{S}}_k\delta_i + \tfrac{1}{2}\Big[-\sum_n b_{i,n}q_{kn} + \sum_{n\in i} w_n'(\mathbf{x}_n^\top\mathbf{J}_k)h_n + \sum_n w_n''(\mathbf{x}_n^\top\delta_i)(\mathbf{x}_n^\top\mathbf{J}_k)h_n - \sum_n b_{i,n}(\mathbf{x}_n^\top\tilde{\mathbf{a}})(\mathbf{x}_n^\top\mathbf{J}_k) - \tilde{\mathbf{a}}^\top\dot{\mathrm{S}}_k\delta_i\Big],$$

$$s_{i,\tau} = -\frac{\|\mathrm{y}_i - \mu_i\|^2}{2\phi} + \frac{D_i}{2} \quad \text{(Gaussian)}.$$

The first term of s_{i,ρ_k} is the route through $\hat\theta^\top\dot{\mathrm{S}}_k\hat\theta$. The bracket is the route through $\log|\mathbf{F}|$: the trace term changes with $\mathbf{c}$ through $\mathbf{F}^{-1}$ (first sum), through the weights of curve i and their change with $\hat\eta$ (second and third sums), and through $\mathbf{J}_k$, whose derivative is $\partial\mathbf{J}_k/\partial c_i = -\mathbf{V}_p\mathbf{X}^\top\mathrm{diag}(\mathbf{b}_i)\mathbf{X}\mathbf{J}_k - \mathbf{V}_p\dot{\mathrm{S}}_k\delta_i$ (last two terms). For Gaussian responses, only the first sum remains; it varies over curves because their designs $\mathbf{X}_i$ differ (functional covariates). The scale enters through the curve's share of the residual sum of squares and of N. The cost is $O(GNp + KNp^2)$ per fit. The penalty and M_p do not scale with $\mathbf{c}$, so the scores are centered as in S4.1(e), and the estimators (IJ, OS, exact curve jackknife, regularized) and the two ways of adding the term to CL2 are those of S4.1(e). For Poisson and binary responses, OS evaluates the leave-out gradient at the converged penalized IRLS solution without curve i.

Derivation: non-linear variants. Because $\mathrm{Var}(\log\hat\lambda)$ of the surface penalty is large and far from normal under dependence, $\mathbf{J}\Delta_{i,\rho}$ can be a poor approximation of the change of $\hat\theta$. The non-linear variants replace it by $\hat\theta_{(-i)}(\hat\rho + \Delta_{i,\rho}) - \hat\theta_{(-i)}(\hat\rho)$, refitted by penalized IRLS at fixed smoothing parameters, for the one-step and the exact jackknife changes.

Result. In the main-factorial settings with dependent errors, the term raises REML + CL2's coverage of $\beta(s,t)$ from 0.93 to 0.94 ($G = 40$) and from 0.95 to 0.96 ($G = 100$), at 1.07–1.09 times the width, with interval score ratios

0.996–1.009. The within-setting rank correlation between a replicate's standard-error inflation and CL2's share of missed grid points in that replicate is -0.24 at $G = 40$ and -0.11 at $G = 100$. As a diagnostic, a refit with the smoothing parameters fixed at the setting's geometric-mean REML value, computed from the same replicates, is narrower than CL2 and covers better (0.94, interval score ratio 0.88 at $G = 40$); it changes both the typical amount of smoothing and its variability, so it does not separate the two.

On the plasmode data, the term raises REML + CL2's coverage of β by 0.0–1.8 pp per combination at interval score ratios 0.96–1.11.

Reason. The term raises coverage but not the interval score: the added width is not concentrated in the replicates where CL2 misses (the rank correlation above is negative or near zero), so coverage rises at about the cost of the added width. These results do not exclude a variance contribution to CL2's shortfall; they show that this first-order term does not target it.

Table 46: Smoothing-parameter terms for REML + CL2, main-factorial settings with dependent errors at the default signal (200 replicates): coverage, then width / interval score relative to CL2. mgcv's term: mgcv's smoothing-parameter correction added to CL2. λ fixed: refit at the setting's geometric-mean REML smoothing parameters, computed from the same replicates (a diagnostic).

estimand	G	CL2	CL2 + mgcv's term	CL2 + curve-robust λ term	λ fixed (diagnostic)
β(s,t)	40	0.929 1.00 / 1.000	0.936 1.02 / 0.984	0.945 1.09 / 0.996	0.945 0.99 / 0.880
β(s,t)	100	0.946 1.00 / 1.000	0.951 1.02 / 0.993	0.958 1.07 / 1.009	0.955 1.00 / 0.936
E(Y | X)	40	0.929 1.00 / 1.000	0.933 1.01 / 0.991	0.938 1.04 / 0.992	0.941 1.01 / 0.951
E(Y | X)	100	0.943 1.00 / 1.000	0.945 1.01 / 0.997	0.947 1.02 / 1.000	0.949 1.01 / 0.978

S5 Misspecified truths and basis size

S5.1 Truths outside the spline space

Three truths are scored against the exact functions, not their basis projections (Gaussian, $R^2 = 0.5$, $D = 61$, 200 replicates): T0 the study's smooth $\beta(s,t)$ (control); T1 = T0 plus a two-dimensional Gaussian bump (SD 0.05, center (0.3, 0.6)); T2 = T0 plus $\sum_{k=12}^{16} \sqrt{2}\cos(k\pi s)\sin(\pi t)$, directions the covariates barely excite. Each feature carries 20% of the β-term signal. Settings: truth × errors and G {independent errors with $G = 100$, smooth errors with $G = 100$, smooth errors with $G = 40$} × basis {default; 4× finer knots: surface 23×31, $k = 39$ along t}. The bump region is the set of grid points within two SDs of the bump center.

For the control T0, scoring against the exact function instead of its projection changes nothing of note: with the default basis, REML + CL2 covers β at 0.93–0.94 under smooth errors. With the default basis, approximation bias breaks every interval under T2 (REML + CL2 covers β, over the three error and G settings, at 0.30–0.56, NCV + CL2 at 0.27–0.33) and inside T1's bump (REML + CL2 0.17–0.58). With the basis with 4× finer knots and dependent (smooth) errors, REML + CL2 recovers: β 0.93–0.95, bump region 0.93–0.94. NCV + CL2 stays biased (T2, smooth errors: 0.41–0.79). REML's coverage with the finer basis comes at the cost of noisy estimates: its mean squared error for β under smooth errors is 11.8–66.9 against 0.12–39.47 for NCV.

Under independent errors, the finer basis does not restore REML + CL2 (T2 β 0.86). Coverage of $\alpha(t)$ and $\gamma(t)$ is unaffected (REML + CL2 0.93–0.98).

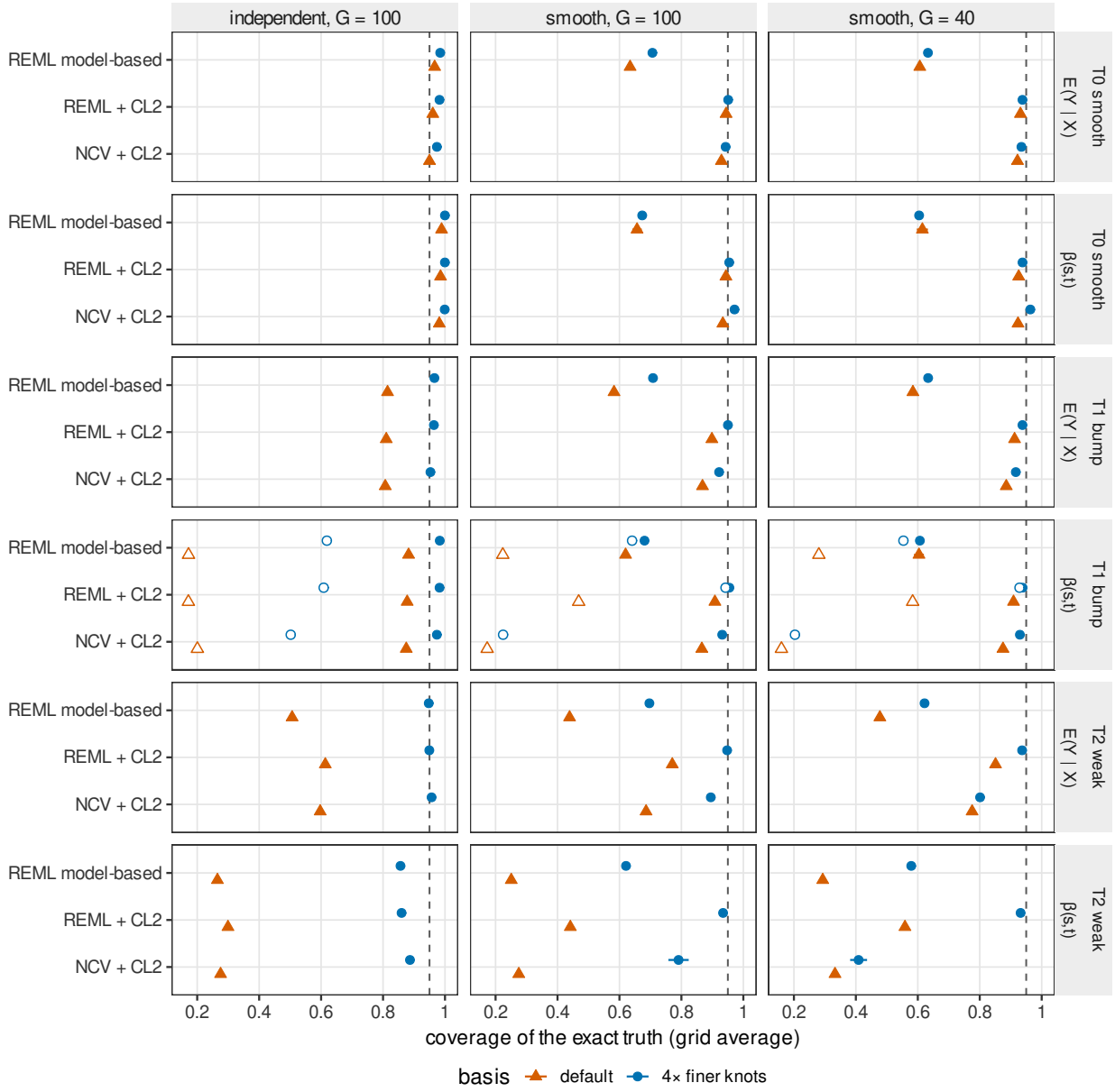


Figure 26: Misspecified truths: coverage of the exact $\beta(s,t)$ and mean by truth (rows), errors and G (columns), basis (color and symbol) and interval; open symbols in the T1 row: coverage inside the bump region. Bars: ±2 MC SE; dashed: nominal 0.95.

Table 47: Misspecified truths: feature amplitude, the feature's share of $\int \beta^2$, and the relative L_2 error and residual signal share after projecting β onto the default and surface bases with 4× finer knots.

truth	amplitude	share $\int\beta^2$	rel. L2 err. default	resid. signal default	rel. L2 err. 4× finer knots	resid. signal 4× finer knots
T0 smooth	0.00	0.000	0.000	0.000	0.000	0.000
T1 bump	9.29	0.206	0.261	0.023	0.004	0.000
T2 weak	3.75	0.925	0.891	0.222	0.097	0.001

Table 48: Misspecified truths: coverage (MC SE) of the exact truth per estimand, truth, setting, basis and interval.

estimand	truth	setting	basis	REML model-based	REML + CL2	REML + CL2 (freq.)	NCV + CL2
α(t)	T0 smooth	iid, G = 100	default	0.956 (0.004)	0.956 (0.004)	0.941 (0.005)	0.945 (0.005)
α(t)	T0 smooth	iid, G = 100	4× finer knots	0.978 (0.002)	0.975 (0.002)	0.940 (0.004)	0.967 (0.004)
α(t)	T0 smooth	smooth, G = 100	default	0.641 (0.011)	0.950 (0.005)	0.948 (0.005)	0.941 (0.006)
α(t)	T0 smooth	smooth, G = 100	4× finer knots	0.739 (0.010)	0.954 (0.004)	0.942 (0.005)	0.944 (0.007)
α(t)	T0 smooth	smooth, G = 40	default	0.614 (0.013)	0.937 (0.007)	0.933 (0.007)	0.919 (0.008)
α(t)	T0 smooth	smooth, G = 40	4× finer knots	0.640 (0.012)	0.938 (0.006)	0.926 (0.007)	0.932 (0.009)
α(t)	T1 bump	iid, G = 100	default	0.953 (0.004)	0.954 (0.004)	0.941 (0.005)	0.947 (0.005)
α(t)	T1 bump	iid, G = 100	4× finer knots	0.978 (0.002)	0.974 (0.003)	0.942 (0.004)	0.966 (0.004)
α(t)	T1 bump	smooth, G = 100	default	0.645 (0.011)	0.949 (0.005)	0.947 (0.005)	0.937 (0.006)
α(t)	T1 bump	smooth, G = 100	4× finer knots	0.739 (0.010)	0.955 (0.004)	0.942 (0.005)	0.943 (0.007)
α(t)	T1 bump	smooth, G = 40	default	0.616 (0.013)	0.935 (0.007)	0.932 (0.007)	0.919 (0.008)
α(t)	T1 bump	smooth, G = 40	4× finer knots	0.641 (0.012)	0.938 (0.006)	0.926 (0.007)	0.927 (0.009)
α(t)	T2 weak	iid, G = 100	default	0.842 (0.012)	0.945 (0.007)	0.937 (0.007)	0.941 (0.007)
α(t)	T2 weak	iid, G = 100	4× finer knots	0.975 (0.003)	0.973 (0.003)	0.939 (0.004)	0.968 (0.004)
α(t)	T2 weak	smooth, G = 100	default	0.634 (0.013)	0.946 (0.006)	0.944 (0.006)	0.936 (0.006)
α(t)	T2 weak	smooth, G = 100	4× finer knots	0.732 (0.010)	0.955 (0.004)	0.941 (0.005)	0.941 (0.007)
α(t)	T2 weak	smooth, G = 40	default	0.597 (0.013)	0.936 (0.007)	0.932 (0.008)	0.929 (0.008)
α(t)	T2 weak	smooth, G = 40	4× finer knots	0.629 (0.013)	0.936 (0.006)	0.926 (0.007)	0.939 (0.009)
β(s,t)	T0 smooth	iid, G = 100	default	0.989 (0.001)	0.986 (0.001)	0.940 (0.003)	0.982 (0.001)
β(s,t)	T0 smooth	iid, G = 100	4× finer knots	1.000 (0.000)	1.000 (0.000)	0.942 (0.002)	0.999 (0.000)
β(s,t)	T0 smooth	smooth, G = 100	default	0.657 (0.008)	0.944 (0.003)	0.935 (0.003)	0.934 (0.004)
β(s,t)	T0 smooth	smooth, G = 100	4× finer knots	0.673 (0.006)	0.954 (0.002)	0.939 (0.002)	0.972 (0.003)
β(s,t)	T0 smooth	smooth, G = 40	default	0.615 (0.009)	0.926 (0.004)	0.915 (0.004)	0.924 (0.005)
β(s,t)	T0 smooth	smooth, G = 40	4× finer knots	0.604 (0.006)	0.938 (0.003)	0.923 (0.003)	0.963 (0.003)
β(s,t)	T1 bump	iid, G = 100	default	0.882 (0.001)	0.878 (0.001)	0.823 (0.002)	0.875 (0.002)
β(s,t)	T1 bump	iid, G = 100	4× finer knots	0.983 (0.000)	0.982 (0.000)	0.907 (0.001)	0.974 (0.001)
β(s,t)	T1 bump	smooth, G = 100	default	0.620 (0.006)	0.908 (0.003)	0.899 (0.003)	0.866 (0.003)
β(s,t)	T1 bump	smooth, G = 100	4× finer knots	0.681 (0.005)	0.955 (0.002)	0.939 (0.002)	0.932 (0.002)
β(s,t)	T1 bump	smooth, G = 40	default	0.604 (0.008)	0.909 (0.003)	0.898 (0.004)	0.875 (0.004)
β(s,t)	T1 bump	smooth, G = 40	4× finer knots	0.607 (0.006)	0.938 (0.003)	0.924 (0.003)	0.930 (0.003)
β(s,t)	T2 weak	iid, G = 100	default	0.265 (0.002)	0.299 (0.002)	0.273 (0.002)	0.275 (0.005)
β(s,t)	T2 weak	iid, G = 100	4× finer knots	0.856 (0.002)	0.860 (0.002)	0.769 (0.003)	0.887 (0.004)
β(s,t)	T2 weak	smooth, G = 100	default	0.251 (0.002)	0.441 (0.003)	0.433 (0.004)	0.275 (0.004)
β(s,t)	T2 weak	smooth, G = 100	4× finer knots	0.621 (0.005)	0.935 (0.002)	0.920 (0.002)	0.791 (0.016)
β(s,t)	T2 weak	smooth, G = 40	default	0.293 (0.003)	0.559 (0.004)	0.548 (0.004)	0.333 (0.004)
β(s,t)	T2 weak	smooth, G = 40	4× finer knots	0.579 (0.005)	0.932 (0.003)	0.918 (0.003)	0.409 (0.014)
γ(t)	T0 smooth	iid, G = 100	default	0.960 (0.004)	0.950 (0.005)	0.937 (0.005)	0.946 (0.005)
γ(t)	T0 smooth	iid, G = 100	4× finer knots	0.976 (0.002)	0.973 (0.003)	0.936 (0.004)	0.966 (0.004)
γ(t)	T0 smooth	smooth, G = 100	default	0.655 (0.012)	0.948 (0.005)	0.945 (0.006)	0.935 (0.006)
γ(t)	T0 smooth	smooth, G = 100	4× finer knots	0.734 (0.011)	0.950 (0.005)	0.939 (0.005)	0.935 (0.007)
γ(t)	T0 smooth	smooth, G = 40	default	0.621 (0.012)	0.946 (0.006)	0.941 (0.006)	0.927 (0.007)
γ(t)	T0 smooth	smooth, G = 40	4× finer knots	0.643 (0.013)	0.932 (0.007)	0.921 (0.007)	0.929 (0.010)
γ(t)	T1 bump	iid, G = 100	default	0.956 (0.004)	0.949 (0.004)	0.937 (0.005)	0.944 (0.005)
γ(t)	T1 bump	iid, G = 100	4× finer knots	0.976 (0.002)	0.975 (0.002)	0.936 (0.004)	0.968 (0.004)
γ(t)	T1 bump	smooth, G = 100	default	0.657 (0.012)	0.950 (0.005)	0.947 (0.005)	0.936 (0.006)
γ(t)	T1 bump	smooth, G = 100	4× finer knots	0.736 (0.011)	0.950 (0.005)	0.937 (0.005)	0.939 (0.007)
γ(t)	T1 bump	smooth, G = 40	default	0.630 (0.013)	0.947 (0.006)	0.942 (0.006)	0.927 (0.007)
γ(t)	T1 bump	smooth, G = 40	4× finer knots	0.640 (0.013)	0.932 (0.007)	0.922 (0.007)	0.924 (0.010)
γ(t)	T2 weak	iid, G = 100	default	0.839 (0.012)	0.946 (0.008)	0.935 (0.009)	0.930 (0.009)
γ(t)	T2 weak	iid, G = 100	4× finer knots	0.975 (0.003)	0.974 (0.003)	0.935 (0.004)	0.966 (0.004)
γ(t)	T2 weak	smooth, G = 100	default	0.622 (0.013)	0.944 (0.007)	0.940 (0.007)	0.933 (0.007)
γ(t)	T2 weak	smooth, G = 100	4× finer knots	0.729 (0.011)	0.950 (0.005)	0.937 (0.005)	0.931 (0.008)
γ(t)	T2 weak	smooth, G = 40	default	0.604 (0.013)	0.945 (0.007)	0.940 (0.007)	0.936 (0.007)
γ(t)	T2 weak	smooth, G = 40	4× finer knots	0.635 (0.013)	0.934 (0.007)	0.922 (0.007)	0.931 (0.010)
E(Y | X)	T0 smooth	iid, G = 100	default	0.967 (0.001)	0.961 (0.002)	0.939 (0.002)	0.950 (0.002)
E(Y | X)	T0 smooth	iid, G = 100	4× finer knots	0.985 (0.001)	0.982 (0.001)	0.938 (0.001)	0.974 (0.001)
E(Y | X)	T0 smooth	smooth, G = 100	default	0.634 (0.005)	0.944 (0.002)	0.940 (0.002)	0.929 (0.003)
E(Y | X)	T0 smooth	smooth, G = 100	4× finer knots	0.706 (0.003)	0.951 (0.002)	0.939 (0.002)	0.943 (0.003)
E(Y | X)	T0 smooth	smooth, G = 40	default	0.606 (0.005)	0.931 (0.003)	0.926 (0.003)	0.922 (0.004)
E(Y | X)	T0 smooth	smooth, G = 40	4× finer knots	0.633 (0.004)	0.938 (0.002)	0.926 (0.002)	0.935 (0.004)
E(Y | X)	T1 bump	iid, G = 100	default	0.815 (0.001)	0.810 (0.002)	0.781 (0.002)	0.807 (0.002)
E(Y | X)	T1 bump	iid, G = 100	4× finer knots	0.966 (0.001)	0.964 (0.001)	0.914 (0.001)	0.953 (0.001)
E(Y | X)	T1 bump	smooth, G = 100	default	0.582 (0.004)	0.898 (0.002)	0.894 (0.002)	0.868 (0.002)
E(Y | X)	T1 bump	smooth, G = 100	4× finer knots	0.708 (0.003)	0.950 (0.001)	0.938 (0.002)	0.922 (0.003)
E(Y | X)	T1 bump	smooth, G = 40	default	0.584 (0.005)	0.912 (0.003)	0.906 (0.003)	0.886 (0.004)
E(Y | X)	T1 bump	smooth, G = 40	4× finer knots	0.633 (0.004)	0.938 (0.002)	0.926 (0.002)	0.916 (0.004)
E(Y | X)	T2 weak	iid, G = 100	default	0.506 (0.002)	0.614 (0.003)	0.591 (0.003)	0.597 (0.004)
E(Y | X)	T2 weak	iid, G = 100	4× finer knots	0.948 (0.001)	0.949 (0.001)	0.896 (0.002)	0.957 (0.002)
E(Y | X)	T2 weak	smooth, G = 100	default	0.439 (0.003)	0.771 (0.003)	0.765 (0.003)	0.686 (0.003)
E(Y | X)	T2 weak	smooth, G = 100	4× finer knots	0.697 (0.003)	0.948 (0.002)	0.937 (0.002)	0.895 (0.007)
E(Y | X)	T2 weak	smooth, G = 40	default	0.478 (0.004)	0.851 (0.003)	0.844 (0.004)	0.775 (0.004)
E(Y | X)	T2 weak	smooth, G = 40	4× finer knots	0.622 (0.004)	0.937 (0.002)	0.926 (0.002)	0.801 (0.005)

Table 49: Misspecified truths: MSE and squared-bias share of the MSE for the REML and NCV estimates.

estimand	truth	setting	basis	mse_REML	mse_NCV	bias_share_REML	bias_share_NCV
β(s,t)	T0 smooth	iid, G = 100	default	0.0412	0.0269	0.0439	0.2129
β(s,t)	T0 smooth	iid, G = 100	4× finer knots	0.0895	0.0409	0.0427	0.2603
β(s,t)	T0 smooth	smooth, G = 100	default	0.7524	0.1177	0.0075	0.1284
β(s,t)	T0 smooth	smooth, G = 100	4× finer knots	11.7721	0.1183	0.0026	0.1722
β(s,t)	T0 smooth	smooth, G = 40	default	2.7504	0.2633	0.0094	0.1114
β(s,t)	T0 smooth	smooth, G = 40	4× finer knots	52.7704	0.2551	-0.0011	0.1687
β(s,t)	T1 bump	iid, G = 100	default	0.3312	0.3636	0.8356	0.6942
β(s,t)	T1 bump	iid, G = 100	4× finer knots	0.2338	0.2371	0.4852	0.5810
β(s,t)	T1 bump	smooth, G = 100	default	1.1114	0.5116	0.2153	0.6274
β(s,t)	T1 bump	smooth, G = 100	4× finer knots	12.5268	0.4226	0.0024	0.6029
β(s,t)	T1 bump	smooth, G = 40	default	3.1364	0.6563	0.0820	0.5801
β(s,t)	T1 bump	smooth, G = 40	4× finer knots	54.3339	0.6555	-0.0011	0.5126
β(s,t)	T2 weak	iid, G = 100	default	39.4160	38.2757	0.9942	0.9867
β(s,t)	T2 weak	iid, G = 100	4× finer knots	7.7238	9.8557	0.8074	0.5261
β(s,t)	T2 weak	smooth, G = 100	default	38.1191	39.3778	0.9588	0.9904
β(s,t)	T2 weak	smooth, G = 100	4× finer knots	20.5659	18.9438	0.1660	0.4700
β(s,t)	T2 weak	smooth, G = 40	default	40.5430	39.8773	0.8820	0.9836
β(s,t)	T2 weak	smooth, G = 40	4× finer knots	66.8557	39.4673	0.0453	0.8759
E(Y | X)	T0 smooth	iid, G = 100	default	0.0084	0.0081	0.0425	0.1337
E(Y | X)	T0 smooth	iid, G = 100	4× finer knots	0.0138	0.0113	0.0455	0.2120
E(Y | X)	T0 smooth	smooth, G = 100	default	0.0494	0.0340	0.0049	0.1024
E(Y | X)	T0 smooth	smooth, G = 100	4× finer knots	0.1209	0.0374	0.0031	0.1464
E(Y | X)	T0 smooth	smooth, G = 40	default	0.1415	0.0832	0.0048	0.0964
E(Y | X)	T0 smooth	smooth, G = 40	4× finer knots	0.4706	0.0900	0.0006	0.1455
E(Y | X)	T1 bump	iid, G = 100	default	0.0306	0.0311	0.7026	0.6733
E(Y | X)	T1 bump	iid, G = 100	4× finer knots	0.0185	0.0177	0.1959	0.3172
E(Y | X)	T1 bump	smooth, G = 100	default	0.0719	0.0629	0.2754	0.4329
E(Y | X)	T1 bump	smooth, G = 100	4× finer knots	0.1270	0.0498	0.0088	0.2677
E(Y | X)	T1 bump	smooth, G = 40	default	0.1671	0.1151	0.1259	0.3041
E(Y | X)	T1 bump	smooth, G = 40	4× finer knots	0.4819	0.1070	0.0018	0.2242
E(Y | X)	T2 weak	iid, G = 100	default	0.1408	0.1354	0.8539	0.8460
E(Y | X)	T2 weak	iid, G = 100	4× finer knots	0.0354	0.0340	0.2890	0.1854
E(Y | X)	T2 weak	smooth, G = 100	default	0.1742	0.1655	0.6219	0.7483
E(Y | X)	T2 weak	smooth, G = 100	4× finer knots	0.1471	0.1115	0.0253	0.2664
E(Y | X)	T2 weak	smooth, G = 40	default	0.3001	0.2349	0.3586	0.5629
E(Y | X)	T2 weak	smooth, G = 40	4× finer knots	0.5287	0.2455	0.0086	0.5093

S5.2 Basis size with representable truths

With nested larger bases (2× and 4× finer knots; the truths stay representable), REML's surface EDF under smooth errors rises from 27–42 (default) to 113–176 (4× finer knots) across families, NCV's from 12–21 to 18–41. REML + CL2 stays near nominal (0.95–0.96 with 4× finer knots) but its intervals widen with REML's variance: mean width for Gaussian $\beta(s,t)$ grows from 2.81 to 13.15. With 4× finer knots and smooth errors, the NCV/REML ratio of mean squared errors for β falls to 0.01.

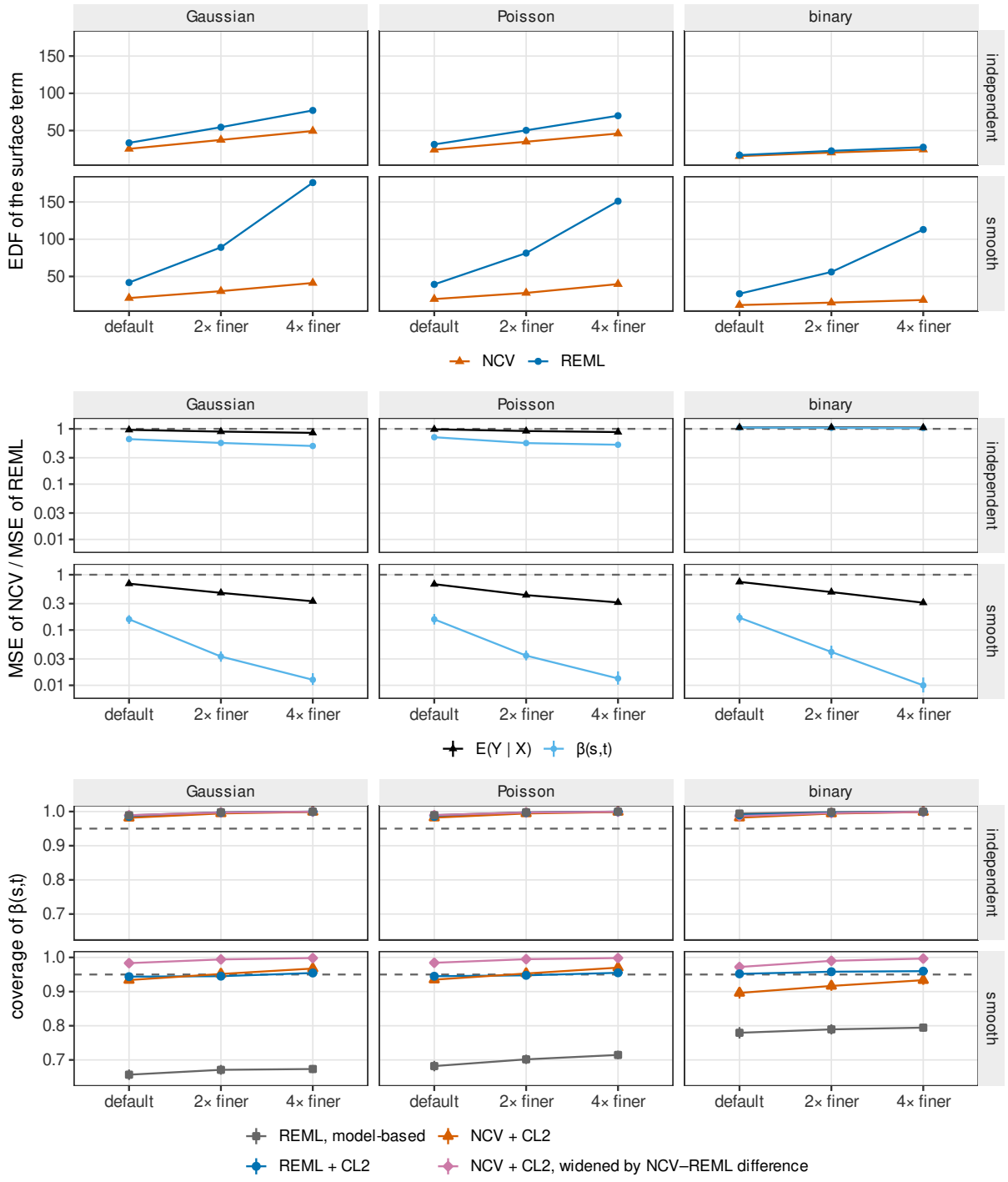


Figure 27: Basis size ($G = 100$, default signal, smooth truth), by errors and family: EDF of the surface term under REML and NCV (top), NCV/REML ratio of mean squared errors with 95% bootstrap intervals (middle; log scale; dashed: equal error) and coverage of $\beta(s,t)$ (bottom; bars: ±2 MC SE; dashed: nominal 0.95).

Table 50: Basis size: coverage, interval score and mean width of $\beta(s,t)$ intervals per interval construction, and the NCV/REML MSE ratio. NCV wid.: NCV + CL2 widened by the NCV–REML difference.

			coverage				interval score				mean width				MSE ratio
family	error	basis	NCV + CL2	NCV wid.	REML + CL2	REML model	NCV + CL2	NCV wid.	REML + CL2	REML model	NCV + CL2	NCV wid.	REML + CL2	REML model	NCV/REML
Gaussian	independent	default	0.982	0.989	0.986	0.989	0.764	0.819	1.016	1.023	0.724	0.796	0.982	0.996	0.652
Gaussian	independent	2× finer knots	0.994	0.997	0.997	0.998	1.044	1.141	1.547	1.566	1.029	1.132	1.539	1.560	0.556
Gaussian	independent	4× finer knots	0.999	0.999	1.000	1.000	1.358	1.471	2.154	2.183	1.355	1.469	2.153	2.182	0.490
Gaussian	smooth	default	0.934	0.983	0.944	0.657	1.480	2.418	3.493	7.226	1.094	2.314	2.814	1.374	0.156
Gaussian	smooth	2× finer knots	0.952	0.994	0.945	0.671	1.557	5.187	7.803	15.488	1.266	5.150	6.244	3.110	0.033
Gaussian	smooth	4× finer knots	0.967	0.998	0.954	0.673	1.695	10.327	15.510	29.378	1.479	10.313	13.149	6.407	0.013
Poisson	independent	default	0.982	0.990	0.986	0.989	0.438	0.467	0.573	0.578	0.413	0.454	0.553	0.563	0.705
Poisson	independent	2× finer knots	0.994	0.997	0.997	0.998	0.577	0.627	0.854	0.867	0.568	0.622	0.849	0.864	0.553
Poisson	independent	4× finer knots	0.999	0.999	1.000	1.000	0.747	0.805	1.173	1.192	0.745	0.804	1.172	1.192	0.515
Poisson	smooth	default	0.935	0.984	0.945	0.682	0.801	1.345	1.941	3.741	0.597	1.294	1.567	0.804	0.156
Poisson	smooth	2× finer knots	0.953	0.995	0.948	0.702	0.833	2.806	4.212	7.704	0.687	2.790	3.427	1.804	0.034
Poisson	smooth	4× finer knots	0.970	0.998	0.955	0.715	0.910	5.445	8.180	13.982	0.806	5.436	6.942	3.663	0.013
binary	independent	default	0.982	0.987	0.990	0.994	1.300	1.318	1.410	1.464	1.227	1.274	1.372	1.444	1.055
binary	independent	2× finer knots	0.994	0.996	0.997	0.998	1.638	1.676	1.856	1.943	1.611	1.659	1.844	1.937	1.046
binary	independent	4× finer knots	0.999	0.999	0.999	1.000	2.020	2.060	2.325	2.439	2.015	2.057	2.323	2.439	1.029
binary	smooth	default	0.896	0.972	0.952	0.779	2.545	3.742	5.227	7.995	1.512	3.465	4.305	2.648	0.167
binary	smooth	2× finer knots	0.917	0.990	0.958	0.789	2.533	7.752	11.569	16.642	1.702	7.659	9.790	5.961	0.040
binary	smooth	4× finer knots	0.933	0.997	0.960	0.795	2.593	16.870	25.640	35.337	1.896	16.838	21.803	13.305	0.010

S6 Data and code

Recommended analysis. The procedures recommended in Section 6 of the paper are implemented in the software of the study (refund at commit 79a346fb, patched mgcv 1.9-5; below) and will be the defaults of forthcoming refund releases. With these defaults and the bases of the study, the analysis reads as follows. The evaluated interval for $\alpha(t)$ is that of the full intercept curve, the scalar intercept plus the centered `Intercept(yindex)` term.

```
library(refund)
bs_t <- list(bs = "ps", k = 12, m = c(2, 1))
bs_ff <- list(bs = "ps", k = c(8, 10), m = list(c(2, 1), c(2, 1)))
form <- Y ~ ff(X, xind = s, splinepars = bs_ff) + z
# inference: REML fit, CL2 with Satterthwaite critical values (the defaults)
fit_reml <- pffr(form, yind = t, data = d, bs.yindex = bs_t, bs.int = bs_t)
ci_reml <- coef(fit_reml, ci = "pointwise")
# shape of the surface: curve-blocked NCV estimate
fit_ncv <- pffr(form, yind = t, data = d, bs.yindex = bs_t, bs.int = bs_t,
                method = "NCV")
est_ncv <- coef(fit_ncv)
# covariate with few principal components: ffpc() term with a small K
fit_pc <- pffr(Y ~ ffpc(X, xind = s, splinepars = bs_t,
                        decomppars = list(pve = 0.95, useSymm = TRUE)) + z,
               yind = t, data = d, bs.yindex = bs_t, bs.int = bs_t)
ci_pc <- coef(fit_pc, ci = "pointwise")
# several curves per subject: cluster by subject
fit_rm <- pffr(form, yind = t, data = d, bs.yindex = bs_t, bs.int = bs_t,
               cluster = subject)
```

Software. The study library ran R 4.6.1 with 101 packages (full list with versions in `paper/data/restart-lib-versions.txt`); the key versions are in the next table. refund is the development branch `pffr-refactor` on GitHub (refunders/refund). The study build is commit 79a346fb (reported version 0.1-40), installed by its immutable commit with `remotes::install_github("refunders/refund@79a346fb")`.

mgcv is a pre-release build of version 1.9-5, in two builds. Both exact source builds are deposited in the code archive (`software/mgcv_1.9-5.tar.gz`, unpatched; `software/mgcv_1.9-5-patched.tar.gz`, patched), together with the patch (`software/mgcv-1.9-5-ncv-vc-and-cxwyd.patch`); their SHA-256 checksums are in the table below. The unpatched build ran the main factorial and the heteroskedastic and low-rank-covariate extension settings. The patched build ran the basis-size extension (study settings 112–123), the settings with 20 curves and with a zero surface (study settings 124–152), the whole plasmode study and all later experiments (covariance components and Satterthwaite degrees of freedom, Appendices A.6 and A.7 of the paper, S1.7; the smoothing-parameter terms and NCV-centered corrections, S4; the comparators, S3.2 and S3.3; the misspecified truths, S5.1; the principal component term, S1.9).

The patch changes $\mathbf{V}_c$ of NCV fits (in the unpatched build it holds only the smoothing-parameter correction and is not a valid covariance; with the patch, it is $\mathbf{V}_p$ plus that correction) and two `bam()` paths (NCV, and non-Gaussian discrete fits with AR(1) errors). The REML, NCV and CL2 computations evaluated here use none of these; the study uses $\mathbf{V}_p$ as the model-based covariance of NCV fits throughout, so only the row "... with $\hat{\lambda}$ correction" in S4.2, which uses $\mathbf{V}_c$ of the NCV fit, depends on the patch. Runs on the patched build reproduced the stored results of the unpatched build to about 10^{-9}. Functional GEE used fastFGEE 0.2.2 from CRAN. No container or `renv` lockfile is provided; the package list and checksums are the record of the computing environment.

Reproducing one synthetic setting. From the root directory of the code archive, with R 4.6.1, single-threaded BLAS, mgcv installed from the deposited source build and refund from commit 79a346fb, the following runs the 200 replicates of study setting 12 (Gaussian responses, smooth errors, $G = 100$, default signal) and summarizes them:

```
export OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1
export REFUND_SHA=79a346fb8cc1220a2a6fd2dc8dd3e33166a21061
R CMD INSTALL software/mgcv_1.9-5.tar.gz
Rscript -e 'remotes::install_github("refunders/refund@79a346fb")'
for run in 1 2; do RESTART_CELLS=12 SLURM_ARRAY_TASK_ID=1 Rscript scripts/run-array.R 100 $run results-12; done
Rscript -e 'source("scripts/setup.R"); readr::write_csv(summarise_study("results-12")$cells, "cells-12.csv")'
```

`scripts/run-array.R` (arguments: chunk size, half of the replicates, output directory) runs the tasks of the selected setting. `REFUND_SHA` declares the refund commit to the study's software check, because a GitHub installation does not record it. The last line summarizes the setting's replicates; `analysis/run-analysis.R` writes the summary tables for the whole study, from which `writeup/RESULTS.qmd` and this supplement compute their exhibits. We checked this procedure: one task run with `scripts/run-array.R` completes, and the summary of the 200 stored replicates of setting 12 reproduces that setting's rows of the study's summary table, mean squared errors identical to all digits.

Table 51: Key package versions of the study library.

Package	Version
R	4.6.1
dplyr	1.2.1
gamm4	0.3-0
ggplot2	4.0.3
lme4	2.0-6
Matrix	1.7-6
mgcv	1.9-5
nlme	3.1-171
refund	0.1-40

Table 52: SHA-256 checksums of the mgcv source archives and the patch.

file	sha256
mgcv 1.9-5, patched build	ad56a0a02f8cf2ab84abe6f0d5271ff6662764e7e5d4eb4afff24143e728cb5d
mgcv 1.9-5, unpatched build	b20330037cdcd78f4c2018a3d64aed946c3ab20de9bd09c8677070d3b9df6fb4
patch	4684be40034776fae7bb402e8c324eee0f11d737032d9ff6cbfc350fb78be306

Exhibits and files. All exhibits are computed from summary files of the study's code archive: the simulation from `summaries/final/` (coverage, mean squared error, relative error, EDF, timing) and the studentized-error and design-effect tables computed from the replicate records; the plasmode study from `summaries/plasmode/`; NCV-centered intervals widened by the NCV–REML difference from `summaries/hybrid/`; Appendix A.6 of the paper (covariance components) from `ablation/summaries/`; S1.9 from `ffpc-pilot/ffpc-lrz-summary.csv` and `ffpc-pilot/ffpc-pilot-summary.csv`; Appendix A.7 of the paper, S1.7, S1.10 and the Satterthwaite results of S1.2 and S2.1 from `satterthwaite/summaries/`; the 20-curve and zero-surface settings (S1.2, S1.8) from `small-g-null/summaries/`; S3.2 from `comparators/summaries/`; S3.3 from `competitors/fgee/summaries/`; S4.1 (d)–(g) from `ncv-rbc/summaries/` (including the `combo2-*` files

for (e)–(g)) and `lambda-correction/summaries-ncv*/`; S4.2 from `ncv-jack-recheck/` and `ncv-jack-recheck/summaries/`; S4.3 from `lambda-correction/summaries/` and `summaries-plasmode/`; S5.1 from `misspec/summaries/`.

Data. The DTI, running, gait, 8-lead ECG, ocean, weather and electricity data are public (sources in Appendix B.2 of the paper). The two cardiology datasets are patient and volunteer data that cannot be shared; only aggregate results are reported, so their plasmode results cannot be reproduced from public data. The replicate records behind the summary files, without those of the cardiology datasets, are available at https://doi.org/10.5281/zenodo.23916388.

Use of generative AI. All code and text, including this supplement, were generated by large language models (details in the paper's statement on the use of generative AI); the author takes full responsibility for the content.

S7 Values shown in figures

The tables of this section list the values plotted in the figures of the paper and of S1, S2, S3.3 and S4. REML model: model-based REML interval; NCV widened: NCV + CL2 widened by the NCV–REML difference; q05: 5% quantile of pointwise coverage.

Table 53: Paper, simulation, all settings with dependent errors, 40 or 100 curves and a non-zero surface: median over settings of the relative error, grid-averaged coverage, q05, relative half-width and detection rate; interval score relative to REML + CL2 (IS ratio; geometric mean over settings).

estimand	interval	settings	rel. error	coverage	q05	half-width	detection	IS ratio
β(s,t)	REML model	69	0.668	0.671	0.575	0.566	0.833	1.956
	REML + CL2	69	0.668	0.943	0.915	1.352	0.542	1.000
	NCV + CL2	69	0.209	0.924	0.840	0.370	0.948	0.409
E(Y | X)	REML model	83	0.278	0.652	0.560	0.267		1.950
	REML + CL2	83	0.278	0.941	0.910	0.577		1.000
	NCV + CL2	83	0.227	0.918	0.845	0.420		0.850
α(t)	REML model	69	0.314	0.655	0.583	0.313		1.965
	REML + CL2	69	0.314	0.947	0.920	0.650		1.000
	NCV + CL2	69	0.280	0.924	0.857	0.568		0.995
γ(t)	REML model	75	0.222	0.663	0.587	0.225	0.989	1.958
	REML + CL2	75	0.222	0.940	0.915	0.461	0.937	1.000
	NCV + CL2	75	0.218	0.920	0.860	0.418	0.958	1.033

Table 54: Paper, zero surface, default signal: pointwise false-positive rate (nominal 0.05) of the model-based REML interval (model), REML + CL2 with the normal (z) and the Satterthwaite (Satt.) critical value and NCV + CL2; root-mean-square size of the REML and NCV estimates relative to the non-zero surface of the corresponding setting.

			false-positive rate				size	
family	errors	G	model	z	Satt.	NCV + CL2	REML	NCV
Gaussian	smooth	40	0.399	0.076	0.052	0.130	0.771	0.113
		100	0.361	0.058	0.049	0.151	0.387	0.064
	independent	40	0.024	0.095	0.075	0.096	0.020	0.032
		100	0.009	0.050	0.046	0.047	0.012	0.019
binary	smooth	40	0.250	0.071	0.048	0.111	2.255	0.452
		100	0.228	0.054	0.046	0.118	1.109	0.229
	independent	40	0.009	0.056	0.029	0.060	0.093	0.157
		100	0.015	0.042	0.031	0.048	0.059	0.091

Table 55: Paper, plasmode study, $\beta(s,t)$, all subjects: median relative error of the REML and NCV estimates; REML + CL2 coverage, q05, relative half-width and detection rate; NCV + CL2 coverage, detection rate and interval score relative to REML + CL2 (IS ratio) (means, medians or geometric means over truth and residual sources). The last three datasets have dependent source curves.

	rel. error		REML + CL2				NCV + CL2		
dataset	REML	NCV	cov.	q05	half-width	det.	cov.	det.	IS ratio
ECG strain	0.717	0.693	0.944	0.901	1.46	0.424	0.904	0.461	1.038
AF trial	0.865	0.714	0.938	0.879	1.82	0.372	0.789	0.533	1.190
running	0.825	0.661	0.919	0.873	1.85	0.637	0.734	0.735	1.497
DTI	0.633	0.469	0.927	0.876	1.17	0.638	0.846	0.772	1.034
gait	1.007	0.765	0.939	0.903	2.39	0.546	0.816	0.650	1.283
ECG 8-lead	2.427	0.806	0.942	0.873	4.88	0.216	0.739	0.568	0.747
ocean	1.035	0.667	0.925	0.887	2.25	0.450	0.708	0.528	1.430
weather	1.007	0.678	0.882	0.833	1.98	0.321	0.608	0.538	1.838
electricity	3.418	0.815	0.927	0.881	5.72	0.076	0.611	0.512	0.863

Table 56: Paper, Appendix A.6, covariance components, $\beta(s,t)$: grid-averaged coverage of nominal 95% pointwise intervals, main factorial settings at the default signal level.

family	error	G	model	CR1	CL2, shortcut	CL2	CL2, freq.	CL2, t(G-1)	CL2, fixed λ
Gaussian	independent	40	0.992	0.983	0.988	0.989	0.930	0.991	0.989
		100	0.989	0.983	0.985	0.986	0.940	0.987	0.986
	rough stationary	40	0.638	0.889	0.924	0.927	0.915	0.935	0.939
		100	0.672	0.929	0.940	0.940	0.931	0.944	0.949
	smooth	40	0.615	0.886	0.923	0.926	0.915	0.934	0.940
		100	0.657	0.931	0.943	0.944	0.935	0.947	0.951
Poisson	independent	40	0.993	0.984	0.988	0.989	0.928	0.991	0.990
		100	0.989	0.983	0.985	0.986	0.937	0.987	0.986
	rough stationary	40	0.684	0.887	0.924	0.927	0.911	0.935	0.942
		100	0.698	0.928	0.941	0.942	0.930	0.945	0.950
	smooth	40	0.658	0.886	0.925	0.928	0.914	0.936	0.943
		100	0.682	0.931	0.944	0.945	0.935	0.948	0.953
binary	independent	40	0.984	0.969	0.972	0.973	0.869	0.976	0.990
		100	0.994	0.989	0.990	0.990	0.917	0.991	0.993
	rough stationary	40	0.800	0.906	0.934	0.937	0.907	0.945	0.958
		100	0.819	0.941	0.950	0.951	0.929	0.954	0.961
	smooth	40	0.749	0.889	0.922	0.926	0.901	0.934	0.947
		100	0.779	0.941	0.951	0.952	0.933	0.954	0.964

Table 57: Paper, Appendix A.6, covariance components, mean: grid-averaged coverage of nominal 95% pointwise intervals, main factorial settings at the default signal level.

family	error	G	model	CR1	CL2, shortcut	CL2	CL2, freq.	CL2, t(G-1)	CL2, fixed λ
Gaussian	independent	40	0.968	0.943	0.956	0.958	0.926	0.964	0.960
		100	0.967	0.954	0.960	0.961	0.939	0.963	0.962
	rough stationary	40	0.624	0.895	0.928	0.931	0.925	0.939	0.937
		100	0.646	0.930	0.941	0.942	0.937	0.945	0.946
	smooth	40	0.606	0.895	0.929	0.931	0.926	0.939	0.939
		100	0.634	0.932	0.943	0.944	0.940	0.947	0.948
Poisson	independent	40	0.968	0.942	0.956	0.958	0.922	0.964	0.960
		100	0.966	0.953	0.959	0.960	0.935	0.962	0.961
	rough stationary	40	0.657	0.893	0.927	0.930	0.922	0.938	0.939
		100	0.664	0.929	0.942	0.943	0.937	0.946	0.947
	smooth	40	0.642	0.896	0.930	0.934	0.926	0.941	0.942
		100	0.653	0.932	0.944	0.945	0.940	0.948	0.950
binary	independent	40	0.952	0.911	0.923	0.925	0.857	0.933	0.943
		100	0.963	0.944	0.949	0.949	0.898	0.952	0.960
	rough stationary	40	0.747	0.894	0.922	0.927	0.909	0.935	0.946
		100	0.752	0.929	0.938	0.940	0.927	0.943	0.952
	smooth	40	0.704	0.884	0.917	0.921	0.906	0.929	0.941
		100	0.722	0.929	0.939	0.941	0.930	0.944	0.952

Table 58: Paper, Appendix A.6, covariance components, $\alpha(t)$ and $\gamma(t)$: grid-averaged coverage of nominal 95% pointwise intervals, main factorial settings at the default signal level.

			α(t)				γ(t)			
family	error	G	model	CR1	CL2	fixed λ	model	CR1	CL2	fixed λ
Gaussian	independent	40	0.956	0.932	0.947	0.950	0.959	0.927	0.951	0.953
		100	0.956	0.946	0.956	0.958	0.960	0.942	0.950	0.951
	rough stationary	40	0.632	0.909	0.940	0.945	0.653	0.906	0.938	0.943
		100	0.659	0.940	0.951	0.952	0.663	0.925	0.938	0.940
	smooth	40	0.614	0.906	0.937	0.941	0.621	0.910	0.946	0.950
		100	0.641	0.941	0.950	0.951	0.655	0.938	0.948	0.951
Poisson	independent	40	0.960	0.942	0.953	0.957	0.955	0.924	0.948	0.950
		100	0.954	0.945	0.951	0.953	0.963	0.945	0.955	0.957
	rough stationary	40	0.655	0.910	0.935	0.940	0.667	0.900	0.937	0.945
		100	0.668	0.943	0.950	0.951	0.677	0.921	0.937	0.939
	smooth	40	0.641	0.907	0.933	0.938	0.647	0.903	0.943	0.948
		100	0.665	0.940	0.948	0.951	0.662	0.933	0.945	0.948
binary	independent	40	0.903	0.845	0.859	0.867	0.931	0.872	0.890	0.929
		100	0.943	0.919	0.924	0.939	0.945	0.914	0.923	0.949
	rough stationary	40	0.712	0.870	0.904	0.929	0.724	0.880	0.920	0.951
		100	0.749	0.919	0.926	0.942	0.740	0.912	0.924	0.946
	smooth	40	0.668	0.855	0.885	0.925	0.691	0.874	0.922	0.950
		100	0.702	0.910	0.920	0.936	0.730	0.925	0.939	0.953

Table 59: Paper, Appendix A.6, covariance components: paired coverage differences in percentage points, averaged over families and over the mean and $\beta(s,t)$, by errors and G; in brackets the range of the per-setting paired Monte Carlo SEs (each from the replicate-level differences of one setting and estimand).

	independent		dependent	
contrast	G = 40	G = 100	G = 40	G = 100
CL2 minus CR1	+1.0 (0.03–0.05)	+0.4 (0.01–0.03)	+3.7 (0.09–0.19)	+1.2 (0.04–0.07)
CL2 minus CL2, shortcut	+0.1 (0.01–0.01)	+0.1 (0.00–0.01)	+0.3 (0.01–0.02)	+0.1 (0.01–0.01)
Bayesian minus frequentist, CL2	+6.0 (0.09–0.45)	+4.4 (0.06–0.32)	+1.4 (0.02–0.12)	+1.0 (0.02–0.10)
Bayesian minus frequentist, CR1	+8.1 (0.10–0.50)	+5.2 (0.06–0.35)	+2.1 (0.02–0.14)	+1.3 (0.02–0.10)
t(G-1) minus normal	+0.4 (0.02–0.03)	+0.2 (0.01–0.01)	+0.8 (0.02–0.04)	+0.3 (0.01–0.02)
fixed minus reselected λ, CL2	+0.7 (0.01–0.48)	+0.3 (0.01–0.11)	+1.4 (0.08–0.32)	+0.8 (0.05–0.16)
fixed minus reselected λ, model-based	-0.1 (0.01–0.40)	-0.1 (0.01–0.08)	-0.3 (0.17–0.52)	-0.6 (0.10–0.43)

Table 60: Paper, Appendix A.6, critical values for REML + CL2: grid-averaged coverage (q05) and width relative to the normal critical value, by dependence, number of curves G, family and estimand; means over the error processes of the main factorial at the default signal level, 200 replicates. Satterthwaite: per-point degrees of freedom for the sandwich part; incl. Vp - Ve: degrees of freedom for the full Bayesian CL2 variance.

dependence	G	family	estimand	z	t(G-1)	Satterthwaite	Satterthwaite incl. Vp - Ve
independent	40	Gaussian	E(Y | X)	0.958 (0.93) 1.00	0.964 (0.94) 1.03	0.973 (0.95) 1.10	0.968 (0.94) 1.06
			β(s,t)	0.989 (0.97) 1.00	0.991 (0.97) 1.03	0.995 (0.98) 1.10	0.991 (0.97) 1.03
			α(t)	0.947 (0.91) 1.00	0.953 (0.92) 1.03	0.956 (0.92) 1.05	0.954 (0.92) 1.03
			γ(t)	0.951 (0.92) 1.00	0.959 (0.93) 1.03	0.969 (0.94) 1.11	0.963 (0.94) 1.07
		Poisson	E(Y | X)	0.958 (0.93) 1.00	0.964 (0.94) 1.03	0.974 (0.94) 1.12	0.967 (0.94) 1.07
			β(s,t)	0.989 (0.96) 1.00	0.991 (0.97) 1.03	0.995 (0.98) 1.11	0.991 (0.97) 1.03
			α(t)	0.953 (0.93) 1.00	0.958 (0.93) 1.03	0.961 (0.94) 1.05	0.957 (0.93) 1.03
			γ(t)	0.948 (0.92) 1.00	0.953 (0.92) 1.03	0.967 (0.94) 1.12	0.961 (0.93) 1.07
		binary	E(Y | X)	0.925 (0.82) 1.00	0.933 (0.84) 1.03	0.946 (0.86) 1.10	0.935 (0.84) 1.05
			β(s,t)	0.973 (0.92) 1.00	0.976 (0.92) 1.03	0.982 (0.94) 1.10	0.975 (0.92) 1.02
			α(t)	0.859 (0.67) 1.00	0.868 (0.68) 1.03	0.872 (0.69) 1.04	0.865 (0.68) 1.02
			γ(t)	0.890 (0.79) 1.00	0.900 (0.80) 1.03	0.918 (0.83) 1.11	0.904 (0.81) 1.05
	100	Gaussian	E(Y | X)	0.961 (0.93) 1.00	0.963 (0.94) 1.01	0.967 (0.94) 1.04	0.965 (0.94) 1.02
			β(s,t)	0.986 (0.97) 1.00	0.987 (0.97) 1.01	0.989 (0.97) 1.04	0.987 (0.97) 1.01
			α(t)	0.956 (0.93) 1.00	0.959 (0.94) 1.01	0.959 (0.94) 1.01	0.959 (0.94) 1.01
			γ(t)	0.950 (0.92) 1.00	0.953 (0.93) 1.01	0.959 (0.93) 1.04	0.957 (0.93) 1.03
		Poisson	E(Y | X)	0.960 (0.93) 1.00	0.962 (0.94) 1.01	0.967 (0.94) 1.04	0.965 (0.94) 1.03
			β(s,t)	0.986 (0.96) 1.00	0.987 (0.97) 1.01	0.989 (0.97) 1.04	0.987 (0.97) 1.01
			α(t)	0.951 (0.92) 1.00	0.953 (0.92) 1.01	0.954 (0.92) 1.01	0.953 (0.92) 1.01
			γ(t)	0.955 (0.92) 1.00	0.958 (0.93) 1.01	0.965 (0.93) 1.05	0.962 (0.93) 1.03
		binary	E(Y | X)	0.949 (0.89) 1.00	0.952 (0.89) 1.01	0.957 (0.90) 1.04	0.953 (0.89) 1.02
			β(s,t)	0.990 (0.96) 1.00	0.991 (0.96) 1.01	0.993 (0.97) 1.04	0.991 (0.96) 1.01
			α(t)	0.924 (0.78) 1.00	0.926 (0.78) 1.01	0.927 (0.78) 1.01	0.926 (0.78) 1.01
			γ(t)	0.923 (0.86) 1.00	0.927 (0.86) 1.01	0.932 (0.88) 1.04	0.928 (0.86) 1.02
dependent	40	Gaussian	E(Y | X)	0.931 (0.90) 1.00	0.939 (0.91) 1.03	0.953 (0.93) 1.10	0.951 (0.93) 1.09
			β(s,t)	0.926 (0.90) 1.00	0.934 (0.90) 1.03	0.950 (0.92) 1.10	0.946 (0.92) 1.09
			α(t)	0.938 (0.91) 1.00	0.947 (0.92) 1.03	0.952 (0.93) 1.05	0.952 (0.93) 1.05
			γ(t)	0.942 (0.92) 1.00	0.948 (0.93) 1.03	0.961 (0.94) 1.11	0.959 (0.94) 1.10
		Poisson	E(Y | X)	0.932 (0.90) 1.00	0.940 (0.91) 1.03	0.954 (0.93) 1.12	0.951 (0.93) 1.11
			β(s,t)	0.928 (0.90) 1.00	0.936 (0.91) 1.03	0.953 (0.93) 1.11	0.948 (0.92) 1.09
			α(t)	0.934 (0.91) 1.00	0.942 (0.92) 1.03	0.945 (0.92) 1.05	0.944 (0.92) 1.05
			γ(t)	0.940 (0.91) 1.00	0.947 (0.92) 1.03	0.963 (0.94) 1.12	0.961 (0.94) 1.11
		binary	E(Y | X)	0.924 (0.89) 1.00	0.932 (0.90) 1.03	0.946 (0.92) 1.10	0.942 (0.91) 1.08
			β(s,t)	0.932 (0.90) 1.00	0.939 (0.91) 1.03	0.954 (0.93) 1.11	0.948 (0.92) 1.08
			α(t)	0.894 (0.79) 1.00	0.904 (0.81) 1.03	0.909 (0.81) 1.05	0.907 (0.81) 1.04
			γ(t)	0.921 (0.89) 1.00	0.931 (0.90) 1.03	0.945 (0.92) 1.11	0.942 (0.92) 1.09
	100	Gaussian	E(Y | X)	0.943 (0.91) 1.00	0.946 (0.92) 1.01	0.951 (0.92) 1.04	0.951 (0.92) 1.03
			β(s,t)	0.942 (0.92) 1.00	0.945 (0.92) 1.01	0.951 (0.92) 1.04	0.950 (0.92) 1.03
			α(t)	0.951 (0.93) 1.00	0.953 (0.94) 1.01	0.953 (0.94) 1.02	0.953 (0.94) 1.01
			γ(t)	0.943 (0.92) 1.00	0.947 (0.92) 1.01	0.953 (0.93) 1.04	0.952 (0.93) 1.04
		Poisson	E(Y | X)	0.944 (0.92) 1.00	0.947 (0.92) 1.01	0.953 (0.93) 1.05	0.952 (0.93) 1.04
			β(s,t)	0.943 (0.92) 1.00	0.946 (0.92) 1.01	0.953 (0.93) 1.04	0.952 (0.93) 1.04

Table 60: Paper, Appendix A.6, critical values for REML + CL2: grid-averaged coverage (q05) and width relative to the normal critical value, by dependence, number of curves G, family and estimand; means over the error processes of the main factorial at the default signal level, 200 replicates. Satterthwaite: per-point degrees of freedom for the sandwich part; incl. Vp - Ve: degrees of freedom for the full Bayesian CL2 variance. *(continued)*

dependence	G	family	estimand	z	t(G-1)	Satterthwaite	Satterthwaite incl. Vp - Ve
			α(t)	0.949 (0.93) 1.00	0.952 (0.93) 1.01	0.952 (0.93) 1.01	0.952 (0.93) 1.01
			γ(t)	0.941 (0.91) 1.00	0.943 (0.92) 1.01	0.951 (0.93) 1.05	0.951 (0.93) 1.04
		binary	E(Y | X)	0.940 (0.91) 1.00	0.943 (0.92) 1.01	0.949 (0.92) 1.04	0.947 (0.92) 1.03
			β(s,t)	0.951 (0.93) 1.00	0.954 (0.94) 1.01	0.960 (0.94) 1.04	0.957 (0.94) 1.03
			α(t)	0.923 (0.87) 1.00	0.927 (0.87) 1.01	0.928 (0.87) 1.01	0.927 (0.87) 1.01
			γ(t)	0.931 (0.90) 1.00	0.935 (0.91) 1.01	0.941 (0.92) 1.04	0.939 (0.91) 1.03

Table 61: Paper, Appendix C, computing time in seconds per fit or per covariance, by method and data source (A: local benchmark, Gaussian, $G = 100$, $D = 61$; B: comparator runs, Gaussian synthetic settings and plasmode datasets with independent source curves; C: main-factorial settings; D: plasmode data). n: number of timed fits or covariances.

method	data	n	median	25%	75%	90%	max
REML fit	A	90	2.02	1.85	2.66	4.04	5.14
curve-blocked NCV fit	A	90	8.56	7.18	10.60	13.60	17.10
AR(1) working model (profiled)	A	90	39.00	35.10	42.20	51.50	58.70
CL2 covariance (REML fit)	A	90	0.95	0.92	1.13	1.27	1.56
REML fit	B	3600	2.39	1.42	3.92	7.37	36.70
GLS, FPCA residual covariance	B	3600	5.93	3.50	8.49	16.30	67.70
GLS, raw residual covariance	B	3600	5.88	3.45	8.37	17.50	83.70
pcre fit	B	3600	20.90	9.99	96.20	244.00	1360.00
curve bootstrap (199 refits)	B	1798	507.00	315.00	801.00	1610.00	6640.00
CL2 on a GLS fit	B	7200	0.12	0.07	0.18	0.32	1.09
CL2 on a pcre fit	B	1799	1.40	0.73	2.80	4.30	11.00
smoothing-parameter term, REML	C	3600	1.16	0.49	1.35	1.53	3.81
smoothing-parameter term, REML	D	26000	1.17	0.74	1.68	3.48	7.17
smoothing-parameter term, NCV	C	1200	2.10	1.12	3.70	3.95	5.85
smoothing-parameter term, NCV	D	13000	2.48	1.36	3.66	8.61	38.10
smoothing-parameter term, NCV (Poisson, binary)	C	2400	5.25	2.29	11.60	12.20	13.90

Table 62: S1.2: q05 (minimum and median over main-factorial settings) and minimum grid-averaged coverage, by dependence, G, estimand and interval.

dependence	G	estimand	interval	q05 min	q05 median	min coverage
independent	40	E(Y | X)	NCV + CL2	0.835	0.900	0.923
			NCV widened	0.850	0.905	0.935
			REML + CL2	0.825	0.925	0.925
			REML model	0.875	0.940	0.952
		β(s,t)	NCV + CL2	0.925	0.945	0.973
			NCV widened	0.930	0.960	0.979
			REML + CL2	0.915	0.965	0.973
			REML model	0.945	0.975	0.984
	100	E(Y | X)	NCV + CL2	0.860	0.900	0.939
			NCV widened	0.870	0.910	0.945
			REML + CL2	0.885	0.930	0.949
			REML model	0.910	0.935	0.962
		β(s,t)	NCV + CL2	0.925	0.940	0.976
			NCV widened	0.945	0.951	0.986
			REML + CL2	0.955	0.965	0.979
			REML model	0.965	0.970	0.983
dependent	40	E(Y | X)	NCV + CL2	0.685	0.827	0.842
			NCV widened	0.860	0.907	0.930
			REML + CL2	0.887	0.900	0.921
			REML model	0.515	0.560	0.601
		β(s,t)	NCV + CL2	0.600	0.861	0.785
			NCV widened	0.885	0.950	0.948
			REML + CL2	0.891	0.897	0.923
			REML model	0.520	0.580	0.609
	100	E(Y | X)	NCV + CL2	0.740	0.863	0.876
			NCV widened	0.865	0.920	0.933
			REML + CL2	0.910	0.915	0.940
			REML model	0.535	0.570	0.631
		β(s,t)	NCV + CL2	0.775	0.861	0.896
			NCV widened	0.920	0.950	0.971
			REML + CL2	0.911	0.915	0.940
			REML model	0.540	0.596	0.653

Table 63: S1.2: main factorial by number of curves G = 20, 40, 100 (all signal levels; dep.: rough stationary and smooth errors; indep.: independent errors): grid-averaged coverage (mean over settings), median relative error of the estimate (median over settings of the median over replicates; the model-based and CL2 intervals of a fit share the estimate) and detection rate (median over settings; $\beta(s,t)$ and $\gamma(t)$ only), per family, dependence, estimand and interval.

				coverage			relative error			detection		
family	errors	estimand	interval	20	40	100	20	40	100	20	40	100
Gaussian	indep.	E(Y | X)	REML model	0.966	0.967	0.966	0.193	0.135	0.091			
			REML + CL2	0.949	0.957	0.960	0.193	0.135	0.091			
			NCV + CL2	0.941	0.947	0.950	0.193	0.133	0.088			
			NCV widened	0.949	0.954	0.958	0.193	0.133	0.088			
		β(s,t)	REML model	0.992	0.991	0.988	0.185	0.146	0.109	0.925	0.959	0.982
			REML + CL2	0.987	0.987	0.985	0.185	0.146	0.109	0.926	0.960	0.983
			NCV + CL2	0.983	0.982	0.981	0.159	0.116	0.083	0.943	0.971	0.993
			NCV widened	0.988	0.989	0.989	0.159	0.116	0.083	0.939	0.967	0.990
		α(t)	REML model	0.952	0.955	0.956	0.256	0.181	0.116			
			REML + CL2	0.934	0.945	0.953	0.256	0.181	0.116			
			NCV + CL2	0.932	0.939	0.945	0.259	0.182	0.118			
			NCV widened	0.937	0.942	0.949	0.259	0.182	0.118			
		γ(t)	REML model	0.956	0.959	0.961	0.189	0.132	0.089	0.966	0.998	1.000
			REML + CL2	0.932	0.946	0.950	0.189	0.132	0.089	0.962	0.998	1.000
			NCV + CL2	0.927	0.937	0.944	0.191	0.133	0.090	0.964	0.997	1.000
			NCV widened	0.932	0.941	0.950	0.191	0.133	0.090	0.963	0.997	1.000
	dep.	E(Y | X)	REML model	0.576	0.615	0.640	0.570	0.358	0.215			
			REML + CL2	0.909	0.931	0.943	0.570	0.358	0.215			

Table 63: S1.2: main factorial by number of curves G = 20, 40, 100 (all signal levels; dep.: rough stationary and smooth errors; indep.: independent errors): grid-averaged coverage (mean over settings), median relative error of the estimate (median over settings of the median over replicates; the model-based and CL2 intervals of a fit share the estimate) and detection rate (median over settings; $\beta(s,t)$ and $\gamma(t)$ only), per family, dependence, estimand and interval. *(continued)*

family	errors	estimand	interval	20	40	100	20	40	100	20	40	100
			NCV + CL2	0.875	0.912	0.928	0.402	0.270	0.178			
			NCV widened	0.943	0.954	0.957	0.402	0.270	0.178			
		β(s,t)	REML model	0.597	0.625	0.663	1.188	0.680	0.353	0.636	0.779	0.916
			REML + CL2	0.912	0.926	0.942	1.188	0.680	0.353	0.325	0.525	0.788
			NCV + CL2	0.882	0.918	0.932	0.301	0.216	0.152	0.842	0.922	0.973
			NCV widened	0.975	0.982	0.983	0.301	0.216	0.152	0.412	0.623	0.845
		α(t)	REML model	0.580	0.623	0.650	0.585	0.385	0.232			
			REML + CL2	0.904	0.937	0.950	0.585	0.385	0.232			
			NCV + CL2	0.864	0.904	0.937	0.542	0.361	0.227			
			NCV widened	0.920	0.931	0.946	0.542	0.361	0.227			
		γ(t)	REML model	0.589	0.637	0.657	0.435	0.284	0.177	0.879	0.968	0.998
			REML + CL2	0.917	0.942	0.943	0.435	0.284	0.177	0.600	0.849	0.975
			NCV + CL2	0.872	0.912	0.925	0.380	0.274	0.172	0.688	0.888	0.983
			NCV widened	0.927	0.937	0.938	0.380	0.274	0.172	0.597	0.870	0.981
Poisson	indep.	E(Y | X)	REML model	0.968	0.967	0.964	0.156	0.103	0.067			
			REML + CL2	0.950	0.956	0.958	0.156	0.103	0.067			
			NCV + CL2	0.934	0.943	0.945	0.160	0.103	0.065			
			NCV widened	0.945	0.951	0.955	0.160	0.103	0.065			
		β(s,t)	REML model	0.993	0.992	0.987	0.189	0.145	0.108	0.919	0.955	0.979
			REML + CL2	0.989	0.988	0.982	0.189	0.145	0.108	0.921	0.956	0.980
			NCV + CL2	0.982	0.983	0.979	0.162	0.118	0.082	0.936	0.968	0.990
			NCV widened	0.989	0.989	0.988	0.162	0.118	0.082	0.930	0.964	0.986
		α(t)	REML model	0.956	0.961	0.952	0.270	0.189	0.123			
			REML + CL2	0.939	0.952	0.950	0.270	0.189	0.123			
			NCV + CL2	0.920	0.933	0.931	0.281	0.194	0.126			
			NCV widened	0.935	0.946	0.942	0.281	0.194	0.126			
		γ(t)	REML model	0.955	0.955	0.959	0.200	0.134	0.085	0.962	0.995	1.000
			REML + CL2	0.931	0.945	0.949	0.200	0.134	0.085	0.959	0.994	1.000
			NCV + CL2	0.918	0.933	0.942	0.206	0.137	0.088	0.961	0.994	1.000
			NCV widened	0.928	0.940	0.947	0.206	0.137	0.088	0.959	0.994	1.000
	dep.	E(Y | X)	REML model	0.650	0.657	0.663	0.470	0.280	0.160			
			REML + CL2	0.907	0.932	0.943	0.470	0.280	0.160			
			NCV + CL2	0.874	0.908	0.925	0.351	0.217	0.130			
			NCV widened	0.934	0.950	0.955	0.351	0.217	0.130			
		β(s,t)	REML model	0.685	0.682	0.695	0.865	0.571	0.322	0.676	0.817	0.930
			REML + CL2	0.912	0.928	0.943	0.865	0.571	0.322	0.451	0.631	0.837
			NCV + CL2	0.903	0.923	0.933	0.289	0.206	0.141	0.836	0.922	0.972
			NCV widened	0.969	0.980	0.982	0.289	0.206	0.141	0.518	0.707	0.879
		α(t)	REML model	0.645	0.656	0.665	0.599	0.394	0.243			
			REML + CL2	0.906	0.938	0.949	0.599	0.394	0.243			
			NCV + CL2	0.864	0.903	0.924	0.549	0.375	0.236			
			NCV widened	0.923	0.943	0.948	0.549	0.375	0.236			
		γ(t)	REML model	0.662	0.660	0.674	0.425	0.277	0.167	0.873	0.969	0.996
			REML + CL2	0.914	0.937	0.941	0.425	0.277	0.167	0.643	0.866	0.976
			NCV + CL2	0.862	0.904	0.924	0.397	0.277	0.166	0.704	0.899	0.983
			NCV widened	0.919	0.933	0.936	0.397	0.277	0.166	0.613	0.877	0.982
binary	indep.	E(Y | X)	REML model	0.939	0.958	0.965	0.539	0.386	0.258			
			REML + CL2	0.903	0.937	0.954	0.539	0.386	0.258			
			NCV + CL2	0.907	0.930	0.941	0.566	0.393	0.260			
			NCV widened	0.926	0.940	0.948	0.566	0.393	0.260			
		β(s,t)	REML model	0.957	0.987	0.993	0.449	0.340	0.246	0.517	0.711	0.850
			REML + CL2	0.935	0.979	0.990	0.449	0.340	0.246	0.595	0.744	0.862
			NCV + CL2	0.947	0.975	0.981	0.440	0.317	0.229	0.585	0.760	0.886
			NCV widened	0.959	0.982	0.987	0.440	0.317	0.229	0.548	0.747	0.879
		α(t)	REML model	0.885	0.930	0.950	0.805	0.592	0.389			
			REML + CL2	0.847	0.899	0.936	0.805	0.592	0.389			
			NCV + CL2	0.850	0.896	0.922	0.829	0.597	0.405			
			NCV widened	0.872	0.909	0.929	0.829	0.597	0.405			
		γ(t)	REML model	0.928	0.937	0.946	0.598	0.433	0.300	0.313	0.582	0.851
			REML + CL2	0.875	0.906	0.929	0.598	0.433	0.300	0.393	0.633	0.866
			NCV + CL2	0.874	0.893	0.917	0.632	0.461	0.304	0.399	0.629	0.862
			NCV widened	0.893	0.907	0.922	0.632	0.461	0.304	0.363	0.613	0.858

Table 63: S1.2: main factorial by number of curves G = 20, 40, 100 (all signal levels; dep.: rough stationary and smooth errors; indep.: independent errors): grid-averaged coverage (mean over settings), median relative error of the estimate (median over settings of the median over replicates; the model-based and CL2 intervals of a fit share the estimate) and detection rate (median over settings; $\beta(s,t)$ and $\gamma(t)$ only), per family, dependence, estimand and interval. *(continued)*

family	errors	estimand	interval	20	40	100	20	40	100	20	40	100
	dep.	E(Y | X)	REML model	0.722	0.731	0.742	1.352	0.868	0.530			
			REML + CL2	0.895	0.925	0.942	1.352	0.868	0.530			
			NCV + CL2	0.831	0.861	0.895	0.971	0.704	0.468			
			NCV widened	0.936	0.937	0.942	0.971	0.704	0.468			
		β(s,t)	REML model	0.774	0.782	0.800	2.428	1.479	0.853	0.326	0.450	0.627
			REML + CL2	0.912	0.934	0.951	2.428	1.479	0.853	0.161	0.237	0.412
			NCV + CL2	0.783	0.844	0.913	0.743	0.577	0.359	0.399	0.593	0.779
			NCV widened	0.954	0.962	0.976	0.743	0.577	0.359	0.097	0.232	0.470
		α(t)	REML model	0.680	0.703	0.735	2.001	1.182	0.734			
			REML + CL2	0.852	0.900	0.928	2.001	1.182	0.734			
			NCV + CL2	0.832	0.853	0.873	1.340	1.001	0.740			
			NCV widened	0.932	0.920	0.919	1.340	1.001	0.740			
		γ(t)	REML model	0.721	0.719	0.741	1.340	0.845	0.521	0.389	0.547	0.769
			REML + CL2	0.895	0.922	0.936	1.340	0.845	0.521	0.182	0.278	0.514
			NCV + CL2	0.815	0.822	0.867	1.044	0.811	0.518	0.167	0.286	0.539
			NCV widened	0.913	0.899	0.911	1.044	0.811	0.518	0.069	0.177	0.466

Table 64: S1.3: heteroskedastic Gaussian errors next to the homoskedastic smooth process, coverage per interval. var(t): variance along t; var(z): variance by the covariate; var(t,z): both.

error	G	estimand	REML model	REML + CL2	NCV + CL2	NCV widened	AR(1)
smooth	40	β(s,t)	0.615	0.926	0.924	0.983	
		γ(t)	0.621	0.946	0.927	0.946	
		E(Y | X)	0.606	0.931	0.922	0.959	
	100	β(s,t)	0.657	0.944	0.934	0.983	0.993
		γ(t)	0.655	0.948	0.935	0.946	0.967
		E(Y | X)	0.634	0.944	0.929	0.959	0.972
var(z)	40	β(s,t)	0.619	0.939	0.925	0.985	
		γ(t)	0.490	0.926	0.898	0.931	
		E(Y | X)	0.588	0.934	0.912	0.955	
	100	β(s,t)	0.660	0.949	0.935	0.984	0.992
		γ(t)	0.494	0.938	0.920	0.945	0.879
		E(Y | X)	0.606	0.944	0.923	0.955	0.949
var(t)	40	β(s,t)	0.619	0.926	0.918	0.979	
		γ(t)	0.632	0.944	0.925	0.945	
		E(Y | X)	0.614	0.932	0.919	0.957	
	100	β(s,t)	0.662	0.944	0.933	0.983	0.991
		γ(t)	0.661	0.947	0.932	0.943	0.967
		E(Y | X)	0.642	0.944	0.928	0.959	0.972
var(t,z)	40	β(s,t)	0.629	0.940	0.924	0.983	
		γ(t)	0.502	0.925	0.893	0.930	
		E(Y | X)	0.596	0.936	0.911	0.955	
	100	β(s,t)	0.666	0.950	0.934	0.984	0.990
		γ(t)	0.510	0.940	0.920	0.944	0.889
		E(Y | X)	0.615	0.944	0.922	0.955	0.951

Table 65: S1.3: misregistration settings (high signal), coverage of the observed-clock estimands; binary and Poisson settings score the mean only.

family	G	truth	estimand	REML model	REML + CL2	NCV + CL2	NCV widened	NCV widened (freq.)	AR(1)
binary	40	smooth	E(Y | X)	0.945	0.944	0.919	0.932	0.884	
		rough	E(Y | X)	0.924	0.925	0.892	0.908	0.854	
	100	smooth	E(Y | X)	0.943	0.954	0.931	0.941	0.902	
		rough	E(Y | X)	0.918	0.932	0.909	0.919	0.878	
Gaussian	40	smooth	β(s,t)	0.761	0.949	0.937	0.974	0.956	
			E(Y | X)	0.729	0.938	0.919	0.946	0.934	
		rough	β(s,t)	0.739	0.945	0.820	0.928	0.901	
			E(Y | X)	0.706	0.934	0.885	0.932	0.918	
	100	smooth	β(s,t)	0.753	0.951	0.937	0.976	0.959	0.875
			E(Y | X)	0.730	0.942	0.930	0.952	0.941	0.844
		rough	β(s,t)	0.729	0.944	0.863	0.934	0.908	0.860
			E(Y | X)	0.706	0.937	0.911	0.938	0.926	0.838
Poisson	40	smooth	E(Y | X)	0.634	0.902	0.792	0.910	0.900	
		rough	E(Y | X)	0.614	0.899	0.773	0.910	0.902	
	100	smooth	E(Y | X)	0.571	0.910	0.801	0.922	0.915	
		rough	E(Y | X)	0.558	0.906	0.780	0.918	0.912	

Table 66: S1.5: NCV/REML MSE ratio, geometric mean over the main-factorial settings of each error process (all G and signal levels), 95% bootstrap intervals.

family	estimand	independent	rough stationary	smooth
Gaussian	E(Y | X)	0.97 [0.96, 0.98]	0.66 [0.64, 0.68]	0.64 [0.62, 0.65]
	β(s,t)	0.73 [0.69, 0.78]	0.15 [0.13, 0.16]	0.12 [0.10, 0.14]
Poisson	E(Y | X)	0.97 [0.93, 1.01]	0.59 [0.53, 0.66]	0.58 [0.52, 0.64]
	β(s,t)	0.69 [0.66, 0.72]	0.17 [0.14, 0.19]	0.15 [0.13, 0.17]
binary	E(Y | X)	1.06 [1.04, 1.07]	0.74 [0.72, 0.76]	0.68 [0.66, 0.70]
	β(s,t)	1.05 [0.96, 1.15]	0.21 [0.18, 0.24]	0.17 [0.14, 0.19]

Table 67: S1.5: mean EDF per term of the REML and NCV fits, main factorial at $G = 100$, default signal.

family	term	error	REML	NCV	REML/NCV
Gaussian	β(s,t)	indep.	33.50	25.48	1.31
		rough	41.09	21.35	1.92
		smooth	41.84	21.03	1.99
	α(t)	indep.	8.33	8.09	1.03
		rough	8.41	7.33	1.15
		smooth	8.46	7.26	1.17
	γ(t)	indep.	9.02	8.39	1.07
		rough	9.12	7.49	1.22
		smooth	9.18	7.44	1.23
Poisson	β(s,t)	indep.	31.42	24.30	1.29
		rough	38.71	20.07	1.93
		smooth	39.37	19.64	2.00
	α(t)	indep.	7.97	7.53	1.06
		rough	8.05	6.69	1.20
		smooth	8.09	6.63	1.22
	γ(t)	indep.	8.68	8.15	1.07
		rough	8.75	7.25	1.21
		smooth	8.82	7.26	1.21
binary	β(s,t)	indep.	17.07	15.72	1.09
		rough	25.39	11.99	2.12
		smooth	26.78	11.60	2.31
	α(t)	indep.	4.47	4.69	0.95
		rough	5.03	2.74	1.83
		smooth	5.06	2.83	1.79
	γ(t)	indep.	5.21	5.33	0.98
		rough	5.72	3.69	1.55
		smooth	5.93	3.70	1.60

Table 68: S1.5: SD over replicates of the selected log smoothing parameters of the surface term (mean over its two penalties), main factorial at the default signal.

family	error	G	REML	NCV
Gaussian	indep.	40	0.128	0.885
		100	0.097	0.661
	rough	40	1.421	1.372
		100	1.168	0.972
	smooth	40	1.508	1.014
		100	1.281	1.063
Poisson	indep.	40	0.130	0.756
		100	0.106	0.894
	rough	40	1.367	1.448
		100	1.157	1.128
	smooth	40	1.479	1.106
		100	1.268	1.045
binary	indep.	40	0.906	1.425
		100	0.267	1.123
	rough	40	1.502	4.257
		100	1.130	1.850
	smooth	40	1.742	4.672
		100	1.340	1.912

Table 69: S1.7: REML + CL2 with the normal (z) and the Satterthwaite (Satt.) critical value in the extension settings: grid-averaged coverage per block, number of curves and estimand (mean over the block's settings; β includes $f(x,t)$ of the smooth-effect settings; misregistered count and binary settings score the mean only), and the mean width ratio Satt./z.

			E(Y | X)		β(s,t)		α(t)		γ(t)		
block	G	settings	z	Satt.	z	Satt.	z	Satt.	z	Satt.	width
misregistration	40	6	0.924	0.948	0.947	0.967	0.946	0.956	0.922	0.945	1.11
misregistration	100	6	0.930	0.940	0.948	0.956	0.951	0.954	0.930	0.941	1.04
dense grid	100	9	0.947	0.955	0.954	0.962	0.943	0.947	0.944	0.953	1.03
rough truth	100	14	0.944	0.953	0.948	0.957	0.945	0.948	0.936	0.946	1.04
smooth effect	100	9	0.945	0.960	0.955	0.964			0.941	0.951	1.06
extended families	100	8	0.950	0.958	0.962	0.968	0.948	0.951	0.948	0.957	1.03
exact AR(1)	100	1	0.942	0.950	0.940	0.949	0.955	0.957	0.937	0.947	1.03
sign-changing	100	3	0.942	0.950	0.947	0.956	0.944	0.948	0.940	0.950	1.03
misreg. (AR(1) fit)	100	1	0.937	0.946	0.944	0.953	0.949	0.951	0.928	0.940	1.03
heteroskedastic	40	3	0.934	0.955	0.935	0.958	0.935	0.946	0.932	0.956	1.09
heteroskedastic	100	3	0.944	0.952	0.948	0.956	0.948	0.951	0.942	0.952	1.03
low-rank covariate	100	6	0.949	0.956	0.970	0.976	0.942	0.945	0.943	0.953	1.03
larger bases	100	12	0.962	0.968	0.976	0.980	0.950	0.953	0.954	0.962	1.03

Table 70: S2.1: q05 per dataset and estimand, all subjects: mean and minimum over each dataset's combinations, per interval.

			mean				minimum			
dataset	source	estimand	REML model	REML + CL2	NCV + CL2	NCV wid.	REML model	REML + CL2	NCV + CL2	NCV wid.
ECG strain	indep. source	β(s,t)	0.219	0.901	0.809	0.862	0.151	0.895	0.775	0.840
		E(Y | X)	0.238	0.888	0.820	0.876	0.190	0.880	0.765	0.850
AF trial	indep. source	β(s,t)	0.363	0.879	0.485	0.772	0.360	0.870	0.455	0.745
		E(Y | X)	0.385	0.896	0.782	0.856	0.385	0.895	0.760	0.845
running	indep. source	β(s,t)	0.268	0.873	0.441	0.766	0.255	0.860	0.300	0.690
		E(Y | X)	0.268	0.887	0.770	0.875	0.260	0.880	0.655	0.860
DTI	indep. source	β(s,t)	0.475	0.876	0.586	0.806	0.445	0.845	0.316	0.715
		E(Y | X)	0.469	0.892	0.789	0.871	0.465	0.890	0.705	0.840
gait	indep. source	β(s,t)	0.231	0.903	0.547	0.833	0.225	0.900	0.512	0.810
		E(Y | X)	0.243	0.910	0.833	0.895	0.240	0.910	0.830	0.895
ECG 8-lead	indep. source	β(s,t)	0.282	0.873	0.449	0.840	0.265	0.870	0.301	0.740
		E(Y | X)	0.329	0.907	0.714	0.887	0.325	0.905	0.635	0.860
ocean	dep. source	β(s,t)	0.224	0.887	0.281	0.794	0.210	0.880	0.055	0.720
		E(Y | X)	0.234	0.899	0.815	0.891	0.230	0.895	0.755	0.880
weather	dep. source	β(s,t)	0.233	0.833	0.415	0.715	0.200	0.796	0.252	0.670
		E(Y | X)	0.342	0.865	0.728	0.809	0.335	0.835	0.662	0.785
electricity	dep. source	β(s,t)	0.266	0.881	0.398	0.835	0.230	0.820	0.075	0.695
		E(Y | X)	0.295	0.887	0.756	0.899	0.285	0.880	0.505	0.830

Table 71: S2.1: coverage with all subjects minus coverage with 40 subjects, in percentage points (mean of paired differences over truth × residual combinations).

dataset	source	estimand	REML model	REML + CL2	NCV + CL2	NCV widened
ECG strain	indep. source	β(s,t)	1.9	1.0	3.5	0.6
		E(Y | X)	1.6	0.8	1.9	0.3
AF trial	indep. source	β(s,t)	4.5	1.7	18.6	3.1
		E(Y | X)	3.4	1.2	4.7	0.9
running	indep. source	β(s,t)	6.0	1.7	12.4	1.3
		E(Y | X)	4.4	1.1	3.0	0.2
DTI	indep. source	β(s,t)	1.3	1.0	8.9	1.8
		E(Y | X)	1.9	1.4	4.4	1.3
gait	indep. source	β(s,t)	4.8	0.4	18.5	2.0
		E(Y | X)	3.5	0.7	5.8	0.5
ECG 8-lead	indep. source	β(s,t)	0.2	0.6	8.3	0.3
		E(Y | X)	2.2	0.8	3.1	0.6
ocean	dep. source	β(s,t)	3.2	-0.4	14.1	1.0
		E(Y | X)	2.8	0.3	2.4	-0.3
weather	dep. source	β(s,t)	3.8	4.7	21.3	5.9
		E(Y | X)	3.8	0.8	4.1	0.7
electricity	dep. source	β(s,t)	7.2	1.4	1.6	0.8
		E(Y | X)	5.2	0.5	0.4	0.6

Table 72: S2.1: plasmode study, curve flips: grid-averaged coverage of REML + CL2 with the normal (z) and the Satterthwaite (Satt.) critical value per dataset and number of subjects G (mean over the dataset's combinations of truth and residual source), and the width ratio Satt./z for $\beta(s,t)$.

		E(Y | X)		β(s,t)		α(t)		γ(t)		
dataset	G	z	Satt.	z	Satt.	z	Satt.	z	Satt.	width
ECG strain	40	0.936	0.964	0.934	0.959	0.949	0.967	0.943	0.977	1.12
	78	0.944	0.960	0.944	0.957	0.949	0.960	0.951	0.967	1.06
AF trial	40	0.934	0.955	0.920	0.947	0.923	0.938	0.936	0.949	1.12
	118	0.946	0.953	0.938	0.947	0.943	0.948	0.946	0.949	1.04
running	40	0.922	0.948	0.902	0.931	0.927	0.950	0.926	0.951	1.12
	90	0.933	0.944	0.919	0.932	0.939	0.950	0.933	0.943	1.05
DTI	40	0.918	0.944	0.917	0.943	0.936	0.956	0.933	0.951	1.11
	92	0.932	0.943	0.927	0.938	0.937	0.947	0.932	0.937	1.04
gait	40	0.936	0.958	0.935	0.958	0.892	0.925	0.946	0.959	1.11
	138	0.943	0.950	0.939	0.946	0.930	0.939	0.932	0.938	1.03
ECG 8-lead	40	0.944	0.960	0.936	0.963	0.956	0.967	0.951	0.961	1.15
	100	0.952	0.959	0.942	0.955	0.956	0.960	0.960	0.963	1.07
ocean	40	0.932	0.958	0.928	0.956	0.931	0.954	0.933	0.954	1.14
	116	0.935	0.943	0.925	0.936	0.943	0.947	0.947	0.952	1.05
weather	40	0.916	0.940	0.835	0.863	0.939	0.963	0.890	0.921	1.12
	73	0.924	0.938	0.882	0.899	0.927	0.943	0.907	0.927	1.06
electricity	40	0.922	0.943	0.914	0.941	0.859	0.879	0.966	0.977	1.13
	102	0.927	0.936	0.927	0.941	0.904	0.912	0.950	0.953	1.05

Table 73: S3.3: plasmode datasets (midpoint truth, REML residuals), $\beta(s,t)$: grid-averaged coverage of nominal 95% pointwise intervals (Model, CL2: REML model-based and REML + CL2; Indep., AR(1), FPCA: fGEE Wald intervals from its covariance under that working correlation), and paired MSE ratios with 95% replicate-bootstrap intervals (fGEE with independence working correlation).

		coverage					MSE ratio	
dataset	G	Model	CL2	Indep.	AR(1)	FPCA	fGEE / REML	fGEE / NCV
ECG strain	78	0.56	0.94	0.87	0.83	0.87	0.87 [0.85, 0.90]	0.94 [0.92, 0.96]
	40	0.53	0.93	0.81	0.79	0.78	0.74 [0.71, 0.76]	0.95 [0.92, 0.97]
AF trial	118	0.67	0.94	0.78	0.78	0.69	0.39 [0.21, 0.70]	1.35 [0.75, 2.47]
	40	0.62	0.92	0.59	0.64	0.67	0.88 [0.35, 1.48]	4.72 [1.92, 9.29]
running	90	0.52	0.92	0.75	0.75	0.70	0.54 [0.51, 0.59]	1.05 [0.97, 1.15]
	40	0.47	0.91	0.69	0.73	0.71	0.59 [0.52, 0.67]	2.04 [1.61, 2.59]
DTI	92	0.64	0.93	0.85	0.84	0.82	0.66 [0.63, 0.69]	1.08 [1.05, 1.12]
	40	0.63	0.92	0.76	0.76	0.72	0.67 [0.63, 0.71]	1.25 [1.16, 1.37]
gait	138	0.52	0.94	0.83	0.87	0.82	0.61 [0.55, 0.66]	1.09 [0.99, 1.21]
	40	0.47	0.94	0.74	0.80	0.76	0.53 [0.46, 0.59]	2.09 [1.63, 2.75]
ECG 8-lead	100	0.64	0.94	0.81	0.83	0.83	0.45 [0.36, 0.55]	2.42 [1.66, 3.68]
	40	0.64	0.94	0.77	0.79	0.75	0.61 [0.50, 0.75]	5.25 [3.73, 7.48]

Table 74: S4.1: one-step bias correction, $\beta(s,t)$, 200 fresh replicates per setting: coverage, q05, width relative to REML + CL2 (geometric mean), detection rate and median relative error of the center.

group	interval	settings	coverage	q05	width rel.	detection	rel. error
binary	NCV + CL2	6	0.848	0.659	0.358	0.598	0.498
	bias correction	6	0.873	0.763	0.435	0.587	0.547
	REML + CL2	6	0.941	0.914	1.000	0.267	1.199
independent errors	NCV + CL2	13	0.959	0.877	0.913	0.914	0.155
	bias correction	13	0.951	0.909	0.975	0.916	0.166
	REML + CL2	13	0.968	0.907	1.000	0.913	0.165
rough truth, Gaussian/Poisson	NCV + CL2	5	0.849	0.609	0.509	0.922	0.294
	bias correction	5	0.899	0.784	0.617	0.902	0.306
	REML + CL2	5	0.940	0.914	1.000	0.746	0.416
smooth truth, Gaussian/Poisson	NCV + CL2	8	0.930	0.872	0.380	0.933	0.193
	bias correction	8	0.932	0.901	0.467	0.910	0.228
	REML + CL2	8	0.937	0.911	1.000	0.656	0.509

Table 75: S4.1: plasmode, $\beta(s,t)$, all subjects, mean over truth × residual combinations: coverage (q05) and width relative to REML + CL2 for the four NCV-centered intervals (50 replicates), and REML + CL2 (100 replicates). Datasets with independent source curves first.

dataset	NCV + CL2	+ λ term	bias correction	bias correction + λ term	REML + CL2
ECG strain	0.89 (0.76) 0.84	0.92 (0.81) 0.94	0.91 (0.82) 0.94	0.94 (0.87) 1.09	0.94 (0.89)
AF trial	0.81 (0.53) 0.53	0.89 (0.70) 0.77	0.84 (0.66) 0.63	0.94 (0.81) 1.04	0.94 (0.86)
running	0.72 (0.42) 0.52	0.78 (0.48) 0.71	0.76 (0.48) 0.59	0.83 (0.55) 0.88	0.91 (0.84)
DTI	0.84 (0.54) 0.64	0.88 (0.64) 0.76	0.88 (0.68) 0.75	0.93 (0.77) 0.95	0.93 (0.86)
gait	0.81 (0.53) 0.58	0.88 (0.66) 0.82	0.84 (0.61) 0.67	0.92 (0.75) 1.06	0.94 (0.89)
ECG 8-lead	0.76 (0.47) 0.32	0.87 (0.64) 0.53	0.80 (0.56) 0.38	0.92 (0.75) 0.73	0.94 (0.88)
ocean	0.72 (0.30) 0.34	0.78 (0.38) 0.43	0.78 (0.39) 0.41	0.84 (0.47) 0.56	0.93 (0.87)
weather	0.53 (0.30) 0.33	0.61 (0.44) 0.51	0.60 (0.43) 0.42	0.66 (0.53) 0.69	0.88 (0.82)
electricity	0.62 (0.37) 0.21	0.70 (0.46) 0.31	0.63 (0.38) 0.24	0.73 (0.50) 0.41	0.93 (0.88)

Table 76: S4.1: plasmode, $\beta(s,t)$, all subjects, mean over truth × residual combinations: coverage (q05) and SE/SD for the smoothing-parameter term without and with the cross term, alone and combined with the one-step bias correction (50 replicates), and REML + CL2 (100 replicates). Datasets with independent source curves first.

dataset	+ λ term	+ λ term, cross	bias corr. + λ term	bias corr. + λ term, cross	REML + CL2
ECG strain	0.92 (0.80) 1.02	0.92 (0.81) 1.05	0.94 (0.86) 1.06	0.94 (0.87) 1.10	0.94 (0.89) 1.01
AF trial	0.88 (0.65) 1.12	0.89 (0.70) 1.25	0.92 (0.79) 1.27	0.94 (0.81) 1.44	0.94 (0.86) 0.96
running	0.77 (0.47) 0.95	0.78 (0.48) 1.06	0.81 (0.53) 1.07	0.83 (0.55) 1.20	0.91 (0.84) 0.90
DTI	0.87 (0.61) 1.09	0.88 (0.64) 1.15	0.92 (0.75) 1.16	0.93 (0.77) 1.24	0.93 (0.86) 0.99
gait	0.87 (0.64) 1.07	0.88 (0.66) 1.13	0.91 (0.74) 1.21	0.92 (0.75) 1.31	0.94 (0.89) 0.99
ECG 8-lead	0.86 (0.62) 1.29	0.87 (0.64) 1.38	0.90 (0.73) 1.52	0.92 (0.75) 1.64	0.94 (0.88) 0.94
ocean	0.76 (0.36) 1.03	0.78 (0.38) 1.10	0.83 (0.45) 1.11	0.84 (0.47) 1.20	0.93 (0.87) 0.94
weather	0.59 (0.41) 0.94	0.61 (0.44) 1.08	0.65 (0.50) 1.07	0.66 (0.53) 1.23	0.88 (0.82) 0.93
electricity	0.68 (0.44) 0.86	0.70 (0.46) 0.99	0.71 (0.47) 0.96	0.73 (0.50) 1.12	0.93 (0.88) 0.90

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